

Testing treatment effect homogeneity under finite conditional moments

CausalSmith*

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Abstract

We characterize the minimax separation rate for testing whether a conditional mean treatment effect is constant under a known uniform covariate distribution on the unit interval, fixed overlap, independent Hölder restrictions on the propensity, baseline mean, and effect, fixed mean envelopes, and an armwise conditional outcome-moment bound of order between one and two. Separation is measured by the population L_2 distance of the effect from its average. Matching upper and lower bounds identify two rate mechanisms: effect approximation under outcome tails and the interaction of rough propensity and baseline functions. An explicit test combines outcome clipping, histogram projection corrections, independent evaluation blocks, and profiling over the unknown common effect. It controls size over the entire composite null at every stipulated sample size and attains uniform power at the stated separation whenever that separation lies below the model maximum, with public thresholds ensuring this condition. At matched smoothness, every moment order below two yields a polynomially larger critical radius than the bounded-outcome benchmark, whose minimax risk equals that of a signed-binary submodel. Supplying the entire true continuous propensity yields a matching frontier over a broader continuous-nuisance class retaining uniform design, overlap, effect smoothness, mean caps, and the conditional moment envelope. Comparing the two experiments on the original class identifies the exact region where the unknown-propensity experiment attains the supplied-propensity separation order.

1 Introduction

This paper establishes matching minimax separation bounds for conditional mean treatment-effect homogeneity under finite conditional outcome moments and independently specified nuisance smoothness. The statistical target is constancy of the difference between the treated and untreated conditional means, with an unknown common effect. Under a potential-outcome representation satisfying consistency and conditional independence of treatment assignment, this contrast is the conditional average treatment effect. We measure heterogeneity by its population L_2 distance from its average, which equals the standard deviation of the conditional effect under the stipulated covariate distribution. The resulting frontier quantifies how outcome tails and assignment information jointly determine the smallest uniformly detectable departure from homogeneity.

The model has a known uniform covariate distribution on $[0, 1]$, a propensity between $1/4$ and $3/4$, and separate Hölder restrictions on the propensity, baseline conditional mean, and

*Machine-generated by the CausalSmith research pipeline.

mean effect. Write p , the conditional moment order, for an exponent satisfying $1 < p \leq 2$; write α and β , the propensity and baseline smoothness exponents, for numbers in $(0, 1]$; and write γ , the effect smoothness exponent, for a number in $[1/4, 1]$. These exponents are public. The Hölder radii are fixed at 20, the baseline mean and effect have supremum norms at most $1/2$, and each treatment arm has conditional raw p -moment at most 10, uniformly in the covariate up to null sets. These restrictions define the class in [Definition 1](#). Testing errors are assessed uniformly over that class: size covers every admissible constant effect and nuisance function, while power covers every law separated by the stipulated population distance, as specified in [Definition 5](#).

Our main result gives the critical separation radius up to positive constants depending on the public tuple. Define $q = (p - 1)/p$, the moment-derived exponent, and $S = \alpha + \beta$, the sum of nuisance smoothness exponents. The two separation exponents are

$$E_0(p) = \frac{2\gamma q}{2\gamma + q}, \quad E_4(p) = \frac{2S}{1 + 2\alpha + \beta/q + S/(2\gamma)}.$$

For sample size $n \geq 2$, the capped critical radius has order

$$n^{-\min\{E_0(p), E_4(p)\}},$$

as established in [Theorem 4](#). The first exponent balances effect approximation with outcome-moment control. The second incorporates the interaction of propensity and baseline regularity. Their exact equality boundary is determined by

$$F(p) = \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{S}{2\gamma},$$

the phase-comparison quantity: effect smoothness determines the rate when $F(p) \geq 1$, and rough confounding determines it when $F(p) < 1$. The bounds are pure powers of sample size, including at equality, with constants locally uniform on compact subsets of the admissible exponent domain.

The attaining procedure operates directly on the original records. It clips outcomes, forms histogram scores with ordered-pair projection corrections, and takes the inner product of scores from two independent evaluation blocks. Profiling this quadratic statistic over the admissible common-effect interval accommodates the composite null. The public construction in [Algorithm 1](#) specifies all ranks, clipping levels, bias and covariance bounds, and rejection cutoffs. Its rejection probability is at most 0.0075 throughout the null for every $n \geq 2$. Its type-II error is at most 0.0075 at the stated upper separation when that separation lies below the model's maximal heterogeneity distance; fully evaluable sample-size thresholds ensure a nonempty power domain. These guarantees appear in [Theorems 1](#) and [2](#). Matching converses apply to all admissible randomized tests observing the complete records.

The matched outcome comparison retains the same design, overlap, smoothness, and mean envelopes while imposing bounded outcomes. Its separation exponent is the original exponent evaluated at $p = 2$. Every admissible $p < 2$ strictly reduces that exponent, so the finite-moment critical radius exceeds its bounded counterpart by a polynomial factor. At $p = 2$, the ratio remains bounded across sample sizes. Moreover, mean-preserving sign randomization gives exact equality of bounded and signed-binary minimax risks and capped radii, as shown in [Proposition 1](#). The full-record witnesses in [Theorem 3](#) connect the slower finite-moment frontier to growing outcome marks: for each fixed tuple with $p < 2$, close finite-moment null and

alternative mixtures coexist with eventual total-variation separation of every normalized finite prior pair supported on the bounded null and bounded alternatives at the same lower separation.

Supplying the entire true continuous propensity gives a complementary frontier. Over the continuous-carrier class in [Definition 22](#), the propensity and conditional arm means may have unrestricted moduli of continuity, while uniform design, overlap, effect Hölder smoothness, mean caps, and the conditional moment envelope are retained. A clipped inverse-propensity test attains the matching order $n^{-E_0(p)}$ on this broader class, with whole-null calibration and a public nonempty-power threshold, by [Theorem 5](#). Comparing supplied and unknown propensity on the original shared class then yields an exact preservation boundary: their critical-radius ratio remains bounded over $n \geq 2$ precisely when $F(p) \geq 1$. For $F(p) < 1$, supplying the propensity improves separation by a factor of order $n^{E_0(p)-E_4(p)}$. This comparison, including its equality case, follows from the matched frontiers in [Theorems 6 and 7](#).

These results connect econometric homogeneity testing with quadratic nonparametric testing and higher-order causal methods. [Crump et al. \[2008, Section III\]](#) develop sieve Wald tests of conditional mean-effect homogeneity. Recent orthogonal-score procedures provide asymptotic calibration and power under their respective outcome and nuisance-learning conditions [[Lu and Song, 2026](#), Assumptions 1–5, Theorems 3.1–3.3 and 4.1]; [[Lapenta et al., 2026](#), Assumptions 3.3–3.4, Proposition 3.1]. Directional testing also explicitly pursues uniform asymptotic calibration, with power tied to alignment of a learned direction and the population signal [[Dhawan et al., 2026](#), Theorem 1, Corollaries 1–2]. Our contribution concerns finite-sample whole-null calibration and uniform population separation over the primitive class specified above. The approximation–variability balance has antecedents in quadratic goodness-of-fit theory [[Ingster and Sapatinas, 2009](#), Sections 2–4, Theorems 1–2], while projection corrections connect to higher-order conditional-effect estimation [[Kennedy et al., 2024](#), Sections 3–4].

The paper proceeds through related work, setup and assumptions, the matched separation rates and outcome-tail comparison, the calibrated original-record construction, and the supplied-propensity frontiers and assignment-information comparison. A discussion considers interpretation and extensions. The appendices develop causal identification and benchmark equivalence, calibration and attainment, full-record lower bounds, supplied-propensity arguments and comparison algebra, followed by proofs of the main results.

2 Related work

The closest literature tests whether conditional average treatment effects are constant across covariate-defined subpopulations. [Crump et al. \[2008, Section III\]](#) develop a sieve Wald approach that compares series estimates of the two conditional outcome means and removes the constant component when testing homogeneity. Their analysis combines smoothness and sieve-growth restrictions with conditions on conditional variances and third moments of outcome disturbances to obtain asymptotic calibration and consistency against specified smooth alternatives. This establishes an important precedent for treating homogeneity as a global restriction on conditional means. The present paper studies that restriction through finite-sample control over the entire composite null and a population separation criterion that is uniform over the stipulated model class; the corresponding guarantees are stated in [Theorems 1 and 2](#).

Orthogonal conditional moments provide a second route to the same substantive question. [Lu and Song \[2026, Assumptions 1–5, Proposition 2, and Theorems 3.1–3.3 and 4.1\]](#) construct a marked empirical process from a doubly robust score and centered covariate indicators. Their

theory specifies series estimators for the propensity score and outcome regressions, with primitive smoothness and series-growth conditions ensuring a feasible-to-oracle approximation uniform over the process index. The outcome conditions include finite second moments and uniformly bounded conditional variances. The resulting tests have asymptotic Gaussian-process calibration, multiplier-bootstrap validity, consistency against fixed alternatives, and nontrivial power along specified local alternatives approaching the null at the inverse square root of sample size. Uniformity over the process index supports these distributional approximations. Our separation criterion instead takes a supremum over all admissible population laws whose effect function is sufficiently far from its average, as specified in [Definition 5](#).

[Lapenta et al. \[2026\]](#), Assumptions 3.3–3.4, Proposition 3.1, and Section 3.3] integrate orthogonal unconditional moments to obtain an omnibus test that accommodates flexible nuisance estimation. Their formulation separates the controls used for identification from the covariates used to assess heterogeneity. Under unconfoundedness, the stated theory assumes bounded outcomes and nuisance estimators converging faster than the inverse fourth root of sample size in population L_2 , together with boundedness and overlap conditions on the fitted functions. It establishes an asymptotic null distribution, consistency against fixed alternatives, and a bootstrap implementation that retains the fitted nuisance functions across replications. These conditions make explicit the nuisance-learning and outcome envelopes supporting their calibration. Here, the primitive restrictions in [Definition 1](#) determine the public tuning and finite-sample bounds directly.

A closely related scalar target is the variance of the conditional average treatment effect. [Yu \[2026\]](#), Assumptions 3–4, Algorithm 1, and Theorem 1] address its zero boundary using intra-fold sample splitting: distinct evaluation subsamples share the same out-of-fold nuisance estimates, producing cancellation of nuisance-bias terms. The stated asymptotic normality result uses finite fourth outcome moments, a nondegeneracy condition, and convergence faster than the inverse fourth root of sample size for the propensity, outcome-regression, and conditional-effect estimators. Under the uniform covariate marginal in [Assumption 1](#), the squared population L_2 distance of the effect from its average also equals its variance. Our testing risk therefore measures separation using a closely related population quantity, with calibration and power assessed uniformly over the model’s conditional-moment envelope.

Earlier variance-target work studies estimation and confidence intervals rather than minimax separation. [Levy et al. \[2021\]](#), Sections 2.1–2.3] define the variance of the full conditional treatment-effect function and develop TMLE and cross-validated TMLE estimators; their efficiency conditions include control of an outcome-regression remainder that is not removed merely by knowing the treatment mechanism. [Sanchez-Becerra \[2023\]](#), Sections 3–4, Theorems 6–7] instead work in experiments with known assignment probabilities and a predictive first stage. They establish efficiency for the full variance under additional learning conditions and uniform asymptotic coverage for a fold-specific proxy variance under their stated fourth-moment and regularity conditions. Our target is the same full population variance under the uniform design, but the criterion is finite-sample testing risk over Hölder and conditional-moment classes; it does not replace the variance by a fold-specific proxy or assume a learned first stage.

Directional testing offers a complementary way to concentrate power. [Dhawan et al. \[2026\]](#), Section 2.2.1, Assumptions 1–5, Theorem 1, and Corollaries 1–2] learn a promising score direction on one sample and evaluate a debiased score on an independent sample. Their general theory incorporates conditional score-moment bounds, variance conditions, uniform nuisance-estimation requirements, and the additional moment conditions stated in their asymptotic theorem. It de-

livers uniform asymptotic null calibration and a power guarantee requiring sufficient correlation between the learned direction and the population signal. Thus, uniform asymptotic inference is already an explicit objective in this literature. The present analysis adds finite-sample whole-null calibration and uniform separation guarantees indexed by primitive moment and smoothness restrictions. The distinction concerns both the error criterion and the alternatives over which power is guaranteed.

The quadratic structure connects these causal tests to nonparametric goodness-of-fit theory. [Ingster and Sapatinas \[2009, Sections 2–4 and Theorems 1–2\]](#) analyze minimax separation in multivariate regression using increasing-dimensional U-statistics and ellipsoidal smoothness classes. [Comminges and Dalalyan \[2013, Sections 2.3–2.4 and Theorems 1–2\]](#) study composite regression nulls defined by quadratic functionals, obtaining sharp asymptotic testing results for nonnegative diagonal functionals under their stated conditions. These analyses explain how approximation error and the stochastic cost of a quadratic statistic jointly determine detectable signal size. In the present setting, that balance also incorporates unknown treatment assignment and conditional outcome tails. The comparisons in [Theorems 4 and 7](#) characterize their consequences for critical separation radii within the stipulated classes.

The projection corrections also have a close methodological antecedent in higher-order causal estimation. [Kennedy et al. \[2024, Sections 3–4\]](#) derive minimax bounds for pointwise conditional-effect estimation in Hölder models and construct a higher-order local polynomial R-learner, with attainment under the stated additional conditions. Its second-order corrections use projections to control nuisance-product bias, linking the accuracy of causal estimation to separate smoothness restrictions on treatment assignment and outcome means. Our corrected scores in [Definition 21](#) and [Algorithm 1](#) use related projection ideas for a global homogeneity test. The population targets and loss criteria distinguish the two analyses: pointwise estimation measures recovery of the effect function at a covariate value, whereas [Definitions 5 and 6](#) assess rejection errors for constancy against population-separated alternatives.

Outcome-tail handling connects the construction to robust U-statistics. [Joly and Lugosi \[2016, Section 2 and Theorem 3\]](#) use median-of-means aggregation to estimate U-statistic expectations; their canonical-statistic result permits moment orders strictly above one and at most two. This supplies a precise comparator for finite-moment control of nonlinear statistics. Our construction combines clipping, projection corrections, and independent evaluation blocks, with explicit bias and covariance bounds in [Algorithm 1](#). Together with the full-record converse in [Theorem 3](#), the analysis relates the attainable separation scale to the conditional outcome-moment restriction.

Other heterogeneity procedures address distinct scientific targets. [Dukes et al. \[2026, Sections 2–4\]](#) study quantitative and qualitative effect heterogeneity through contrasts tied to the population value of treatment rules, with asymptotic testing guarantees under their regularity conditions. [Jain and Luedtke \[2026, Sections 2–3 and Theorem 3.3\]](#) target equality of conditional potential-outcome distributions and establish asymptotic bootstrap calibration and consistency against fixed alternatives under their stated conditions. [Ashouri and Henderson \[2026, Sections 2–3\]](#) assess the predictive advantage of fitted models that allow heterogeneous effects over models imposing a constant effect. Policy-value contrasts, conditional distributional equality, and predictive performance each give heterogeneity a specific operational meaning. The contribution developed here concerns constancy of the conditional mean contrast under fixed overlap, a known uniform covariate marginal, stipulated smoothness restrictions, and a conditional raw-moment envelope. Within that scope, it combines whole-null finite-sample calibration with

separation-rate comparisons that quantify the roles of outcome tails and supplied propensity information.

3 Setup and assumptions

Throughout, generic positive constants may depend on the fixed public exponents and numerical envelopes, while remaining independent of the sample size n . We write $\|f\|_\infty = \sup_{x \in [0,1]} |f(x)|$ and $\|f\|_2 = (\int f^2 d\lambda)^{1/2}$, where λ is Lebesgue probability measure on $[0, 1]$. The relations $\lesssim$ and $\asymp$ denote inequalities up to such constants and their two-sided counterpart, respectively. Sample indices range over $i = 1, \dots, n$, treatment-arm indices over $a \in \{0, 1\}$, and all sample sizes satisfy $n \in \mathbb{N}$ with $n \geq 2$.

3.1 Records, public exponents, and conditional means

An original record is $O = (X, A, Y)$, with covariate $X \in [0, 1]$, treatment indicator $A \in \{0, 1\}$, and observed outcome $Y \in \mathbb{R}$. An original-record law P is a Borel probability measure on this product space; the restrictions defining the statistical model are collected in [Definition 1](#). Its covariate marginal is denoted by P_X .

The public exponent tuple $v = (p, \alpha, \beta, \gamma)$ consists of the conditional moment order p , the propensity smoothness exponent α , the baseline smoothness exponent β , and the effect smoothness exponent γ . The admissible public domain $\mathcal{V}$ is

$$\mathcal{V} = \{(p, \alpha, \beta, \gamma) \in \mathbb{R}^4 : 1 < p \leq 2, \quad 0 < \alpha \leq 1, \quad 0 < \beta \leq 1, \quad 1/4 \leq \gamma \leq 1\}.$$

For benchmark comparisons, the matched smoothness tuple $w = (\alpha, \beta, \gamma)$ belongs to the domain $\mathcal{W}$, where

$$\mathcal{W} = (0, 1] \times (0, 1] \times [1/4, 1].$$

These exponents are public parameters indexing the classes over which testing errors are controlled.

For $s \in (0, 1]$, the Hölder ball $\mathcal{H}^s(20)$ uses the convention

$$[f]_s = \sup_{\substack{x, z \in [0, 1] \\ x \neq z}} \frac{|f(x) - f(z)|}{|x - z|^s}, \quad \mathcal{H}^s(20) = \{f \in C([0, 1]) : \|f\|_\infty \leq 20, [f]_s \leq 20\},$$

where $C([0, 1])$ denotes continuous real functions with the uniform topology. Thus the convention at $s = 1$ is Lipschitz continuity. We write $[f]_\lambda$ for the equivalence class of a function under Lebesgue almost-everywhere equality; the same convention applies to sums of functions and to each conditional mean.

Let $Q_{a,P}(dy | x)$ be a conditional outcome probability kernel given $X = x$ and $A = a$. The propensity equivalence class $\mathcal{E}_P$ is the class of $x \mapsto P(A = 1 | X = x)$, and the conditional arm-mean classes $\mathcal{G}_{0,P}$ and $\mathcal{G}_{1,P}$ are the classes of

$$x \mapsto \int y Q_{0,P}(dy | x) \quad \text{and} \quad x \mapsto \int y Q_{1,P}(dy | x),$$

respectively. Under the restrictions below, these classes have continuous representatives: the propensity e_P , the baseline mean $m_{0,P}$, and the treated mean $m_{1,P}$, as specified in [Definition 1](#). Their canonical mean contrast is $\tau_P = m_{1,P} - m_{0,P}$. Continuity and the full support of Lebesgue measure make each representative unique, so pointwise restrictions on these functions have an unambiguous meaning.

3.2 Primitive restrictions

The design conditions fix the distribution over which heterogeneity is measured and ensure that both treatment arms are represented throughout the covariate interval.

Assumption 1. `[ass:uniform]` (Uniform covariate distribution) *Let λ denote Lebesgue probability measure on $[0, 1]$. The covariate marginal P_X satisfies*

$$P_X = \lambda.$$

The known uniform random design condition supplies a fixed integration measure for the population target and for approximation errors, following the random-design goodness-of-fit formulation [Ingster and Sapatinas, 2009]. Treatment assignment is further restricted by a fixed overlap envelope.

Assumption 2. `[ass:overlap]` (Fixed propensity overlap) *The continuous propensity representative e_P satisfies*

$$e_P([0, 1]) \subseteq [1/4, 3/4].$$

Uniform overlap keeps both conditional arm distributions observable across the entire interval. The overlap form follows the conditional-effect model in Kennedy et al. [2024, Section 2]; the numerical bounds $1/4$ and $3/4$ are fixed normalizations stipulated for the present class.

Smoothness is indexed separately for treatment assignment, the baseline mean, and the mean contrast. We begin with regularity of the propensity.

Assumption 3. `[ass:propensity-smoothness]` (Propensity Hölder smoothness) *Let $\mathcal{H}^\alpha(20)$ denote the continuous functions on $[0, 1]$ with supremum norm and α -Hölder constant at most 20, and let $[f]_\lambda$ denote the equivalence class of f under almost-everywhere equality for Lebesgue probability measure λ . The propensity equivalence class $\mathcal{E}_P$ satisfies*

$$\mathcal{E}_P \in \{[f]_\lambda : f \in \mathcal{H}^\alpha(20)\}.$$

In our model, independent propensity Hölder smoothness controls variation in assignment probabilities through its own exponent, with radius 20 fixing the scale of that variation. Baseline outcome variation receives a separate regularity restriction.

Assumption 4. `[ass:baseline-smoothness]` (Baseline mean smoothness) *Let $\mathcal{H}^\beta(20)$ denote the continuous functions on $[0, 1]$ with supremum norm and β -Hölder constant at most 20, and let $[g]_\lambda$ denote the equivalence class of g under almost-everywhere equality for Lebesgue probability measure λ . The baseline conditional arm-mean equivalence class $\mathcal{G}_{0,P}$ satisfies*

$$\mathcal{G}_{0,P} \in \{[g]_\lambda : g \in \mathcal{H}^\beta(20)\}.$$

In our model, the independent baseline Hölder condition controls the conditional mean in the untreated arm, allowing its regularity to differ from propensity regularity. The third exponent governs the conditional mean contrast directly.

Assumption 5. `[ass:effect-smoothness]` (Mean effect smoothness) *Let $\mathcal{H}^\gamma(20)$ denote the continuous functions on $[0, 1]$ with supremum norm and γ -Hölder constant at most 20, and let $[f]_\lambda$ denote the equivalence class of f under almost-everywhere equality for Lebesgue probability measure λ . With $m_{0,P}$ the continuous baseline arm-mean representative, the treated conditional arm-mean equivalence class $\mathcal{G}_{1,P}$ satisfies*

$$\mathcal{G}_{1,P} \in \{[m_{0,P} + t]_\lambda : t \in \mathcal{H}^\gamma(20)\}.$$

Independent effect Hölder smoothness describes the treated mean as the baseline mean plus a regular contrast. This separates the target regularity from the two nuisance regularities. By comparison, the pointwise conditional-effect model of [Kennedy et al. \[2024, Sections 2–3\]](#) summarizes propensity and baseline smoothness through an average exponent. The Hölder radius 20 used here is a fixed normalization of the present model.

We also fix the scales of the baseline mean and the mean effect. The first envelope bounds the baseline level.

Assumption 6. `[ass:baseline-cap]` (Baseline mean bound) *The continuous baseline arm-mean representative $m_{0,P}$ satisfies*

$$\|m_{0,P}\|_\infty \leq 1/2.$$

The bounded conditional baseline mean condition uses the fixed normalization 1/2, placing baseline levels on a common scale across the indexed models. This numerical envelope is stipulated here. A corresponding envelope fixes the scale of the contrast.

Assumption 7. `[ass:effect-cap]` (Uniform mean effect bound) *The mean effect τ_P , the canonical difference between the treated and baseline conditional means, satisfies*

$$\|\tau_P\|_\infty \leq \frac{1}{2}.$$

The bounded conditional effect condition uses a second stipulated normalization 1/2, keeping the target on the same scale across smoothness and moment classes. Together, the two mean envelopes imply $\|m_{1,P}\|_\infty \leq 1$.

Outcome tails are controlled separately from conditional mean levels. The following restriction bounds a raw conditional moment in each arm.

Assumption 8. `[ass:raw-moment]` (Uniform conditional moment bound) *Let λ denote Lebesgue probability measure on $[0, 1]$, and let $Q_{a,P}(dy \mid x)$ denote the conditional outcome probability kernel in treatment arm a . The conditional raw moment of order p satisfies*

$$\max_{a \in \{0,1\}} \operatorname{ess\,sup}_{x \sim \lambda} \int |y|^p Q_{a,P}(dy \mid x) \leq 10.$$

This conditional raw-moment envelope is specific to the present analysis. It controls outcome magnitudes uniformly across covariate values, up to null sets, and across treatment arms. The range $1 < p \leq 2$ ensures integrability of the arm means while allowing the moment order to index the strength of tail control.

The original-record model $\mathcal{M}_v$ in [Definition 1](#) collects these design, smoothness, mean-envelope, and moment restrictions.

Definition 1. `[def:model]` (Original-record model $\mathcal{M}_v$) *The model $\mathcal{M}_v$ is the set of original-record probability laws P satisfying the following restrictions, indexed by $v = (p, \alpha, \beta, \gamma) \in \mathcal{V}$, where $\mathcal{V}$ denotes the admissible domain of public moment and primitive smoothness tuples:*

- (Covariate law.) [Assumption 1](#).
- (Overlap.) [Assumption 2](#).
- (Propensity smoothness.) [Assumption 3](#).

- (*Baseline smoothness.*) [Assumption 4](#).
- (*Effect smoothness.*) [Assumption 5](#).
- (*Baseline bound.*) [Assumption 6](#).
- (*Effect bound.*) [Assumption 7](#).
- (*Conditional moment bound.*) [Assumption 8](#).

Here P_X denotes the covariate marginal of P . For a function f , write $[f]_\lambda$ for its equivalence class under Lebesgue almost-everywhere equality. Let $\mathcal{E}_P$ denote the propensity equivalence class, and let $\mathcal{G}_{0,P}$ and $\mathcal{G}_{1,P}$ denote the baseline and treated conditional arm-mean equivalence classes. Their continuous representatives are defined by

$$\mathcal{E}_P = [e_P]_\lambda, \quad \mathcal{G}_{0,P} = [m_{0,P}]_\lambda, \quad \mathcal{G}_{1,P} = [m_{1,P}]_\lambda, \quad e_P, m_{0,P}, m_{1,P} \in C([0, 1]),$$

where $C([0, 1])$ is the space of continuous real functions on $[0, 1]$ with the uniform topology. The canonical mean effect is

$$\tau_P = m_{1,P} - m_{0,P}.$$

3.3 Causal interpretation and the homogeneity target

A potential-outcome representation connects the canonical mean contrast to a conditional average treatment effect. Let $Y(0)$ and $Y(1)$ denote the baseline and treated potential-outcome coordinates. The completion $\mathbb{Q}_P$ in [Definition 3](#) combines the observed arm distributions with assignment that is conditionally independent of these coordinates given the covariate.

Assumption 9. [ass:null-constancy] (Constant mean effect restriction) *There exists $c \in [-1/2, 1/2]$ such that*

$$\tau_P(x) = c \quad \text{for every } x \in [0, 1].$$

Under this completion, consistency and conditional independence of assignment give

$$\mathbb{E}_{\mathbb{Q}_P}[Y(1) - Y(0) \mid X = x] = \tau_P(x) \quad \text{for } \lambda\text{-almost every } x.$$

The observed-record marginal is P , so the same conditional mean contrast admits the potential-outcome interpretation associated with consistency and unconfounded assignment [[Rubin, 1974](#), Section 3]; [[Rosenbaum and Rubin, 1983](#), Section 2]. The auxiliary justification for this completion is given in [Lemma 1](#).

To measure heterogeneity, define the average contrast $\bar{\tau}_P$ and its population distance from constancy $d(P)$ by

$$\bar{\tau}_P = \int_{[0,1]} \tau_P(x) d\lambda(x), \quad d(P) = \left(\int_{[0,1]} \{\tau_P(x) - \bar{\tau}_P\}^2 d\lambda(x) \right)^{1/2}.$$

Equivalently, $d(P) = \inf_{b \in \mathbb{R}} \|\tau_P - b\|_2$. The uniform design fixes the population weighting, and continuity makes zero distance equivalent to constancy on the entire interval. Homogeneity therefore permits an unknown common effect c in the public mean envelope.

Definition 2. [def:null] (Constant-effect null $H_0(v)$)

$$H_0(v) = \{P \in \mathcal{M}_v : \exists c \in [-1/2, 1/2] \forall x \in [0, 1], \tau_P(x) = c\}.$$

The model restrictions and null constancy condition are given in [Assumptions 1 to 9](#).

The unknown-constant conditional-average-effect homogeneity condition treats the common effect level as a nuisance parameter, matching the mean-homogeneity target of [[Crump et al., 2008](#)]. The constant-effect null $H_0(v)$ in [Definition 2](#) imposes this condition within the original-record model.

Definition 3. [def:causal-completion] (Potential-outcome completion $\mathbb{Q}_P$) The potential-outcome completion $\mathbb{Q}_P$ is defined using Lebesgue probability measure λ on $[0, 1]$ and the conditional outcome probability kernels $Q_{a,P}(dy | x)$ in arms $a \in \{0, 1\}$ by

$$\mathbb{Q}_P(dx, dy_0, dy_1, da) = \lambda(dx) Q_{0,P}(dy_0 | x) Q_{1,P}(dy_1 | x) e_P(x)^a (1 - e_P(x))^{1-a}.$$

Writing $Y(0)$ and $Y(1)$ for the baseline and treated potential-outcome coordinates, respectively, the observed outcome satisfies

$$Y = (1 - A)Y(0) + AY(1).$$

Changes to the conditional outcome kernels on λ -null sets leave $\mathbb{Q}_P$ unchanged.

Thus the null consists precisely of model laws with $d(P) = 0$. Under the causal completion, its target is constancy of the conditional average treatment effect.

3.4 The original-record testing experiment

Testing uses an independent sample $\mathcal{D}_n = ((X_i, A_i, Y_i))_{i=1}^n$ from P , together with an independent public uniform variable U on $[0, 1]$. An original-record randomized test is a jointly Borel rejection-probability map ϕ with domain and range

$$\phi : ([0, 1] \times \{0, 1\} \times \mathbb{R})^n \times [0, 1] \longrightarrow [0, 1].$$

Its expected rejection probability is denoted by $R_n(P, \phi)$, the expectation over both the sample and the public randomization. The experiment and this expectation are specified together in [Definition 4](#).

Definition 4. [def:original-record-experiment] (Original-record experiment $S_{n,P}$) The original-record experiment is the probability law

$$S_{n,P} = P^{\otimes n} \otimes \lambda$$

on

$$([0, 1] \times \{0, 1\} \times \mathbb{R})^n \times [0, 1],$$

where P is the original-record law and λ is Lebesgue probability measure on $[0, 1]$. Each original record has coordinates $X \in [0, 1]$, $A \in \{0, 1\}$, and $Y \in \mathbb{R}$, representing the covariate, treatment, and observed outcome. Write

$$\mathcal{D}_n = ((X_i, A_i, Y_i))_{i=1}^n$$

for the independent sample and U for the independent public uniform randomization variable. For a randomized rejection-probability map ϕ , its rejection expectation is

$$R_n(P, \phi) = \int \phi(\mathcal{D}_n, u) S_{n,P}(d\mathcal{D}_n, du).$$

Tests may depend on the public tuple and sample size. Their size is controlled uniformly over all laws in the constant-effect null, including every admissible common effect level and nuisance function.

Let D_v denote the original-model distance supremum,

$$D_v = \sup_{P \in \mathcal{M}_v} d(P),$$

and let r be a separation satisfying $0 < r < D_v$. The minimax type-II error $B_n(v, r)$ in [Definition 5](#) compares tests of size at most $1/10$ against every model law whose population heterogeneity distance is at least r .

Definition 5. `[def:testing-risk]` (Minimax testing risk $B_n(v, r)$)

$$B_n(v, r) = \inf_{\phi: \sup_{P \in H_0(v)} \mathbf{R}_n(P, \phi) \leq 1/10} \sup_{P \in \mathcal{M}_v: d(P) \geq r} \{1 - \mathbf{R}_n(P, \phi)\}.$$

The parameters satisfy:

- **(Sample size.)** $n \geq 2$.
- **(Public tuple.)** $v \in \mathcal{V}$, the admissible domain of moment and primitive smoothness tuples.
- **(Separation.)** $0 < r < D_v$, where D_v is the supremum of $d(P)$ over $\mathcal{M}_v$.

Here $\mathcal{M}_v$ is the original-record model, $H_0(v)$ consists of its laws with an unknown constant mean effect, and $d(P)$ is the population Lebesgue L_2 distance of the effect from its average. The infimum ranges over randomized rejection-probability maps ϕ , and $\mathbf{R}_n(P, \phi)$ is their expected rejection probability in the original-record experiment.

The size constraint and the alternative supremum give the risk a uniform interpretation over the stipulated model. To summarize the separation at which this risk reaches $1/10$, we use the capped critical radius $r_n^*(v)$ in [Definition 6](#).

Definition 6. `[def:critical-radius]` (Capped critical radius $r_n^*(v)$)

$$r_n^*(v) = \inf \left(\{r \in (0, D_v) : B_n(v, r) \leq 1/10\} \cup \{D_v\} \right).$$

Here $B_n(v, r)$ is the original-model minimax type-II error under the size constraint $1/10$, and D_v is the supremum of the population effect distance over the original model.

Including the model supremum makes the defining set nonempty for every admissible sample size and public tuple. When the risk threshold is achieved at an interior separation, the radius records the infimum of such separations; otherwise, the cap supplies the value D_v .

3.5 Matched bounded and signed-binary benchmarks

The benchmark family retains the same design, overlap, smoothness exponents, and mean envelopes while imposing public outcome support restrictions. Its bounded member uses the following unit envelope.

Assumption 10. [ass:bounded-outcome] (Almost sure outcome bound) *The observed outcome Y satisfies*

$$P(|Y| \leq 1) = 1.$$

The bounded-outcome condition places every observed outcome within a public unit envelope, a fixed-envelope version of bounded-outcome regularity used in homogeneity testing [Lapenta et al., 2026]. The bounded model $\mathcal{M}_w^b$ in Definition 7 combines this support restriction with the primitive restrictions at moment order two.

Definition 7. [def:bounded-model] (Bounded-outcome model $\mathcal{M}_w^b$) *The bounded-outcome model $\mathcal{M}_w^b$ is defined for the matched smoothness tuple $w = (\alpha, \beta, \gamma)$ in $\mathcal{W}$, the admissible domain of matched smoothness tuples, by*

$$\mathcal{M}_w^b = \{P \in \mathcal{M}_{(2,\alpha,\beta,\gamma)} : P(|Y| \leq 1) = 1\}, \quad w = (\alpha, \beta, \gamma) \in \mathcal{W}.$$

Its restrictions are given in Assumptions 1 to 8 and 10.

The signed-binary member specializes the support further.

Assumption 11. [ass:binary-outcome] (Signed binary outcome support) *The observed outcome Y satisfies*

$$P(Y \in \{-1, 1\}) = 1.$$

Signed-binary support is the affine recoding $Y = 2B - 1$ of a conventional binary outcome $B \in \{0, 1\}$, the outcome convention used in the binary conditional-effect model of Kennedy et al. [2024, Section 2]. Either support restriction implies a conditional second raw moment bounded by one almost everywhere, and hence satisfies the moment envelope at $p = 2$. The signed-binary model $\mathcal{M}_w^{\text{bin}}$ in Definition 8 is consequently indexed by the matched smoothness tuple.

Definition 8. [def:binary-model] (Signed-binary model $\mathcal{M}_w^{\text{bin}}$) *The signed-binary model $\mathcal{M}_w^{\text{bin}}$ is defined for the matched smoothness tuple $w = (\alpha, \beta, \gamma)$ in $\mathcal{W}$, the admissible domain of matched smoothness tuples, by*

$$\mathcal{M}_w^{\text{bin}} = \{P \in \mathcal{M}_{(2,\alpha,\beta,\gamma)} : P(Y \in \{-1, 1\}) = 1\}, \quad w = (\alpha, \beta, \gamma) \in \mathcal{W}.$$

Its restrictions are given in Assumptions 1 to 8 and 11.

These definitions give

$$\mathcal{M}_w^{\text{bin}} \subseteq \mathcal{M}_w^b \subseteq \mathcal{M}_{(2,\alpha,\beta,\gamma)}.$$

Both benchmarks preserve the unknown common effect as a nuisance parameter. Their nulls, denoted by $H_0^b(w)$ and $H_0^{\text{bin}}(w)$, are defined in Definitions 9 and 10.

Definition 9. [def:bounded-null] (Bounded constant-effect null $H_0^b(w)$)

$$H_0^b(w) = \left\{P \in \mathcal{M}_w^b : \exists c \in [-1/2, 1/2] \forall x \in [0, 1], \tau_P(x) = c\right\}.$$

The model restrictions and null constancy condition are given in Assumptions 1 to 10.

Definition 10. [def:binary-null] (Signed-binary constant-effect null $H_0^{\text{bin}}(w)$)

$$H_0^{\text{bin}}(w) = \left\{P \in \mathcal{M}_w^{\text{bin}} : \exists c \in [-1/2, 1/2] \forall x \in [0, 1], \tau_P(x) = c\right\}.$$

The model restrictions and null constancy condition are given in Assumptions 1 to 9 and 11.

For their separation domains, define the bounded and signed-binary distance suprema D_w^b and D_w^{bin} by

$$D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P), \quad D_w^{\text{bin}} = \sup_{P \in \mathcal{M}_w^{\text{bin}}} d(P).$$

The corresponding minimax type-II errors $B_n^b(w, r)$ and $B_n^{\text{bin}}(w, r)$, defined in [Definitions 11](#) and [12](#), use the same original-record experiment, population distance, and size threshold.

Definition 11. `[def:bounded-risk]` (Bounded-model testing risk) *The bounded-model minimax type-II error $B_n^b(w, r)$ is defined by*

$$B_n^b(w, r) = \inf_{\phi: \sup_{P \in H_0^b(w)} R_n(P, \phi) \leq 1/10} \sup_{P \in \mathcal{M}_w^b: d(P) \geq r} \{1 - R_n(P, \phi)\},$$

where the tests ϕ , rejection probabilities $R_n(P, \phi)$, and heterogeneity distances $d(P)$ are as in [Definition 5](#), and the bounded model and null are those of [Definitions 7](#) and [9](#). The parameter domain is:

- **(Sample size.)** $n \in \mathbb{N}$ with $n \geq 2$.
- **(Smoothness.)** $w = (\alpha, \beta, \gamma) \in \mathcal{W}$, where the admissible smoothness domain is

$$\mathcal{W} = \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : 0 < \alpha \leq 1, \quad 0 < \beta \leq 1, \quad 1/4 \leq \gamma \leq 1\}.$$

- **(Separation.)** $0 < r < D_w^b$, where the bounded-model distance supremum is

$$D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P).$$

Definition 12. `[def:binary-risk]` (Signed-binary minimax testing risk) *The signed-binary minimax type-II error $B_n^{\text{bin}}(w, r)$ is defined by*

$$B_n^{\text{bin}}(w, r) = \inf_{\phi: \sup_{P \in H_0^{\text{bin}}(w)} R_n(P, \phi) \leq 1/10} \sup_{P \in \mathcal{M}_w^{\text{bin}}: d(P) \geq r} \{1 - R_n(P, \phi)\},$$

where the tests and their expected rejection probabilities $R_n(P, \phi)$ follow [Definition 5](#), and the signed-binary model and null are specified in [Definitions 8](#) and [10](#). Here λ denotes Lebesgue probability measure on $[0, 1]$, and the population heterogeneity distance is

$$d(P) = \left(\int_{[0,1]} \left\{ \tau_P(x) - \int_{[0,1]} \tau_P(z) d\lambda(z) \right\}^2 d\lambda(x) \right)^{1/2}.$$

The public parameter domain is:

- **(Sample size.)** $n \in \mathbb{N}$ with $n \geq 2$.
- **(Smoothness.)** $w = (\alpha, \beta, \gamma) \in \mathcal{W}$, where

$$\mathcal{W} = (0, 1] \times (0, 1] \times [1/4, 1].$$

- **(Separation.)** $0 < r < D_w^{\text{bin}}$, where the supremum of legal signed-binary heterogeneity distances is

$$D_w^{\text{bin}} = \sup_{P \in \mathcal{M}_w^{\text{bin}}} d(P).$$

Finally, the benchmark critical radii $r_n^{*,\text{b}}(w)$ and $r_n^{*,\text{bin}}(w)$ in [Definitions 13](#) and [14](#) apply the same capping convention to the respective risks.

Definition 13. `[def:bounded-radius]` (Bounded-model critical separation radius) *The capped bounded-model critical separation radius $r_n^{*,\text{b}}(w)$ is defined for $n \in \mathbb{N}$ and $w = (\alpha, \beta, \gamma) \in \mathbb{R}^3$. Write $d(P)$ for the population Lebesgue L_2 distance of the mean effect under P from its average, and define the bounded-model cap by*

$$D_w^{\text{b}} = \sup_{P \in \mathcal{M}_w^{\text{b}}} d(P),$$

where $\mathcal{M}_w^{\text{b}}$ is the bounded model in [Definition 7](#). With $B_n^{\text{b}}(w, r)$, the bounded-model testing risk defined in [Definition 11](#), set

$$r_n^{*,\text{b}}(w) = \inf \left(\left\{ r \in (0, D_w^{\text{b}}) : B_n^{\text{b}}(w, r) \leq \frac{1}{10} \right\} \cup \{D_w^{\text{b}}\} \right).$$

Definition 14. `[def:binary-radius]` (Capped signed-binary critical radius) *The capped signed-binary critical separation radius $r_n^{*,\text{bin}}(w)$, for sample size $n \in \mathbb{N}$ and primitive smoothness tuple $w = (\alpha, \beta, \gamma) \in \mathbb{R}^3$, is*

$$r_n^{*,\text{bin}}(w) = \inf \left(\left\{ r \in (0, D_w^{\text{bin}}) : B_n^{\text{bin}}(w, r) \leq \frac{1}{10} \right\} \cup \{D_w^{\text{bin}}\} \right),$$

where D_w^{bin} is the supremum of the population heterogeneity distances over the signed-binary model in [Definition 8](#), and $B_n^{\text{bin}}(w, r)$ is the signed-binary testing risk in [Definition 12](#).

Throughout the benchmark comparisons, these quantities are evaluated at $n \geq 2$ and $w \in \mathcal{W}$. Holding the primitive smoothness tuple fixed aligns the population target and error thresholds across the original, bounded, and signed-binary families, providing a common basis for assessing the role of outcome tails.

4 Separation rates and the cost of outcome tails

Holding the primitive smoothness tuple fixed gives a direct comparison of the precision attainable under the finite conditional moment restriction and the bounded-outcome benchmark. The comparison concerns the population heterogeneity distance and the capped critical radii defined in [Definitions 6](#) and [13](#). Two exponents determine the original-record separation rate: one depends on effect smoothness and moment order, while the other also reflects the smoothness of the propensity and baseline mean.

We collect the scales and their finite-sample multipliers before stating the comparison. Write C_v^{att} and N_v^{att} for the fully evaluable attainment multiplier and nonempty-power threshold specified in [Theorem 2](#); evaluation at $(2, w)$ gives their matched bounded counterparts. The following definition combines these attainment quantities with the lower-bound multipliers.

Definition 15. [def:sharp-frontier-scales] (Separation scales $\rho_n(v)$ and $\rho_n^b(w)$) *The separation scales and lower multipliers are*

$$\rho_n(v) = n^{-E(p)}, \quad \rho_n^b(w) = n^{-E(2)}, \quad c_w^b = c_{(2,w)},$$

$$c_v = \begin{cases} c_0^e, & F(p) \geq 1, \\ \min\{k_v, c_0^e N_{\text{low}}(v)^{-(E_0(p)-E_4(p))}\}, & F(p) < 1. \end{cases}$$

For an admissible public tuple $v = (p, \alpha, \beta, \gamma) \in \mathcal{V}$ and its matched smoothness tuple $w = (\alpha, \beta, \gamma)$, the quantities in these formulas are defined by

$$\begin{aligned} q &= \frac{p-1}{p}, \quad S = \alpha + \beta, \quad D_p = 1 + 2\alpha + \frac{\beta}{q} + \frac{S}{2\gamma}, \\ E_0(p) &= \frac{2\gamma q}{2\gamma + q}, \quad E_4(p) = \frac{2S}{D_p}, \quad F(p) = \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{S}{2\gamma}, \quad E(p) = \min\{E_0(p), E_4(p)\}, \\ H_{\text{fine}} &= 2^{40}, \quad N_{\text{low}}(v) = \left\lceil \max\left\{2, 32^{D_p \gamma / (2S)}\right\} \right\rceil, \\ k_v &= 2^{-17} (2H_{\text{fine}})^{-S}, \quad c_0^e = \frac{4^{-\gamma}}{16\sqrt{3}}. \end{aligned}$$

The arguments p in E_0, E_4, F, E vary with (α, β, γ) fixed; $c_{(2,w)}$ is c_v evaluated at $v = (2, \alpha, \beta, \gamma)$.

The upper multipliers and sample-size thresholds are the fully evaluable attainment quantities:

$$C_v = C_v^{\text{att}}, \quad N_v = N_v^{\text{att}}, \quad C_w^b = C_{(2,w)}^{\text{att}}, \quad N_w^b = N_{(2,w)}^{\text{att}}.$$

The test $\phi_{n,v}^*$ is the total original-record test specified by the explicit score construction, and its bounded-model counterpart is

$$\phi_{n,w}^{*,b} = \phi_{n,(2,w)}^*.$$

The original-model scale $\rho_n(v)$ and the matched bounded scale $\rho_n^b(w)$ summarize the sample-size exponents. Their lower multipliers, c_v and c_w^b , and upper multipliers, C_v and C_w^b , retain the finite-sample content of the comparison. The associated public thresholds N_v and N_w^b place the upper separations below a fixed attainable heterogeneity distance. Thus the power statements below are attached to an explicit nonempty separation domain.

To measure the cost of outcome tails at the same smoothness tuple, we use the ratio of the capped radii.

Definition 16. [def:comparison-handle] (Capped-radius ratio $\Delta_n(v)$)

$$\Delta_n(v) = \frac{r_n^*(v)}{r_n^{*,b}(w)}, \quad w = (\alpha, \beta, \gamma).$$

The numerator and denominator are the capped critical separation radii for the original and bounded original-record experiments, respectively, at the same sample size n and matched smoothness tuple w .

The ratio $\Delta_n(v)$ in Definition 16 compares experiments with the same mean-effect target and error thresholds. We distinguish persistent enlargement of the original-model radius from a comparison that remains bounded over all sample sizes.

Definition 17. [def:tail-region] (Tail region $\mathcal{T}$) The tail region $\mathcal{T}$ consists of admissible public tuples $v = (p, \alpha, \beta, \gamma)$, with moment exponent p and primitive smoothness exponents α, β, γ , for which

$$\mathcal{T} = \left\{ v \in \mathcal{V} : p < 2, \lim_{n \rightarrow \infty} \Delta_n(v) = \infty \right\},$$

where $\mathcal{V}$ is the admissible tuple domain and $\Delta_n(v)$ is the original-to-bounded capped-radius ratio at matched smoothness.

Definition 18. [def:equality-region] (Equality region $\mathcal{E}$) The equality region $\mathcal{E}$ is defined using $\mathcal{V}$, the admissible domain of public moment and smoothness tuples, by

$$\mathcal{E} = \left\{ v \in \mathcal{V} : \sup_{n \geq 2} \Delta_n(v) < \infty \right\},$$

where $\Delta_n(v)$ is the original-to-bounded capped-radius ratio at matched smoothness.

The tail region $\mathcal{T}$ and equality region $\mathcal{E}$ describe the behavior of this matched comparison. The equality region concerns bounded radius ratios; its definition permits different finite-sample constants.

Before comparing rates, the bounded benchmark can be identified exactly with its signed-binary submodel. Let λ denote Lebesgue probability measure on $[0, 1]$, and let $d_0 = 1/(8\sqrt{2})$ be the fixed positive witness distance supplied by [Lemma 1](#). The next result shows that both benchmarks have the same attainable heterogeneity distances and the same minimax testing risks.

Proposition 1. [prop:bounded-submodel-comparison] (Bounded and binary submodel comparison) Let $v = (p, \alpha, \beta, \gamma)$ be an admissible parameter tuple as in [Definition 1](#), and let $w = (\alpha, \beta, \gamma)$. Use the model classes in [Definitions 1, 7 and 8](#). Let λ denote Lebesgue probability measure on $[0, 1]$, and define the heterogeneity distance, its model suprema, and the witness distance by

$$d(P) = \left[\int_{[0,1]} \left(\tau_P(x) - \int_{[0,1]} \tau_P(z) \lambda(dz) \right)^2 \lambda(dx) \right]^{1/2},$$

$$D_w^{\text{bin}} = \sup_{P \in \mathcal{M}_w^{\text{bin}}} d(P), \quad D_w^{\text{b}} = \sup_{P \in \mathcal{M}_w^{\text{b}}} d(P), \quad D_v = \sup_{P \in \mathcal{M}_v} d(P), \quad d_0 = \frac{1}{8\sqrt{2}}.$$

Then

$$\mathcal{M}_w^{\text{bin}} \subseteq \mathcal{M}_w^{\text{b}} \subseteq \mathcal{M}_v, \quad D_w^{\text{bin}} = D_w^{\text{b}} = D_v \geq d_0.$$

For every integer $n \geq 2$ and every real r satisfying $0 < r < D_v$, the testing risks of [Definitions 5, 11 and 12](#) satisfy

$$B_n^{\text{bin}}(w, r) = B_n^{\text{b}}(w, r) \leq B_n(v, r).$$

For every integer $n \geq 2$, the capped critical radii of [Definitions 6, 13 and 14](#) satisfy

$$0 < r_n^{*, \text{bin}}(w) = r_n^{*, \text{b}}(w) \leq r_n^*(v).$$

For every $P \in \mathcal{M}_{(2, \alpha, \beta, \gamma)}$, its signed-binary conversion belongs to $\mathcal{M}_w^{\text{bin}}$. This conversion retains the distribution of (X, A) and uses conditional signed-binary outcomes whose probabilities of +1 in arms 0 and 1, respectively, are

$$\frac{1 + m_{0,P}(x)}{2}, \quad \frac{1 + m_{1,P}(x)}{2},$$

with the complementary probabilities assigned to -1 . It preserves the propensity e_P , the baseline mean $m_{0,P}$, the effect τ_P , and the heterogeneity distance $d(P)$. The converted law satisfies the null constancy condition of [Assumption 9](#) if and only if the original law does.

For every $n \in \mathbb{N}$, every test ϕ in the sense of [Definition 5](#), and every $P \in \mathcal{M}_w^b$, independently convert each original record by retaining (X, A) and drawing an outcome sign with conditional probabilities

$$\Pr(+1 \mid X, A, Y) = \frac{1+Y}{2}, \quad \Pr(-1 \mid X, A, Y) = \frac{1-Y}{2}.$$

Integrating this conversion randomization into ϕ gives a rejection-probability map on the original experiment. Its expected rejection probability under P equals the expected rejection probability of ϕ under the signed-binary conversion of P .

The sign randomization in [Proposition 1](#) preserves the conditional outcome means, and hence the effect function, its heterogeneity distance, and null membership. Integrating the randomization into a test transfers its rejection probabilities to the bounded experiment. The resulting equality of minimax risks holds throughout the common nonempty separation domain $0 < r < D_v$, and equality of capped radii holds for every $n \geq 2$. This gives an exact decision-theoretic benchmark for the mean-effect target.

The lower-bound comparison uses finite mixtures of original-record laws. Their public tuning selects a copula-frame construction in the rough-confounding regime once its sample-size threshold is reached, and an opposite paired-tent construction otherwise. The following selection rule fixes the priors used in the main result; their support laws and complete-record comparison bounds are developed in [Lemmas 5 to 8](#) and [Algorithm 5](#).

The selected null and alternative priors $\pi_{0,n,v}$ and $\pi_{1,n,v}$ in [Algorithm 6](#) induce the complete independent-record mixtures $\mathbb{P}_{0,n,v}$ and $\mathbb{P}_{1,n,v}$. Their rare-mark probability $\varepsilon_{n,v}$ and outcome magnitude $L_{n,v}$ describe how the lower witnesses use the finite moment allowance.

For the next statement, $R_n(P, \phi)$ denotes expected rejection in the original-record experiment, $Q_{a,P}(dy \mid x)$ denotes the conditional outcome probability kernel in arm a , and TV denotes the supremum of the absolute difference in event probabilities. The threshold $N(v)$ identifies the sample sizes at which every bounded-model finite prior pair at the original-model lower separation has mixture total variation at least one half. The theorem combines the matched separation bounds, the power guarantees, and this tail-essential comparison.

The exponent comparison in [Theorem 4](#) explains the two statistical regimes. When $F(p) \geq 1$, the effect-smoothness exponent $E_0(p)$ determines the separation scale: moment order and effect smoothness govern its sample-size dependence. When $F(p) < 1$, the rough-confounding exponent $E_4(p)$ determines the scale, bringing the propensity and baseline smoothness exponents into the rate. The identity for $E_4(p) - E_0(p)$ makes $F(p) = 1$ their exact equality boundary. Across both regimes, the strict inequality $E(p) < E(2)$ for $p < 2$ translates the finite moment allowance into a polynomial enlargement of the matched critical radius.

The two denominators reflect different balances. At histogram rank J , approximating the effect contributes the usual Hölder term of order $J^{-\gamma}$; clipping at level T contributes a moment-tail term of order T^{1-p} , while the clipped quadratic variability grows with T^{2-p} . Balancing these terms gives $E_0(p)$. When the nuisance functions are rough, the ordered-pair correction removes the first-order propensity error, leaving a product of the fine-scale propensity and baseline approximation errors. Its rank cost depends on $2\alpha + \beta/q$: propensity regularity enters twice through treatment weighting and projection correction, whereas the baseline appears once and is amplified by the tail factor $1/q$. The additional term $S/(2\gamma)$ records the coarse resolution

needed to retain the effect signal. Correction-specific clipping prevents the fine increments from paying the main cutoff at every resolution, which is what makes the multiresolution covariance sum compatible with $E_4(p)$. These statements summarize the verified mean and covariance bounds in [Lemmas 2 and 3](#); their full calculations are in the attainment appendix.

For example, $v = (3/2, 1, 1, 1)$ has $F(p) = 8$, so the effect channel governs with $E = E_0 = 2/7$. In contrast, $v = (3/2, 1/20, 1/20, 1)$ has $F(p) = 2/5$, so rough confounding governs with $E = E_4 = 2/13$. The first tuple uses smooth nuisance functions and the second deliberately makes them much rougher than the effect. These statistical regimes are distinct from the procedure's representation choice in [Algorithm 1](#), which uses the conditions $S \geq \gamma$ or $\alpha \geq q$ to choose between a single correction and a multiresolution expansion.

The regimes can be summarized by substituting the formulas in [Definition 15](#). In the following table, $q = (p - 1)/p$ and $S = \alpha + \beta$, with the primitive smoothness tuple held fixed.

Model and regime	Condition	Separation exponent
Original, effect smoothness	$F(p) > 1$	$E_0(p) = \frac{2\gamma q}{2\gamma + q}$
Original, equality boundary	$F(p) = 1$	$E_0(p) = E_4(p) = \frac{2\gamma q}{2\gamma + q}$
Original, rough confounding	$F(p) < 1$	$E_4(p) = \frac{2S}{1 + 2\alpha + \beta/q + S/(2\gamma)}$
Bounded, effect smoothness or boundary	$F(2) \geq 1$	$E_0(2) = \frac{2\gamma}{4\gamma + 1}$
Bounded, rough confounding	$F(2) < 1$	$E_4(2) = \frac{2S}{1 + 2S + S/(2\gamma)}$

Here $F(2) = 2S + S/(2\gamma)$. Each row describes the power of n in the corresponding scale; the finite-sample radius bounds retain the multipliers and capping convention of [Theorem 4](#). The original and bounded exponents are selected separately at their respective moment orders.

For every fixed admissible tuple, the two-sided bounds on $\Delta_n(v)$ quantify the cost of tails through $n^{E(2)-E(p)}$. The moment-order-two face has equal original and bounded exponents and a uniformly bounded ratio over $n \geq 2$, together with the ordering $\Delta_n(v) \geq 1$. Every admissible tuple with $p < 2$ instead belongs to the tail region, where the ratio diverges. This distinction between moment orders is separate from the equality boundary $F(p) = 1$, which identifies the meeting of the two rate mechanisms within a given moment class.

The finite-sample cap remains part of the interpretation. The radius inequalities hold for every $n \geq 2$, while the power guarantees for the attaining tests $\phi_{n,v}^*$ and $\phi_{n,w}^{*,b}$ from [Definition 15](#) apply when their upper separations lie below the corresponding model distance suprema. The public thresholds ensure this condition through the fixed witness distance d_0 . Size is controlled over the whole null at every $n \geq 2$. On compact subsets of the admissible exponent domains, common lower and upper multipliers and a common bound on the public thresholds make these comparisons locally uniform.

Finally, the tail-essential witnesses identify the statistical content of the enlargement. For $p < 2$, their outcome magnitudes grow with sample size, while the selected null and separated alternative mixtures remain within total variation one half in the complete-record experiment. At the same original-model lower separation, every bounded-model finite prior pair has mixture total variation at least one half once the tuple-dependent threshold is reached. Thus the witnesses connect the slower finite-moment rate to outcome tails while preserving the stipulated primitive restrictions and the mean-effect target. The full support-law constructions and the quantified

finite-prior comparison appear in the proof appendix through [Definitions 29](#) and [30](#), [Algorithm 5](#), and [Theorem 3](#).

5 A calibrated test from the original records

The upper separation guarantee in [Theorem 4](#) is attained by a procedure whose inputs are the original records and the public moment and smoothness tuple. The construction combines histogram projection, clipped outcomes, and ordered-pair corrections within two independent evaluation blocks. Minimization over the candidate constant then accommodates the composite null of a homogeneous mean effect. We first specify the projection geometry and score ingredients, and then give the complete calculation.

At a positive integer rank J , the histogram projection averages functions over equal-width cells. Its endpoint convention is part of the procedure.

Definition 19. `[def:projection]` (Histogram projection Π_J) *For a positive integer J , define the histogram cells, feature vector, and projection kernel by*

$$I_{j,J} = \begin{cases} [(j-1)/J, j/J), & 1 \leq j < J, \\ [(J-1)/J, 1], & j = J, \end{cases} \quad F_J(x) = (\sqrt{J} \mathbf{1}_{I_{j,J}}(x))_{j=1}^J,$$

$$\Pi_J(x, z) = \langle F_J(x), F_J(z) \rangle.$$

The corresponding projection of an integrable function f is

$$(\Pi_J f)(x) = \int_0^1 \Pi_J(x, z) f(z) dz.$$

The normalized feature vector $F_J(x)$ represents cellwise constant functions in Euclidean coordinates, and the projection kernel $\Pi_J(x, z)$ selects pairs in the same cell. An internal bin endpoint belongs to the cell on its right, while the final cell contains 1. These conventions assign a value to every record, including records at cell boundaries.

To obtain independent score vectors, partition the independent sample into two deterministic blocks of equal size.

Definition 20. `[def:blocks]` (Evaluation blocks $\mathcal{I}_1, \mathcal{I}_2$) *The two evaluation blocks and their common size are*

$$s_n = \lfloor n/2 \rfloor, \quad \mathcal{I}_1 = \{1, \dots, s_n\}, \quad \mathcal{I}_2 = \{s_n + 1, \dots, 2s_n\}.$$

The evaluation blocks $\mathcal{I}_1, \mathcal{I}_2$ each contain s_n records. When the sample size is odd, the last record is left outside these blocks. Independence between the blocks permits their score inner product to estimate a squared population mean.

Within each block, a single-record score is paired with an ordered-pair correction. Both depend affinely on the candidate constant $c \in [-1/2, 1/2]$. The following ingredients use a coarse histogram rank J_c , whose public selection is specified in [Algorithm 1](#).

Definition 21. `[def:score-family]` (Score families $H_{b,T}$ and $U_{b,G,T}$) *For $n \geq 4$, the single-record and ordered-pair score vectors are*

$$H_{b,T}(c) = \frac{1}{s_n} \sum_{i \in \mathcal{I}_b} F_{J_c}(X_i) A_i \{\ell_T(Y_i) - c\},$$

$$U_{b,G,T}(c) = \frac{1}{s_n(s_n - 1)} \sum_{\substack{i,j \in \mathcal{I}_b \\ i \neq j}} F_{J_c}(X_i) A_i G(X_i, X_j) \{\ell_T(Y_j) - c A_j\}.$$

Here $b \in \{1, 2\}$, $G : [0, 1]^2 \rightarrow \mathbb{R}$ is the pair-score kernel, and the clipping function is

$$\ell_T(y) = \max\{-T, \min\{y, T\}\}.$$

For $n = 2, 3$, set

$$H_{b,T}(c) = U_{b,G,T}(c) = 0.$$

The clipping function $\ell_T(y)$ bounds each outcome contribution before averaging. The single-record score $H_{b,T}(c)$ uses treated records, while the ordered-pair score $U_{b,G,T}(c)$ combines the treatment indicator of one record with the clipped outcome and treatment indicator of a distinct record. The correction kernel G supplies the projection used in this pairing. The explicit dependence on c makes profiling a finite quadratic calculation.

The construction uses two resolutions for different tasks. The coarse rank M retains the effect signal tested by the cross-block inner product, while the fine rank K controls the product of propensity and baseline approximation errors left after the ordered-pair correction. A single correction suffices when $S \geq \gamma$ or $\alpha \geq q$. Otherwise the dyadic increments between M and K receive separate cutoffs T_j : coarse increments can tolerate more clipping variability, while fine increments use smaller cutoffs to keep their covariance contribution summable. This representation condition determines how the same statistic is assembled; it is separate from the phase condition $F(p) = 1$, which determines which separation exponent is smaller.

The complete procedure now selects these ranks and clipping thresholds, subtracts the pair corrections, and computes the bias and covariance bounds used for calibration. The fine histogram rank J_f governs the correction resolution. Each selection depends on the sample size and public tuple.

Algorithm 1 (Explicit score and ledgers $Z_b(c), \mathbf{b}_{n,v}, \Lambda_{n,v}$)

Inputs are the original records and the public tuple $v = (p, \alpha, \beta, \gamma)$; $H_{b,T}(c)$ and $U_{b,G,T}(c)$ denote the blockwise single-record histogram score and ordered-pair correction score, respectively.

1. Define the public exponents

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad t = \frac{2-p}{p-1},$$

$$E_0 = \frac{2\gamma q}{2\gamma + q}, \quad E_4 = \frac{2S}{1 + 2\alpha + \beta/q + S/(2\gamma)}, \quad E = \min\{E_0, E_4\}.$$

For $n = 2, 3$, set

$$J_c = J_f = 1, \quad T_j = 1 \text{ for every cutoff}, \quad Z_b(c) = 0,$$

$$\mathbf{b}_{n,v} = \Lambda_{n,v} = h_{n,v} = 0, \quad \phi_{n,v}^* = 0.$$

The remaining steps apply for $n \geq 4$.

2. Let s_n be the evaluation-block size. Set

$$s = s_n, \quad a = s^{-E}, \quad M = J_c = 2^{\lceil \log_2(a^{-1/\gamma}) \rceil},$$

$$K = J_f = 2^{\lceil \log_2(\max\{M, a^{-1/S}\}) \rceil}.$$

Define upper dyadic rounding and the main clipping quantities by

$$\mathcal{R}(x) = 2^{\lceil \log_2 x \rceil}, \quad x > 0, \quad T_0 = \mathcal{R}(a^{-1/(p-1)}), \quad V_0 = 32T_0^{2-p}.$$

3. If $S \geq \gamma$ or $\alpha \geq q$, set

$$Z_b(c) = H_{b,T_0}(c) - U_{b,\Pi_K,T_0}(c),$$

$$B = 400K^{-S} + 20T_0^{1-p}, \quad L_1 = 16\sqrt{V_0}, \quad L_2 = 4\sqrt{KV_0}.$$

4. Otherwise, set

$$\lambda_0 = \frac{(p-1)^2}{4p}, \quad L = \log_2(K/M), \quad R_j = 2^j M \quad (0 \leq j \leq L),$$

$$D_j = \Pi_{R_j} - \Pi_{R_{j-1}}, \quad 1 \leq j \leq L.$$

For each increment define

$$T_j = \mathcal{R} \left(\max \left\{ 1, \min \left\{ a^{-1/(p-1)}, \left[\frac{R_j^{-\alpha}}{a(R_j/K)^{\lambda_0}} \right]^{1/(p-1)} \right\} \right\} \right), \quad V_j = 32T_j^{2-p},$$

$$a_j = 40R_{j-1}^{-\alpha}, \quad d_j = 110R_{j-1}^{-\min\{\alpha,\beta,\gamma\}} + 20T_j^{1-p}.$$

Here L is an integer, and empty sums equal zero. Set

$$Z_b(c) = H_{b,T_0}(c) - U_{b,\Pi_M,T_0}(c) - \sum_{j=1}^L U_{b,D_j,T_j}(c),$$

$$B = 400K^{-S} + 20T_0^{1-p} + 10 \sum_{j=1}^L a_j T_j^{1-p},$$

$$L_1 = 16\sqrt{V_0} + 8 \sum_{j=1}^L (d_j + a_j \sqrt{V_j}),$$

$$L_2 = 4 \left(\sqrt{MV_0} + \sum_{j=1}^L \sqrt{R_j V_j} \right).$$

5. In either branch, define the public bias ledger, covariance ledger, and rejection cutoff

$$\mathfrak{b}_{n,v} = B, \quad \Lambda_{n,v} = \frac{L_1^2}{s} + \frac{2L_2^2}{s(s-1)},$$

$$h_{n,v} = \mathcal{R} \left(2B^2 + 2^{16}\sqrt{M} \Lambda_{n,v} \right).$$

Use the public rank and outcome grids

$$\{2^j : 0 \leq j \leq \lceil \log_2(2s^2) \rceil\}, \quad \{2^j : 0 \leq j \leq \lceil \log_2(2s) \rceil\},$$

respectively. The cutoff is the upper endpoint of the two-point dyadic grid bracketing its unrounded formula.

6. Form the quadratic statistic

$$W(c) = \langle Z_1(c), Z_2(c) \rangle.$$

Evaluate its minimum over $[-1/2, 1/2]$ at the two endpoints and, when the quadratic coefficient is positive and the vertex is interior, at the vertex. Output

$$\phi_{n,v}^* = \mathbf{1} \left\{ \min_{|c| \leq 1/2} W(c) > h_{n,v} \right\}.$$

Every sum is finite. Histogram and ordered-pair scores take their stipulated values at bin endpoints and are evaluated directly from the original records, without division by observed cell counts.

The corrected block score $Z_b(c)$ uses either a single fine-rank correction or a sum of dyadic projection increments. In the latter construction, each increment receives its own clipping threshold; the main clipping threshold is T_0 . This allocation enters both the public bias ledger $\mathbf{b}_{n,v}$ and the public covariance ledger $\Lambda_{n,v}$. The rejection cutoff $h_{n,v}$ combines their contributions and is rounded upward.

The statistical comparison and the computational selection have distinct roles. The comparison of E_0 and E_4 , as defined in [Definition 15](#) and [Algorithm 1](#), determines the exponent used for tuning. The condition $S \geq \gamma$ or $\alpha \geq q$ selects the single-correction representation; when $S < \gamma$ and $\alpha < q$, the procedure uses the multiresolution representation. Both representations use the same minimum exponent E .

All calculations in [Algorithm 1](#) are finite. The public dyadic grids specify the rank and outcome selections, and the upper-dyadic rule specifies the rejection cutoff. Scores are normalized by the block size and number of ordered pairs. Consequently, empty histogram cells and observations at bin endpoints have stipulated values throughout the calculation. For $n = 2, 3$, the zero-score convention and zero rejection rule complete the procedure.

The cross-block statistic $W(c)$ is quadratic because each block score is affine in c . Its minimum is therefore obtained by comparing the endpoints and, when applicable, the interior vertex. Rejection requires the statistic to exceed its cutoff for every admissible constant, which gives the profiling step its role in testing the whole null.

For the calibration statement, write $\psi_{n,v} := \phi_{n,v}^*$ for the rule in [Algorithm 1](#), and define the population score mean $\theta_P(c)$ by

$$\theta_P(c) := \mathbb{E}_{P^{\otimes n}}[Z_b(c)].$$

Let λ be Lebesgue probability measure on $[0, 1]$, and use

$$d(P) := \left\{ \int_0^1 \left(\tau_P(x) - \int_0^1 \tau_P(t) d\lambda(t) \right)^2 d\lambda(x) \right\}^{1/2}, \quad D_v := \sup_{P \in \mathcal{M}_v} d(P),$$

for the heterogeneity distance and its model supremum. Set $C := 128$ for the simultaneous profiling multiplier, and define the calibrated separation by

$$R_{\text{att}}(n, v) := \frac{16}{3} \left\{ 16J_c^{-\gamma} + 5\mathbf{b}_{n,v} + 1024J_c^{1/4} \sqrt{\Lambda_{n,v}} \right\}.$$

The following result connects these public quantities to uniform size control and power.

Theorem 1. [thm:whole-null-calibration-resolved] (Whole-null calibration and power) *Let $v = (p, \alpha, \beta, \gamma)$ be an admissible public tuple as in Definition 1, and let $n \geq 2$ be an integer. Use the ranks, cutoffs, scores, bias ledger $\mathbf{b}_{n,v}$, and covariance ledger $\Lambda_{n,v}$ of Algorithm 1. Write $\psi_{n,v}$ for the original-record test constructed there, and define its rejection expectation by*

$$R_n(P, \psi_{n,v}) := \mathbb{E}[\psi_{n,v}(\text{sample}, \text{seed})],$$

where the sample consists of n independent records with law P , and the seed is independent and uniform on the unit interval. Define the population score mean by

$$\theta_P(c) := \mathbb{E}_{P^{\otimes n}}[Z_b(c)].$$

Let λ denote Lebesgue probability measure on $[0, 1]$, and define the heterogeneity distance and its model supremum by

$$d(P) := \left\{ \int_0^1 \left(\tau_P(x) - \int_0^1 \tau_P(t) d\lambda(t) \right)^2 d\lambda(x) \right\}^{1/2}, \quad D_v := \sup_{P \in \mathcal{M}_v} d(P).$$

For every $P \in \mathcal{M}_v$, each affine coefficient of each block score is square-integrable, and its scalar projection in any direction $u \in \mathbb{R}^{J_c}$ has variance at most

$$\Lambda_{n,v} \|u\|_2^2.$$

For each block, the absolute covariance between projections of its two affine coefficients in directions $u, z \in \mathbb{R}^{J_c}$ is at most

$$\Lambda_{n,v} \|u\|_2 \|z\|_2.$$

The simultaneous profiling event for $W(c)$ specified in Algorithm 1, with profiling multiplier $C := 128$, is measurable and has probability at least 0.9925 under $P^{\otimes n}$. If $\Lambda_{n,v} = 0$, then, almost surely, simultaneously for every $|c| \leq 1/2$,

$$W(c) = \|\theta_P(c)\|_2^2.$$

For every $P \in H_0(v)$, with the null class defined in Definition 2,

$$\inf_{|c| \leq 1/2} \|\theta_P(c)\|_2 \leq \mathbf{b}_{n,v}, \quad R_n(P, \psi_{n,v}) \leq 0.0075.$$

Define the calibrated separation by

$$R_{\text{att}}(n, v) := \frac{16}{3} \left\{ 16J_c^{-\gamma} + 5\mathbf{b}_{n,v} + 1024J_c^{1/4} \sqrt{\Lambda_{n,v}} \right\}.$$

If $n \geq 4$ and $R_{\text{att}}(n, v) < D_v$, then every $P \in \mathcal{M}_v$ satisfying $d(P) \geq R_{\text{att}}(n, v)$ obeys

$$1 - R_n(P, \psi_{n,v}) \leq 0.0075.$$

For $n = 2$ or $n = 3$, $\psi_{n,v}$ equals zero at every sample and seed.

Under a homogeneous mean effect, an admissible constant makes the population score norm at most the bias ledger. Profiling retains this comparison while the covariance bounds control fluctuations simultaneously over the candidate constants. Independence between the evaluation blocks gives the population identity $\mathbb{E}[W(c)] = \|\theta_P(c)\|_2^2$. The calibrated separation combines coarse-projection approximation, original-mean bias, and stochastic fluctuation. For $n \geq 4$ with $R_{\text{att}}(n, v) < D_v$, effects at or above that separation are detected with type-II error at most 0.0075; the size bound applies throughout the null for every $n \geq 2$.

The appendix develops the three parts of this calibration. The original-mean analysis in [Lemma 2](#) controls the bias introduced by projection and clipping. The coefficient covariance bounds are supplied by [Lemma 3](#), and [Lemma 4](#) controls the simultaneous profiling operation. Their combination yields the attainable separation result in [Theorem 2](#), connecting the explicit procedure to the power scale $\rho_n(v)$ in [Definition 15](#) and the upper separation bound in [Theorem 4](#).

6 Supplied propensity scores and assignment information

The calibrated procedure in [Algorithm 1](#) obtains assignment information from the original records. We now consider an experiment in which the entire true continuous propensity is supplied together with those records. This experiment supports two complementary conclusions: an attaining test over a broader class with continuous propensity and conditional mean functions, and an exact comparison of testing sensitivity on the original model.

A supplied-function test ϕ^e is a jointly Borel map

$$\phi^e : C([0, 1]) \times ([0, 1] \times \{0, 1\} \times \mathbb{R})^n \times [0, 1] \longrightarrow [0, 1].$$

The function space carries the uniform topology. The remaining arguments are the independent original-record sample and an independent public uniform randomization variable. Under a law P , the supplied function is e_P ; the formal risk definition below specifies the corresponding rejection expectation.

6.1 The continuous-carrier experiment

Let $\mathcal{V}$ denote the admissible domain of public moment and smoothness tuples in [Definition 1](#), and let λ be Lebesgue probability measure on $[0, 1]$. The broader model retains uniform design, overlap, effect smoothness, the mean caps, and the conditional moment envelope. Its propensity and both conditional arm means have continuous representatives. The following definition specifies these restrictions and the maximum heterogeneity distance.

Definition 22. [\[def:oracle-broad-model\]](#) (Continuous-carrier model $\mathcal{M}_v^{\text{e,br}}$) *The class $\mathcal{M}_v^{\text{e,br}}$ consists of the original-record laws satisfying the six conditions below and admitting continuous representatives of their propensity and both conditional arm means. Here $v = (p, \alpha, \beta, \gamma) \in \mathcal{V}$ is an admissible public tuple, λ is Lebesgue probability measure on $[0, 1]$, and $Q_{a,P}(dy \mid x)$ is the conditional outcome probability kernel in treatment arm a .*

- **(Uniform covariates.)** *The covariate marginal satisfies*

$$P_X = \lambda,$$

as stipulated in [Assumption 1](#).

- **(Overlap.)** The continuous propensity representative satisfies

$$e_P([0, 1]) \subseteq [1/4, 3/4],$$

as stipulated in [Assumption 2](#).

- **(Effect smoothness.)** The continuous mean effect satisfies

$$\tau_P = m_{1,P} - m_{0,P} \in \mathcal{H}^\gamma(20),$$

where $\mathcal{H}^\gamma(20)$ consists of continuous functions on $[0, 1]$ whose sup norm and γ -Hölder constant are at most 20, as stipulated in [Assumption 5](#).

- **(Baseline cap.)** The baseline mean satisfies

$$\|m_{0,P}\|_\infty \leq 1/2,$$

as stipulated in [Assumption 6](#).

- **(Effect cap.)** The mean effect satisfies

$$\|\tau_P\|_\infty \leq 1/2,$$

as stipulated in [Assumption 7](#).

- **(Raw moment.)** The conditional outcome kernels satisfy

$$\max_{a \in \{0,1\}} \operatorname{ess\,sup}_{x \sim \lambda} \int |y|^p Q_{a,P}(dy \mid x) \leq 10,$$

as stipulated in [Assumption 8](#).

For a function h , write $[h]_\lambda$ for its equivalence class under Lebesgue almost-everywhere equality. Let $\mathcal{E}_P$ denote the propensity equivalence class and let $\mathcal{G}_{0,P}, \mathcal{G}_{1,P}$ denote the conditional arm-mean equivalence classes. The continuous-representative requirement is

$$\exists f, g_0, g_1 \in C([0, 1]) : \quad \mathcal{E}_P = [f]_\lambda, \quad \mathcal{G}_{0,P} = [g_0]_\lambda, \quad \mathcal{G}_{1,P} = [g_1]_\lambda.$$

Here $C([0, 1])$ is the space of continuous real functions on the unit interval with the uniform topology. These representatives are the existing functionals $e_P, m_{0,P}, m_{1,P}$. Membership depends on p and γ , with the full tuple v retained as the public index.

Writing $d(P)$ for the population Lebesgue L_2 distance of τ_P from its average, define

$$D_v^{\text{e,br}} = \sup_{P \in \mathcal{M}_v^{\text{e,br}}} d(P).$$

The continuous-carrier class $\mathcal{M}_v^{\text{e,br}}$ in [Definition 22](#) permits continuous propensity and baseline mean functions with unrestricted moduli of continuity. Effect smoothness controls the approximation of heterogeneity, while overlap bounds the inverse-propensity weights. The conditional moment envelope controls outcome clipping uniformly across covariate values and treatment arms. The caps bound the scale of the conditional means and effect. Within this class, the null retains the unknown constant effect used in the original experiment.

Definition 23. [def:oracle-broad-null] (Constant-effect null $H_0^{\text{e,br}}(v)$) *The constant-effect null is*

$$H_0^{\text{e,br}}(v) = \left\{ P \in \mathcal{M}_v^{\text{e,br}} : \exists c \in [-1/2, 1/2] \forall x \in [0, 1], \tau_P(x) = c \right\}.$$

Membership in the continuous-carrier model imposes the restrictions in [Assumptions 1, 2 and 5](#) to [8](#). The unknown-constant effect condition is that of [Assumption 9](#).

Size is controlled uniformly over the constant-effect null $H_0^{\text{e,br}}(v)$ in [Definition 23](#). For a supplied-function test, write $R_n^{\text{e}}(P, \phi^{\text{e}})$ for its rejection expectation when it receives e_P . The following risk and radius measure the separation required to attain type-II error at most 1/10, subject to the same bound on size.

Definition 24. [def:oracle-broad-risk] (Supplied-propensity risk $B_n^{\text{e,br}}(v, r)$) *The supplied-propensity minimax type-II error is*

$$B_n^{\text{e,br}}(v, r) = \inf_{\phi^{\text{e}}: \sup_{P \in H_0^{\text{e,br}}(v)} R_n^{\text{e}}(P, \phi^{\text{e}}) \leq 1/10} \sup_{P \in \mathcal{M}_v^{\text{e,br}}: d(P) \geq r} \{1 - R_n^{\text{e}}(P, \phi^{\text{e}})\},$$

for parameters satisfying

- **(Sample size.)** $n \geq 2$.
- **(Public tuple.)** $v \in \mathcal{V}$, the admissible domain of moment and primitive smoothness tuples.
- **(Separation.)** $0 < r < D_v^{\text{e,br}}$, where $D_v^{\text{e,br}}$ is the supremum of the population effect distance over $\mathcal{M}_v^{\text{e,br}}$.

Here $d(P)$ is the population Lebesgue L_2 distance of the mean effect from its average. The admissible tests ϕ^{e} are precisely the jointly Borel maps taking a supplied continuous function, an original-record sample, and a public randomization variable to a rejection probability in $[0, 1]$. Their rejection expectation is

$$R_n^{\text{e}}(P, \phi^{\text{e}}) = \int \phi^{\text{e}}(e_P, \mathcal{D}_n, u) P^{\otimes n}(d\mathcal{D}_n) du.$$

The sample $\mathcal{D}_n$ consists of n independent original records with law P ; the integration in u is over an independent uniform variable on $[0, 1]$. The supplied function e_P is the whole true continuous propensity.

The broader-class risk $B_n^{\text{e,br}}(v, r)$ in [Definition 24](#) uses the original outcome records throughout. Its capped radius includes the model supremum as a saturation value, making the sensitivity measure well defined at every stipulated sample size.

Definition 25. [def:oracle-broad-radius] (Capped radius $r_n^{*,\text{e,br}}(v)$) *The capped supplied-propensity critical radius is*

$$r_n^{*,\text{e,br}}(v) = \inf \left(\left\{ r \in (0, D_v^{\text{e,br}}) : B_n^{\text{e,br}}(v, r) \leq 1/10 \right\} \cup \left\{ D_v^{\text{e,br}} \right\} \right).$$

Here $D_v^{\text{e,br}}$ is the supremum of the population effect distance over the continuous-carrier model and serves as the saturation value in the capped-radius convention.

The capped radius $r_n^{*,e,br}(v)$ and its saturation value $D_v^{e,br}$ are defined in [Definitions 22](#) and [25](#). To attain the separation rate, the test uses clipped inverse-propensity outcomes and the histogram features of [Definition 19](#). Removing the constant histogram direction targets deviations from an unknown constant effect. The procedure then takes the inner product of two independent block averages.

Algorithm 2 (Supplied-propensity test $\phi_{n,v}^{*,e}$)

Inputs are $n \geq 2$ original records, the public tuple $v = (p, \alpha, \beta, \gamma)$, and a supplied continuous propensity e .

1. Set the block size, moment exponent, separation exponent, and target scale to

$$s = \lfloor n/2 \rfloor, \quad q = \frac{p-1}{p}, \quad E_0 = \frac{2\gamma q}{2\gamma + q}, \quad a = s^{-E_0}.$$

Use the evaluation blocks

$$\mathcal{I}_1 = \{1, \dots, s\}, \quad \mathcal{I}_2 = \{s+1, \dots, 2s\}.$$

2. Select the clipping cutoff and histogram rank

$$T = 2^{\lceil \log_2(a^{-1/(p-1)}) \rceil}, \quad J = 2^{\lceil \log_2(a^{-1/\gamma}) \rceil}.$$

Define outcome clipping and the overlap-clipped propensity by

$$\ell_T(y) = \max\{-T, \min\{y, T\}\}, \quad \tilde{e}(x) = \max\{1/4, \min\{e(x), 3/4\}\}.$$

For every legal supplied true propensity, $\tilde{e} = e$.

3. Form the original-record inverse-propensity scores

$$R_i = \frac{A_i \ell_T(Y_i)}{\tilde{e}(X_i)} - \frac{(1 - A_i) \ell_T(Y_i)}{1 - \tilde{e}(X_i)}.$$

Let F_J be the normalized histogram feature vector. Its constant direction and the centered block vectors are

$$u = J^{-1/2}(1, \dots, 1)^T, \quad V_b = \frac{1}{s} \sum_{i \in \mathcal{I}_b} (F_J(X_i) - u) R_i, \quad b \in \{1, 2\}.$$

These vectors represent the block scores obtained by applying the constant-centered histogram projection $\Pi_J - 1$.

4. Set the public clipping-bias and covariance bounds to

$$b = 20T^{1-p}, \quad \Lambda = \frac{80T^{2-p}}{s}.$$

Output the rejection rule

$$\phi_{n,v}^{*,e} = \mathbf{1}\left\{\langle V_1, V_2 \rangle > 2b^2 + 1024\sqrt{J}\Lambda\right\}.$$

Its calibration and necessity properties are given by its singleton and canonical quadratic projections and the normalized common-propensity paired-tent converse.

The supplied-propensity rule $\phi_{n,v}^{*,e}$ in [Algorithm 2](#) uses public clipping-bias and covariance bounds to set its rejection threshold. The tuning depends on the moment order and effect smoothness. The next result gives matching bounds for the broader-class critical radius, uniform calibration over its entire null, and power whenever the stated separation lies below the saturation value.

The conditional-mean identity in [Theorem 5](#) explains the role of supplied assignment information: inverse weighting recovers the mean effect directly, and constant centering isolates its heterogeneity. Clipping controls the contribution of outcome tails, while effect smoothness controls histogram approximation. Their balance yields the supplied-propensity separation scale $\rho_n^e(v)$ defined in that result. The matching lower bound uses null and alternative laws with the same constant propensity and close complete-record distributions. Thus the stated separation order remains necessary even when assignment information is identical across the comparison laws.

The radius bounds apply to every $n \geq 2$, with the cap accounting for finite-sample saturation. The procedure’s power guarantee applies when its upper separation is strictly below the broader-model supremum. The public sample-size threshold in [Theorem 5](#) ensures this condition by placing that separation at most halfway to the fixed witness distance. These statements distinguish an all-sample radius bound from a power guarantee over a nonempty range of alternatives.

6.2 Comparison on the original model

To measure the value of supplied assignment information on a fixed statistical class, we now use the original model of [Definition 1](#). Its quantitative propensity and baseline smoothness restrictions are retained in both experiments being compared. The supplied-function test carrier remains the same, while the null and alternative suprema are taken over the original model. The following definitions make this shared-class comparison precise.

Definition 26. [def:oracle-risk] (Supplied-propensity risk $B_n^e(v, r)$)

$$B_n^e(v, r) = \inf_{\phi^e: \sup_{P \in H_0(v)} R_n^e(P, \phi^e) \leq 1/10} \sup_{P \in \mathcal{M}_v: d(P) \geq r} \{1 - R_n^e(P, \phi^e)\}.$$

The parameters satisfy:

- **(Sample size.)** $n \geq 2$.
- **(Public tuple.)** $v \in \mathcal{V}$, the admissible domain of moment and primitive smoothness tuples.
- **(Separation.)** $0 < r < D_v$, where D_v is the supremum of $d(P)$ over $\mathcal{M}_v$.

Here $\mathcal{M}_v$ is the original-record model, $H_0(v)$ consists of its laws with an unknown constant mean effect, and $d(P)$ is the population Lebesgue L_2 distance of the effect from its average. The infimum ranges over randomized rejection-probability maps ϕ^e receiving the whole true continuous propensity, and $R_n^e(P, \phi^e)$ is their expected rejection probability with that propensity supplied.

The shared-class risk $B_n^e(v, r)$ in [Definition 26](#) uses the original-model null for its size constraint. Its radius therefore uses the original-model supremum as the saturation value.

Definition 27. [def:oracle-radius] (Supplied-propensity radius $r_n^{*,e}(v)$)

$$r_n^{*,e}(v) = \inf \left(\{r \in (0, D_v) : B_n^e(v, r) \leq 1/10\} \cup \{D_v\} \right).$$

Here $B_n^e(v, r)$ is the supplied-propensity minimax type-II error on the original model under the size constraint $1/10$, and D_v is the supremum of the population effect distance over that model.

The shared-class capped radius $r_n^{*,e}(v)$ of Definition 27 is compatible with the broader-class attainment result. The original model is contained in the continuous-carrier class, and the finite lower-bound witnesses in Theorem 5 belong to both classes. The next result records the matching separation bounds and attainment on the original model, providing the supplied-propensity benchmark for the radius comparison.

The compatibility in Theorem 6 rests on both attainment and necessity. The supplied-propensity procedure controls size and power on the original class, and common-propensity witnesses establish the corresponding lower separation bound within that class. Consequently, the original-model comparison uses matched statistical classes in its numerator and denominator. Its finite-sample power condition is expressed relative to the original-model supremum; the broader-class result uses its own supremum.

We measure preservation of separation order by a uniformly bounded ratio of the unknown- and supplied-propensity capped radii. The following region includes every admissible tuple for which that ratio remains bounded across sample sizes.

Definition 28. [def:preservation-region] (Propensity preservation region $\mathcal{R}$)

$$\mathcal{R} = \left\{ v \in \mathcal{V} : \sup_{n \geq 2} \frac{r_n^*(v)}{r_n^{*,e}(v)} < \infty \right\}.$$

Here $\mathcal{V}$ is the admissible domain of public moment and primitive smoothness tuples. The radii $r_n^*(v)$ and $r_n^{*,e}(v)$ are the capped critical separation radii on the same original-record model with unknown and supplied true continuous propensity, respectively.

The preservation region $\mathcal{R}$ in Definition 28 follows from comparing the matched rates in Theorems 4 and 6. The scalar comparison quantity $F(p)$ defined in Definition 15 determines whether the separation exponents coincide. The next result gives the exact boundary, including its equality case and bounds for every capped finite-sample radius ratio.

The boundary in Theorem 7 expresses the value of assignment information through the difference between the two separation exponents. When $F(p) \geq 1$, the unknown-propensity experiment attains the supplied-propensity separation order on the same original class. At $F(p) = 1$, the ratio remains between positive, sample-size-independent constants. When $F(p) < 1$, supplying the true propensity improves sensitivity by a polynomial factor whose exponent is stated explicitly in the result.

These comparisons concern orders of critical separation, with finite-sample constants and saturation incorporated into the capped radii. They hold for every $n \geq 2$, and their constants admit uniform bounds on compact subsets of the admissible parameter domain. Together with the broader-class attainment result, they distinguish testing with supplied assignment information from the exact region where the original unknown-propensity experiment attains the same separation order. Requirements for a particular estimated-propensity procedure are outside this experiment comparison.

7 Discussion and extensions

The separation comparisons provide two benchmarks for the sensitivity of homogeneity testing: one varies the outcome envelope at matched primitive smoothness, and the other varies the assignment information available to the test on a shared model class. The first comparison, in [Theorem 4](#), measures the cost of allowing outcomes subject to a finite conditional moment bound relative to bounded outcomes. The second, in [Theorem 7](#), measures the value of supplying the true propensity function. Together, they characterize how outcome tails and assignment information affect the smallest detectable departure from a constant conditional mean effect.

Matching primitive smoothness makes the outcome comparison interpretable. The propensity and baseline smoothness exponents describe nuisance complexity, whereas the effect smoothness exponent describes the complexity of the departure being tested. The bounded benchmark in [Definition 7](#) retains these three restrictions. Consequently, the radius ratio in [Definition 16](#) compares outcome envelopes while keeping the stipulated effect and nuisance complexity fixed. The regions in [Definitions 17](#) and [18](#) distinguish a diverging separation penalty from a uniformly bounded radius ratio. These are benchmarks for worst-case sensitivity over the specified classes: they identify when the conditional moment envelope changes the order of the testing problem.

The assignment comparison separates the complexity of the effect from the information needed to handle the nuisance functions. On the shared original class, [Theorem 7](#) identifies the exact region where the unknown-propensity experiment attains the supplied-propensity separation order. The broader-class result in [Theorem 5](#) gives a complementary interpretation. Its model, defined in [Definition 22](#), permits continuous propensity and baseline mean functions subject to the retained overlap and mean envelopes, together with the conditional moment and effect smoothness restrictions. Attainment over this class supports a broader nuisance scope when the true propensity is supplied. The resulting separation order is governed by the moment order and effect smoothness, clarifying the role of assignment information in accommodating nuisance variation.

The population target also has a direct variance interpretation. Writing the distance from a constant effect as

$$d(P) = \left[\int_0^1 \left\{ \tau_P(x) - \int_0^1 \tau_P(z) dz \right\}^2 dx \right]^{1/2},$$

the uniform covariate design in [Assumption 1](#) gives

$$d(P)^2 = \text{Var}_P\{\tau_P(X)\}.$$

Thus separation is measured in the standard deviation of the conditional mean treatment effect across covariate values. Under the causal completion in [Definition 3](#), this contrast equals the conditional expectation of the potential-outcome difference. The target therefore quantifies heterogeneity in conditional mean effects, with the covariate distribution determining the population weighting.

Public finite-sample quantities connect these comparisons to calibrated rejection rules. The construction in [Algorithm 1](#) specifies the projection ranks, clipping levels, bias and covariance bounds, and rejection cutoff. Whole-null calibration in [Theorem 1](#) accounts for the unknown constant effect through profiling. Combined with the attainment guarantees in [Theorem 4](#), the public sample-size thresholds identify when the stated power separation lies inside the model’s nonempty alternative domain. The analogous supplied-propensity guarantees appear in [Theorems 5](#) and [6](#).

The explicit sufficient constants are deliberately conservative. The following values come directly from the displayed formulas; logarithms make their scale readable. Here N_v^{att} is the original-record nonempty-power threshold and N_v^e is its supplied-propensity counterpart.

p	(α, β, γ)	regime	E	$\log_{10} C_v^{\text{att}}$	$\log_{10} N_v^{\text{att}} / \log_{10} N_v^e$
2	(1, 1, 1)	effect	0.400	8.785	25.35 / 12.86
3/2	(1, 1, 1)	effect	0.286	8.786	35.49 / 18.00
3/2	(1/20, 1/20, 1)	rough	0.154	9.934	73.37 / 18.00

For the first row, for example, the supplied-propensity formula gives $N_v^e = 7,183,754,725,838$. These are analytic sufficient thresholds for the stated finite-sample power guarantee, rather than estimates of the sample size needed in practice. The theorem’s main quantitative content is the minimax power of n and the matching lower bound; practical calibration of the constants would require implementation and simulation beyond the present analysis.

7.1 Limitations and future work

The analysis treats a known uniform covariate design on the unit interval and public moment and smoothness exponents. Extending the construction to an unknown covariate distribution would require handling the population weighting and the estimation of projection geometry together. Higher-dimensional covariates raise a related question about how approximation complexity and nuisance smoothness interact in the separation comparison.

Adaptive selection of the moment and smoothness exponents is another research direction. A useful extension would quantify the cost of choosing clipping levels and projection ranks from the data while maintaining calibration over the entire constant-effect null and establishing nonempty power domains.

Finally, the supplied-function experiment provides a benchmark for studying estimated assignment functions. Such an extension would quantify how propensity approximation error and the dependence introduced by its estimation enter the bias bounds, rejection threshold, and attainable separation. Comparing those guarantees with the exact supplied-function benchmark would clarify the assignment accuracy needed to retain its sensitivity to effect heterogeneity.

Appendices

A Causal interpretation and benchmark equivalence

The population target in [Definitions 1](#) and [2](#) uses continuous representatives of conditional means. Its interpretation therefore requires both invariance to conditional-distribution versions and a causal representation that reproduces the observed law. The following result establishes these properties and supplies legal constant-effect and heterogeneous-effect witnesses for every admissible parameter tuple.

Lemma 1. [[lem:causal-nonempty](#)] (Nonemptiness and causal identification) *Fix an admissible parameter tuple $v = (p, \alpha, \beta, \gamma)$ as in [Definition 1](#). Let λ denote Lebesgue probability measure on $[0, 1]$, and define the heterogeneity distance, its model supremum, and the witness distance by*

$$d(P) = \left\{ \int_0^1 \left(\tau_P(x) - \int_0^1 \tau_P(z) d\lambda(z) \right)^2 d\lambda(x) \right\}^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P), \quad d_0 = \frac{1}{8\sqrt{2}}.$$

The set $\{d(P) : P \in \mathcal{M}_v\}$ is bounded above, and

$$d_0 \leq D_v \leq \frac{1}{2}.$$

For every $P \in \mathcal{M}_v$, with model membership understood as in [Definition 1](#), the continuous propensity, baseline mean, and contrast representatives are determined uniquely by the observed law: every other observational representation of the same probability law has exactly the same e_P , $m_{0,P}$, and τ_P , pointwise on $[0, 1]$. Consequently, the treated mean representative $m_{1,P} = m_{0,P} + \tau_P$ is also uniquely determined, and

$$|m_{1,P}(x) - m_{1,P}(z)| \leq 40|x - z|^{\min\{\beta, \gamma\}} \quad (x, z \in [0, 1]).$$

In the completion $\mathbb{Q}_P$ of [Definition 3](#), let $Y(0)$ and $Y(1)$ denote the baseline and treated potential-outcome coordinates. The distribution under $\mathbb{Q}_P$ of

$$(X, A, (1 - A)Y(0) + AY(1))$$

is P , and

$$(Y(0), Y(1)) \perp A \mid X.$$

Both potential outcomes are integrable under $\mathbb{Q}_P$, and

$$\mathbb{E}_{\mathbb{Q}_P}[Y(1) - Y(0) \mid X] = \tau_P(X) \quad \mathbb{Q}_P\text{-almost surely.}$$

Moreover,

$$d(P) = 0 \quad \Longleftrightarrow \quad P \in H_0(v),$$

where $H_0(v)$ is defined in [Definition 2](#).

The law with uniform covariate distribution, propensity and mean functions

$$e_P(x) = \frac{1}{2} + \frac{\sin(2\pi x)}{10}, \quad m_{0,P}(x) = \frac{\cos(2\pi x)}{8}, \quad \tau_P(x) = \frac{\cos(2\pi x)}{8},$$

and deterministic conditional outcome

$$Y = m_{0,P}(X) + A\tau_P(X)$$

belongs to $\mathcal{M}_v$ and has $d(P) = d_0$. Replacing its contrast function by $\tau_P(x) = 1/8$ for every $x \in [0, 1]$, while retaining the displayed propensity, baseline mean, and deterministic conditional-outcome construction, yields a law in $H_0(v)$.

Proof of [Lemma 1](#). Fix an admissible tuple v . Thus

$$1 < p \leq 2, \quad 0 < \alpha, \beta \leq 1, \quad \frac{1}{4} \leq \gamma \leq 1.$$

We establish the witness construction and distance bounds first, followed by the identification and completion properties.

1. Define the separated witness primitives by

$$e_{\text{sep}}(x) = \frac{1}{2} + \frac{\sin(2\pi x)}{10}, \quad m_{0,\text{sep}}(x) = \frac{\cos(2\pi x)}{8}, \quad \tau_{\text{sep}}(x) = \frac{\cos(2\pi x)}{8}, \quad x \in [0, 1].$$

For $y_* \in \mathbb{R}$, write δ_{y_*} for the unit point mass, defined by

$$\delta_{y_*}(\mathbf{E}) = \mathbf{1}_{\mathbf{E}}(y_*) \quad (\mathbf{E} \in \mathcal{B}(\mathbb{R})),$$

and use the same notation for point masses on the treatment space $\{0, 1\}$. The witness law is

$$P_{\text{sep}}(dx, da, dy) = \lambda(dx) \left[(1 - e_{\text{sep}}(x)) \delta_0(da) \delta_{m_{0,\text{sep}}(x)}(dy) + e_{\text{sep}}(x) \delta_1(da) \delta_{m_{0,\text{sep}}(x) + \tau_{\text{sep}}(x)}(dy) \right].$$

The two weights are nonnegative and sum to one, so this construction is a probability law with covariate marginal λ . Its arm kernels and means satisfy

$$Q_{0,P_{\text{sep}}}(dy | x) = \delta_{m_{0,\text{sep}}(x)}(dy), \quad Q_{1,P_{\text{sep}}}(dy | x) = \delta_{m_{0,\text{sep}}(x) + \tau_{\text{sep}}(x)}(dy),$$

$$\int y Q_{0,P_{\text{sep}}}(dy | x) = m_{0,\text{sep}}(x), \quad \int y Q_{1,P_{\text{sep}}}(dy | x) = m_{0,\text{sep}}(x) + \tau_{\text{sep}}(x).$$

The sine and cosine bounds give

$$\frac{1}{4} \leq e_{\text{sep}}(x) \leq \frac{3}{4}, \quad |m_{0,\text{sep}}(x)| = |\tau_{\text{sep}}(x)| \leq \frac{1}{8}.$$

Their increment bounds are

$$|e_{\text{sep}}(x) - e_{\text{sep}}(z)| \leq \frac{\pi}{5} |x - z| \leq 20 |x - z|,$$

$$|m_{0,\text{sep}}(x) - m_{0,\text{sep}}(z)| = |\tau_{\text{sep}}(x) - \tau_{\text{sep}}(z)| \leq \frac{\pi}{4} |x - z| \leq 20 |x - z|.$$

For every exponent $\sigma_H \in (0, 1]$,

$$|x - z| \leq |x - z|^{\sigma_H} \quad (x, z \in [0, 1]).$$

Consequently these continuous functions satisfy the supremum and increment bounds in [Assumptions 3 to 5](#) at their respective exponents.

For the moment restriction, the baseline and effect caps imply

$$|m_{0,\text{sep}}(x)| \leq 1, \quad |m_{0,\text{sep}}(x) + \tau_{\text{sep}}(x)| \leq |m_{0,\text{sep}}(x)| + |\tau_{\text{sep}}(x)| \leq 1.$$

Since $p > 0$, both deterministic arm kernels therefore satisfy

$$\int |y|^p Q_{a_{\text{arm}}, P_{\text{sep}}}(dy | x) \leq 1 \leq 10 \quad (a_{\text{arm}} \in \{0, 1\}, x \in [0, 1]).$$

This verifies every restriction in [Definition 1](#), and hence

$$P_{\text{sep}} \in \mathcal{M}_v.$$

2. For each $P \in \mathcal{M}_v$, define its average contrast by

$$\bar{\tau}_P = \int_0^1 \tau_P(x) d\lambda(x).$$

By [Assumption 7](#),

$$|\bar{\tau}_P| \leq \int_0^1 |\tau_P(x)| d\lambda(x) \leq \frac{1}{2}, \quad |\tau_P(x) - \bar{\tau}_P| \leq 1.$$

The bounded functions below are integrable. Expanding the square and using $\lambda([0, 1]) = 1$ gives

$$\begin{aligned} \int_0^1 (\tau_P(x) - \bar{\tau}_P)^2 d\lambda(x) &= \int_0^1 \tau_P(x)^2 d\lambda(x) - 2\bar{\tau}_P \int_0^1 \tau_P(x) d\lambda(x) + \bar{\tau}_P^2 \\ &= \int_0^1 \tau_P(x)^2 d\lambda(x) - \bar{\tau}_P^2. \end{aligned}$$

Moreover,

$$\int_0^1 \tau_P(x)^2 d\lambda(x) \leq \int_0^1 \frac{1}{4} d\lambda(x) = \frac{1}{4}.$$

Thus

$$0 \leq d(P) \leq \frac{1}{2} \quad (P \in \mathcal{M}_v).$$

The distance set is therefore bounded above. It is nonempty by the witness from the preceding step, so its supremum satisfies

$$D_v \leq \frac{1}{2}.$$

For the witness, substitution in the cosine integrals gives

$$\bar{\tau}_{P_{\text{sep}}} = \frac{1}{16\pi} \int_0^{2\pi} \cos \vartheta d\vartheta = \frac{1}{16\pi} [\sin \vartheta]_0^{2\pi} = 0$$

and

$$\begin{aligned} \int_0^1 \tau_{\text{sep}}(x)^2 d\lambda(x) &= \frac{1}{128\pi} \int_0^{2\pi} \cos^2 \vartheta d\vartheta \\ &= \frac{1}{128\pi} \left[\frac{\vartheta}{2} + \frac{\sin \vartheta \cos \vartheta}{2} \right]_0^{2\pi} = \frac{1}{128}. \end{aligned}$$

Since

$$d_0 > 0, \quad d_0^2 = \frac{1}{128},$$

taking the nonnegative square root yields

$$d(P_{\text{sep}}) = d_0.$$

Membership of this distance in the bounded distance set then gives

$$d_0 = d(P_{\text{sep}}) \leq D_v \leq \frac{1}{2}.$$

3. Define the constant witness by

$$e_{\text{const}} = e_{\text{sep}}, \quad m_{0,\text{const}} = m_{0,\text{sep}}, \quad \tau_{\text{const}}(x) = \frac{1}{8} \quad (x \in [0, 1]),$$

$$P_{\text{const}}(dx, da, dy) = \lambda(dx) \left[(1 - e_{\text{const}}(x)) \delta_0(da) \delta_{m_{0,\text{const}}(x)}(dy) + e_{\text{const}}(x) \delta_1(da) \delta_{m_{0,\text{const}}(x)+1/8}(dy) \right].$$

The constant contrast satisfies

$$|\tau_{\text{const}}(x)| = \frac{1}{8} \leq \frac{1}{2}, \quad |\tau_{\text{const}}(x) - \tau_{\text{const}}(z)| = 0 \leq 20|x - z|^\gamma.$$

The preceding deterministic construction again gives uniform design, the displayed primitives, and the required arm means. The propensity and baseline restrictions have already been verified. The same moment argument applies because

$$|m_{0,\text{const}}(x)| \leq \frac{1}{2}, \quad |m_{0,\text{const}}(x) + \tau_{\text{const}}(x)| \leq 1.$$

Thus $P_{\text{const}} \in \mathcal{M}_v$, and its contrast is everywhere the admissible constant $1/8$. By [Definition 2](#),

$$P_{\text{const}} \in H_0(v).$$

4. Fix $P \in \mathcal{M}_v$, and consider any other observational representation of the same probability law. Denote its continuous primitives and arm kernels by

$$\begin{aligned} \tilde{e}_P, \tilde{m}_{0,P}, \tilde{\tau}_P &\in C([0, 1]), \quad \tilde{m}_{1,P} = \tilde{m}_{0,P} + \tilde{\tau}_P, \\ \tilde{Q}_{a_{\text{arm}},P}(\cdot | x) &\in \mathcal{P}(\mathbb{R}) \quad (a_{\text{arm}} \in \{0, 1\}, x \in [0, 1]), \end{aligned}$$

where $\mathcal{P}(\mathbb{R})$ denotes the Borel probability measures on $\mathbb{R}$. Define the two conditional record kernels by

$$\begin{aligned} K_P(x, da, dy) &= (1 - e_P(x)) \delta_0(da) Q_{0,P}(dy | x) + e_P(x) \delta_1(da) Q_{1,P}(dy | x), \\ \tilde{K}_P(x, da, dy) &= (1 - \tilde{e}_P(x)) \delta_0(da) \tilde{Q}_{0,P}(dy | x) + \tilde{e}_P(x) \delta_1(da) \tilde{Q}_{1,P}(dy | x). \end{aligned}$$

The second representation has covariate marginal λ , because its probability law is P . Their observational factorizations therefore give

$$P(dx, da, dy) = \lambda(dx) K_P(x, da, dy) = \lambda(dx) \tilde{K}_P(x, da, dy).$$

Uniqueness of probability kernels in this factorization implies

$$\tilde{K}_P(x, \cdot) = K_P(x, \cdot) \quad \text{for } \lambda\text{-almost every } x.$$

Evaluating the equality on $\{1\} \times \mathbb{R}$ yields

$$\tilde{e}_P(x) = e_P(x) \quad \text{for } \lambda\text{-almost every } x.$$

Continuous functions agreeing λ -almost everywhere agree throughout $[0, 1]$: a nonzero difference at any point would, by continuity, remain nonzero on a relative neighborhood of positive Lebesgue measure. Hence

$$\tilde{e}_P = e_P \quad \text{pointwise on } [0, 1].$$

For every Borel $E \subseteq \mathbb{R}$, evaluation on the two arm slices now gives, at almost every x ,

$$\begin{aligned}(1 - e_P(x))\tilde{Q}_{0,P}(E | x) &= (1 - e_P(x))Q_{0,P}(E | x), \\ e_P(x)\tilde{Q}_{1,P}(E | x) &= e_P(x)Q_{1,P}(E | x).\end{aligned}$$

Both weights are strictly positive by [Assumption 2](#). Cancelling them proves

$$\tilde{Q}_{0,P}(\cdot | x) = Q_{0,P}(\cdot | x), \quad \tilde{Q}_{1,P}(\cdot | x) = Q_{1,P}(\cdot | x) \quad \text{for } \lambda\text{-almost every } x.$$

The arm-mean version identities consequently imply

$$\begin{aligned}\tilde{m}_{0,P}(x) &= \int y \tilde{Q}_{0,P}(dy | x) = \int y Q_{0,P}(dy | x) = m_{0,P}(x), \\ \tilde{m}_{1,P}(x) &= \int y \tilde{Q}_{1,P}(dy | x) = \int y Q_{1,P}(dy | x) = m_{1,P}(x)\end{aligned} \quad \text{for } \lambda\text{-almost every } x.$$

Continuity upgrades both equalities to pointwise equalities. Subtraction then gives

$$\tilde{\tau}_P = \tilde{m}_{1,P} - \tilde{m}_{0,P} = m_{1,P} - m_{0,P} = \tau_P \quad \text{pointwise on } [0, 1].$$

5. The smoothness restrictions in [Assumptions 4](#) and [5](#), together with $m_{1,P} = m_{0,P} + \tau_P$, give

$$\begin{aligned}|m_{1,P}(x) - m_{1,P}(z)| &= |(m_{0,P}(x) - m_{0,P}(z)) + (\tau_P(x) - \tau_P(z))| \\ &\leq |m_{0,P}(x) - m_{0,P}(z)| + |\tau_P(x) - \tau_P(z)| \\ &\leq 20|x - z|^\beta + 20|x - z|^\gamma \\ &\leq 40|x - z|^{\min\{\beta, \gamma\}}.\end{aligned}$$

The last inequality uses $0 \leq |x - z| \leq 1$ and $\min\{\beta, \gamma\} > 0$. It includes $x = z$, when all displayed increments and powers vanish.

6. In the completion of [Definition 3](#), define

$$Y_{\text{obs}} = (1 - A)Y(0) + AY(1).$$

For any nonnegative Borel function

$$f_{\text{obs}} : [0, 1] \times \{0, 1\} \times \mathbb{R} \longrightarrow [0, \infty],$$

integration over the completion gives

$$\begin{aligned}&\int f_{\text{obs}}(X, A, Y_{\text{obs}}) d\mathbb{Q}_P \\ &= \int_0^1 \int_{\mathbb{R}} \int_{\mathbb{R}} [(1 - e_P(x))f_{\text{obs}}(x, 0, y_0) + e_P(x)f_{\text{obs}}(x, 1, y_1)] Q_{1,P}(dy_1 | x) Q_{0,P}(dy_0 | x) d\lambda(x) \\ &= \int_0^1 \left[(1 - e_P(x)) \int_{\mathbb{R}} f_{\text{obs}}(x, 0, y) Q_{0,P}(dy | x) + e_P(x) \int_{\mathbb{R}} f_{\text{obs}}(x, 1, y) Q_{1,P}(dy | x) \right] d\lambda(x) \\ &= \int f_{\text{obs}} dP.\end{aligned}$$

Indeed, in each arm the unused potential-outcome kernel integrates to one; the last equality is the observational factorization with uniform covariate marginal. Equality for all such functions proves

$$\mathcal{L}_{\mathbb{Q}_P}(X, A, Y_{\text{obs}}) = P.$$

7. Integrating out the probability kernels first gives

$$\mathcal{L}_{\mathbb{Q}_P}(X) = \lambda.$$

The conditional law of the remaining coordinates is the defining product kernel:

$$\begin{aligned} \mathcal{L}_{\mathbb{Q}_P}(A, Y(0), Y(1) \mid X = x) \\ = ((1 - e_P(x))\delta_0 + e_P(x)\delta_1) \otimes Q_{0,P}(\cdot \mid x) \otimes Q_{1,P}(\cdot \mid x) \quad \text{for } \lambda\text{-almost every } x. \end{aligned}$$

Projecting this identity onto treatment and onto the potential-outcome pair yields

$$\begin{aligned} \mathcal{L}_{\mathbb{Q}_P}(A \mid X = x) &= (1 - e_P(x))\delta_0 + e_P(x)\delta_1, \\ \mathcal{L}_{\mathbb{Q}_P}(Y(0), Y(1) \mid X = x) &= Q_{0,P}(\cdot \mid x) \otimes Q_{1,P}(\cdot \mid x). \end{aligned}$$

Thus the joint conditional law equals the product of these two conditional marginals. The conditional-independence criterion gives

$$(Y(0), Y(1)) \perp A \mid X.$$

8. For $y \in \mathbb{R}$, the elementary bound needed for integrability is

$$|y| \leq 1 + |y|^p.$$

If $|y| \leq 1$, this follows from $|y|^p \geq 0$. If $|y| > 1$, the condition $p > 1$ gives

$$|y| \leq |y|^p \leq 1 + |y|^p.$$

Hence [Assumption 8](#) implies, for each $a_{\text{arm}} \in \{0, 1\}$,

$$\int_{\mathbb{R}} |y| Q_{a_{\text{arm}},P}(dy \mid x) \leq 1 + \int_{\mathbb{R}} |y|^p Q_{a_{\text{arm}},P}(dy \mid x) \leq 11 \quad \text{for } \lambda\text{-almost every } x.$$

Integrating the assignment kernel and the unused potential-outcome kernel, both of total mass one, gives

$$\begin{aligned} \int |Y(a_{\text{arm}})| d\mathbb{Q}_P &= \int_0^1 \int_{\mathbb{R}} |y| Q_{a_{\text{arm}},P}(dy \mid x) d\lambda(x) \\ &\leq \int_0^1 11 d\lambda(x) = 11 < \infty. \end{aligned}$$

This proves integrability for $Y(0)$ and $Y(1)$.

9. For each arm, integration against the joint conditional law from [Item 7](#) reduces to the corresponding arm mean, since the other two factors are probability measures. The conditional-expectation formula for an integrable function therefore gives

$$\mathbb{E}_{\mathbb{Q}_P}[Y(a_{\text{arm}}) \mid X] = \int_{\mathbb{R}} y Q_{a_{\text{arm}},P}(dy \mid X) \quad \mathbb{Q}_P\text{-almost surely, } a_{\text{arm}} \in \{0, 1\}.$$

Because X has law λ , the observational arm-mean version identities hold also at X , almost surely under $\mathbb{Q}_P$:

$$\int y Q_{0,P}(dy \mid X) = m_{0,P}(X), \quad \int y Q_{1,P}(dy \mid X) = m_{0,P}(X) + \tau_P(X).$$

Linearity of conditional expectation, justified by the integrability established above, now yields

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}_P}[Y(1) - Y(0) \mid X] &= \mathbb{E}_{\mathbb{Q}_P}[Y(1) \mid X] - \mathbb{E}_{\mathbb{Q}_P}[Y(0) \mid X] \\ &= (m_{0,P}(X) + \tau_P(X)) - m_{0,P}(X) = \tau_P(X)\end{aligned}\quad \mathbb{Q}_P\text{-almost surely.}$$

10. Finally, suppose $d(P) = 0$. The defining integral is nonnegative, so

$$\int_0^1 (\tau_P(x) - \bar{\tau}_P)^2 d\lambda(x) = 0.$$

Its integrand is nonnegative and bounded by one, as established in [Item 2](#). The zero-integral criterion therefore gives

$$(\tau_P(x) - \bar{\tau}_P)^2 = 0 \quad \text{and hence} \quad \tau_P(x) = \bar{\tau}_P \quad \text{for } \lambda\text{-almost every } x.$$

Continuity and the full support of λ upgrade this to

$$\tau_P(x) = \bar{\tau}_P \quad (x \in [0, 1]).$$

The average bound in [Item 2](#) gives

$$-\frac{1}{2} \leq \bar{\tau}_P \leq \frac{1}{2}.$$

Together with $P \in \mathcal{M}_v$, this is exactly the membership condition in [Definition 2](#).

Conversely, if $P \in H_0(v)$, that definition supplies a constant

$$c_{\text{null}} \in [-1/2, 1/2], \quad \tau_P(x) = c_{\text{null}} \quad (x \in [0, 1]).$$

Since λ is a probability measure,

$$\bar{\tau}_P = c_{\text{null}}, \quad d(P) = \left\{ \int_0^1 (c_{\text{null}} - c_{\text{null}})^2 d\lambda(x) \right\}^{1/2} = 0.$$

Thus

$$d(P) = 0 \quad \Longleftrightarrow \quad P \in H_0(v),$$

completing the proof. □

The uniqueness assertion makes every pointwise restriction on the propensity and mean functions a property of the observed law. The uniform design and overlap conditions in [Assumptions 1](#) and [2](#) give the conditional arm distributions a common Lebesgue almost-everywhere interpretation; continuity makes their canonical mean representatives unambiguous across the whole interval. In particular, the treated mean inherits the regularity stated in [Lemma 1](#) from the separate baseline and contrast restrictions.

Version invariance also applies to the potential-outcome completion. As specified in [Definition 3](#), changing either conditional outcome distribution on a Lebesgue-null set leaves $\mathbb{Q}_P$

unchanged. The completion combines the observed arm distributions with conditionally independent assignment and reproduces the original-record marginal. Its integrable potential-outcome difference has conditional mean $\tau_P(X)$, so $d(P)$ measures variation in conditional average treatment effects under this completion. The equivalence between zero distance and membership in [Definition 2](#) connects the population criterion to the pointwise constant-effect restriction.

The two witnesses in [Lemma 1](#) give the testing problem a nonempty null and a positive separation domain for every admissible tuple. The heterogeneous witness attains the same fixed distance d_0 throughout the parameter domain, while the constant-contrast witness permits a nonzero common effect. Together with the upper bound on D_v , these properties give the cap in [Definition 6](#) a finite, positive population scale.

A.1 Mean preservation and the benchmark experiment

The bounded and signed-binary models in [Definitions 7](#) and [8](#) retain the primitive restrictions on assignment and conditional means. Their comparison in [Proposition 1](#) uses two related operations: a model-level conversion specified by conditional arm means, and a record-level randomization available for bounded observations. At the model level, the converted outcome has probabilities

$$\Pr(+1 \mid X = x, A = a) = \frac{1 + m_{a,P}(x)}{2}, \quad \Pr(-1 \mid X = x, A = a) = \frac{1 - m_{a,P}(x)}{2}, \quad a \in \{0, 1\}.$$

These probabilities describe the signed-binary conversion in [Proposition 1](#). It retains the propensity, both conditional arm means, and hence the contrast and its distance from constancy. The same result establishes that the bounded, signed-binary, and original models have equal distance suprema.

For a bounded original record, the conversion can instead use its observed outcome directly. To express this randomization on the full real-outcome test carrier, define the measurable sign map

$$s(y, t) = 2\mathbf{1}\left\{t \leq \frac{1 + \max\{-1, \min\{y, 1\}\}}{2}\right\} - 1, \quad (y, t) \in \mathbb{R} \times [0, 1].$$

For $|y| \leq 1$, a uniform t gives the probabilities $(1 + y)/2$ and $(1 - y)/2$ for the two signs, with conditional mean y . The truncation in this definition specifies the map on the entire original-record carrier. Its restriction to the bounded model is precisely the record-level conversion in [Proposition 1](#).

Write $\mathbf{V} = (V_1, \dots, V_n)$ for independent uniform variables on $[0, 1]$, independent of the sample and of the public seed U in [Definition 4](#). The randomized signed-binary sample is

$$\mathcal{D}_n^s(\mathbf{V}) = ((X_i, A_i, s(Y_i, V_i)))_{i=1}^n.$$

For a test ϕ on the carrier of [Definition 5](#), its integrated rejection-probability map is

$$\tilde{\phi}(\mathcal{D}_n, u) = \int_{[0,1]^n} \phi(((X_i, A_i, s(Y_i, t_i)))_{i=1}^n, u) \lambda^{\otimes n}(d\mathbf{t}), \quad \mathbf{t} = (t_1, \dots, t_n).$$

Thus $\tilde{\phi}$ is a jointly Borel map from

$$([0, 1] \times \{0, 1\} \times \mathbb{R})^n \times [0, 1] \quad \text{to} \quad [0, 1].$$

It uses the original-record experiment and its public seed; the additional sign randomization is averaged into the rejection probability.

For $P \in \mathcal{M}_w^b$, let P^s denote its signed-binary conversion from [Proposition 1](#). The rejection-probability preservation assertion there takes the explicit form

$$R_n(P, \tilde{\phi}) = R_n(P^s, \phi).$$

This identity expresses the statistical role of mean-preserving randomization: rejection probabilities under each bounded law transfer exactly to those under its signed-binary conversion, including laws in the constant-effect null and laws at a specified heterogeneity distance.

[Proposition 1](#) consequently gives the exact benchmark identity

$$B_n^{\text{bin}}(w, r) = B_n^b(w, r), \quad n \geq 2, \quad 0 < r < D_v,$$

for the risks in [Definitions 11](#) and [12](#). With the common distance supremum and the capping conventions in [Definitions 13](#) and [14](#), the same comparison gives equal capped critical radii for every $n \geq 2$. The signed-binary benchmark therefore represents the bounded mean-effect testing problem at the level of attainable distances, minimax risks, and critical radii.

B Bias, covariance, profiling, and attainment

The calibration in [Theorem 1](#) rests on three population quantities: the mean of the original-record score, the covariance of its affine coefficients, and the fluctuation of the cross-block inner product uniformly over candidate constants. This appendix organizes those quantities around the public construction in [Algorithm 1](#). The resulting bounds connect the finite score calculation to the explicit separation guarantee.

B.1 Calibration and population decomposition

The deterministic calibration handle collects the score construction, its affine coefficients, the public bounds, and the profiling operation. Keeping these components together makes their dependence on the sample size and public exponent tuple explicit.

Algorithm 3 (Deterministic calibration handle)

The calibration handle, indexed by a sample size $n \in \mathbb{N}$ and a public exponent tuple $v = (p, \alpha, \beta, \gamma) \in \mathbb{R}^4$, is the ordered collection of the following seven components. Its branch, ranks, cutoffs, ledgers, and public grids are the deterministic selections from n and v in [Algorithm 1](#).

1. The tuning collection

$$(M, K, T_0, (T_j)_{j=1}^L), \quad M = J_c, \quad K = J_f,$$

with the number of increments L specified in [Algorithm 1](#).

2. The block-score family, taking a block index, a real candidate constant c , and a dataset of n original records to a vector in $\mathbb{R}^M$. For $n \geq 4$, it is

$$Z_b(c) = \begin{cases} H_{b,T_0}(c) - U_{b,\Pi_K,T_0}(c), & \text{single-correction branch,} \\ H_{b,T_0}(c) - U_{b,\Pi_M,T_0}(c) - \sum_{j=1}^L U_{b,D_j,T_j}(c), & \text{multiresolution branch,} \end{cases}$$

where the block and kernel conventions are those of [Definitions 20](#) and [21](#) and [Algorithm 1](#). For $n < 4$, $Z_b(c) = 0$.

3. The two affine-coefficient maps, indexed by the block and a binary coefficient selector, with values

$$Z_{b,0} = Z_b(0), \quad Z_{b,A} = Z_b(0) - Z_b(1).$$

Thus, for every real c and every dataset,

$$Z_b(c) = Z_{b,0} - cZ_{b,A}.$$

4. The deterministic ledger triple

$$(\mathbf{b}_{n,v}, \Lambda_{n,v}, h_{n,v}),$$

whose bias, covariance, and rejection-cutoff selections are exactly those of [Algorithm 1](#).

5. The dataset-to-real profiling map

$$\inf_{c \in [-1/2, 1/2]} W(c), \quad W(c) = \langle Z_1(c), Z_2(c) \rangle,$$

using the two-block indexing convention of [Algorithm 1](#).

6. The original-record rejection rule specified in [Algorithm 1](#), namely, for $n \geq 4$,

$$\mathbf{1} \left\{ \inf_{c \in [-1/2, 1/2]} W(c) > h_{n,v} \right\},$$

and the zero rejection rule for $n < 4$.

7. The ordered triple of finite public rank, outcome-cutoff, and rejection-cutoff grids selected in [Algorithm 1](#). The first two grids depend on n , and the third depends on n and v .

The affine coefficient $Z_{b,A}$ records the score's response to the candidate constant, while $Z_{b,0}$ records its outcome contribution. Their joint population behavior is therefore sufficient to describe the entire profiled statistic. The mean handle below retains both coefficients, their singleton and canonical pair projections, and all four combinations of single-record and pair contributions.

Algorithm 4 (Original score mean handle)

The original score mean handle is the ordered pair of multiresolution and single-correction handles associated with the two score members of [Algorithm 3](#). Its parameters are an original-record law P , ranks $J, K, L \in \mathbb{N}$, a main cutoff $T_{\text{main}} \in \mathbb{R}$, and increment cutoffs $T_j^{\text{inc}} \in \mathbb{R}$ indexed by $1 \leq j \leq L$. Each branch retains, in order, a mean map indexed by $c \in \mathbb{R}$, coefficient means, singleton projections, canonical pair projections, four population contractions for each coefficient pair, and a sample statistic indexed by n and c .

The multiresolution branch uses coarse rank $J_c = J$ and fine rank $J_f = 2^L J$; the single-correction branch uses

$$J_c = J, \quad J_f = K.$$

Both branches use T_{main} ; the multiresolution branch also uses the increment cutoffs T_j^{inc} , $1 \leq j \leq L$. Define the projected propensity and propensity weight by

$$e_K = \Pi_{J_f} e_P, \quad w_K(x) = e_P(x) \{1 - e_K(x)\}.$$

The retained mean map is

$$\int_0^1 F_{J_c}(x) [w_K(x) \{\tau_P(x) - c\} + \{e_P(x) - e_K(x)\} m_{0,P}(x)] dx$$

plus the exact multiresolution or single-correction clipping-tail remainder, respectively, specified in [Lemma 2](#).

This retained map equals the population expectation of the corresponding corrected block score for every $c \in \mathbb{R}$ and either evaluation block under the following conditions:

- **(Model.)** The public tuple v is admissible and $P \in \mathcal{M}_v$, as specified in [Definition 1](#).
- **(Sample and coarse rank.)** $n \geq 4$ and $J > 0$.
- **(Main cutoff.)** $T_{\text{main}} \geq 1$.
- **(Multiresolution cutoffs.)** In the multiresolution branch, $T_j^{\text{inc}} \geq 1$ for every integer $1 \leq j \leq L$.
- **(Single-correction refinement.)** In the single-correction branch, $K > 0$ and $J \mid K$.

For each affine coefficient $r \in \{0, 1\}$, let h_r and g_r be the selected branch's raw single-record and symmetric pair kernels from [Algorithm 3](#). All expectations below use independent original records with law P . Define the centered pair-kernel singleton projection by

$$g_{r,1}(o) = \mathbb{E}[g_r(o, O')] - \mathbb{E}[g_r(O, O')].$$

The retained coefficient means, singleton projections, and canonical pair projections are, respectively,

$$\begin{aligned} & \mathbb{E}[h_r(O)] + \mathbb{E}[g_r(O, O')], \\ & h_r(o) - \mathbb{E}[h_r(O)] + 2g_{r,1}(o), \\ & g_r(o, z) - \mathbb{E}[g_r(O, O')] - g_{r,1}(o) - g_{r,1}(z). \end{aligned}$$

For every $r, r' \in \{0, 1\}$, the retained ordered quadruple of population contractions is defined by

$$\begin{aligned} h_r \times h_{r'} &= \mathbb{E}\langle h_r(O_1), h_{r'}(O_2) \rangle, \\ h_r \times g_{r'} &= \mathbb{E}\langle h_r(O_1), g_{r'}(O_2, O_3) \rangle, \\ g_r \times h_{r'} &= \mathbb{E}\langle g_r(O_1, O_2), h_{r'}(O_3) \rangle, \\ g_r \times g_{r'} &= \mathbb{E}\langle g_r(O_1, O_2), g_{r'}(O_3, O_4) \rangle, \end{aligned}$$

where O_1, O_2, O_3, O_4 are independent with law P .

For each candidate constant $c \in \mathbb{R}$, set

$$h_c = h_0 - ch_1, \quad g_c = g_0 - cg_1.$$

For $n \geq 4$, use the block size $s_n = \lfloor n/2 \rfloor$ and the two deterministic blocks $\mathcal{I}_1, \mathcal{I}_2$ of [Definition 20](#). Define

$$Z_b(c) = \frac{1}{s_n} \sum_{i \in \mathcal{I}_b} h_c(O_i) + \frac{1}{s_n(s_n - 1)} \sum_{\substack{i, j \in \mathcal{I}_b \\ i \neq j}} g_c(O_i, O_j), \quad b \in \{1, 2\}.$$

The retained sample output is the single function $c \mapsto W(c)$, given by

$$\begin{aligned} W(c) &= \langle Z_1(c), Z_2(c) \rangle \\ &= \frac{1}{s_n^2} \sum_{\substack{i \in \mathcal{I}_1 \\ j \in \mathcal{I}_2}} \langle h_c(O_i), h_c(O_j) \rangle \\ &\quad + \frac{1}{s_n^2(s_n - 1)} \sum_{\substack{i \in \mathcal{I}_1 \\ j, k \in \mathcal{I}_2, j \neq k}} \langle h_c(O_i), g_c(O_j, O_k) \rangle \\ &\quad + \frac{1}{s_n^2(s_n - 1)} \sum_{\substack{i, j \in \mathcal{I}_1, i \neq j \\ k \in \mathcal{I}_2}} \langle g_c(O_i, O_j), h_c(O_k) \rangle \\ &\quad + \frac{1}{s_n^2(s_n - 1)^2} \sum_{\substack{i, j \in \mathcal{I}_1, i \neq j \\ k, l \in \mathcal{I}_2, k \neq l}} \langle g_c(O_i, O_j), g_c(O_k, O_l) \rangle. \end{aligned}$$

For $n < 4$, its defining value is

$$W(c) = 0.$$

The two mixed affine-coefficient contributions are retained through their sum in $W(c)$.

The conventions in [Algorithms 3](#) and [4](#) use the same one-based blocks $\mathcal{I}_1, \mathcal{I}_2$. In the mean calculation, T_{main} denotes the procedure's main cutoff T_0 , while T_j^{inc} denotes its increment cutoff T_j for $1 \leq j \leq L$. Thus the two cutoff families and the two block scores have distinct names without a positional shift. At the public selections, the coarse rank is $J = M$ and the multiresolution fine rank is $2^L J = K$.

For clarity about the symmetric pair convention, write $o = (x, a, y)$ and $z = (x', a', y')$. For a correction function G and cutoff T , define its signed, symmetrized contribution by

$$k_{G, T, c}(o, z) := -\frac{1}{2} \left[F_M(x) a G(x, x') \{ \ell_T(y') - ca' \} + F_M(x') a' G(x', x) \{ \ell_T(y) - ca \} \right].$$

The single-record contribution is

$$h_c(o) := F_M(x)a\{\ell_{T_0}(y) - c\}.$$

In the single-correction branch, $g_c := k_{\Pi_K, T_0, c}$. In the multiresolution branch, using the increment indices of [Algorithm 1](#),

$$g_c := k_{\Pi_M, T_0, c} + \sum_{j=1}^L k_{D_j, T_j, c}.$$

The affine coefficients in [Algorithm 4](#) are thus $h_0 = h_c|_{c=0}$, $h_1 = h_c|_{c=0} - h_c|_{c=1}$, and the corresponding two coefficients of g_c . Symmetrization preserves the ordered-pair average because that average contains both orientations of every distinct pair.

The four contractions in [Algorithm 4](#) retain the single-single, single-pair, pair-single, and pair-pair contributions separately. Their independent-record arguments match the four finite sums defining $W(c)$. Both orientations of the mixed contribution are retained, including their dependence on the two affine coefficients.

B.2 Original-mean bias

The population calculation separates a weighted effect contrast, a baseline approximation term, and clipping tails. To specify the latter explicitly, let λ denote Lebesgue probability measure on $[0, 1]$, and define the conditional clipping discrepancy and its unconditional conditional mean by

$$\delta_{a,T}(x) := \mathbb{E}_P[\ell_T(Y) - Y \mid X = x, A = a], \quad r_T(x) := e_P(x)\delta_{1,T}(x) + \{1 - e_P(x)\}\delta_{0,T}(x).$$

These quantities are defined up to λ -almost-everywhere equality; their integrated contributions use that convention. Under the mean handle's conditions, the clipping-tail remainders for the two branches are

$$\begin{aligned} \mathcal{T}_{\text{single}} &:= \int_0^1 F_M(x) [e_P(x)\delta_{1,T_0}(x) - e_K(x)r_{T_0}(x)] d\lambda(x), \\ \mathcal{T}_{\text{multi}} &:= \int_0^1 F_M(x) \left[e_P(x)\delta_{1,T_0}(x) - (\Pi_M e_P)(x)r_{T_0}(x) - \sum_{j=1}^L (D_j e_P)(x)r_{T_j}(x) \right] d\lambda(x). \end{aligned}$$

Here $(D_j e_P)(x) := \int_0^1 D_j(x, z)e_P(z) d\lambda(z)$, and the increment indices follow [Algorithm 1](#). Each remainder is independent of the candidate constant.

The retained mean in [Algorithm 4](#) therefore identifies the baseline approximation contribution through $(e_P - e_K)m_{0,P}$, while the clipping remainder records the outcome-tail contribution at each selected resolution. The public bound B in [Algorithm 1](#) aggregates these terms. The next statement controls that bound uniformly over the candidate constants and relates the score mean to heterogeneity.

Lemma 2. [lem:original-score-mean-ledger] (Original score mean bounds) *Let $v = (p, \alpha, \beta, \gamma)$ be an admissible tuple and $P \in \mathcal{M}_v$, as specified in [Definition 1](#), and let $n \geq 4$ be an integer. Use the explicit score ledger's coarse rank M , fine rank K , corrected block scores $Z_b(c)$, and deterministic bias bound B .*

Let λ denote Lebesgue probability measure on $[0, 1]$, and define the projected propensity and its associated weight by

$$e_K(x) = (\Pi_K e_P)(x), \quad w_K(x) = e_P(x)(1 - e_K(x)).$$

Define the population heterogeneity distance by

$$d(P) = \left(\int_0^1 \left[\tau_P(x) - \int_0^1 \tau_P(z) d\lambda(z) \right]^2 d\lambda(x) \right)^{1/2}.$$

For every block index b and every $c \in \mathbb{R}$, $Z_b(c)$ is integrable under the original independent-record law $P^{\otimes n}$. Its population mean is

$$\theta_P(c) = \mathbb{E}_{P^{\otimes n}}[Z_b(c)].$$

For every block index b and every $c \in \mathbb{R}$ with $|c| \leq \frac{1}{2}$,

$$\left\| \theta_P(c) - \int_0^1 w_K(x) (\tau_P(x) - c) F_M(x) d\lambda(x) \right\|_2 \leq B, \quad \|\theta_P(c)\|_2 \geq \frac{3}{16}d(P) - 16M^{-\gamma} - B.$$

Furthermore, if $P \in H_0(v)$ as defined in [Definition 2](#), then for every block index b and every $c_0 \in \mathbb{R}$ satisfying $\tau_P(x) = c_0$ for all $x \in [0, 1]$,

$$\|\theta_P(c_0)\|_2 \leq B.$$

Proof of [Lemma 2](#). 1. Fix a block index b . The clipped outcomes, treatment indicators, histogram features, and histogram kernels are bounded measurable functions for every fixed choice of the public ranks and cutoffs. In each branch of [Algorithm 1](#), $Z_b(c)$ is a finite linear combination of products of these functions. Consequently, for every $c \in \mathbb{R}$,

$$\int \|Z_b(c)\|_2 dP^{\otimes n} < \infty, \quad \theta_P(c) = \int Z_b(c) dP^{\otimes n}.$$

The mean calculations below have the same right-hand side for both blocks.

2. The ranks in [Algorithm 1](#) are positive powers of two, and the definition of the fine rank gives

$$0 < M \leq K, \quad M \mid K.$$

Indeed, the dyadic exponent of K is at least that of M , so their quotient is an integer power of two. The cutoffs satisfy

$$T_0 \geq 1, \quad T_j \geq 1 \quad (1 \leq j \leq L)$$

in the branch where the increment cutoffs occur.

We record the histogram identities needed for the mean calculation. For every positive integer J and measurable integrable function $f : [0, 1] \rightarrow \mathbb{R}$, define

$$\text{coef}_J(f) = \int_0^1 f(x) F_J(x) d\lambda(x) \in \mathbb{R}^J.$$

The cells in [Definition 19](#) partition $[0, 1]$, with the stipulated convention at the right end-point, and each has measure $1/J$. Thus, for $x \in I_{j,J}$,

$$(\Pi_J f)(x) = J \int_{I_{j,J}} f(z) d\lambda(z), \quad (\text{coef}_J(f))_j = \sqrt{J} \int_{I_{j,J}} f(z) d\lambda(z).$$

In particular, if $\mathfrak{c} \in [0, \infty)$ and $|f| \leq \mathfrak{c}$ almost everywhere, then

$$\left| (\text{coef}_J(f))_j \right| \leq \frac{\mathfrak{c}}{\sqrt{J}}, \quad \|\text{coef}_J(f)\|_2^2 \leq \sum_{j=1}^J \frac{\mathfrak{c}^2}{J} = \mathfrak{c}^2.$$

Hence

$$\|\text{coef}_J(f)\|_2 \leq \mathfrak{c}.$$

Cell averaging also preserves almost-everywhere bounds and constants.

For a function f with Hölder exponent $\mathfrak{s} \in (0, 1]$ and Hölder constant at most 20, the diameter of a cell gives, for $x \in I_{j,J}$,

$$\begin{aligned} |f(x) - (\Pi_J f)(x)| &\leq J \int_{I_{j,J}} |f(x) - f(z)| d\lambda(z) \\ &\leq 20J \int_{I_{j,J}} |x - z|^\mathfrak{s} d\lambda(z) \leq 20J^{-\mathfrak{s}}. \end{aligned}$$

If $M \mid J$, the fine cells partition each coarse cell. Averaging their integrals therefore yields

$$\Pi_M(\Pi_J f) = \Pi_M f, \quad \Pi_J(\Pi_J f) = \Pi_J f.$$

Moreover, the support of $\Pi_J(x, z)$ places x and z in the same coarse cell, so

$$\Pi_J(x, z)F_M(x) = \Pi_J(z, x)F_M(z).$$

For a bounded measurable function $\varphi_{\text{bd}} : [0, 1] \rightarrow \mathbb{R}$ and a measurable integrable function $\varphi_{\text{int}} : [0, 1] \rightarrow \mathbb{R}$, Fubini's theorem consequently gives

$$\begin{aligned} \text{coef}_M(\varphi_{\text{bd}} \Pi_J \varphi_{\text{int}}) &= \int_0^1 \int_0^1 \varphi_{\text{bd}}(x) \Pi_J(x, z) \varphi_{\text{int}}(z) F_M(x) d\lambda(z) d\lambda(x) \\ &= \int_0^1 \varphi_{\text{int}}(z) F_M(z) \left[\int_0^1 \Pi_J(z, x) \varphi_{\text{bd}}(x) d\lambda(x) \right] d\lambda(z) \\ &= \text{coef}_M((\Pi_J \varphi_{\text{bd}}) \varphi_{\text{int}}). \end{aligned}$$

The double integral is absolutely integrable because φ_{bd} , the kernel, and the features are bounded.

- For every positive integer J divisible by M and every $c \in \mathbb{R}$, define the original mean vector

$$\theta_{P,J}^{\text{orig}}(c) = \text{coef}_M(e_P(1 - \Pi_J e_P)(\tau_P - c) + (e_P - \Pi_J e_P)m_{0,P}).$$

Also define the weighted effect vector

$$\theta_P^{\text{wt}}(c) = \text{coef}_M(w_K(\tau_P - c)).$$

Their difference at the fine rank is

$$\theta_{P,K}^{\text{orig}}(c) - \theta_P^{\text{wt}}(c) = \text{coef}_M((e_P - e_K)m_{0,P}).$$

Idempotence and the self-adjointness identity established above imply

$$\Pi_K(e_P - e_K) = 0, \quad \text{coef}_M((e_P - e_K)\Pi_K m_{0,P}) = 0.$$

Subtracting the latter zero vector gives the residual identity

$$\text{coef}_M((e_P - e_K)m_{0,P}) = \text{coef}_M((e_P - e_K)(m_{0,P} - \Pi_K m_{0,P})).$$

The smoothness restrictions in [Assumptions 3](#) and [4](#), together with the Hölder estimate above, give

$$|e_P - e_K| \leq 20K^{-\alpha}, \quad |m_{0,P} - \Pi_K m_{0,P}| \leq 20K^{-\beta}.$$

Since $S = \alpha + \beta$ in [Algorithm 1](#),

$$|(e_P - e_K)(m_{0,P} - \Pi_K m_{0,P})| \leq 400K^{-S}.$$

Applying the coefficient norm bound proves, for every real c ,

$$\|\theta_{P,K}^{\text{orig}}(c) - \theta_P^{\text{wt}}(c)\|_2 \leq 400K^{-S}.$$

4. First consider the single-correction branch of [Algorithm 1](#). We derive the mean identity for any positive J divisible by M , so that it can also be used at successive ranks in the other branch.

Use the clipping function ℓ_T and the score formulas defined in [Definition 21](#). For $T \geq 1$, $x \in [0, 1]$, and $\mathfrak{a} \in \{0, 1\}$, define the arm-specific and pooled clipping remainders by

$$r_T^{(\mathfrak{a})}(x) = \int_{\mathbb{R}} (y - \ell_T(y)) Q_{\mathfrak{a},P}(dy \mid x), \quad r_T(x) = e_P(x)r_T^{(1)}(x) + (1 - e_P(x))r_T^{(0)}(x).$$

At a covariate where the integrand in an arm-specific integral is nonintegrable, assign that integral the value zero. The moment restriction makes these integrands integrable for almost every covariate.

Indeed, admissibility gives $p > 1$, and

$$|y| \leq 1 + |y|^p.$$

Thus [Assumption 8](#) supplies integrability of the original outcome in each conditional arm almost everywhere. For $|y| \leq T$, the clipping remainder vanishes; for $|y| > T$,

$$|y - \ell_T(y)| \leq |y| = |y|^p |y|^{1-p} \leq |y|^p T^{1-p}.$$

Integration therefore gives, almost everywhere,

$$|r_T^{(\mathfrak{a})}(x)| \leq T^{1-p} \int_{\mathbb{R}} |y|^p Q_{\mathfrak{a},P}(dy \mid x) \leq 10T^{1-p}.$$

Taking the propensity-weighted average yields

$$|r_T(x)| \leq e_P(x) 10T^{1-p} + (1 - e_P(x)) 10T^{1-p} = 10T^{1-p}.$$

For $c \in \mathbb{R}$, define

$$f_{P,c}^{\text{mark}}(x) = m_{0,P}(x) + e_P(x)(\tau_P(x) - c).$$

The conditional arm means specified in [Definition 1](#) give, almost everywhere,

$$\mathbb{E}_P[\ell_T(Y) - cA \mid X = x] = f_{P,c}^{\text{mark}}(x) - r_T(x),$$

$$\mathbb{E}_P[A(\ell_T(Y) - c) \mid X = x] = e_P(x)(m_{0,P}(x) + \tau_P(x) - c - r_T^{(1)}(x)).$$

By [Definition 20](#), $s_n \geq 2$, and each block contains s_n records and $s_n(s_n - 1)$ ordered pairs of distinct records. Identical distribution cancels the single-record normalization, and independence cancels the ordered-pair normalization. Conditioning and [Assumption 1](#) consequently give

$$\begin{aligned} \mathbb{E}_{P^{\otimes n}}[H_{b,T}(c)] &= \text{coef}_M(e_P(m_{0,P} + \tau_P - c)) - \text{coef}_M(e_P r_T^{(1)}), \\ \mathbb{E}_{P^{\otimes n}}[U_{b,\Pi_J,T}(c)] &= \text{coef}_M(e_P \Pi_J f_{P,c}^{\text{mark}}) - \text{coef}_M(e_P \Pi_J r_T). \end{aligned}$$

Self-adjointness moves the projection in the first correction term:

$$\text{coef}_M(e_P \Pi_J f_{P,c}^{\text{mark}}) = \text{coef}_M((\Pi_J e_P) f_{P,c}^{\text{mark}}).$$

The resulting original-outcome difference is exactly

$$\begin{aligned} &\text{coef}_M(e_P(m_{0,P} + \tau_P - c) - (\Pi_J e_P) f_{P,c}^{\text{mark}}) \\ &= \text{coef}_M(e_P(1 - \Pi_J e_P)(\tau_P - c) + (e_P - \Pi_J e_P)m_{0,P}) = \theta_{P,J}^{\text{orig}}(c). \end{aligned}$$

Define, for these J and T ,

$$\theta_{P,J}^{\text{tail}}(T) = -\text{coef}_M(e_P r_T^{(1)}) + \text{coef}_M(e_P \Pi_J r_T).$$

We have proved the exact identity

$$\mathbb{E}_{P^{\otimes n}}[H_{b,T}(c) - U_{b,\Pi_J,T}(c)] = \theta_{P,J}^{\text{orig}}(c) + \theta_{P,J}^{\text{tail}}(T).$$

Since $0 \leq e_P \leq 1$, and cell averaging preserves the almost-everywhere bound on r_T ,

$$|e_P r_T^{(1)}| \leq 10T^{1-p} \quad \text{almost everywhere,} \quad |e_P \Pi_J r_T| \leq 10T^{1-p} \quad \text{everywhere.}$$

The coefficient norm bound and the triangle inequality give

$$\begin{aligned} \|\theta_{P,J}^{\text{tail}}(T)\|_2 &\leq \|\text{coef}_M(e_P r_T^{(1)})\|_2 + \|\text{coef}_M(e_P \Pi_J r_T)\|_2 \\ &\leq 10T^{1-p} + 10T^{1-p} = 20T^{1-p}. \end{aligned}$$

In the single-correction branch, take $J = K$ and $T = T_0$. Combining this estimate with the original-mean error bound gives

$$\begin{aligned} \|\theta_P(c) - \theta_P^{\text{wt}}(c)\|_2 &\leq \|\theta_{P,K}^{\text{orig}}(c) - \theta_P^{\text{wt}}(c)\|_2 + \|\theta_{P,K}^{\text{tail}}(T_0)\|_2 \\ &\leq 400K^{-S} + 20T_0^{1-p} = B. \end{aligned}$$

This calculation applies to every $c \in \mathbb{R}$.

5. Now consider the multiresolution branch. Because M and K are ordered powers of two, the increment chain in [Algorithm 1](#) satisfies

$$L = \log_2(K/M) \in \mathbb{N}_0, \quad R_0 = M, \quad R_j = 2R_{j-1}, \quad R_L = 2^L M = K.$$

For a measurable integrable $f : [0, 1] \rightarrow \mathbb{R}$, the action of its increment kernel is

$$(D_j f)(x) = \int_0^1 D_j(x, z) f(z) d\lambda(z) = (\Pi_{R_j} f)(x) - (\Pi_{R_{j-1}} f)(x), \quad 1 \leq j \leq L.$$

Linearity of the ordered-pair score gives

$$U_{b, D_j, T_j}(c) = U_{b, \Pi_{R_j}, T_j}(c) - U_{b, \Pi_{R_{j-1}}, T_j}(c).$$

The exact single-correction identity established above applies at both ranks. Also, subtracting the definitions of the two tail vectors gives

$$\theta_{P, R_j}^{\text{tail}}(T_j) - \theta_{P, R_{j-1}}^{\text{tail}}(T_j) = \text{coef}_M(e_P D_j r_{T_j}).$$

Consequently,

$$\mathbb{E}_{P^{\otimes n}}[U_{b, D_j, T_j}(c)] = \theta_{P, R_j}^{\text{orig}}(c) - \theta_{P, R_{j-1}}^{\text{orig}}(c) - \text{coef}_M(e_P D_j r_{T_j}).$$

The original means telescope:

$$\sum_{j=1}^L \left(\theta_{P, R_{j-1}}^{\text{orig}}(c) - \theta_{P, R_j}^{\text{orig}}(c) \right) = \theta_{P, M}^{\text{orig}}(c) - \theta_{P, K}^{\text{orig}}(c).$$

Taking the expectation of the multiresolution score therefore yields

$$\theta_P(c) = \theta_{P, K}^{\text{orig}}(c) + \theta_{P, M}^{\text{tail}}(T_0) + \sum_{j=1}^L \text{coef}_M(e_P D_j r_{T_j}).$$

For each increment, self-adjointness at its two ranks gives

$$\text{coef}_M(e_P D_j r_{T_j}) = \text{coef}_M((D_j e_P) r_{T_j}).$$

The propensity Hölder estimate and $R_j = 2R_{j-1}$ imply

$$\begin{aligned} |D_j e_P| &\leq |e_P - \Pi_{R_j} e_P| + |e_P - \Pi_{R_{j-1}} e_P| \\ &\leq 20R_j^{-\alpha} + 20R_{j-1}^{-\alpha} \leq 40R_{j-1}^{-\alpha} = a_j. \end{aligned}$$

Together with the pooled tail bound, this gives

$$|(D_j e_P) r_{T_j}| \leq 10a_j T_j^{1-p} \quad \text{almost everywhere,} \quad \|\text{coef}_M(e_P D_j r_{T_j})\|_2 \leq 10a_j T_j^{1-p}.$$

Hence the complete tail vector satisfies

$$\left\| \theta_{P, M}^{\text{tail}}(T_0) + \sum_{j=1}^L \text{coef}_M(e_P D_j r_{T_j}) \right\|_2 \leq 20T_0^{1-p} + 10 \sum_{j=1}^L a_j T_j^{1-p}.$$

Combining this with the original-mean error bound proves

$$\|\theta_P(c) - \theta_P^{\text{wt}}(c)\|_2 \leq 400K^{-S} + 20T_0^{1-p} + 10 \sum_{j=1}^L a_j T_j^{1-p} = B.$$

When $L = 0$, $K = M$, both increment sums are empty, and the same identities reduce to the single correction at rank M . Thus, in either branch, we have established for every real c

$$\left\| \theta_P(c) - \int_0^1 w_K(x)(\tau_P(x) - c)F_M(x) d\lambda(x) \right\|_2 \leq B.$$

6. Fix now $c \in \mathbb{R}$ with $|c| \leq 1/2$, and define

$$f_{P,c}^{\text{cell}}(x) = (\Pi_M \tau_P)(x) - c, \quad \theta_P^{\text{cell}}(c) = \text{coef}_M(w_K f_{P,c}^{\text{cell}}).$$

By [Assumption 2](#) and preservation of bounds under cell averaging,

$$\frac{1}{4} \leq e_P(x), e_K(x) \leq \frac{3}{4}, \quad 0 \leq w_K(x) = e_P(x)(1 - e_K(x)) \leq \frac{9}{16}.$$

Since e_K is constant on each fine cell,

$$(\Pi_K w_K)(x) = e_K(x)(1 - e_K(x)).$$

For every real number in $[1/4, 3/4]$, the product with its complement lies in $[3/16, 1/4]$. Indeed,

$$e_K(1 - e_K) - \frac{3}{16} = \left(e_K - \frac{1}{4}\right) \left(\frac{3}{4} - e_K\right) \geq 0, \quad \frac{1}{4} - e_K(1 - e_K) = \left(e_K - \frac{1}{2}\right)^2 \geq 0.$$

Projection nesting and preservation of bounds now yield

$$\Pi_M w_K = \Pi_M(\Pi_K w_K), \quad \frac{3}{16} \leq (\Pi_M w_K)(x) \leq \frac{1}{4}.$$

Define the coarse-cell average weights by

$$\bar{w}_{j,M} = M \int_{I_{j,M}} w_K(x) d\lambda(x), \quad j \in \{1, \dots, M\}.$$

The function $f_{P,c}^{\text{cell}}$ is constant on each coarse cell, so the coordinate formula gives

$$(\theta_P^{\text{cell}}(c))_j = \bar{w}_{j,M} (\text{coef}_M(f_{P,c}^{\text{cell}}))_j, \quad \bar{w}_{j,M} \geq \frac{3}{16}.$$

Squaring and summing proves

$$\|\theta_P^{\text{cell}}(c)\|_2^2 \geq \left(\frac{3}{16}\right)^2 \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2^2,$$

and therefore

$$\|\theta_P^{\text{cell}}(c)\|_2 \geq \frac{3}{16} \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2.$$

The effect cap in [Assumption 7](#) also gives the bounds used to integrate these vectors:

$$|\Pi_M \tau_P| \leq \frac{1}{2}, \quad |f_{P,c}^{\text{cell}}| \leq 1, \quad |\tau_P - c| \leq 1.$$

7. To compare the unweighted coarse coefficients with heterogeneity, define

$$f_{P,c}^{\text{raw}}(x) = \tau_P(x) - c, \quad \bar{\tau}_P = \int_0^1 \tau_P(x) d\lambda(x).$$

Preservation of constants and cell integrals gives

$$\Pi_M f_{P,c}^{\text{raw}} = f_{P,c}^{\text{cell}}, \quad \text{coef}_M(f_{P,c}^{\text{raw}}) = \text{coef}_M(f_{P,c}^{\text{cell}}).$$

The histogram coordinate formula reconstructs the projection:

$$(\Pi_M f_{P,c}^{\text{raw}})(x) = \langle \text{coef}_M(f_{P,c}^{\text{raw}}), F_M(x) \rangle.$$

Integrating this identity against $f_{P,c}^{\text{raw}}$ gives

$$\int_0^1 f_{P,c}^{\text{raw}}(x) (\Pi_M f_{P,c}^{\text{raw}})(x) d\lambda(x) = \|\text{coef}_M(f_{P,c}^{\text{raw}})\|_2^2.$$

Disjointness of the cells and their measure $1/M$ likewise give

$$\int_0^1 (\Pi_M f_{P,c}^{\text{raw}})(x)^2 d\lambda(x) = \sum_{j=1}^M (\text{coef}_M(f_{P,c}^{\text{raw}}))_j^2 = \|\text{coef}_M(f_{P,c}^{\text{raw}})\|_2^2.$$

Expanding the squared projection residual consequently proves the energy identity

$$\int_0^1 f_{P,c}^{\text{raw}}(x)^2 d\lambda(x) = \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2^2 + \int_0^1 (f_{P,c}^{\text{raw}}(x) - (\Pi_M f_{P,c}^{\text{raw}})(x))^2 d\lambda(x).$$

The Hölder estimate applied using [Assumption 5](#) gives

$$|f_{P,c}^{\text{raw}}(x) - (\Pi_M f_{P,c}^{\text{raw}})(x)| = |\tau_P(x) - (\Pi_M \tau_P)(x)| \leq 20M^{-\gamma}.$$

Since λ is a probability measure,

$$\int_0^1 (f_{P,c}^{\text{raw}} - \Pi_M f_{P,c}^{\text{raw}})^2 d\lambda \leq (20M^{-\gamma})^2.$$

On the other hand, expansion around the mean gives

$$\begin{aligned} \int_0^1 (\tau_P(x) - c)^2 d\lambda(x) &= \int_0^1 \tau_P(x)^2 d\lambda(x) - 2c\bar{\tau}_P + c^2 \\ &= d(P)^2 + (\bar{\tau}_P - c)^2 \geq d(P)^2. \end{aligned}$$

Combining these displays yields

$$d(P)^2 \leq \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2^2 + (20M^{-\gamma})^2 \leq \left(\|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2 + 20M^{-\gamma} \right)^2.$$

Both sides inside the last square are nonnegative, so

$$d(P) - 20M^{-\gamma} \leq \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2.$$

8. The difference between the weighted effect vector and its coarse-effect approximation is

$$\theta_P^{\text{wt}}(c) - \theta_P^{\text{cell}}(c) = \text{coef}_M(w_K(\tau_P - \Pi_M \tau_P)).$$

The weight and Hölder bounds imply

$$|w_K(x)(\tau_P(x) - (\Pi_M \tau_P)(x))| \leq \frac{9}{16} 20M^{-\gamma} = \frac{45}{4} M^{-\gamma}.$$

Thus

$$\|\theta_P^{\text{wt}}(c) - \theta_P^{\text{cell}}(c)\|_2 \leq \frac{45}{4} M^{-\gamma}.$$

The triangle inequality, together with the diagonal and distance bounds, gives

$$\begin{aligned} \frac{3}{16}(d(P) - 20M^{-\gamma}) &\leq \frac{3}{16} \|\text{coef}_M(f_{P,c}^{\text{cell}})\|_2 \\ &\leq \|\theta_P^{\text{cell}}(c)\|_2 \\ &\leq \|\theta_P^{\text{wt}}(c)\|_2 + \frac{45}{4} M^{-\gamma}. \end{aligned}$$

Since

$$\frac{3}{16} 20 + \frac{45}{4} = 15,$$

we obtain the weighted separation estimate

$$\frac{3}{16} d(P) - 15M^{-\gamma} \leq \|\theta_P^{\text{wt}}(c)\|_2.$$

9. To transfer this estimate to the score mean, write

$$\theta_P^{\text{wt}}(c) = \theta_P(c) - (\theta_P(c) - \theta_P^{\text{wt}}(c)).$$

The triangle inequality and the established mean-error bound give

$$\|\theta_P^{\text{wt}}(c)\|_2 \leq \|\theta_P(c)\|_2 + \|\theta_P(c) - \theta_P^{\text{wt}}(c)\|_2 \leq \|\theta_P(c)\|_2 + B.$$

Consequently,

$$\|\theta_P(c)\|_2 \geq \frac{3}{16} d(P) - 15M^{-\gamma} - B \geq \frac{3}{16} d(P) - 16M^{-\gamma} - B,$$

where the last inequality uses $M^{-\gamma} \geq 0$.

10. Finally, let $P \in H_0(v)$, as specified in [Definition 2](#), and let $c_0 \in \mathbb{R}$ satisfy $\tau_P(x) = c_0$ for every $x \in [0, 1]$. Then

$$\theta_P^{\text{wt}}(c_0) = \int_0^1 w_K(x)(\tau_P(x) - c_0) F_M(x) d\lambda(x) = 0.$$

The mean-error bound was established for every real candidate constant. Applying it at c_0 therefore gives

$$\|\theta_P(c_0)\|_2 = \|\theta_P(c_0) - \theta_P^{\text{wt}}(c_0)\|_2 \leq B.$$

All calculations apply to either block, which completes the proof. □

The mean bounds supply the population distinction used by the test. Under a constant effect, evaluating at that constant gives a score mean of norm at most B . Under heterogeneity, every admissible candidate constant has a score mean bounded below by a multiple of $d(P)$, after accounting for coarse approximation and original-mean bias. Uniformity in the candidate constant is what makes these bounds compatible with profiling.

B.3 Singleton, pair, and cross-level covariance bounds

The singleton projection in [Algorithm 4](#) combines the centered single-record contribution with twice the pair-kernel singleton projection. Its canonical pair projection retains the component centered with respect to either record. These two components have separate public bounds, L_1 and L_2 , in [Algorithm 1](#).

For the multiresolution construction, both bounds aggregate the main correction and every retained increment. In particular, L_1 contains the sum of the increment-specific smoothness and clipping contributions, while L_2 contains the sum of their rank-weighted second-moment bounds. The squared sums in

$$\Lambda_{n,v} = \frac{L_1^2}{s} + \frac{2L_2^2}{s(s-1)}$$

retain the cross-level contributions within each component. The covariance statement below controls both affine coefficients and their covariance in arbitrary, potentially different, directions.

Lemma 3. [lem:original-score-covariance-ledger] (Original score covariance bounds) *Let $v = (p, \alpha, \beta, \gamma)$ be an admissible tuple of moment and smoothness exponents, and let $P \in \mathcal{M}_v$, as specified in [Definition 1](#). Let $n \geq 2$ be an integer. Write the affine block score as*

$$Z_b(c) = Z_{b,0} - cZ_{b,A}.$$

Under the independent-record sample law $P^{\otimes n}$, for every evaluation block b and every $u \in \mathbb{R}^M$, both $u^T Z_{b,0}$ and $u^T Z_{b,A}$ are square integrable and satisfy

$$\text{Var}_{P^{\otimes n}}(u^T Z_{b,0}) \leq \Lambda_{n,v} \|u\|_2^2, \quad \text{Var}_{P^{\otimes n}}(u^T Z_{b,A}) \leq \Lambda_{n,v} \|u\|_2^2.$$

Moreover, for every evaluation block b and every $u, z \in \mathbb{R}^M$,

$$|\text{Cov}_{P^{\otimes n}}(u^T Z_{b,0}, z^T Z_{b,A})| \leq \Lambda_{n,v} \|u\|_2 \|z\|_2.$$

Proof of [Lemma 3](#). 1. Suppose first that $n < 4$. Since $n \geq 2$, the zero-score convention in [Algorithms 1](#) and [3](#) gives

$$Z_{b,0} = Z_{b,A} = 0, \quad \Lambda_{n,v} = 0.$$

Both scalar coefficients are square integrable, and their variances and cross-covariance vanish. This proves all the asserted bounds in this case.

2. Henceforth $n \geq 4$. Use the ranks, cutoffs, and ledgers of [Algorithm 1](#), and the blocks of [Definition 20](#). In particular,

$$s = s_n = \lfloor n/2 \rfloor \geq 2, \quad M > 0, \quad K > 0, \quad M \mid K, \quad T_0 \geq 1,$$

and, in the multiresolution branch,

$$R_j = 2^j M, \quad M \mid R_{j-1}, \quad T_j \geq 1 \quad (1 \leq j \leq L).$$

The divisibility follows because the selected ranks are powers of two and $K \geq M$.

Fix $u \in \mathbb{R}^M$. Write

$$\mathcal{O} = [0, 1] \times \{0, 1\} \times \mathbb{R}, \quad O_i = (X_i, A_i, Y_i) \in \mathcal{O},$$

and let O, O' be independent records with law P . For $x \in [0, 1]$, define the scalar histogram feature by

$$f_u(x) = u^T F_M(x).$$

The disjoint histogram cells in [Definition 19](#), together with the uniform covariate law, give

$$\int_0^1 f_u(x)^2 dx = \sum_{j=1}^M u_j^2 = \|u\|_2^2, \quad \mathbb{E}[f_u(X)^2] = \|u\|_2^2.$$

Use a coefficient selector $r \in \{0, 1\}$, with

$$Z_{b,r}^{\text{coef}} = \begin{cases} Z_{b,0}, & r = 0, \\ Z_{b,A}, & r = 1. \end{cases}$$

For every real cutoff $T \geq 1$, define

$$\mathbf{m}_{r,T}(o) = \begin{cases} \ell_T(Y(o)), & r = 0, \\ A(o), & r = 1, \end{cases} \quad V_T = 32T^{2-p}, \quad o \in \mathcal{O}.$$

Thus $V_{T_0} = V_0$ and $V_{T_j} = V_j$. For a pair kernel G from the score construction, define

$$\begin{aligned} \mathcal{E}_{r,u}[G, T](o, o') &= f_u(X(o))A(o)G(X(o), X(o'))\mathbf{m}_{r,T}(o'), \\ \mathcal{P}_{r,u}[G, T](o, o') &= \frac{\mathcal{E}_{r,u}[G, T](o, o') + \mathcal{E}_{r,u}[G, T](o', o)}{2}, \quad o, o' \in \mathcal{O}. \end{aligned}$$

The selected scalar single-record and pair summands are

$$\begin{aligned} h_{r,u}(o) &= f_u(X(o))A(o)\mathbf{m}_{r,T_0}(o), \\ g_{r,u}(o, o') &= \begin{cases} -\mathcal{P}_{r,u}[\Pi_K, T_0](o, o'), & \text{single-correction branch,} \\ -\mathcal{P}_{r,u}[\Pi_M, T_0](o, o') - \sum_{j=1}^L \mathcal{P}_{r,u}[D_j, T_j](o, o'), & \text{multiresolution branch.} \end{cases} \end{aligned}$$

Here $A(o)^2 = A(o)$, so the $r = 1$ single-record summand agrees with the treatment coefficient. Package these summands into

$$k_{r,u}(o, o') = \frac{h_{r,u}(o) + h_{r,u}(o')}{2} + g_{r,u}(o, o').$$

All these functions are measurable and bounded for the fixed finite tuning, hence square integrable.

For either block, symmetrization preserves the ordered-pair sum, and each index occurs $s - 1$ times in each position. Consequently, the coefficient definitions in [Algorithm 3](#) and the score formulas in [Definition 21](#) give

$$u^T Z_{b,r}^{\text{coef}} = \frac{1}{s} \sum_{i \in \mathcal{I}_b} h_{r,u}(O_i) + \frac{1}{s(s-1)} \sum_{\substack{i,j \in \mathcal{I}_b \\ i \neq j}} g_{r,u}(O_i, O_j) = \frac{1}{s(s-1)} \sum_{\substack{i,j \in \mathcal{I}_b \\ i \neq j}} k_{r,u}(O_i, O_j).$$

Empty increment sums have value zero throughout.

3. We first establish the variance identity for this pair-average representation. For any measurable symmetric square-integrable kernel $\mathcal{H} : \mathcal{O}^2 \rightarrow \mathbb{R}$, define

$$\begin{aligned}\overline{\mathcal{H}} &= \mathbb{E}[\mathcal{H}(O, O')], & \mathcal{H}^{(1)}(o) &= \mathbb{E}[\mathcal{H}(o, O')] - \overline{\mathcal{H}}, \\ \mathcal{H}^{(2)}(o, o') &= \mathcal{H}(o, o') - \overline{\mathcal{H}} - \mathcal{H}^{(1)}(o) - \mathcal{H}^{(1)}(o'),\end{aligned}$$

and

$$\mathcal{A}_b(\mathcal{H}) = \frac{1}{s(s-1)} \sum_{\substack{i, j \in \mathcal{I}_b \\ i \neq j}} \mathcal{H}(O_i, O_j).$$

Conditional integration gives

$$\mathbb{E}[\mathcal{H}^{(1)}(O)] = 0, \quad \mathbb{E}[\mathcal{H}^{(2)}(O, O') \mid O] = \mathbb{E}[\mathcal{H}^{(2)}(O, O') \mid O'] = 0.$$

Expanding the square of $\mathcal{H} = \overline{\mathcal{H}} + \mathcal{H}^{(1)} + \mathcal{H}^{(1)} + \mathcal{H}^{(2)}$ and integrating therefore yields

$$\mathbb{E}[\mathcal{H}(O, O')^2] = \overline{\mathcal{H}}^2 + 2\mathbb{E}[\mathcal{H}^{(1)}(O)^2] + \mathbb{E}[\mathcal{H}^{(2)}(O, O')^2].$$

Indeed, the two singleton terms have zero product expectation by independence, and their products with the canonical term have zero expectation by conditional integration.

Counting the occurrences of each singleton term gives

$$\mathcal{A}_b(\mathcal{H}) - \overline{\mathcal{H}} = \frac{2}{s} \sum_{i \in \mathcal{I}_b} \mathcal{H}^{(1)}(O_i) + \mathcal{A}_b(\mathcal{H}^{(2)}).$$

The singleton sum has variance

$$\text{Var}\left(\frac{2}{s} \sum_{i \in \mathcal{I}_b} \mathcal{H}^{(1)}(O_i)\right) = \frac{4}{s} \text{Var}(\mathcal{H}^{(1)}(O)).$$

For distinct indices within each pair, conditional centering and independence imply

$$\text{Cov}\left(\mathcal{H}^{(2)}(O_i, O_j), \mathcal{H}^{(2)}(O_{i'}, O_{j'})\right) = \begin{cases} \mathbb{E}[\mathcal{H}^{(2)}(O, O')^2], & (i', j') = (i, j) \text{ or } (j, i), \\ 0, & \text{otherwise.} \end{cases}$$

The zero case includes pairs sharing exactly one index: integrate first in a coordinate that occurs in only one pair. There are $s(s-1)$ ordered pairs and two contributing ordered pairs for each of them. Thus

$$\text{Var}(\mathcal{A}_b(\mathcal{H}^{(2)})) = \frac{2}{s(s-1)} \mathbb{E}[\mathcal{H}^{(2)}(O, O')^2].$$

The same conditional-centering argument gives

$$\text{Cov}\left(\frac{2}{s} \sum_{i \in \mathcal{I}_b} \mathcal{H}^{(1)}(O_i), \mathcal{A}_b(\mathcal{H}^{(2)})\right) = 0.$$

Each evaluation block has marginal law $P^{\otimes s}$ under $P^{\otimes n}$, so

$$\text{Var}_{P^{\otimes n}}(\mathcal{A}_b(\mathcal{H})) = \frac{4}{s} \text{Var}_P(\mathcal{H}^{(1)}) + \frac{2}{s(s-1)} \mathbb{E}[\mathcal{H}^{(2)}(O, O')^2].$$

4. We next bound the singleton channel. Admissibility gives $1 < p \leq 2$. For $T \geq 1$, clipping satisfies the pointwise bounds

$$|y - \ell_T(y)| \leq |y|^p T^{1-p}, \quad \ell_T(y)^2 \leq |y|^p T^{2-p}.$$

Integrating under the conditional moment restriction in [Assumption 8](#) gives, for each treatment arm and almost every covariate value,

$$\left| \int \{y - \ell_T(y)\} Q_{a,P}(dy | x) \right| \leq 10T^{1-p}, \quad \int \ell_T(y)^2 Q_{a,P}(dy | x) \leq 10T^{2-p}.$$

Define the conditional mark mean, for $x \in [0, 1]$, by

$$\bar{\mathbf{m}}_{r,T}(x) = e_P(x) \int \mathbf{m}_{r,T}(x, 1, y) Q_{1,P}(dy | x) + \{1 - e_P(x)\} \int \mathbf{m}_{r,T}(x, 0, y) Q_{0,P}(dy | x).$$

The preceding bounds and $0 \leq e_P \leq 1$ imply

$$\mathbb{E}[\mathbf{m}_{0,T}(O)^2 | X = x] \leq 10T^{2-p} \leq V_T.$$

For the treatment mark,

$$\mathbb{E}[\mathbf{m}_{1,T}(O)^2 | X = x] = e_P(x) \leq 1 \leq V_T, \quad \bar{\mathbf{m}}_{1,T}(x) = e_P(x).$$

In particular, for both selectors,

$$\mathbb{E}[\mathbf{m}_{r,T}(O)^2 | X = x] \leq V_T, \quad |\bar{\mathbf{m}}_{r,T}(x)| \leq \sqrt{V_T} \quad \text{almost everywhere,}$$

where the second inequality is conditional Cauchy-Schwarz.

Since $|A| \leq 1$, the single-record summand obeys

$$h_{r,u}(O)^2 \leq 2f_u(X)^2 \mathbf{m}_{r,T_0}(O)^2.$$

Conditioning on X and using the feature isometry yields

$$\mathbb{E}[h_{r,u}(O)^2] \leq 2V_0 \|u\|_2^2 \leq B_h^2 \|u\|_2^2, \quad B_h = 2\sqrt{V_0}.$$

5. For any selected projection or difference kernel G , and any integrable function $\varphi : [0, 1] \rightarrow \mathbb{R}$, use the operator notation

$$(G\varphi)(x) = \int_0^1 G(x, x') \varphi(x') dx'.$$

Nested histogram cells imply

$$G(x, x') f_u(x') = G(x, x') f_u(x)$$

for $G = \Pi_K, \Pi_M$, and for every $G = D_j$. Integrating the two symmetrized legs separately therefore gives

$$\mathbb{E}[\mathcal{P}_{r,u}[G, T](o, O')] = \frac{1}{2} f_u(X(o)) [A(o)(G\bar{\mathbf{m}}_{r,T})(X(o)) + \mathbf{m}_{r,T}(o)(Ge_P)(X(o))].$$

For a positive histogram rank J_{aux} divisible by M , cell averaging gives

$$|\Pi_{J_{\text{aux}}} \bar{\mathbf{m}}_{r,T}| \leq \sqrt{V_T}, \quad |\Pi_{J_{\text{aux}}} e_P| \leq 1.$$

Consequently,

$$|\mathbb{E}[\mathcal{P}_{r,u}[\Pi_{J_{\text{aux}}}, T](o, O')]| \leq \frac{1}{2} |f_u(X(o))| \left(\sqrt{V_T} + |\mathbf{m}_{r,T}(o)| \right).$$

Using

$$\frac{1}{4} \left(\sqrt{V_T} + |\mathbf{m}_{r,T}(o)| \right)^2 \leq \frac{1}{2} (V_T + \mathbf{m}_{r,T}(o)^2)$$

and then conditioning on X , we obtain

$$\mathbb{E} \left[\left\{ \mathbb{E}[\mathcal{P}_{r,u}[\Pi_{J_{\text{aux}}}, T](O, O') \mid O] \right\}^2 \right] \leq V_T \|u\|_2^2.$$

The augmented kernel has singleton projection

$$k_{r,u}^{(1)}(o) = \frac{h_{r,u}(o) - \mathbb{E}[h_{r,u}(O)]}{2} + g_{r,u}^{(1)}(o).$$

Centering decreases second moment, so the triangle inequality in $L_2(P)$ gives

$$2\|k_{r,u}^{(1)}\|_{L_2(P)} \leq \sqrt{\mathbb{E}[h_{r,u}(O)^2]} + 2\sqrt{\mathbb{E}[\{\mathbb{E}[g_{r,u}(O, O') \mid O]\}^2]}.$$

Here

$$\text{Var}_P(k_{r,u}^{(1)}) = \|k_{r,u}^{(1)}\|_{L_2(P)}^2$$

because its mean is zero.

In the single-correction branch, the projection estimate with $J_{\text{aux}} = K$ and $T = T_0$ gives

$$\mathbb{E} \left[\left\{ \mathbb{E}[g_{r,u}(O, O') \mid O] \right\}^2 \right] \leq B_{g,\text{single}}^2 \|u\|_2^2, \quad B_{g,\text{single}} = \sqrt{V_0}.$$

It follows that

$$4 \text{Var}_P(k_{r,u}^{(1)}) \leq (B_h + 2B_{g,\text{single}})^2 \|u\|_2^2 = (4\sqrt{V_0})^2 \|u\|_2^2 \leq (16\sqrt{V_0})^2 \|u\|_2^2 = L_1^2 \|u\|_2^2.$$

6. In the multiresolution branch, define

$$s_0 = \min\{\alpha, \beta, \gamma\} \in (0, 1], \quad f_{\text{mean}}(x) = m_{0,P}(x) + e_P(x)\tau_P(x), \quad x \in [0, 1].$$

The smoothness and boundedness restrictions in [Assumptions 2](#) to [7](#) give

$$|f_{\text{mean}}(x)| \leq 2$$

and

$$\begin{aligned} |f_{\text{mean}}(x) - f_{\text{mean}}(x')| &\leq |m_{0,P}(x) - m_{0,P}(x')| + |e_P(x)| |\tau_P(x) - \tau_P(x')| \\ &\quad + |e_P(x) - e_P(x')| |\tau_P(x')| \\ &\leq 20|x - x'|^\beta + 15|x - x'|^\gamma + 20|x - x'|^\alpha \\ &\leq 55|x - x'|^{s_0}. \end{aligned}$$

For $x = x'$ both sides vanish; for $0 < |x - x'| \leq 1$, the last inequality follows because each exponent is at least s_0 .

To apply the histogram approximation bound with constant 20, define

$$\tilde{f}_{\text{mean}}(x) = \frac{20}{55} f_{\text{mean}}(x).$$

Then

$$\|\tilde{f}_{\text{mean}}\|_{\infty} \leq \frac{40}{55} \leq 20, \quad |\tilde{f}_{\text{mean}}(x) - \tilde{f}_{\text{mean}}(x')| \leq 20|x - x'|^{s_0}.$$

Averaging over the histogram cell containing x , whose diameter is at most J_{aux}^{-1} , gives, for every positive integer J_{aux} ,

$$|\Pi_{J_{\text{aux}}} \tilde{f}_{\text{mean}}(x) - \tilde{f}_{\text{mean}}(x)| \leq 20J_{\text{aux}}^{-s_0}.$$

Thus, for $1 \leq j \leq L$,

$$\begin{aligned} |D_j f_{\text{mean}}(x)| &= \frac{55}{20} \left| \Pi_{R_j} \tilde{f}_{\text{mean}}(x) - \Pi_{R_{j-1}} \tilde{f}_{\text{mean}}(x) \right| \\ &\leq \frac{55}{20} \left(20R_j^{-s_0} + 20R_{j-1}^{-s_0} \right) \leq 110R_{j-1}^{-s_0}. \end{aligned}$$

For $T \geq 1$, define on all of $[0, 1]$

$$\mathbf{t}_T(x) = f_{\text{mean}}(x) - \bar{\mathbf{m}}_{0,T}(x).$$

The conditional mean versions in [Definition 1](#) imply, almost everywhere,

$$\begin{aligned} \mathbf{t}_T(x) &= e_P(x) \int \{y - \ell_T(y)\} Q_{1,P}(dy \mid x) \\ &\quad + \{1 - e_P(x)\} \int \{y - \ell_T(y)\} Q_{0,P}(dy \mid x). \end{aligned}$$

The clipping-tail estimate therefore gives

$$\|\mathbf{t}_T\|_{L_{\infty}(\lambda)} \leq 10T^{1-p}, \quad |D_j \mathbf{t}_{T_j}(x)| \leq 20T_j^{1-p}.$$

For the propensity, the same cell-averaging argument gives

$$|\Pi_{J_{\text{aux}}} e_P(x) - e_P(x)| \leq 20J_{\text{aux}}^{-\alpha},$$

hence

$$|D_j e_P(x)| \leq 20R_j^{-\alpha} + 20R_{j-1}^{-\alpha} \leq 40R_{j-1}^{-\alpha} = a_j.$$

Combining the smoothness and tail bounds yields

$$|D_j \bar{\mathbf{m}}_{0,T_j}(x)| \leq 110R_{j-1}^{-s_0} + 20T_j^{1-p} = d_j.$$

Substitution into the conditional pair formula gives the outcome-coefficient envelope

$$|\mathbb{E}[\mathcal{P}_{0,u}[D_j, T_j](o, O')]| \leq \frac{1}{2} |f_u(X(o))| (d_j + a_j |\bar{\mathbf{m}}_{0,T_j}(o)|).$$

Since

$$\frac{1}{4}(d_j + a_j |\mathbf{m}_{0,T_j}|)^2 \leq \frac{1}{2}(d_j^2 + a_j^2 \mathbf{m}_{0,T_j}^2),$$

conditioning on X gives

$$\mathbb{E} \left[\left\{ \mathbb{E}[\mathcal{P}_{0,u}[D_j, T_j](O, O') \mid O] \right\}^2 \right] \leq \frac{1}{2}(d_j^2 + a_j^2 V_j) \|u\|_2^2.$$

For the treatment coefficient, $\bar{\mathbf{m}}_{1,T_j} = e_P$ and $|\mathbf{m}_{1,T_j}| \leq 1$, so

$$|\mathbb{E}[\mathcal{P}_{1,u}[D_j, T_j](o, O')]| \leq a_j |f_u(X(o))|.$$

Applying the same square-envelope inequality with constant term $2a_j$ and zero mark term yields

$$\mathbb{E} \left[\left\{ \mathbb{E}[\mathcal{P}_{1,u}[D_j, T_j](O, O') \mid O] \right\}^2 \right] \leq 2a_j^2 \|u\|_2^2.$$

All a_j, d_j are nonnegative and $V_j \geq 1$. The two coefficient estimates fit the common bound because

$$\frac{1}{2}(d_j^2 + a_j^2 V_j) \leq \{2(d_j + a_j \sqrt{V_j})\}^2, \quad 2a_j^2 \leq \{2(d_j + a_j \sqrt{V_j})\}^2.$$

Define the nonnegative increment sum and conditional-pair budget by

$$\mathcal{S}_{\text{inc}} = \sum_{j=1}^L (d_j + a_j \sqrt{V_j}), \quad B_{g,\text{multi}} = \sqrt{V_0} + 2\mathcal{S}_{\text{inc}}.$$

The initial projection estimate at rank M , followed by the triangle inequality for the finite increment sum in $L_2(P)$, gives

$$\begin{aligned} \|\mathbb{E}[g_{r,u}(O, O') \mid O]\|_{L_2(P)} &\leq \sqrt{V_0} \|u\|_2 + \sum_{j=1}^L 2(d_j + a_j \sqrt{V_j}) \|u\|_2 \\ &= B_{g,\text{multi}} \|u\|_2. \end{aligned}$$

This includes $L = 0$, when the sum is zero. The singleton estimate for the augmented kernel now gives

$$\begin{aligned} 4 \text{Var}_P(k_{r,u}^{(1)}) &\leq (B_h + 2B_{g,\text{multi}})^2 \|u\|_2^2 \\ &= (4\sqrt{V_0} + 4\mathcal{S}_{\text{inc}})^2 \|u\|_2^2 \\ &\leq (16\sqrt{V_0} + 8\mathcal{S}_{\text{inc}})^2 \|u\|_2^2 = L_1^2 \|u\|_2^2. \end{aligned}$$

7. We next bound the canonical channel, again treating the two branches separately. The definitions of the projections give

$$\overline{k_{r,u}} = \mathbb{E}[h_{r,u}(O)] + \overline{g_{r,u}},$$

and cancellation of the single-record terms gives

$$k_{r,u}^{(2)}(o, o') = g_{r,u}^{(2)}(o, o') \quad \text{almost everywhere.}$$

The pair-energy identity therefore implies

$$\begin{aligned}\mathbb{E}[k_{r,u}^{(2)}(O, O')^2] &= \mathbb{E}[g_{r,u}^{(2)}(O, O')^2] \\ &\leq \overline{g_{r,u}}^2 + 2\mathbb{E}[g_{r,u}^{(1)}(O)^2] + \mathbb{E}[g_{r,u}^{(2)}(O, O')^2] \\ &= \mathbb{E}[g_{r,u}(O, O')^2].\end{aligned}$$

To bound this raw pair energy, the histogram kernels have exact row identities

$$\begin{aligned}\int_0^1 \Pi_{J_{\text{aux}}}(x, x')^2 dx' &= J_{\text{aux}} \quad (J_{\text{aux}} \in \mathbb{N}, J_{\text{aux}} > 0), \\ \int_0^1 \Pi_{R_j}(x, x') \Pi_{R_{j-1}}(x, x') dx' &= R_{j-1},\end{aligned}$$

and hence

$$\int_0^1 D_j(x, x')^2 dx' = R_j + R_{j-1} - 2R_{j-1} = R_{j-1} \quad (1 \leq j \leq L).$$

The first identity follows because a projection row equals its rank on one cell of reciprocal length. For the cross-product identity, the finer cell lies inside the coarser cell, so integrating their product gives the coarser rank.

Conditioning on the covariate of O' , the mark bound gives, for every selected G, T ,

$$\mathbb{E}[G(x, X(O'))^2 \mathbf{m}_{r,T}(O')^2] \leq V_T \int_0^1 G(x, x')^2 dx'.$$

Thus, integrating first in O' and using $A(O)^2 \leq 1$,

$$\begin{aligned}\mathbb{E}[\mathcal{E}_{r,u}[\Pi_{J_{\text{aux}}}, T](O, O')^2] &\leq J_{\text{aux}} V_T \mathbb{E}[f_u(X)^2] = J_{\text{aux}} V_T \|u\|_2^2, \\ \mathbb{E}[\mathcal{E}_{r,u}[D_j, T_j](O, O')^2] &\leq R_{j-1} V_j \|u\|_2^2.\end{aligned}$$

Symmetrization satisfies

$$\mathcal{P}_{r,u}[G, T](o, o')^2 \leq \frac{\mathcal{E}_{r,u}[G, T](o, o')^2 + \mathcal{E}_{r,u}[G, T](o', o)^2}{2}.$$

The two legs have equal integrated energy under $P \otimes P$. Consequently,

$$\begin{aligned}\mathbb{E}[\mathcal{P}_{r,u}[\Pi_{J_{\text{aux}}}, T](O, O')^2] &\leq J_{\text{aux}} V_T \|u\|_2^2, \\ \mathbb{E}[\mathcal{P}_{r,u}[D_j, T_j](O, O')^2] &\leq R_{j-1} V_j \|u\|_2^2 \leq R_j V_j \|u\|_2^2.\end{aligned}$$

In the single-correction branch, this gives

$$\begin{aligned}\mathbb{E}[k_{r,u}^{(2)}(O, O')^2] &\leq \mathbb{E}[g_{r,u}(O, O')^2] \\ &\leq K V_0 \|u\|_2^2 \\ &\leq (4\sqrt{K V_0})^2 \|u\|_2^2 = L_2^2 \|u\|_2^2.\end{aligned}$$

In the multiresolution branch, the triangle inequality in $L_2(P \otimes P)$ gives

$$\|g_{r,u}\|_{L_2(P \otimes P)} \leq \left(\sqrt{MV_0} + \sum_{j=1}^L \sqrt{R_j V_j} \right) \|u\|_2.$$

Every summand is nonnegative, including the empty-sum case. Therefore

$$\begin{aligned} \mathbb{E}[k_{r,u}^{(2)}(O, O')^2] &\leq \left(\sqrt{MV_0} + \sum_{j=1}^L \sqrt{R_j V_j} \right)^2 \|u\|_2^2 \\ &\leq \left[4 \left(\sqrt{MV_0} + \sum_{j=1}^L \sqrt{R_j V_j} \right) \right]^2 \|u\|_2^2 \\ &= L_2^2 \|u\|_2^2. \end{aligned}$$

8. Apply the pair-average variance identity to $k_{r,u}$. Since $s \geq 2$, both sample factors are positive, and the two projection estimates imply

$$\begin{aligned} \text{Var}_{P^{\otimes n}}(u^T Z_{b,r}^{\text{coef}}) &= \frac{4}{s} \text{Var}_P(k_{r,u}^{(1)}) + \frac{2}{s(s-1)} \mathbb{E}[k_{r,u}^{(2)}(O, O')^2] \\ &\leq \frac{L_1^2}{s} \|u\|_2^2 + \frac{2L_2^2}{s(s-1)} \|u\|_2^2 \\ &= \Lambda_{n,v} \|u\|_2^2. \end{aligned}$$

The finite-sum representation also gives

$$u^T Z_{b,r}^{\text{coef}} \in L_2(P^{\otimes n}).$$

Taking $r = 0$ and $r = 1$ proves square integrability and the two variance bounds.

9. Finally, the covariance ledger is nonnegative:

$$\Lambda_{n,v} = \frac{L_1^2}{s} + \frac{2L_2^2}{s(s-1)} \geq 0 \quad (n \geq 4),$$

and its value is zero for $n = 2, 3$. For arbitrary $u, z \in \mathbb{R}^M$, Cauchy–Schwarz for the centered square-integrable coefficients gives

$$\begin{aligned} |\text{Cov}_{P^{\otimes n}}(u^T Z_{b,0}, z^T Z_{b,A})| &\leq \sqrt{\text{Var}_{P^{\otimes n}}(u^T Z_{b,0})} \sqrt{\text{Var}_{P^{\otimes n}}(z^T Z_{b,A})} \\ &\leq \sqrt{\Lambda_{n,v} \|u\|_2^2} \sqrt{\Lambda_{n,v} \|z\|_2^2} \\ &= (\sqrt{\Lambda_{n,v}})^2 \|u\|_2 \|z\|_2 \\ &= \Lambda_{n,v} \|u\|_2 \|z\|_2. \end{aligned}$$

The square-root simplification applies also when either direction or the ledger is zero. $\square$

The directional form gives a common covariance bound for the outcome coefficient and the constant-effect coefficient. The mixed bound retains their dependence within a block, including the dependence contributed by the different correction levels. Independence between the two evaluation blocks then provides the setting for the cross-block profiling statement.

B.4 Simultaneous affine profiling and calibration

The profiling multiplier $C = 128$ and the population mean $\theta_P(c)$ were introduced with [Theorem 1](#). The following bound applies to the entire interval of candidate constants on one measurable event.

Lemma 4. [lem:uniform-affine-profile-bound] (Uniform affine profile bound) *Let $v = (p, \alpha, \beta, \gamma)$ be an admissible public tuple as in [Definition 1](#), and let $n \geq 2$ be an integer. For every $P \in \mathcal{M}_v$, consider the explicit affine score construction, with block scores $Z_1(c)$ and $Z_2(c)$, coarse histogram rank J_c , and public covariance ledger $\Lambda_{n,v}$. Define the score inner product and its population target by*

$$W(c) = \langle Z_1(c), Z_2(c) \rangle, \quad \theta_P(c) = \mathbb{E}_{P^{\otimes n}}[Z_1(c)].$$

The event on which, simultaneously for every $c \in [-1/2, 1/2]$,

$$|W(c) - \|\theta_P(c)\|_2^2| \leq 128 \left\{ \sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 + \sqrt{J_c} \Lambda_{n,v} \right\}$$

is measurable and has probability at least 0.9925 under $P^{\otimes n}$. Moreover, if $\Lambda_{n,v} = 0$, then $P^{\otimes n}$ -almost surely, simultaneously for every $c \in [-1/2, 1/2]$,

$$W(c) = \|\theta_P(c)\|_2^2.$$

Proof of [Lemma 4](#). Fix an admissible v , an integer $n \geq 2$, and $P \in \mathcal{M}_v$. Write

$$\mu = P^{\otimes n}, \quad M = J_c.$$

All expectations below are under μ .

1. If $n < 4$, then $n \in \{2, 3\}$. The construction in [Algorithm 1](#) gives, for every sample and every $c \in \mathbb{R}$,

$$Z_1(c) = Z_2(c) = 0, \quad \theta_P(c) = 0, \quad W(c) = 0, \quad \Lambda_{n,v} = 0.$$

Consequently, the asserted profile event is the entire sample space, with probability one, and the equality assertion holds for every sample. Henceforth suppose $n \geq 4$.

2. Define the profile discrepancy and its simultaneous event by

$$F_{\text{prof}}(c) = |W(c) - \|\theta_P(c)\|_2^2| - 128 \left(\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 + \sqrt{M} \Lambda_{n,v} \right),$$

$$\mathcal{G}_{\text{prof}} = \{F_{\text{prof}}(c) \leq 0 \text{ for every } c \in [-1/2, 1/2]\}.$$

For fixed c , the discrepancy is a measurable function of the sample. For each fixed sample, it is continuous in c , because the scores and their population means are affine. Choose a countable dense set with

$$\mathcal{C}_{\text{dense}} \subseteq [-1/2, 1/2], \quad \overline{\mathcal{C}_{\text{dense}}} = [-1/2, 1/2].$$

Continuity gives

$$\mathcal{G}_{\text{prof}} = \bigcap_{c \in \mathcal{C}_{\text{dense}}} \{F_{\text{prof}}(c) \leq 0\}.$$

Indeed, for each sample, the set where the discrepancy is nonpositive is closed in the profiling interval. Thus $\mathcal{G}_{\text{prof}}$ is measurable.

We also record the positivity needed for the tail estimates and the zero-ledger assertion. The public formulas in [Algorithm 1](#) give

$$s = s_n = \lfloor n/2 \rfloor \geq 2, \quad M \geq 1, \quad T_0 \geq 1, \quad V_0 = 32T_0^{2-p} > 0.$$

In the single-resolution branch, $L_1 = 16\sqrt{V_0}$. In the multiresolution branch, the positive ranks and cutoffs give

$$a_j \geq 0, \quad d_j \geq 0, \quad V_j > 0 \quad (1 \leq j \leq L),$$

and hence

$$\sum_{j=1}^L (d_j + a_j \sqrt{V_j}) \geq 0, \quad L_1 = 16\sqrt{V_0} + 8 \sum_{j=1}^L (d_j + a_j \sqrt{V_j}) \geq 16\sqrt{V_0} > 0.$$

These inequalities include the empty-sum case $L = 0$. In either branch,

$$\Lambda_{n,v} = \frac{L_1^2}{s} + \frac{2L_2^2}{s(s-1)} \geq \frac{L_1^2}{s} > 0.$$

Together with the small-sample branch, this establishes, for admissible v and $n \geq 2$,

$$\Lambda_{n,v} = 0 \quad \Longleftrightarrow \quad n < 4.$$

The zero-ledger assertion therefore follows from the pointwise equalities in the first case.

3. For $b \in \{1, 2\}$, define the affine coefficients by

$$Z_{b,0} = Z_b(0), \quad Z_{b,A} = Z_b(0) - Z_b(1), \quad Z_b(c) = Z_{b,0} - cZ_{b,A} \quad (c \in \mathbb{R}).$$

The common block-mean identity supplied by [Lemma 2](#) is

$$\mathbb{E}_\mu[Z_b(c)] = \theta_P(c) \quad (b \in \{1, 2\}, \quad c \in \mathbb{R}).$$

Define the population coefficients and centered coefficient errors by

$$\begin{aligned} \bar{Z}_0 &= \theta_P(0), & \bar{Z}_A &= \theta_P(0) - \theta_P(1), \\ \mathbf{E}_{b,\iota} &= Z_{b,\iota} - \bar{Z}_\iota \quad (b \in \{1, 2\}, \quad \iota \in \{0, A\}), & \mathbf{E}_b(c) &= \mathbf{E}_{b,0} - c\mathbf{E}_{b,A}. \end{aligned}$$

Linearity of expectation gives

$$\begin{aligned} \mathbb{E}_\mu[Z_{b,\iota}] &= \bar{Z}_\iota, & \mathbb{E}_\mu[\mathbf{E}_{b,\iota}] &= 0, \\ \theta_P(c) &= \bar{Z}_0 - c\bar{Z}_A, & Z_b(c) &= \theta_P(c) + \mathbf{E}_b(c). \end{aligned}$$

Introduce the fixed population subspace

$$\mathcal{S}_P = \text{span}\{\bar{Z}_0, \bar{Z}_A\} \subseteq \mathbb{R}^M.$$

Its orthogonal projection $\text{pr}_{\mathcal{S}_P} : \mathbb{R}^M \rightarrow \mathcal{S}_P$ is characterized by

$$\text{pr}_{\mathcal{S}_P} \mathbf{v}_{\text{dir}} \in \mathcal{S}_P, \quad \mathbf{v}_{\text{dir}} - \text{pr}_{\mathcal{S}_P} \mathbf{v}_{\text{dir}} \in \mathcal{S}_P^\perp \quad (\mathbf{v}_{\text{dir}} \in \mathbb{R}^M).$$

For $b \in \{1, 2\}$ and $\iota, j \in \{0, A\}$, define the bad events

$$\begin{aligned} \mathcal{P}_{b,\iota} &= \left\{ \|\text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota}\|_2 > 40\sqrt{\Lambda_{n,v}} \right\}, \\ \mathcal{Q}_{\iota,j} &= \left\{ |\langle \mathbf{E}_{1,\iota}, \mathbf{E}_{2,j} \rangle| > 40\sqrt{M} \Lambda_{n,v} \right\}, \\ \mathcal{F}_{\text{bad}} &= \left(\bigcup_{b \in \{1,2\}} \bigcup_{\iota \in \{0,A\}} \mathcal{P}_{b,\iota} \right) \cup \left(\bigcup_{\iota \in \{0,A\}} \bigcup_{j \in \{0,A\}} \mathcal{Q}_{\iota,j} \right). \end{aligned}$$

The coefficients are measurable, so continuity of projection, norm, and inner product makes all these events measurable.

4. The square integrability and variance bounds in [Lemma 3](#), together with the coefficient means just computed, imply

$$\mathbb{E}_\mu[\langle \mathbf{v}_{\text{dir}}, \mathbf{E}_{b,\iota} \rangle^2] = \text{Var}_\mu(\langle \mathbf{v}_{\text{dir}}, Z_{b,\iota} \rangle) \leq \Lambda_{n,v} \|\mathbf{v}_{\text{dir}}\|_2^2 \quad (\mathbf{v}_{\text{dir}} \in \mathbb{R}^M).$$

Because $\mathcal{S}_P$ is spanned by two vectors,

$$\dim \mathcal{S}_P \leq 2.$$

Choose an orthonormal basis

$$\mathcal{B}_P = \{\mathbf{v}_{\text{dir},1}^{\text{sp}}, \dots, \mathbf{v}_{\text{dir},\dim \mathcal{S}_P}^{\text{sp}}\} \subseteq \mathcal{S}_P.$$

Parseval's identity and the defining property of orthogonal projection give

$$\begin{aligned} \mathbb{E}_\mu \|\text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota}\|_2^2 &= \sum_{\mathbf{v}_{\text{dir}} \in \mathcal{B}_P} \mathbb{E}_\mu \langle \mathbf{v}_{\text{dir}}, \text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota} \rangle^2 \\ &= \sum_{\mathbf{v}_{\text{dir}} \in \mathcal{B}_P} \mathbb{E}_\mu \langle \mathbf{v}_{\text{dir}}, \mathbf{E}_{b,\iota} \rangle^2 \\ &\leq (\dim \mathcal{S}_P) \Lambda_{n,v} \leq 2\Lambda_{n,v}. \end{aligned}$$

If $\mathcal{S}_P = \{0\}$, the basis is empty and these sums equal zero. Markov's inequality applied to the squared projected norm now yields

$$\mu(\mathcal{P}_{b,\iota}) \leq \frac{\mathbb{E}_\mu \|\text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota}\|_2^2}{1600\Lambda_{n,v}} \leq \frac{1}{800}.$$

There are two blocks and two coefficients, so the finite union bound gives

$$\mu \left(\bigcup_{b \in \{1,2\}} \bigcup_{\iota \in \{0,A\}} \mathcal{P}_{b,\iota} \right) \leq \sum_{b \in \{1,2\}} \sum_{\iota \in \{0,A\}} \mu(\mathcal{P}_{b,\iota}) \leq \frac{4}{800}.$$

5. The coefficients in the two blocks depend on disjoint sets of independent records, as specified in [Definition 20](#) and [Algorithm 1](#). Subtracting deterministic means preserves their independence. Define their probability laws on $\mathbb{R}^M$ by

$$\nu_{b,\iota} = \mu \circ \mathbf{E}_{b,\iota}^{-1}.$$

For every $\iota, j \in \{0, A\}$, their joint law is therefore

$$\mu \circ (\mathbf{E}_{1,\iota}, \mathbf{E}_{2,j})^{-1} = \nu_{1,\iota} \otimes \nu_{2,j}.$$

Choose an orthonormal basis of the ambient space,

$$\mathcal{B}_{\text{amb}} = \{\mathbf{v}_{\text{dir},1}^{\text{amb}}, \dots, \mathbf{v}_{\text{dir},M}^{\text{amb}}\} \subseteq \mathbb{R}^M.$$

The directional second-moment bound from the preceding step gives

$$\mathbb{E}_\mu \|\mathbf{E}_{1,\iota}\|_2^2 = \sum_{\mathbf{v}_{\text{dir}} \in \mathcal{B}_{\text{amb}}} \mathbb{E}_\mu \langle \mathbf{v}_{\text{dir}}, \mathbf{E}_{1,\iota} \rangle^2 \leq M \Lambda_{n,v}.$$

Square integrability and

$$|\langle \mathbf{v}_{\text{dir}}, z \rangle|^2 \leq \|\mathbf{v}_{\text{dir}}\|_2^2 \|z\|_2^2 \quad (\mathbf{v}_{\text{dir}}, z \in \mathbb{R}^M)$$

justify iterated integration under the product law. Applying the directional second-moment bound to the second coefficient, for each fixed $\mathbf{v}_{\text{dir}}$, gives

$$\begin{aligned} \mathbb{E}_\mu \langle \mathbf{E}_{1,\iota}, \mathbf{E}_{2,j} \rangle^2 &= \int_{\mathbb{R}^M} \int_{\mathbb{R}^M} \langle \mathbf{v}_{\text{dir}}, z \rangle^2 d\nu_{2,j}(z) d\nu_{1,\iota}(\mathbf{v}_{\text{dir}}) \\ &\leq \int_{\mathbb{R}^M} \Lambda_{n,v} \|\mathbf{v}_{\text{dir}}\|_2^2 d\nu_{1,\iota}(\mathbf{v}_{\text{dir}}) \\ &= \Lambda_{n,v} \mathbb{E}_\mu \|\mathbf{E}_{1,\iota}\|_2^2 \\ &\leq M \Lambda_{n,v}^2. \end{aligned}$$

Since $M \geq 1$ and $\Lambda_{n,v} > 0$, Markov's inequality gives

$$\mu(\mathcal{Q}_{\iota,j}) \leq \frac{\mathbb{E}_\mu \langle \mathbf{E}_{1,\iota}, \mathbf{E}_{2,j} \rangle^2}{1600 M \Lambda_{n,v}^2} \leq \frac{1}{1600}.$$

There are four coefficient pairs, whence

$$\mu \left(\bigcup_{\iota \in \{0,A\}} \bigcup_{j \in \{0,A\}} \mathcal{Q}_{\iota,j} \right) \leq \sum_{\iota \in \{0,A\}} \sum_{j \in \{0,A\}} \mu(\mathcal{Q}_{\iota,j}) \leq \frac{4}{1600}.$$

6. Combining the two union bounds yields

$$\mu(\mathcal{F}_{\text{bad}}) \leq \frac{4}{800} + \frac{4}{1600} = \frac{3}{400}.$$

7. Fix a sample in $\mathcal{F}_{\text{bad}}^c$. By the definitions of the bad events,

$$\|\text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota}\|_2 \leq 40\sqrt{\Lambda_{n,v}}, \quad |\langle \mathbf{E}_{1,\iota}, \mathbf{E}_{2,j} \rangle| \leq 40\sqrt{M} \Lambda_{n,v}.$$

For every $c \in \mathbb{R}$, the affine mean belongs to $\mathcal{S}_P$. Orthogonal projection and Cauchy–Schwarz therefore give

$$\begin{aligned} |\langle \theta_P(c), \mathbf{E}_{b,\iota} \rangle| &= |\langle \theta_P(c), \text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota} \rangle| \\ &\leq \|\theta_P(c)\|_2 \|\text{pr}_{\mathcal{S}_P} \mathbf{E}_{b,\iota}\|_2 \\ &\leq 40\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2. \end{aligned}$$

Now take any $|c| \leq 1/2$. The mean-plus-error expansion is

$$W(c) - \|\theta_P(c)\|_2^2 = \langle \theta_P(c), \mathbf{E}_1(c) + \mathbf{E}_2(c) \rangle + \langle \mathbf{E}_1(c), \mathbf{E}_2(c) \rangle.$$

For the linear term, the triangle inequality yields

$$\begin{aligned} |\langle \theta_P(c), \mathbf{E}_1(c) + \mathbf{E}_2(c) \rangle| &\leq \sum_{b=1}^2 (|\langle \theta_P(c), \mathbf{E}_{b,0} \rangle| + |c| |\langle \theta_P(c), \mathbf{E}_{b,A} \rangle|) \\ &\leq (2 + 2|c|) 40\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 \\ &\leq 120\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2. \end{aligned}$$

For the quadratic term, expansion of the two affine errors gives

$$\begin{aligned} |\langle \mathbf{E}_1(c), \mathbf{E}_2(c) \rangle| &\leq |\langle \mathbf{E}_{1,0}, \mathbf{E}_{2,0} \rangle| + |c| |\langle \mathbf{E}_{1,0}, \mathbf{E}_{2,A} \rangle| \\ &\quad + |c| |\langle \mathbf{E}_{1,A}, \mathbf{E}_{2,0} \rangle| + |c|^2 |\langle \mathbf{E}_{1,A}, \mathbf{E}_{2,A} \rangle| \\ &\leq (1 + 2|c| + |c|^2) 40\sqrt{M} \Lambda_{n,v} \\ &\leq \frac{9}{4} 40\sqrt{M} \Lambda_{n,v} = 90\sqrt{M} \Lambda_{n,v}. \end{aligned}$$

Here $|c|^2 \leq 1/4$. Since both ledger terms are nonnegative,

$$\begin{aligned} |W(c) - \|\theta_P(c)\|_2^2| &\leq 120\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 + 90\sqrt{M} \Lambda_{n,v} \\ &\leq 128 \left(\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 + \sqrt{M} \Lambda_{n,v} \right). \end{aligned}$$

The sample was fixed throughout and c was arbitrary in the profiling interval. Thus

$$\mathcal{F}_{\text{bad}}^c \subseteq \mathcal{G}_{\text{prof}}.$$

8. Finally, measurability of the bad event and the inclusion just proved imply

$$\mu(\mathcal{G}_{\text{prof}}) \geq \mu(\mathcal{F}_{\text{bad}}^c) = 1 - \mu(\mathcal{F}_{\text{bad}}) \geq 1 - \frac{3}{400} = \frac{397}{400} = 0.9925.$$

Together with the measurable-event argument and the zero-ledger case, this proves all the assertions. □

Simultaneous control permits minimization over the candidate constant within the same event. Its fluctuation bound separates a term proportional to the population score norm from a term involving the coarse dimension and covariance ledger. When the covariance ledger vanishes, the simultaneous identity gives the population target exactly.

The public computation retains this affine structure explicitly. Using the one-based block convention of [Algorithm 3](#), set $Z_{b,1} := Z_{b,A}$ and

$$w_{rr'} := \langle Z_{1,r}, Z_{2,r'} \rangle, \quad r, r' \in \{0, 1\}.$$

Then the computed quadratic is

$$W(c) = w_{00} - c(w_{01} + w_{10}) + c^2 w_{11}.$$

Its minimum over $[-1/2, 1/2]$ is obtained by comparing the two endpoints and, when $w_{11} > 0$ and the vertex lies in the interval, the value at

$$c_{\text{vert}} := \frac{w_{01} + w_{10}}{2w_{11}}.$$

This formula preserves both mixed affine-coefficient contributions through their sum. The same calculation applies under the zero-based convention after translating the block indices.

Together, [Lemmas 2 to 4](#) supply the population and fluctuation bounds used in [Theorem 1](#). The cutoff in [Algorithm 1](#) rounds

$$2B^2 + 2^{16}\sqrt{M} \Lambda_{n,v}$$

upward to a dyadic value. The calibration theorem gives size at most 0.0075 throughout the constant-effect null and type-II error at most 0.0075 at its calibrated separation, for $n \geq 4$ when that separation is below the model distance supremum. The zero rejection rule supplies the stated size guarantee at $n = 2, 3$.

The calibrated separation $R_{\text{att}}(n, v)$ in [Theorem 1](#) groups coarse approximation, the bias ledger, and the covariance contribution. The explicit attainment statement below expresses their combined requirement as a power of the sample size with a fully specified multiplier.

B.5 Explicit attainment constants and separation

Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$ from [Definition 1](#), and use the exponents q, S, t, E_0, E_4, E of [Algorithm 1](#). Before stating attainment, define the minimum smoothness s_0 , the clipping allocation exponent λ_0 , the increment summability exponent η_0 , and the geometric-series factor $A(x)$ by

$$s_0 := \min\{\alpha, \beta, \gamma\}, \quad \lambda_0 := \frac{(p-1)^2}{4p}, \quad \eta_0 := \frac{(p-1)(p+6)}{4p}, \quad A(x) := (1-2^{-x})^{-1} \quad (x > 0).$$

Here $A(x)$ denotes a scalar series factor; its argument distinguishes it from the treatment coordinate. Define the grouped bias, summation, singleton, and canonical-pair constants by

$$\begin{aligned} B_* &:= 420 + 800\{A(\alpha) + A(\lambda_0)\}, \\ D_* &:= 1 + \{\exp(1)S \log 2\}^{-1}, \\ L_* &:= 16 + 880A(s_0) + 160D_* + 960A(\alpha), \\ G_* &:= \sqrt{64}\{2^{t/2}A(1/2) + A(\eta_0/2)\}. \end{aligned}$$

The aggregate covariance constant A_* and attainment multiplier C_v^{att} are

$$A_* := \sqrt{2}\{24576 + 64L_*^2 + 16384 + 256G_*^2\}, \quad C_v^{\text{att}} := 16\{16 + 5B_* + 1024\sqrt{A_*}\}.$$

These definitions resolve every constant entering the public attainment multiplier.

Using the heterogeneity distance and model supremum defined in [Theorem 1](#), define the fixed witness distance d_0 and public threshold N_v^{att} by

$$d_0 := \frac{1}{8\sqrt{2}}, \quad N_v^{\text{att}} := \left\lceil \max \left\{ 4, \left(\frac{2C_v^{\text{att}}}{d_0} \right)^{1/E} \right\} \right\rceil.$$

For the rejection expectation below, $\mathcal{D}_n$ denotes n independent original records with law P , and U denotes an independent uniform variable on $[0, 1]$, as in [Definition 4](#). Set

$$R_n(P, \phi) := \mathbb{E}[\phi(\mathcal{D}_n, U)].$$

For an admissible bounded-model tuple $w = (\alpha, \beta, \gamma)$ from [Definition 7](#), define the matched exponent, multiplier, and distance supremum by

$$E_b := \min \left\{ \frac{2\gamma}{4\gamma + 1}, \frac{2(\alpha + \beta)}{1 + 2(\alpha + \beta) + (\alpha + \beta)/(2\gamma)} \right\}, \quad C_{(2,w)}^{\text{att}} := C_{(2,\alpha,\beta,\gamma)}^{\text{att}}, \quad D_w^b := \sup_{P \in \mathcal{M}_w^b} d(P).$$

The following result gives attainment for the original model and its matched bounded submodel, together with public tuning bounds and uniform control of the constants on compact sets of admissible tuples.

Theorem 2. [\[thm:original-record-attainable-rate\]](#) (Original-record attainable separation rate) *For every admissible public tuple $v = (p, \alpha, \beta, \gamma)$ of [Definition 1](#), use the explicit rule $\phi_{n,v}^*$ and its tuning quantities from [Algorithm 1](#) and [Definition 15](#). Write*

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad t = \frac{2-p}{p-1}, \quad E_0 = \frac{2\gamma q}{2\gamma + q}, \quad E_4 = \frac{2S}{1 + 2\alpha + \beta/q + S/(2\gamma)}, \quad E = \min\{E_0, E_4\}.$$

Define the minimum smoothness s_0 , increment summability exponent η_0 , geometric-series factor $A(x)$, and attainment constants by

$$\begin{aligned} s_0 &= \min\{\alpha, \beta, \gamma\}, & \lambda_0 &= \frac{(p-1)^2}{4p}, \quad \eta_0 = \frac{(p-1)(p+6)}{4p}, \\ A(x) &= (1 - 2^{-x})^{-1} \quad (x > 0), \\ B_* &= 420 + 800\{A(\alpha) + A(\lambda_0)\}, \\ D_* &= 1 + \{\exp(1)S \log 2\}^{-1}, \\ L_* &= 16 + 880A(s_0) + 160D_* + 960A(\alpha), \\ G_* &= \sqrt{64}\{2^{t/2}A(1/2) + A(\eta_0/2)\}, \\ A_* &= \sqrt{2}\{24576 + 64L_*^2 + 16384 + 256G_*^2\}, \\ C_v^{\text{att}} &= 16\{16 + 5B_* + 1024\sqrt{A_*}\}. \end{aligned}$$

Define the fixed witness distance d_0 and public attainment threshold N_v^{att} by

$$d_0 = \frac{1}{8\sqrt{2}}, \quad N_v^{\text{att}} = \left\lceil \max \left\{ 4, \left(\frac{2C_v^{\text{att}}}{d_0} \right)^{1/E} \right\} \right\rceil.$$

Let λ be Lebesgue probability measure on $[0, 1]$. Define the heterogeneity distance $d(P)$ and its model supremum D_v by

$$d(P) = \left\{ \int_{[0,1]} \left(\tau_P(x) - \int_{[0,1]} \tau_P(z) d\lambda(z) \right)^2 d\lambda(x) \right\}^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P).$$

Let $\mathcal{D}_n$ be a sample of n independent original records with law P , and let U be an independent uniform randomization variable on $[0, 1]$. For a test ϕ , define its rejection expectation by

$$R_n(P, \phi) = \mathbb{E}[\phi(\mathcal{D}_n, U)].$$

Then $C_v^{\text{att}} > 0$, and the following guarantees hold:

- **(Size and separation.)** For every integer $n \geq 2$,

$$R_n(P, \phi_{n,v}^*) \leq 0.0075 \quad \text{for every } P \in H_0(v),$$

where the null is given in [Definition 2](#). Whenever $C_v^{\text{att}} n^{-E} < D_v$,

$$1 - R_n(P, \phi_{n,v}^*) \leq 0.0075 \quad \text{for every } P \in \mathcal{M}_v \text{ with } d(P) \geq C_v^{\text{att}} n^{-E}.$$

The capped critical radius of [Definition 6](#) satisfies

$$r_n^*(v) \leq C_v^{\text{att}} n^{-E}.$$

- **(Witness threshold.)** For every integer $n \geq N_v^{\text{att}}$,

$$C_v^{\text{att}} n^{-E} < d_0.$$

- **(Tuning bounds.)** For every integer $n \geq 4$, the coarse rank M , fine rank K , block size s , main clipping cutoff T_0 , and increment cutoffs T_j from [Algorithm 1](#) and [Definition 20](#) satisfy

$$M \leq 2s, \quad K \leq 2s^2, \quad 1 \leq T_0 \leq 2s, \quad 1 \leq T_j \leq T_0 \quad (1 \leq j \leq L).$$

- **(Compact uniformity.)** For every compact set K of public tuples in the product topology whose members are all admissible under [Definition 1](#), there exist real constants C, N, e , with $e > 0$, such that, for every $v \in K$,

$$C_v^{\text{att}} \leq C, \quad N_v^{\text{att}} \leq N, \quad e \leq E(v).$$

Here $E(v)$ denotes the displayed minimum exponent evaluated at v .

For every admissible smoothness tuple $w = (\alpha, \beta, \gamma)$ of [Definition 7](#), define the bounded separation exponent E_b , matched attainment multiplier $C_{(2,w)}^{\text{att}}$, and bounded-model distance supremum D_w^b by

$$E_b = \min \left\{ \frac{2\gamma}{4\gamma + 1}, \frac{2(\alpha + \beta)}{1 + 2(\alpha + \beta) + (\alpha + \beta)/(2\gamma)} \right\}, \quad C_{(2,w)}^{\text{att}} = C_{(2,\alpha,\beta,\gamma)}^{\text{att}}, \quad D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P).$$

The original-model exponent evaluated at $(2, \alpha, \beta, \gamma)$ equals E_b . The bounded-model rule $\phi_{n,w}^{*,b}$ from [Definition 15](#), obtained by evaluating the same ledger rule at $(2, \alpha, \beta, \gamma)$, satisfies, for every integer $n \geq 2$,

$$R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad \text{for every } P \in H_0^b(w),$$

with the bounded null as in [Definition 9](#). Whenever $C_{(2,w)}^{\text{att}} n^{-E_b} < D_w^b$,

$$1 - R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad \text{for every } P \in \mathcal{M}_w^b \text{ with } d(P) \geq C_{(2,w)}^{\text{att}} n^{-E_b}.$$

The bounded capped critical radius of [Definition 13](#) satisfies

$$r_n^{*,b}(w) \leq C_{(2,w)}^{\text{att}} n^{-E_b}.$$

The attainment guarantee combines the mean separation in [Lemma 2](#) with the simultaneous fluctuation bound in [Lemma 4](#). Its exponent is the minimum of the two public separation exponents, while its multiplier accounts explicitly for projection, clipping, and covariance contributions. Evaluating the same construction at moment order two gives the bounded-model exponent and multiplier. The power clauses apply when the stated separation is below the corresponding model distance supremum; the capped-radius bounds apply with the capping conventions in [Definitions 6](#) and [13](#).

The tuning bounds place every selected rank and outcome cutoff within the finite public grids of [Algorithm 1](#). Compact uniformity supplies common bounds for the attainment constants and thresholds over compact sets of admissible tuples. Thus the explicit procedure links a finite original-record calculation to a separation guarantee whose dependence on the public moment and smoothness parameters is fully specified.

C Full-record lower bounds and tail essentiality

The lower bounds complement the attainment guarantee in [Theorem 2](#) by constructing null and alternative mixtures that remain close when the entire original record is observed. Two constructions supply the required comparisons: paired tents with constant propensity, and coupled continuous coefficient fields in the rough phase. Their separate bounded versions provide witnesses at moment order two.

We use total variation in the convention

$$\text{TV}(P, Q) = \sup_{E \text{ measurable}} |P(E) - Q(E)|.$$

For probability laws $P \ll Q$, the directed chi-square divergence is

$$\chi^2(P \| Q) = \int \left(\frac{dP}{dQ} - 1 \right)^2 dQ = \int \left(\frac{dP}{dQ} \right)^2 dQ - 1.$$

These divergences compare probability laws on the same measurable space.

A finite prior draws one original-record law and then generates the independent sample from that law. Specifically, for

$$\pi = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad k \geq 1, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1,$$

write

$$\text{supp}_+(\pi) = \{P_j : \omega_j > 0\}, \quad \mathbb{P}_{\pi,n} = \sum_{j=1}^k \omega_j P_j^{\otimes n}.$$

Thus $\mathbb{P}_{\pi,n}$ mixes product laws after the common latent law has been drawn. Repeated support laws have their weights combined. Normalization ensures that the positive-weight support is nonempty.

For the paired-tent comparison, let λ be Lebesgue probability measure on $[0, 1]$, and let $Q_{a,P}(dy \mid x)$ be the conditional outcome probability kernel in treatment arm a . The following result supplies both rare-mark witnesses under a finite conditional moment and signed-binary witnesses in the matched bounded model.

Lemma 5. [lem:paired-tent-full-record-lower] (Paired-tent full-record lower bounds) *Let λ denote Lebesgue probability measure on $[0, 1]$, and let $Q_{a,P}(dy \mid x)$ denote the conditional outcome probability kernel under law P in treatment arm $a \in \{0, 1\}$. Define the heterogeneity distance, its model suprema, and the fixed witness distance by*

$$d(P) = \left\{ \int_0^1 \left(\tau_P(x) - \int_0^1 \tau_P(z) \lambda(dz) \right)^2 \lambda(dx) \right\}^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P), \quad D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P), \quad d_0 = \frac{1}{8\sqrt{3}}$$

where the models are defined in [Definitions 1](#) and [7](#).

Write a normalized finite probability prior as

$$\pi = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1.$$

Define its positive-weight support and full original-record mixture by

$$\text{supp}_+(\pi) = \{P_j : \omega_j > 0\}, \quad \mathbb{P}_{\pi,n} = \sum_{j=1}^k \omega_j P_j^{\otimes n}.$$

For probability laws on the same measurable space, define total variation by

$$\text{TV}(P, Q) = \sup_{E \text{ measurable}} |P(E) - Q(E)|.$$

For every admissible moment and smoothness tuple $v = (p, \alpha, \beta, \gamma)$ as in [Definition 1](#), put

$$q = \frac{p-1}{p}, \quad E_0 = \frac{2\gamma q}{2\gamma + q}, \quad c_v^e = \frac{4^{-\gamma}}{16\sqrt{3}}.$$

There exists a sequence of mark magnitudes $L : \mathbb{N} \rightarrow \mathbb{R}$ with

$$\lim_{n \rightarrow \infty} L(n) = +\infty$$

such that, for every integer $n \geq 2$, there exist a null law P_0 and a normalized finite probability prior π_1 , with at least one strictly positive weight, satisfying:

- **(Null and propensity.)** The law P_0 belongs to $H_0(v)$ from [Definition 2](#), and

$$e_{P_0}(x) = \frac{1}{2} \quad \text{for every } x \in [0, 1].$$

- **(Arm support and moments.)** The magnitude $L(n)$ is positive. For $P = P_0$ and every $P \in \text{supp}_+(\pi_1)$, each arm $a \in \{0, 1\}$ satisfies, for λ -almost every x ,

$$Q_{a,P}(\{0, -L(n), L(n)\} \mid x) = 1, \quad \int_{\mathbb{R}} |y|^p Q_{a,P}(dy \mid x) \leq 1.$$

- **(Alternative placement.)** Every $P \in \text{supp}_+(\pi_1)$ belongs to $\mathcal{M}_v$ from [Definition 1](#), satisfies $e_P(x) = 1/2$ for every $x \in [0, 1]$, and obeys

$$c_v^e n^{-E_0} \leq d(P) < d_0.$$

Moreover, $d_0 \leq D_v$.

- **(Full-record comparison.)** Define the alternative mixture by

$$\bar{P}_1^{(n)} = \mathbb{P}_{\pi_{1,n}}.$$

The same witnesses satisfy

$$\text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}) < \frac{1}{2}.$$

For every such n , the supplied-propensity and original-record minimax testing risks in [Definitions 5](#) and [26](#) satisfy

$$B_n^e(v, c_v^e n^{-E_0}) \geq \frac{2}{5}, \quad B_n(v, c_v^e n^{-E_0}) \geq \frac{2}{5}.$$

For every admissible smoothness tuple $w = (\alpha, \beta, \gamma)$ as in [Definition 7](#), put

$$c_w^b = \frac{4^{-\gamma}}{16\sqrt{3}}.$$

For every integer $n \geq 2$, there exist a signed-binary null law P_0^b and a normalized finite probability prior π_1^b , with at least one strictly positive weight, satisfying:

- **(Signed-binary null.)** The law P_0^b belongs to $H_0^{\text{bin}}(w)$ from [Definition 10](#), and

$$e_{P_0^b}(x) = \frac{1}{2} \quad \text{for every } x \in [0, 1].$$

- **(Signed-binary alternatives.)** Every $P \in \text{supp}_+(\pi_1^b)$ belongs to $\mathcal{M}_w^{\text{bin}}$ from [Definition 8](#), satisfies $e_P(x) = 1/2$ for every $x \in [0, 1]$, and obeys

$$c_w^b n^{-2\gamma/(4\gamma+1)} \leq d(P) < d_0.$$

Moreover, $d_0 \leq D_w^b$.

- **(Full-record comparison.)** Define the signed-binary alternative mixture by

$$\bar{P}_{1,b}^{(n)} = \mathbb{P}_{\pi_{1^b,n}}.$$

The same witnesses satisfy

$$\text{TV}((P_0^b)^{\otimes n}, \bar{P}_{1,b}^{(n)}) < \frac{1}{2}.$$

For every such n , the bounded-model minimax testing risk in [Definition 11](#) satisfies

$$B_n^b(w, c_w^b n^{-2\gamma/(4\gamma+1)}) \geq \frac{2}{5}.$$

Proof of [Lemma 5](#). 1. *Choice of the mark sequence.* Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$.

Define

$$\vartheta_\Delta = \frac{2q}{2\gamma + q}, \quad \kappa_\Delta = \frac{1}{16}, \quad N_{\Delta,n} = 2 \left\lceil n^{\vartheta_\Delta} \right\rceil, \quad h_{\Delta,n} = N_{\Delta,n}^{-1} \quad (n \in \mathbb{N}),$$

where the reciprocal at zero is defined to be zero. Choose the sequence

$$L(n) = \begin{cases} 0, & n = 0, \\ N_{\Delta,n}^{\gamma/(p-1)}, & n \geq 1. \end{cases}$$

Admissibility gives $p - 1 > 0$, $q > 0$, and $0 < \gamma \leq 1$, hence

$$\vartheta_\Delta > 0, \quad \frac{\gamma}{p-1} > 0.$$

For $n \geq 1$, the ceiling inequality gives

$$N_{\Delta,n} \geq n^{\vartheta_\Delta}.$$

Thus $N_{\Delta,n} \rightarrow \infty$, and raising it to the positive power $\gamma/(p-1)$ proves $L(n) \rightarrow \infty$.

Now fix $n \geq 2$. Since $n^{\vartheta_\Delta} \geq 1$,

$$2n^{\vartheta_\Delta} \leq N_{\Delta,n} < 2(n^{\vartheta_\Delta} + 1) \leq 4n^{\vartheta_\Delta}.$$

In particular, $N_{\Delta,n}$ is even and at least two. Define

$$\varepsilon_{\Delta,n} = h_{\Delta,n}^{\gamma/q}.$$

Then

$$0 < h_{\Delta,n} < 1, \quad 0 < \varepsilon_{\Delta,n} < 1, \quad L(n) = h_{\Delta,n}^{-\gamma/(p-1)} > 0.$$

The identities

$$\frac{\gamma}{q} - \frac{\gamma}{p-1} = \gamma, \quad \frac{\gamma}{q} - \frac{p\gamma}{p-1} = 0$$

give

$$\varepsilon_{\Delta,n} L(n) = h_{\Delta,n}^\gamma, \quad \varepsilon_{\Delta,n} L(n)^p = 1.$$

2. *Paired tents and the finite prior.* Define the compactly supported tent on the whole real line by

$$\psi_\Delta(t_\Delta) = \begin{cases} 2 \min\{t_\Delta, 1 - t_\Delta\}, & 0 \leq t_\Delta \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad t_\Delta \in \mathbb{R}.$$

Define the number of pairs, their index set, and their sign vectors by

$$m_{\Delta,n} = \frac{N_{\Delta,n}}{2}, \quad \mathcal{J}_{\Delta,n} = \{0, \dots, m_{\Delta,n} - 1\}, \quad \varsigma \in \{-1, 1\}^{\mathcal{J}_{\Delta,n}}.$$

For $j_\Delta \in \mathcal{J}_{\Delta,n}$ and $x \in \mathbb{R}$, put

$$\psi_{\Delta,j_\Delta}^\varsigma(x) = \varsigma_{j_\Delta} [\psi_\Delta(N_{\Delta,n}x - 2j_\Delta) - \psi_\Delta(N_{\Delta,n}x - (2j_\Delta + 1))],$$

$$f_{\Delta,\varsigma}(x) = \sum_{j_\Delta \in \mathcal{J}_{\Delta,n}} \psi_{\Delta,j_\Delta}^\varsigma(x).$$

These functions are continuous. A nonzero paired bump satisfies

$$\psi_{\Delta,j_\Delta}^\varsigma(x) \neq 0 \implies 2j_\Delta < N_{\Delta,n}x < 2j_\Delta + 2.$$

The open intervals in this display are disjoint for distinct pair indices. Within a pair, the two individual tents also have disjoint nonzero supports. Consequently,

$$|\psi_{\Delta,j_\Delta}^\varsigma(x)| \leq 1, \quad |f_{\Delta,\varsigma}(x)| \leq 1.$$

For the Hölder estimate, the identity

$$\psi_\Delta(t_\Delta) = \max\{0, 2 \min(t_\Delta, 1 - t_\Delta)\}$$

and the Lipschitz inequalities for minimum and maximum imply

$$|\psi_\Delta(t_\Delta) - \psi_\Delta(t'_\Delta)| \leq 2|t_\Delta - t'_\Delta| \quad (t_\Delta, t'_\Delta \in \mathbb{R}).$$

Applying this to the two tents in a pair gives

$$|\psi_{\Delta,j_\Delta}^\varsigma(x) - \psi_{\Delta,j_\Delta}^\varsigma(x')| \leq 4N_{\Delta,n}|x - x'|.$$

Define the active pairs at a point by

$$\mathcal{A}_{\Delta,\varsigma}(x) = \left\{ j_\Delta \in \mathcal{J}_{\Delta,n} : \psi_{\Delta,j_\Delta}^\varsigma(x) \neq 0 \right\}.$$

The support calculation gives

$$|\mathcal{A}_{\Delta,\varsigma}(x)| \leq 1, \quad |\mathcal{A}_{\Delta,\varsigma}(x) \cup \mathcal{A}_{\Delta,\varsigma}(x')| \leq 2.$$

Every summand outside this union vanishes at both points. Therefore

$$\begin{aligned} |f_{\Delta,\varsigma}(x) - f_{\Delta,\varsigma}(x')| &\leq \sum_{j_\Delta \in \mathcal{A}_{\Delta,\varsigma}(x) \cup \mathcal{A}_{\Delta,\varsigma}(x')} \left| \psi_{\Delta,j_\Delta}^\varsigma(x) - \psi_{\Delta,j_\Delta}^\varsigma(x') \right| \\ &\leq 8N_{\Delta,n}|x - x'|. \end{aligned}$$

If $N_{\Delta,n}|x - x'| \leq 1$, then

$$N_{\Delta,n}|x - x'| \leq (N_{\Delta,n}|x - x'|)^\gamma.$$

If $N_{\Delta,n}|x - x'| > 1$, the unit envelope instead gives

$$|f_{\Delta,\varsigma}(x) - f_{\Delta,\varsigma}(x')| \leq 2 \leq 8(N_{\Delta,n}|x - x'|)^\gamma.$$

Both cases yield

$$h_{\Delta,n}^\gamma |f_{\Delta,\varsigma}(x) - f_{\Delta,\varsigma}(x')| \leq 8|x - x'|^\gamma.$$

Give every sign vector the same weight, and define

$$\pi_1 = 2^{-m_{\Delta,n}} \sum_{\varsigma \in \{-1,1\}^{\mathcal{J}_{\Delta,n}}} \delta_{P_\varsigma},$$

where the laws P_ς are constructed below. There are exactly $2^{m_{\Delta,n}}$ sign vectors, so

$$2^{-m_{\Delta,n}} > 0, \quad \sum_{\varsigma} 2^{-m_{\Delta,n}} = 1.$$

The constant sign vector supplies an index, proving that the prior has a strictly positive weight.

3. *The null law and arm moments.* Construct P_0 and P_ς with uniform covariate distribution and propensity $1/2$. Their common control kernel and their treated kernels are

$$\begin{aligned} Q_{0,P_0}(\cdot | x) &= Q_{0,P_\varsigma}(\cdot | x) = \delta_0, \\ Q_{1,P_0}(\cdot | x) &= (1 - \varepsilon_{\Delta,n})\delta_0 + \frac{\varepsilon_{\Delta,n}}{2}\delta_{L(n)} + \frac{\varepsilon_{\Delta,n}}{2}\delta_{-L(n)}, \\ Q_{1,P_\varsigma}(\cdot | x) &= (1 - \varepsilon_{\Delta,n})\delta_0 \\ &\quad + \frac{\varepsilon_{\Delta,n}}{2}(1 + \kappa_\Delta f_{\Delta,\varsigma}(x))\delta_{L(n)} \\ &\quad + \frac{\varepsilon_{\Delta,n}}{2}(1 - \kappa_\Delta f_{\Delta,\varsigma}(x))\delta_{-L(n)}. \end{aligned}$$

Precisely, these kernels specify the record laws through

$$P(dx, \{a\}, dy) = \frac{1}{2} \lambda(dx) Q_{a,P}(dy | x), \quad a \in \{0, 1\}, \quad P \in \{P_0, P_\varsigma\}.$$

The displayed masses sum to one. They are nonnegative because

$$1 \pm \kappa_\Delta f_{\Delta,\varsigma}(x) \geq 1 - \kappa_\Delta = \frac{15}{16} > 0, \quad 1 - \varepsilon_{\Delta,n} > 0.$$

Continuity of the tents supplies the required kernel measurability.

For every $x \in [0, 1]$, direct integration gives

$$e_{P_0}(x) = \frac{1}{2}, \quad m_{0,P_0}(x) = m_{1,P_0}(x) = \tau_{P_0}(x) = 0,$$

and, for $P = P_0$ or $P = P_\varsigma$,

$$Q_{a,P}(\{0, -L(n), L(n)\} | x) = 1, \quad \int |y|^p Q_{a,P}(dy | x) = \begin{cases} 0, & a = 0, \\ \varepsilon_{\Delta,n} L(n)^p = 1, & a = 1. \end{cases}$$

Thus both arm moments are at most one. For P_0 , the propensity lies in $[1/4, 3/4]$, its supremum norm is $1/2$, and its Hölder differences vanish. The baseline and effect have zero supremum norms and zero Hölder differences. Together with uniform design and the moment bound $1 \leq 10$, these facts verify all the restrictions in [Definition 1](#). Its effect is the constant zero, so [Definition 2](#) gives

$$P_0 \in H_0(v).$$

4. *Alternative membership and separation.* The alternative arm means are

$$e_{P_\varsigma}(x) = \frac{1}{2}, \quad m_{0,P_\varsigma}(x) = 0,$$

$$m_{1,P_\varsigma}(x) = \tau_{P_\varsigma}(x) = \varepsilon_{\Delta,n} L(n) \kappa_\Delta f_{\Delta,\varsigma}(x) = \kappa_\Delta h_{\Delta,n}^\gamma f_{\Delta,\varsigma}(x).$$

The geometric estimates imply

$$\|\tau_{P_\varsigma}\|_\infty \leq \kappa_\Delta h_{\Delta,n}^\gamma \leq \frac{1}{16} < \frac{1}{2},$$

$$|\tau_{P_\varsigma}(x) - \tau_{P_\varsigma}(x')| \leq 8\kappa_\Delta |x - x'|^\gamma = \frac{1}{2} |x - x'|^\gamma \leq 20 |x - x'|^\gamma.$$

Thus the effect satisfies its Hölder and supremum restrictions. The constant propensity and zero baseline satisfy their respective smoothness and bound restrictions, and the preceding arm calculation supplies the raw moments. Hence

$$P_\varsigma \in \mathcal{M}_v \quad \text{for every } \varsigma.$$

To compute the centered distance, reflection about $1/2$ gives

$$\int_0^1 \psi_\Delta(t_\Delta) dt_\Delta = 2 \int_0^{1/2} 2t_\Delta dt_\Delta = \frac{1}{2},$$

$$\int_0^1 \psi_\Delta(t_\Delta)^2 dt_\Delta = 2 \int_0^{1/2} 4t_\Delta^2 dt_\Delta = \frac{1}{3}.$$

A change of variables, using the compact support of the tent, yields, for every integer i_Δ with $0 \leq i_\Delta < N_{\Delta,n}$,

$$\int_0^1 \psi_\Delta(N_{\Delta,n}x - i_\Delta) \lambda(dx) = \frac{h_{\Delta,n}}{2},$$

$$\int_0^1 \psi_\Delta(N_{\Delta,n}x - i_\Delta)^2 \lambda(dx) = \frac{h_{\Delta,n}}{3}.$$

The opposite signs cancel the two equal areas in each pair. Disjoint nonzero supports eliminate the cross terms in the square. Consequently,

$$\int f_{\Delta,\varsigma} d\lambda = \sum_{j_\Delta} \varsigma_{j_\Delta} \left(\frac{h_{\Delta,n}}{2} - \frac{h_{\Delta,n}}{2} \right) = 0,$$

$$\int f_{\Delta,\varsigma}^2 d\lambda = m_{\Delta,n} \frac{2h_{\Delta,n}}{3} = \frac{1}{3}.$$

It follows that

$$\int \tau_{P_\varsigma} d\lambda = 0, \quad d(P_\varsigma) = \frac{\kappa_\Delta h_{\Delta,n}^\gamma}{\sqrt{3}}.$$

The upper rank bound supplies

$$h_{\Delta,n}^\gamma \geq (4n^{\vartheta_\Delta})^{-\gamma} = 4^{-\gamma} n^{-E_0},$$

because $\gamma\vartheta_\Delta = E_0$. Therefore

$$d(P_\varsigma) \geq \frac{4^{-\gamma}}{16\sqrt{3}} n^{-E_0} = c_v^e n^{-E_0}.$$

For the strict upper comparison, expand the centered square and use the effect envelope:

$$\begin{aligned} d(P_\varsigma)^2 &= \int \tau_{P_\varsigma}^2 d\lambda - \left(\int \tau_{P_\varsigma} d\lambda \right)^2 \\ &\leq \int \tau_{P_\varsigma}^2 d\lambda \leq \frac{1}{16^2}. \end{aligned}$$

Since $\sqrt{2} < 2$,

$$d(P_\varsigma) \leq \frac{1}{16} < \frac{1}{8\sqrt{2}} = d_0.$$

5. *Comparison of the complete-record laws.* Define the one-record likelihood on $[0, 1] \times \{0, 1\} \times \mathbb{R}$ by

$$\ell_{\Delta, \varsigma}(x, a, y) = \begin{cases} 1 + \kappa_\Delta f_{\Delta, \varsigma}(x), & a = 1, y = L(n), \\ 1 - \kappa_\Delta f_{\Delta, \varsigma}(x), & a = 1, y = -L(n), \\ 1, & \text{otherwise.} \end{cases}$$

Comparison of the conditional atom masses proves

$$P_\varsigma \ll P_0, \quad \frac{dP_\varsigma}{dP_0} = \ell_{\Delta, \varsigma} \quad P_0\text{-almost surely,} \quad \frac{15}{16} \leq \ell_{\Delta, \varsigma} \leq \frac{17}{16}.$$

At each covariate value, summing over the control record and the three treated atoms gives

$$\begin{aligned} &\frac{1}{2} + \frac{1 - \varepsilon_{\Delta, n}}{2} \\ &+ \frac{\varepsilon_{\Delta, n}}{4} [(1 + \kappa_\Delta f_{\Delta, \varsigma}(x))(1 + \kappa_\Delta f_{\Delta, \varsigma'}(x)) + (1 - \kappa_\Delta f_{\Delta, \varsigma}(x))(1 - \kappa_\Delta f_{\Delta, \varsigma'}(x))] \\ &= 1 + \frac{\varepsilon_{\Delta, n} \kappa_\Delta^2}{2} f_{\Delta, \varsigma}(x) f_{\Delta, \varsigma'}(x). \end{aligned}$$

The same support calculation used for the squared tents gives

$$\begin{aligned} \int f_{\Delta, \varsigma} f_{\Delta, \varsigma'} d\lambda &= \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta} \left(\frac{h_{\Delta, n}}{3} + \frac{h_{\Delta, n}}{3} \right) \\ &= \frac{2h_{\Delta, n}}{3} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta}. \end{aligned}$$

Define the overlap coefficient by

$$\varrho_{\Delta, n} = \frac{\varepsilon_{\Delta, n} \kappa_\Delta^2}{6}.$$

Integration over the entire record now yields

$$\int \ell_{\Delta, \varsigma} \ell_{\Delta, \varsigma'} dP_0 = 1 + \frac{\varrho_{\Delta, n}}{m_{\Delta, n}} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta}.$$

Here $m_{\Delta,n} \geq 1$, $0 \leq \varrho_{\Delta,n} \leq 1$, and

$$\left| \sum_{j_{\Delta}} \varsigma_{j_{\Delta}} \varsigma'_{j_{\Delta}} \right| \leq m_{\Delta,n}.$$

Thus every overlap base is nonnegative.

For probability laws $\mu_{\text{alt}} \ll \mu_{\text{ref}}$ with square-integrable density, define the directed chi-square divergence by

$$\chi^2(\mu_{\text{alt}} \parallel \mu_{\text{ref}}) = \int \left(\frac{d\mu_{\text{alt}}}{d\mu_{\text{ref}}} - 1 \right)^2 d\mu_{\text{ref}}.$$

The complete-record mixture and its density are

$$\overline{P}_1^{(n)} = 2^{-m_{\Delta,n}} \sum_{\varsigma} P_{\varsigma}^{\otimes n},$$

$$G_{\Delta}(\mathcal{D}_n) = \frac{d\overline{P}_1^{(n)}}{dP_0^{\otimes n}}(\mathcal{D}_n) = 2^{-m_{\Delta,n}} \sum_{\varsigma} \prod_{i_{\text{rec}}=1}^n \ell_{\Delta,\varsigma}(X_{i_{\text{rec}}}, A_{i_{\text{rec}}}, Y_{i_{\text{rec}}}).$$

These are bounded finite densities, so expansion of their square and independence of the records give

$$\begin{aligned} 1 + \chi^2(\overline{P}_1^{(n)} \parallel P_0^{\otimes n}) &= \int G_{\Delta}^2 dP_0^{\otimes n} \\ &= 2^{-2m_{\Delta,n}} \sum_{\varsigma} \sum_{\varsigma'} \left(1 + \frac{\varrho_{\Delta,n}}{m_{\Delta,n}} \sum_{j_{\Delta}} \varsigma_{j_{\Delta}} \varsigma'_{j_{\Delta}} \right)^n. \end{aligned}$$

Nonnegativity of the bases and $1 + t_{\Delta} \leq e^{t_{\Delta}}$ imply

$$\left(1 + \frac{\varrho_{\Delta,n}}{m_{\Delta,n}} \sum_{j_{\Delta}} \varsigma_{j_{\Delta}} \varsigma'_{j_{\Delta}} \right)^n \leq \exp \left(\frac{n\varrho_{\Delta,n}}{m_{\Delta,n}} \sum_{j_{\Delta}} \varsigma_{j_{\Delta}} \varsigma'_{j_{\Delta}} \right).$$

For each fixed ς , the sum over ς' factors coordinate by coordinate:

$$\begin{aligned} &2^{-m_{\Delta,n}} \sum_{\varsigma'} \exp \left(\frac{n\varrho_{\Delta,n}}{m_{\Delta,n}} \sum_{j_{\Delta}} \varsigma_{j_{\Delta}} \varsigma'_{j_{\Delta}} \right) \\ &= \prod_{j_{\Delta}} \cosh \left(\frac{n\varrho_{\Delta,n}}{m_{\Delta,n}} \varsigma_{j_{\Delta}} \right) \leq \exp \left(\frac{(n\varrho_{\Delta,n})^2}{2m_{\Delta,n}} \right). \end{aligned}$$

The last inequality follows from

$$\cosh(t_{\Delta}) = \sum_{i_{\text{exp}}=0}^{\infty} \frac{t_{\Delta}^{2i_{\text{exp}}}}{(2i_{\text{exp}})!} \leq \sum_{i_{\text{exp}}=0}^{\infty} \frac{(t_{\Delta}^2/2)^{i_{\text{exp}}}}{i_{\text{exp}}!} = e^{t_{\Delta}^2/2},$$

using $(2i_{\text{exp}})! \geq 2^{i_{\text{exp}}} i_{\text{exp}}!$. Averaging also over ς therefore gives

$$1 + \chi^2(\bar{P}_1^{(n)} \| P_0^{\otimes n}) \leq \exp \left(\frac{\kappa_{\Delta}^4}{36} n^2 \varepsilon_{\Delta,n}^2 h_{\Delta,n} \right).$$

Indeed, $2m_{\Delta,n} = N_{\Delta,n} = h_{\Delta,n}^{-1}$. Moreover,

$$h_{\Delta,n} \leq n^{-\vartheta_{\Delta}}, \quad \vartheta_{\Delta} \left(\frac{2\gamma}{q} + 1 \right) = 2,$$

so

$$n^2 \varepsilon_{\Delta,n}^2 h_{\Delta,n} = n^2 h_{\Delta,n}^{2\gamma/q+1} \leq n^2 n^{-2} = 1.$$

Consequently,

$$1 + \chi^2(\bar{P}_1^{(n)} \| P_0^{\otimes n}) \leq e^{\kappa_{\Delta}^4/36} \leq \frac{1}{1 - \kappa_{\Delta}^4/36} < 2.$$

The middle inequality follows by comparing the exponential and geometric series; the last uses $\kappa_{\Delta} = 1/16$.

Since $\int (G_{\Delta} - 1) dP_0^{\otimes n} = 0$, its positive and negative parts have equal integrals. The event $\{G_{\Delta} \geq 1\}$ then gives

$$\text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}) = \frac{1}{2} \int |G_{\Delta} - 1| dP_0^{\otimes n}.$$

Cauchy–Schwarz under the probability measure $P_0^{\otimes n}$ yields

$$\text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}) \leq \frac{1}{2} \sqrt{\chi^2(\bar{P}_1^{(n)} \| P_0^{\otimes n})} < \frac{1}{2}.$$

6. *Supplied-propensity and original-record risks.* By [Lemma 1](#),

$$d_0 \leq D_v.$$

Define the separation used here by

$$r_{\Delta} = c_v^e n^{-E_0}.$$

The positive prior weights and the established placement give

$$0 < r_{\Delta} \leq d(P_{\varsigma}) < d_0 \leq D_v.$$

Take any supplied-propensity test satisfying the size constraint in [Definition 26](#). Define its section at the common propensity and its seed average by

$$\varpi_{\text{seed}} \in [0, 1], \quad \phi_{1/2}(\mathcal{D}_n, \varpi_{\text{seed}}) = \phi^e(\mathcal{D}_n, \varpi_{\text{seed}}; e \equiv 1/2), \quad \bar{\phi}_{1/2}(\mathcal{D}_n) = \int_0^1 \phi_{1/2}(\mathcal{D}_n, \varpi_{\text{seed}}) \lambda(d\varpi_{\text{seed}}).$$

This is a measurable function with values in $[0, 1]$. Integrating its level sets shows that its expectation difference under two probability laws is bounded by their total variation.

Thus

$$\left| 2^{-m_{\Delta,n}} \sum_{\varsigma} \mathbb{R}_n^e(P_{\varsigma}, \phi^e) - \mathbb{R}_n^e(P_0, \phi^e) \right| \leq \text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}) < \frac{1}{2}.$$

Since $P_0 \in H_0(v)$,

$$2^{-m_{\Delta,n}} \sum_{\varsigma} R_n^e(P_{\varsigma}, \phi^e) < \frac{1}{10} + \frac{1}{2} = \frac{3}{5}.$$

If every positive-weight alternative had type-II error strictly below $2/5$, normalization of the weights would instead imply

$$2^{-m_{\Delta,n}} \sum_{\varsigma} R_n^e(P_{\varsigma}, \phi^e) \geq 2^{-m_{\Delta,n}} \sum_{\varsigma} \frac{3}{5} = \frac{3}{5}.$$

Hence some support law has type-II error at least $2/5$. Its model membership and separation imply

$$\sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq r_{\Delta}}} \{1 - R_n^e(P, \phi^e)\} \geq \frac{2}{5}.$$

The constant-zero test makes the admissible test class nonempty, and all rejection probabilities lie in $[0, 1]$. Taking the infimum therefore gives

$$B_n^e(v, r_{\Delta}) \geq \frac{2}{5}.$$

For an original-record test ϕ , use instead

$$\bar{\phi}(\mathcal{D}_n) = \int_0^1 \phi(\mathcal{D}_n, \varpi_{\text{seed}}) \lambda(d\varpi_{\text{seed}}).$$

The identical expectation comparison gives

$$2^{-m_{\Delta,n}} \sum_{\varsigma} R_n(P_{\varsigma}, \phi) < \frac{3}{5}.$$

The same weighted-sum contradiction supplies a support law with type-II error at least $2/5$. Taking the supremum and then the infimum in [Definition 5](#) proves

$$B_n(v, r_{\Delta}) \geq \frac{2}{5}.$$

This completes the first witness package.

7. *The signed-binary construction.* Fix an admissible $w = (\alpha, \beta, \gamma)$ and $n \geq 2$. Define

$$\vartheta_{\Delta}^b = \frac{2}{4\gamma + 1}, \quad N_{\Delta,n}^b = 2 \left\lceil n^{\vartheta_{\Delta}^b} \right\rceil, \quad h_{\Delta,n}^b = (N_{\Delta,n}^b)^{-1},$$

$$m_{\Delta,n}^b = \frac{N_{\Delta,n}^b}{2}, \quad \mathcal{J}_{\Delta,n}^b = \{0, \dots, m_{\Delta,n}^b - 1\}, \quad \delta_{\Delta,n}^b = \kappa_{\Delta}(h_{\Delta,n}^b)^{\gamma}.$$

For $\varsigma \in \{-1, 1\}^{\mathcal{J}_{\Delta,n}^b}$, define

$$f_{\Delta,\varsigma}^b(x) = \sum_{j_{\Delta} \in \mathcal{J}_{\Delta,n}^b} \varsigma_{j_{\Delta}} \left[\psi_{\Delta}(N_{\Delta,n}^b x - 2j_{\Delta}) - \psi_{\Delta}(N_{\Delta,n}^b x - (2j_{\Delta} + 1)) \right], \quad x \in \mathbb{R}.$$

These are the preceding paired-tent constructions evaluated at moment order two, for which $q = 1/2$. In particular,

$$2n^{\vartheta_{\Delta}^b} \leq N_{\Delta,n}^b \leq 4n^{\vartheta_{\Delta}^b}, \quad 0 < h_{\Delta,n}^b < 1,$$

$$|f_{\Delta,\varsigma}^b(x)| \leq 1, \quad (h_{\Delta,n}^b)^{\gamma} |f_{\Delta,\varsigma}^b(x) - f_{\Delta,\varsigma}^b(x')| \leq 8|x - x'|^{\gamma}.$$

To establish legality through the signed-binary conversion, first construct deterministic record laws with uniform design, propensity $1/2$, baseline zero, and outcomes

$$Y = 0 \quad \text{under } P_{\text{det},0}^b, \quad Y = A\delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(X) \quad \text{under } P_{\text{det},\varsigma}^b.$$

Their effects are respectively zero and $\delta_{\Delta,n}^b f_{\Delta,\varsigma}^b$. The displayed geometric estimates give

$$\left\| \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b \right\|_{\infty} \leq \frac{1}{16},$$

$$\left| \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x) - \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x') \right| \leq \frac{1}{2} |x - x'|^{\gamma}.$$

The control second moments are zero, and the treated second moments satisfy

$$|\delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x)|^2 \leq \frac{1}{16^2} \leq 10.$$

Together with the constant propensity and zero baseline, these inequalities put both deterministic laws in $\mathcal{M}_{(2,\alpha,\beta,\gamma)}$; the first has constant zero effect.

Apply the conversion in [Proposition 1](#). It preserves the design, propensity, means, effect, and null constancy. The resulting laws P_0^b and P_{ς}^b have kernels

$$Q_{a,P_0^b}(\cdot | x) = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1} \quad (a \in \{0,1\}),$$

$$Q_{0,P_{\varsigma}^b}(\cdot | x) = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1},$$

$$Q_{1,P_{\varsigma}^b}(\cdot | x) = \frac{1 + \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x)}{2}\delta_1 + \frac{1 - \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x)}{2}\delta_{-1}.$$

Their propensity is identically $1/2$, and

$$P_0^b \in H_0^{\text{bin}}(w), \quad P_{\varsigma}^b \in \mathcal{M}_w^{\text{bin}}, \quad \tau_{P_{\varsigma}^b} = \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b.$$

Define the binary prior by

$$\pi_1^b = 2^{-m_{\Delta,n}^b} \sum_{\varsigma \in \{-1,1\}^{\mathcal{J}_{\Delta,n}^b}} \delta_{P_{\varsigma}^b}.$$

Again,

$$2^{-m_{\Delta,n}^b} > 0, \quad \sum_{\varsigma} 2^{-m_{\Delta,n}^b} = 1,$$

and the constant sign vector gives a positive-weight index.

8. *Signed-binary separation.* The equal-area cancellation and the disjoint-support square calculation give

$$\int f_{\Delta,\varsigma}^b d\lambda = 0, \quad \int (f_{\Delta,\varsigma}^b)^2 d\lambda = \frac{1}{3}.$$

Consequently,

$$d(P_\varsigma^b) = \frac{\kappa_\Delta (h_{\Delta,n}^b)^\gamma}{\sqrt{3}}.$$

Using the upper rank bound and $\gamma\vartheta_\Delta^b = 2\gamma/(4\gamma+1)$,

$$\begin{aligned} d(P_\varsigma^b) &\geq \frac{\kappa_\Delta}{\sqrt{3}} (4n^{\vartheta_\Delta^b})^{-\gamma} \\ &= c_w^b n^{-2\gamma/(4\gamma+1)}. \end{aligned}$$

The effect envelope also gives

$$\begin{aligned} d(P_\varsigma^b)^2 &= \int \tau_{P_\varsigma^b}^2 d\lambda - \left(\int \tau_{P_\varsigma^b} d\lambda \right)^2 \\ &\leq \frac{1}{16^2}, \end{aligned} \quad d(P_\varsigma^b) \leq \frac{1}{16} < d_0.$$

9. *Signed-binary complete-record comparison.* Define, on the whole record space,

$$\ell_{\Delta,\varsigma}^b(x, a, y) = \begin{cases} 1 + \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x), & a = 1, y = 1, \\ 1 - \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x), & a = 1, y = -1, \\ 1, & \text{otherwise.} \end{cases}$$

The kernel probabilities imply

$$P_\varsigma^b \ll P_0^b, \quad \frac{dP_\varsigma^b}{dP_0^b} = \ell_{\Delta,\varsigma}^b \quad P_0^b\text{-almost surely.}$$

Under the null, each treatment–sign combination has conditional probability 1/4. Summing the two control terms and the two treated terms yields

$$\begin{aligned} &\frac{1}{2} + \frac{1}{4} \left[(1 + \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x))(1 + \delta_{\Delta,n}^b f_{\Delta,\varsigma'}^b(x)) \right. \\ &\quad \left. + (1 - \delta_{\Delta,n}^b f_{\Delta,\varsigma}^b(x))(1 - \delta_{\Delta,n}^b f_{\Delta,\varsigma'}^b(x)) \right] \\ &= 1 + \frac{(\delta_{\Delta,n}^b)^2}{2} f_{\Delta,\varsigma}^b(x) f_{\Delta,\varsigma'}^b(x). \end{aligned}$$

Define

$$\varepsilon_{\Delta,n}^b = (h_{\Delta,n}^b)^{2\gamma}, \quad \varrho_{\Delta,n}^b = \frac{\varepsilon_{\Delta,n}^b \kappa_\Delta^2}{6}.$$

Then

$$(\delta_{\Delta,n}^b)^2 = \kappa_\Delta^2 \varepsilon_{\Delta,n}^b,$$

and integration of the paired squared tents gives

$$\int f_{\Delta,\varsigma}^b f_{\Delta,\varsigma'}^b d\lambda = \frac{2h_{\Delta,n}^b}{3} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta}.$$

Thus the complete one-record overlap is

$$\int \ell_{\Delta,\varsigma}^b \ell_{\Delta,\varsigma'}^b dP_0^b = 1 + \frac{\varrho_{\Delta,n}^b}{m_{\Delta,n}^b} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta}.$$

These bases are nonnegative, since

$$0 \leq \varrho_{\Delta,n}^b \leq 1, \quad \left| \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta} \right| \leq m_{\Delta,n}^b.$$

Define the mixture and its density by

$$\begin{aligned} \bar{P}_{1,b}^{(n)} &= 2^{-m_{\Delta,n}^b} \sum_{\varsigma} (P_\varsigma^b)^{\otimes n}, \\ G_\Delta^b(\mathcal{D}_n) &= \frac{d\bar{P}_{1,b}^{(n)}}{d(P_0^b)^{\otimes n}}(\mathcal{D}_n) = 2^{-m_{\Delta,n}^b} \sum_{\varsigma} \prod_{i_{\text{rec}}=1}^n \ell_{\Delta,\varsigma}^b(X_{i_{\text{rec}}}, A_{i_{\text{rec}}}, Y_{i_{\text{rec}}}). \end{aligned}$$

Expansion and independence give

$$1 + \chi^2(\bar{P}_{1,b}^{(n)} \|(P_0^b)^{\otimes n}) = 2^{-2m_{\Delta,n}^b} \sum_{\varsigma} \sum_{\varsigma'} \left(1 + \frac{\varrho_{\Delta,n}^b}{m_{\Delta,n}^b} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta} \right)^n.$$

For fixed ς , the exponential comparison and sign factorization already established give

$$\begin{aligned} &2^{-m_{\Delta,n}^b} \sum_{\varsigma'} \left(1 + \frac{\varrho_{\Delta,n}^b}{m_{\Delta,n}^b} \sum_{j_\Delta} \varsigma_{j_\Delta} \varsigma'_{j_\Delta} \right)^n \\ &\leq \prod_{j_\Delta} \cosh \left(\frac{n\varrho_{\Delta,n}^b}{m_{\Delta,n}^b} \varsigma_{j_\Delta} \right) \leq \exp \left(\frac{(n\varrho_{\Delta,n}^b)^2}{2m_{\Delta,n}^b} \right). \end{aligned}$$

Averaging over the remaining signs and simplifying the exponent yields

$$1 + \chi^2(\bar{P}_{1,b}^{(n)} \|(P_0^b)^{\otimes n}) \leq \exp \left(\frac{\kappa_\Delta^4}{36} n^2 (h_{\Delta,n}^b)^{4\gamma+1} \right).$$

Since

$$h_{\Delta,n}^b \leq n^{-\vartheta_\Delta^b}, \quad \vartheta_\Delta^b(4\gamma+1) = 2,$$

we have

$$n^2 (h_{\Delta,n}^b)^{4\gamma+1} \leq 1,$$

and therefore

$$1 + \chi^2(\bar{P}_{1,b}^{(n)} \|(P_0^b)^{\otimes n}) \leq e^{\kappa_\Delta^4/36} < 2.$$

Applying the zero-integral density-deviation identity and Cauchy-Schwarz to G_Δ^b gives

$$\begin{aligned} \text{TV}((P_0^b)^{\otimes n}, \bar{P}_{1,b}^{(n)}) &= \frac{1}{2} \int |G_\Delta^b - 1| d(P_0^b)^{\otimes n} \\ &\leq \frac{1}{2} \sqrt{\chi^2(\bar{P}_{1,b}^{(n)} \|(P_0^b)^{\otimes n})} < \frac{1}{2}. \end{aligned}$$

10. *Bounded-model placement and risk.* The inclusion supplied by [Proposition 1](#) gives

$$P_\zeta^b \in \mathcal{M}_w^b, \quad P_0^b \in H_0^b(w).$$

For the null, this uses its bounded-model membership together with the preserved constant zero effect. The same comparison, together with [Lemma 1](#) at moment order two, supplies

$$D_w^b = D_{(2,\alpha,\beta,\gamma)} \geq d_0.$$

Define

$$r_\Delta^b = c_w^b n^{-2\gamma/(4\gamma+1)}.$$

The binary alternatives satisfy

$$0 < r_\Delta^b \leq d(P_\zeta^b) < d_0 \leq D_w^b.$$

For any test satisfying the bounded-null size constraint in [Definition 11](#), average its public seed as above. The complete-record total-variation bound gives

$$\left| 2^{-m_{\Delta,n}^b} \sum_{\zeta} R_n(P_\zeta^b, \phi) - R_n(P_0^b, \phi) \right| < \frac{1}{2},$$

hence

$$2^{-m_{\Delta,n}^b} \sum_{\zeta} R_n(P_\zeta^b, \phi) < \frac{3}{5}.$$

If all positive-weight support laws had type-II error below $2/5$, their weighted rejection sum would be at least

$$2^{-m_{\Delta,n}^b} \sum_{\zeta} \frac{3}{5} = \frac{3}{5},$$

a contradiction. Thus a support law has type-II error at least $2/5$. All support laws belong to the bounded model at separation r_Δ^b , so

$$\sup_{\substack{P \in \mathcal{M}_w^b \\ d(P) \geq r_\Delta^b}} \{1 - R_n(P, \phi)\} \geq \frac{2}{5}.$$

The constant-zero test is admissible, and taking the infimum over the bounded-null level class proves

$$B_n^b(w, r_\Delta^b) \geq \frac{2}{5}.$$

The constructed signed-binary null, normalized positive-weight prior, placement, and complete-record comparison establish the second witness package. □

The fixed propensity in [Lemma 5](#) makes the comparison applicable to both observation regimes. Every alternative support law meets the separation requirement, while the mixture remains close to the null in the complete original-record experiment. The growing mark magnitudes coexist with a conditional p -moment bound of one. The signed-binary package supplies a separate bounded comparison at the moment-order-two exponent.

C.1 Marked records and continuous coefficient fields

The rough-phase construction uses a normalized marked table T to specify treatment, outcome sign, and mark occurrence jointly. Keeping its six categories explicit preserves the assignment information and the zero outcomes in the likelihood comparison.

Definition 29. `[def:marked-table]` (Six-category record table T) *The conditional original-record table is*

$$\begin{aligned}\mathsf{T}(A = (1 + \ell)/2, Y = hL \mid X = x) &= \frac{\varepsilon}{4} \{1 + \ell\xi(x) + hv(x) + \ell h\zeta(x)\}, \quad (\ell, h) \in \{-1, 1\}^2, \\ \mathsf{T}(A = (1 + \ell)/2, Y = 0 \mid X = x) &= \frac{1 - \varepsilon}{2} \{1 + \ell\xi(x)\}, \quad \ell \in \{-1, 1\}.\end{aligned}$$

Here ε is the rare-mark probability, $L > 0$ is the mark magnitude, and ξ, v, ζ may be Borel functions of the covariate. The table is evaluated on inputs for which every displayed probability is nonnegative. Likelihood products include all six record categories.

In Definition 29, the treatment sign ℓ and marked-outcome sign h index the four nonzero-outcome categories. The coordinate ξ governs treatment probabilities, v governs marked-outcome signs, and ζ governs their interaction. Summing over the categories gives total probability one and nonzero-outcome probability ε ; the positive mark magnitude is L . The remaining two categories retain the treatment coordinate when the outcome is zero.

Continuous coefficient fields are formed from the finite-overlap frame $(f_i)_{i=0}^K$ below, where K is the fine rank. Its adjacent-cell support gives a continuous construction on the fine partition.

Definition 30. `[def:frame-handle]` (Finite-overlap frame f_i) *The finite-overlap frame at rank $K = J_f$, the fine histogram rank, consists of the functions*

$$f_i(x) = \begin{cases} \cos(\pi(Kx - i)/2), & x \in [i/K, (i + 1)/K], \\ \sin(\pi(Kx - (i - 1))/2), & i > 0, x \in [(i - 1)/K, i/K], \\ 0, & \text{otherwise,} \end{cases} \quad x \in [0, 1], \quad 0 \leq i \leq K.$$

For a coefficient vector

$$a = (a_0, \dots, a_K),$$

the associated frame field is

$$x \longmapsto \sum_{i=0}^K a_i f_i(x).$$

On each fine cell, the two active frame coordinates have squared values summing to one. The prior construction uses these squares to interpolate coarse paired tents, while the frame coordinates themselves carry the random propensity and outcome coefficients. The following algorithm specifies both branches through a finite collection of sign draws.

Algorithm 5 (Copula-frame priors $\pi_{\nu;v,a,u,\varepsilon,L}^{K,M}$)

Inputs are the public tuple v , dyadic ranks $K = J_f$ and $M = J_c$ satisfying $M \geq 2$ and $K \geq 16M$, amplitudes $0 < a, u \leq 1/16$, a rare-mark probability $0 < \varepsilon < 1$, and a mark magnitude $L > 0$.

1. Set the coupling amplitude and triangular tent to

$$\kappa = \frac{1}{16}, \quad b_{\Delta}(t) = 2 \min\{t, 1 - t\}, \quad t \in [0, 1].$$

Draw independent uniform signs

$$\Pr(\sigma_j = 1) = \Pr(\sigma_j = -1) = \frac{1}{2}, \quad 1 \leq j \leq M/2.$$

Define the opposite paired coarse tents by

$$g_{\sigma}(x) = \begin{cases} \sigma_j b_{\Delta}(Mx - (2j - 2)), & x \in [(2j - 2)/M, (2j - 1)/M], \\ -\sigma_j b_{\Delta}(Mx - (2j - 1)), & x \in [(2j - 1)/M, 2j/M], \end{cases} \quad 1 \leq j \leq M/2.$$

Their values at coarse-cell boundaries are zero.

2. On each fine cell define the two nonzero frame coordinates by

$$f_i(x) = \cos(\pi(Kx - i)/2), \quad f_{i+1}(x) = \sin(\pi(Kx - i)/2), \\ x \in [i/K, (i + 1)/K], \quad 0 \leq i < K.$$

Set the other coordinates to zero and form the squared-frame interpolation

$$\tilde{g}_{\sigma}(x) = \sum_{i=0}^K g_{\sigma}(i/K) f_i(x)^2.$$

3. For each branch $\nu \in \{0, 1\}$, draw coefficient pairs independently conditional on σ , with

$$\Pr(\lambda_i = l, \eta_i = h \mid \sigma) = \frac{1 + \nu \kappa g_{\sigma}(i/K) l h}{4}, \quad l, h \in \{-1, 1\}, \quad 0 \leq i \leq K.$$

4. For each coefficient draw, define the conditional-table coordinates

$$\xi(x) = a \sum_{i=0}^K f_i(x) \lambda_i, \quad v(x) = u \sum_{i=0}^K f_i(x) \eta_i, \\ t_{\nu}(x) = -\frac{\nu a u \kappa}{1 - a^2} \tilde{g}_{\sigma}(x), \quad \zeta(x) = \xi(x) v(x) + t_{\nu}(x) (1 - \xi(x)^2).$$

Let P be the original-record law with uniform covariate X and the normalized six-category marked conditional table evaluated at these coordinates, rare-mark probability ε , and mark magnitude L .

5. Output the finite prior obtained by pushing the coefficient distribution forward to original-record laws:

$$\pi_{\nu; v, a, u, \varepsilon, L}^{K, M} = \mathcal{L}_{\nu}(P), \quad \nu \in \{0, 1\}.$$

The copula-frame prior $\pi_{\nu;v,a,u,\varepsilon,L}^{K,M}$ in [Algorithm 5](#) changes the dependence between the fine coefficient signs across branches. Its propensity-coordinate amplitude is a , and its marked-outcome-coordinate amplitude is u . Each coefficient draw determines one original-record law, so the pushforward is a finite probability prior even when several draws determine the same law.

For the model tuning, write b for the baseline amplitude, and denote the resulting null and alternative priors by π_0 and π_1 . The next result gives their support placement under the phase and rank conditions, together with a separate bounded tuning.

Lemma 6. [lem:copula-frame-legality] (Copula-frame legality and moments) *Consider an admissible tuple $v = (p, \alpha, \beta, \gamma)$ as stipulated in [Definition 1](#). Put*

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad F(p) = \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{S}{2\gamma}.$$

Suppose the following conditions hold:

- **(Phase.)** $F(p) < 1$.
- **(Dyadic ranks.)** K and M are powers of two, with $16M \leq K$ and $2 \leq M$.
- **(Rank compatibility.)** $M \leq K^{S/\gamma}$.

Define the construction amplitudes by

$$a = \frac{K^{-\alpha}}{16}, \quad b = \frac{K^{-\beta}}{256}.$$

Let λ denote Lebesgue probability measure on $[0, 1]$, and write $Q_{a,P}(dy \mid x)$ for the conditional outcome kernel in treatment arm a under P . Define the population effect distance by

$$d(P) = \left(\int_0^1 \left| \tau_P(x) - \int_0^1 \tau_P(z) \lambda(dz) \right|^2 \lambda(dx) \right)^{1/2},$$

and let d_0 be the fixed positive distance supplied by the legal model witness.

Use the copula-frame priors of [Algorithm 5](#) with

$$u = \frac{1}{16}, \quad \varepsilon = (16b)^{1/q}, \quad L = \varepsilon^{-1/p}, \quad \pi_0 = \pi_{0;v,a,u,\varepsilon,L}^{K,M}, \quad \pi_1 = \pi_{1;v,a,u,\varepsilon,L}^{K,M}.$$

For every realization of the $M/2$ coarse signs and the $K+1$ fine coefficient-sign pairs, the null-branch law belongs to $H_0(v)$ of [Definition 2](#), and the alternative-branch law belongs to $\mathcal{M}_v$ of [Definition 1](#). Each alternative-branch law satisfies

$$\begin{aligned} \tau_P(x) &= -\frac{2ab\kappa}{1-a^2} \tilde{g}_\sigma(x) \quad \text{for every } x \in [0, 1], & \int_0^1 \tau_P(x) \lambda(dx) &= 0, \\ 2^{-17} K^{-S} &\leq d(P) \leq 2^{-14} K^{-S}, & d(P) &< d_0. \end{aligned}$$

Every law in either branch satisfies

$$\int_{\mathbb{R}} |y|^p Q_{a,P}(dy \mid x) = 1 \quad \text{for every } a \in \{0, 1\} \text{ and } x \in [0, 1].$$

For the bounded construction, let $w = (\alpha, \beta, \gamma)$ be an admissible tuple as stipulated in [Definition 7](#), set $v = (2, \alpha, \beta, \gamma)$, and impose the same phase, dyadic-rank, and rank-compatibility conditions with $p = 2$. Retain the displayed amplitudes a, b , and use the priors of [Algorithm 5](#) with

$$u = 2b, \quad \varepsilon = \frac{1}{2}, \quad L = 1, \quad \pi_0 = \pi_{0;v,a,u,\varepsilon,L}^{K,M}, \quad \pi_1 = \pi_{1;v,a,u,\varepsilon,L}^{K,M}.$$

For every realization of the coarse signs and fine coefficient-sign pairs, the null-branch law belongs to $H_0^b(w)$ of [Definition 9](#), and the alternative-branch law belongs to $\mathcal{M}_w^b$ of [Definition 7](#). Each alternative-branch law satisfies the displayed effect identity, centering identity, and distance bounds. Every law in either bounded branch satisfies

$$\int_{\mathbb{R}} |y|^2 Q_{a,P}(dy \mid x) = \frac{1}{2} \quad \text{for every } a \in \{0, 1\} \text{ and } x \in [0, 1].$$

Proof of Lemma 6. Fix an admissible tuple and ranks satisfying the hypotheses, and fix an arbitrary realization of all coefficient signs. Admissibility gives

$$1 < p \leq 2, \quad 0 < \alpha, \beta \leq 1, \quad \frac{1}{4} \leq \gamma \leq 1.$$

We first treat the finite-moment construction.

1. *Tuning and construction inputs.* The rank assumptions imply $K \geq 32$, so the displayed amplitudes satisfy

$$0 < a \leq \frac{1}{16}, \quad 0 < b \leq \frac{1}{256}, \quad aK^\alpha = \frac{1}{16}, \quad bK^\beta = \frac{1}{256}.$$

Indeed, $K^{-\alpha}, K^{-\beta} \in (0, 1]$, and multiplication by the corresponding positive powers of K cancels the exponents. Also $q > 0$ and $0 < 16b \leq 1/16 < 1$, whence

$$0 < \varepsilon = (16b)^{1/q} < 1, \quad L = \varepsilon^{-1/p} > 0.$$

The tuning identities are

$$\varepsilon L^p = \varepsilon \varepsilon^{-1} = 1, \quad \varepsilon L u = \frac{\varepsilon^{1-1/p}}{16} = \frac{\varepsilon^q}{16} = b.$$

Thus the finite-moment tuning meets the input conditions of [Algorithm 5](#). The second tuning also meets those conditions, since

$$0 < 2b \leq \frac{1}{128} \leq \frac{1}{16}, \quad 0 < \frac{1}{2} < 1, \quad 1 > 0.$$

For that tuning,

$$\varepsilon L u = \frac{1}{2} \cdot 1 \cdot 2b = b, \quad \varepsilon L^2 = \frac{1}{2}.$$

2. *Validity of the conditional table and exact moments.* Use the coordinates and signs of [Algorithm 5](#), and define the two coefficient fields by

$$\mathcal{F}_\lambda(x) = \sum_{i=0}^K \lambda_i f_i(x), \quad \mathcal{F}_\eta(x) = \sum_{i=0}^K \eta_i f_i(x), \quad x \in [0, 1].$$

On a fine cell, write

$$\theta_i(x) = \frac{\pi(Kx - i)}{2}, \quad 0 \leq i < K, \quad x \in [i/K, (i+1)/K].$$

There the only possibly nonzero coordinates are $f_i(x) = \cos \theta_i(x)$ and $f_{i+1}(x) = \sin \theta_i(x)$. Consequently,

$$\sum_{i=0}^K f_i(x)^2 = 1, \quad |\mathcal{F}_\lambda(x)|, |\mathcal{F}_\eta(x)| \leq 2.$$

The coarse tents have absolute value at most one. Hence their squared-frame interpolation satisfies

$$|\tilde{g}_\sigma(x)| \leq \sum_{i=0}^K |g_\sigma(i/K)| f_i(x)^2 \leq \sum_{i=0}^K f_i(x)^2 = 1.$$

For either valid tuning and either branch,

$$|\xi(x)| \leq 2a \leq \frac{1}{8}, \quad |v(x)| \leq 2u \leq \frac{1}{8}, \quad a^2 \leq \frac{1}{256}, \quad au \leq \frac{1}{256}.$$

In the null branch $t_0 = 0$. In the alternative branch, positivity of $1 - a^2$ and the preceding bounds give

$$|t_1(x)| \leq \frac{au\kappa}{1 - a^2} \leq \frac{1}{128},$$

where the last inequality follows from $au \leq 1/256$, $\kappa = 1/16$, and $1 - a^2 \geq 255/256$. Since $0 \leq \xi(x)^2 \leq 1/64$, we also have

$$|1 - \xi(x)^2| \leq 1, \quad |\zeta(x)| \leq |\xi(x)v(x)| + |t_\nu(x)| |1 - \xi(x)^2| \leq \frac{1}{64} + \frac{1}{128} \leq \frac{1}{8}.$$

Thus every unmarked and marked bracket in [Definition 29](#) is positive:

$$1 + \ell\xi(x) \geq \frac{7}{8}, \quad 1 + \ell\xi(x) + hv(x) + \ell h\zeta(x) \geq \frac{5}{8}, \quad \ell, h \in \{-1, 1\}.$$

The frame coordinates are continuous at their cell endpoints, and all three table coordinates are continuous finite expressions in them. The table therefore supplies Borel conditional probability kernels.

Summing the marked probabilities over the outcome sign gives

$$\sum_{h \in \{-1, 1\}} \frac{\varepsilon}{4} \{1 + \ell\xi(x) + hv(x) + \ell h\zeta(x)\} = \frac{\varepsilon}{2} \{1 + \ell\xi(x)\}.$$

Adding the unmarked probability gives $(1 + \ell\xi(x))/2$, and summing over ℓ gives one. In particular, the conditional probability of a mark within either treatment arm is exactly ε . The marked outcomes have absolute value L , and the unmarked outcome is zero. Therefore

$$\int_{\mathbb{R}} |y|^{r_{\text{mom}}} Q_{(1+\ell)/2, P}(dy \mid x) = \varepsilon L^{r_{\text{mom}}}, \quad r_{\text{mom}} \in (0, \infty), \quad \ell \in \{-1, 1\}, \quad x \in [0, 1].$$

The tuning identities from the preceding step give the required value 1 at order p , and the value $1/2$ at order two for the bounded tuning.

Summing the signed marked outcomes within each arm also gives

$$\begin{aligned} e_P(x) &= \frac{1 + \xi(x)}{2}, \\ m_{0,P}(x) &= \varepsilon L \frac{v(x) - \zeta(x)}{1 - \xi(x)} = \varepsilon L \{v(x) - t_\nu(x)(1 + \xi(x))\}, \\ m_{1,P}(x) &= \varepsilon L \frac{v(x) + \zeta(x)}{1 + \xi(x)} = \varepsilon L \{v(x) + t_\nu(x)(1 - \xi(x))\}, \\ \tau_P(x) &= 2\varepsilon L t_\nu(x). \end{aligned}$$

All denominators are positive because $|\xi(x)| \leq 1/8$.

3. *Null membership.* We record the frame estimates needed for the primitive restrictions. For either coefficient field,

$$|\mathcal{F}_\lambda(x)|, |\mathcal{F}_\eta(x)| \leq \sqrt{2}.$$

Indeed, on a fine cell its absolute value is at most $|\cos \theta_i(x)| + |\sin \theta_i(x)|$, whose square is at most two by $\cos^2 \theta_i(x) + \sin^2 \theta_i(x) = 1$ and $(|\cos \theta_i(x)| - |\sin \theta_i(x)|)^2 \geq 0$.

For either field, denoted below by $\mathcal{F} \in \{\mathcal{F}_\lambda, \mathcal{F}_\eta\}$, the sine and cosine Lipschitz inequalities give, when x, z belong to the same fine cell,

$$|\mathcal{F}(x) - \mathcal{F}(z)| \leq 2|\theta_i(x) - \theta_i(z)| = \pi K|x - z|.$$

If $K|x - z| < 1$, the points lie in the same cell or in adjacent cells. In the latter case, split at the common endpoint and add the two cellwise bounds; the two distances sum to $|x - z|$. Equality of the points gives zero, and reversing their order leaves the bound unchanged. Thus

$$|\mathcal{F}(x) - \mathcal{F}(z)| \leq \pi K|x - z| \quad \text{when } K|x - z| < 1.$$

For $0 < s_H \leq 1$, the near case uses $K|x - z| \leq (K|x - z|)^{s_H}$ and $\pi < 4$; the complementary case uses $|\mathcal{F}(x) - \mathcal{F}(z)| \leq 2\sqrt{2} \leq 4$ and $(K|x - z|)^{s_H} \geq 1$. Together these give

$$|\mathcal{F}(x) - \mathcal{F}(z)| \leq 4K^{s_H}|x - z|^{s_H}, \quad 0 < s_H \leq 1, \quad x, z \in [0, 1].$$

The membership calculation applies to valid construction inputs satisfying

$$\varepsilon L u = b, \quad \varepsilon L^p \leq 10.$$

In the null branch, the table identities reduce to

$$e_P(x) = \frac{1 + a\mathcal{F}_\lambda(x)}{2}, \quad m_{0,P}(x) = b\mathcal{F}_\eta(x), \quad \tau_P(x) = 0.$$

The frame envelope and amplitude bounds imply

$$|\xi(x)| \leq a\sqrt{2} \leq \frac{1}{8}, \quad \frac{1}{4} \leq e_P(x) \leq \frac{3}{4}, \quad |m_{0,P}(x)| \leq b\sqrt{2} \leq \frac{1}{2}.$$

The frame Hölder estimates, together with the amplitude-power identities, give

$$\begin{aligned} |e_P(x) - e_P(z)| &\leq 2aK^\alpha|x - z|^\alpha = \frac{1}{8}|x - z|^\alpha \leq 20|x - z|^\alpha, \\ |m_{0,P}(x) - m_{0,P}(z)| &\leq 4bK^\beta|x - z|^\beta = \frac{1}{64}|x - z|^\beta \leq 20|x - z|^\beta. \end{aligned}$$

The effect is continuous, has supremum norm zero, and has zero Hölder differences. These estimates establish [Assumptions 2 to 7](#). The normalized conditional table is integrated against λ , so $P_X = \lambda$, as required by [Assumption 1](#). The exact conditional-moment formula yields

$$\int_{\mathbb{R}} |y|^p Q_{(1+\ell)/2, P}(dy \mid x) = \varepsilon L^p \leq 10,$$

and hence [Assumption 8](#). Finally, $\tau_P = 0$ supplies [Assumption 9](#) with constant zero. Therefore $P \in H_0(v)$ by [Definition 2](#). The finite-moment tuning satisfies the two displayed input conditions by the first step.

4. *Alternative membership.* Define the positive correction amplitude by

$$C_{\text{cop}} = \frac{ab\kappa}{1 - a^2} > 0.$$

The amplitude bounds give

$$1 - a^2 \geq \frac{1}{2}, \quad C_{\text{cop}} \leq 2ab\kappa \leq \frac{1}{32768}.$$

To control its smoothness factors, first observe that $p - 1 \leq 1$ and $q \leq 1/2$. Consequently,

$$2\alpha \leq \frac{2\alpha}{p-1}, \quad 2\beta \leq \frac{\beta}{q}, \quad 2S + \frac{S}{2\gamma} \leq F(p) < 1.$$

If $S \geq \gamma$, the left side would be at least $2\gamma + 1/2 \geq 1$. Thus

$$S < \gamma, \quad \beta \leq \gamma.$$

Rank compatibility and positivity of γ give $M^\gamma \leq K^S$. Using $M \geq 1$ and the amplitude identities, we obtain

$$\begin{aligned} ab &= \frac{K^{-S}}{4096}, \\ abM^\gamma &\leq \frac{K^{-S}K^S}{4096} = \frac{1}{4096}, \\ abM^\beta &\leq abM^\gamma \leq \frac{1}{4096}, \\ a^2bK^\beta &= \frac{a^2}{256} \leq \frac{1}{65536}. \end{aligned}$$

Multiplication by $2\kappa = 1/8$ now gives

$$C_{\text{cop}}M^\gamma \leq \frac{1}{32768}, \quad C_{\text{cop}}M^\beta \leq \frac{1}{32768}, \quad C_{\text{cop}}aK^\beta \leq \frac{1}{524288}.$$

The frame estimates also give

$$|\xi(x)| \leq \frac{1}{8}, \quad |1 + \xi(x)| \leq 2, \quad |\xi(x) - \xi(z)| \leq 4aK^\beta |x - z|^\beta.$$

For valid inputs with $\varepsilon Lu = b$, the alternative table identities become

$$m_{0,P}(x) = b\mathcal{F}_\eta(x) + C_{\text{cop}}\tilde{g}_\sigma(x)(1 + \xi(x)), \quad \tau_P(x) = -2C_{\text{cop}}\tilde{g}_\sigma(x).$$

Their uniform bounds are

$$|m_{0,P}(x)| \leq 2b + 2C_{\text{cop}} \leq \frac{1}{2}, \quad |\tau_P(x)| \leq 2C_{\text{cop}} \leq \frac{1}{2}.$$

The design remains λ , because the alternative conditional table is normalized. Its propensity is the same function $(1 + a\mathcal{F}_\lambda)/2$ as in the null branch, so the preceding overlap and propensity-smoothness estimates apply.

For the remaining smoothness estimates, extend the triangular tent by

$$b_{\Delta,\text{ext}}(r_\Delta) = \max\{0, 2\min(r_\Delta, 1 - r_\Delta)\}, \quad r_\Delta \in \mathbb{R}.$$

This agrees with b_Δ on $[0, 1]$ and vanishes outside that interval. The elementary Lipschitz inequalities for minimum and maximum give

$$|b_{\Delta,\text{ext}}(r_\Delta) - b_{\Delta,\text{ext}}(s_\Delta)| \leq 2|r_\Delta - s_\Delta|, \quad r_\Delta, s_\Delta \in \mathbb{R}.$$

Define each signed pair contribution and its active index set by

$$\begin{aligned} \mathfrak{g}_j(x) &= \sigma_j \{b_{\Delta,\text{ext}}(Mx - (2j - 2)) - b_{\Delta,\text{ext}}(Mx - (2j - 1))\}, \\ \mathcal{J}_\sigma(x) &= \{j \in \{1, \dots, M/2\} : \mathfrak{g}_j(x) \neq 0\}, \quad x \in [0, 1]. \end{aligned}$$

Thus $g_\sigma = \sum_{j=1}^{M/2} \mathfrak{g}_j$. Each contribution satisfies

$$|\mathfrak{g}_j(x) - \mathfrak{g}_j(z)| \leq 4M|x - z|.$$

The interiors of the pair supports are disjoint, so $|\mathcal{J}_\sigma(x)| \leq 1$. Only indices in $\mathcal{J}_\sigma(x) \cup \mathcal{J}_\sigma(z)$ contribute to a difference, and this union has cardinality at most two. Hence

$$|g_\sigma(x) - g_\sigma(z)| \leq 8M|x - z|.$$

On a fine cell, the squared-frame formula is

$$\tilde{g}_\sigma(x) = g_\sigma(i/K) \cos^2 \theta_i(x) + g_\sigma((i+1)/K) \sin^2 \theta_i(x).$$

For x, z in that cell, subtraction and $\cos^2 + \sin^2 = 1$ give

$$\begin{aligned} \tilde{g}_\sigma(x) - \tilde{g}_\sigma(z) &= (g_\sigma((i+1)/K) - g_\sigma(i/K)) \\ &\quad \times (\sin \theta_i(x) - \sin \theta_i(z)) (\sin \theta_i(x) + \sin \theta_i(z)). \end{aligned}$$

The sine Lipschitz inequality and its unit envelope therefore imply

$$|\tilde{g}_\sigma(x) - \tilde{g}_\sigma(z)| \leq \frac{8M}{K} \pi K |x - z| \leq 32M|x - z|.$$

Adjacent cell formulas agree at their common endpoint. Splitting any ordered interval at the intervening fine-grid endpoints and summing the cellwise estimates proves this bound for all $x, z \in [0, 1]$; equal points give zero, and reversing their order preserves the bound.

If $M|x - z| \leq 1$, use $M|x - z| \leq (M|x - z|)^{s_H}$. If $M|x - z| > 1$, use the envelope $|\tilde{g}_\sigma(x) - \tilde{g}_\sigma(z)| \leq 2$. Thus

$$|\tilde{g}_\sigma(x) - \tilde{g}_\sigma(z)| \leq 32M^{s_H}|x - z|^{s_H}, \quad 0 < s_H \leq 1.$$

The same inequality holds for $s_H = 0$, with the convention $r^0 = 1$ for $r \geq 0$, since the left side is at most two.

The baseline product difference decomposes as

$$\begin{aligned} m_{0,P}(x) - m_{0,P}(z) &= b\{\mathcal{F}_\eta(x) - \mathcal{F}_\eta(z)\} \\ &\quad + C_{\text{cop}}[\{\tilde{g}_\sigma(x) - \tilde{g}_\sigma(z)\}(1 + \xi(x)) + \tilde{g}_\sigma(z)\{\xi(x) - \xi(z)\}]. \end{aligned}$$

Applying the displayed estimates to these three terms gives

$$\begin{aligned} |m_{0,P}(x) - m_{0,P}(z)| &\leq (4bK^\beta + 64C_{\text{cop}}M^\beta + 4C_{\text{cop}}aK^\beta)|x - z|^\beta \\ &\leq \left(\frac{4}{256} + \frac{64}{32768} + \frac{4}{524288} \right) |x - z|^\beta \\ &\leq 20|x - z|^\beta. \end{aligned}$$

Similarly,

$$\begin{aligned} |\tau_P(x) - \tau_P(z)| &\leq 64C_{\text{cop}}M^\gamma|x - z|^\gamma \\ &\leq \frac{64}{32768}|x - z|^\gamma \leq 20|x - z|^\gamma. \end{aligned}$$

All these representatives are continuous. Their uniform bounds and Hölder estimates establish the baseline and effect restrictions in [Assumptions 4 to 7](#). The conditional-moment calculation supplies [Assumption 8](#) whenever $\varepsilon L^p \leq 10$. Together with the design, overlap, and propensity restrictions already established, this proves $P \in \mathcal{M}_v$ by [Definition 1](#). In particular it applies to the finite-moment tuning.

5. *Centering of the interpolated tents.* Choose the dyadic exponents and the refinement factor so that

$$\begin{aligned} K &= 2^{k_{\text{rank}}}, \quad M = 2^{m_{\text{rank}}}, \quad k_{\text{rank}}, m_{\text{rank}} \in \mathbb{N}, \quad 1 \leq m_{\text{rank}} \leq k_{\text{rank}}, \\ r_{\text{grid}} &= 2^{k_{\text{rank}} - m_{\text{rank}}} = \frac{K}{M} \in \mathbb{N}, \quad r_{\text{grid}} \geq 16. \end{aligned}$$

The inequality between the dyadic exponents follows from $M \leq K$, and $M \geq 2$ implies that M is even.

Every coarse cell has the same fine-grid sample sum of its unsigned tent. More precisely, support of the extended tent and the change of index $i = i_c r_{\text{grid}} + h_{\text{grid}}$ give

$$\sum_{i=0}^K b_{\Delta, \text{ext}}(Mi/K - i_c) = \sum_{h_{\text{grid}}=0}^{r_{\text{grid}}} b_{\Delta}(h_{\text{grid}}/r_{\text{grid}}), \quad i_c \in \{0, \dots, M-1\}.$$

The two cells in each signed pair therefore have equal sample sums and opposite signs. Consequently,

$$\sum_{i=0}^K g_\sigma(i/K) = 0, \quad g_\sigma(0) = g_\sigma(1) = 0.$$

In particular,

$$\sum_{i=0}^{K-1} g_\sigma(i/K) = 0, \quad \sum_{i=0}^{K-1} g_\sigma((i+1)/K) = 0.$$

Integrating the sine and cosine squares over one fine cell, by an affine change of variable, gives

$$\int_{i/K}^{(i+1)/K} \cos^2 \theta_i(x) dx = \int_{i/K}^{(i+1)/K} \sin^2 \theta_i(x) dx = \frac{1}{2K}.$$

The cell formula therefore yields

$$\begin{aligned} \int_0^1 \tilde{g}_\sigma(x) \lambda(dx) &= \sum_{i=0}^{K-1} \frac{g_\sigma(i/K) + g_\sigma((i+1)/K)}{2K} \\ &= 0. \end{aligned}$$

6. *Second-moment bounds.* We first sharpen the coarse-field difference estimate used for interpolation error. Define the midpoint between the two cells of a pair by

$$x_{j,\text{mid}} = \frac{2j-1}{M}, \quad j \in \{1, \dots, M/2\}.$$

On either side of this point, one of the two extended tents in $\mathfrak{g}_j$ vanishes. The tent Lipschitz estimate thus gives a constant $2M$ when both arguments are on the same side. If $x \leq x_{j,\text{mid}} \leq z$, both tents vanish at the midpoint, and

$$\begin{aligned} |\mathfrak{g}_j(x) - \mathfrak{g}_j(z)| &\leq |\mathfrak{g}_j(x)| + |\mathfrak{g}_j(z)| \\ &\leq 2M(x_{j,\text{mid}} - x) + 2M(z - x_{j,\text{mid}}) = 2M|x - z|. \end{aligned}$$

Reversing the arguments covers the other crossing case. Summing over the at most two active pair indices gives

$$|g_\sigma(x) - g_\sigma(z)| \leq 4M|x - z|, \quad x, z \in [0, 1].$$

The partition identity gives the exact interpolation-error expression

$$\tilde{g}_\sigma(x) - g_\sigma(x) = \sum_{i=0}^K \{g_\sigma(i/K) - g_\sigma(x)\} f_i(x)^2.$$

Whenever $f_i(x) \neq 0$, its adjacent-cell support implies $|i/K - x| \leq 1/K$; terms with $f_i(x) = 0$ vanish. Hence

$$\begin{aligned} |\tilde{g}_\sigma(x) - g_\sigma(x)| &\leq \sum_{i=0}^K \frac{4M}{K} f_i(x)^2 \\ &= \frac{4M}{K} \leq \frac{1}{4}. \end{aligned}$$

For the exact coarse second moment, direct integration gives

$$\int_0^1 b_\Delta(r_\Delta)^2 dr_\Delta = 2 \int_0^{1/2} (2r_\Delta)^2 dr_\Delta = \frac{1}{3}.$$

Each coarse cell therefore contributes $1/(3M)$. The cell interiors are disjoint, the signs square to one, and the even rank supplies all M cells, so

$$\int_0^1 g_\sigma(x)^2 \lambda(dx) = \frac{1}{3}.$$

Pointwise, the error estimate and the elementary square inequality give

$$\begin{aligned} g_\sigma(x)^2 &\leq 2\tilde{g}_\sigma(x)^2 + 2\{\tilde{g}_\sigma(x) - g_\sigma(x)\}^2 \\ &\leq 2\tilde{g}_\sigma(x)^2 + \frac{1}{8}. \end{aligned}$$

Integration against the probability measure λ yields

$$\frac{1}{3} \leq 2 \int_0^1 \tilde{g}_\sigma(x)^2 \lambda(dx) + \frac{1}{8},$$

which implies the lower bound below. The upper bound follows from the unit envelope:

$$\frac{1}{16} \leq \int_0^1 \tilde{g}_\sigma(x)^2 \lambda(dx) \leq \int_0^1 1 \lambda(dx) = 1.$$

7. *Effect identity and distance calibration.* Define the effect multiplier by

$$A_{\text{eff}} = 2C_{\text{cop}} = \frac{2ab\kappa}{1-a^2} > 0.$$

The alternative table calculation gives, pointwise,

$$\tau_P(x) = -A_{\text{eff}}\tilde{g}_\sigma(x) = -\frac{2ab\kappa}{1-a^2}\tilde{g}_\sigma(x).$$

Since the interpolated tent is centered,

$$\int_0^1 \tau_P(x) \lambda(dx) = -A_{\text{eff}} \int_0^1 \tilde{g}_\sigma(x) \lambda(dx) = 0.$$

Substitution into the definition of $d(P)$, followed by factoring the nonnegative multiplier out of the square root, gives

$$d(P) = A_{\text{eff}} \left(\int_0^1 \tilde{g}_\sigma(x)^2 \lambda(dx) \right)^{1/2}.$$

The amplitude product and $\kappa = 1/16$ give

$$2ab\kappa = 2^{-15}K^{-S}.$$

Because $0 \leq a^2 \leq 1/256$, the denominator is positive and lies between $1/2$ and one. Therefore

$$2^{-15}K^{-S} \leq A_{\text{eff}} \leq 2^{-14}K^{-S}.$$

Taking square roots of the second-moment bounds gives

$$\frac{1}{4} \leq \left(\int_0^1 \tilde{g}_\sigma(x)^2 \lambda(dx) \right)^{1/2} \leq 1.$$

Multiplication of the two intervals proves

$$2^{-17}K^{-S} \leq d(P) \leq 2^{-14}K^{-S}.$$

For the strict comparison, [Lemma 1](#) supplies $d_0 = 1/(8\sqrt{2})$. Since $S > 0$ and $K \geq 1$, we have $K^{-S} \leq 1$. Also $\sqrt{2} < 2$, so

$$d(P) \leq 2^{-14}K^{-S} \leq \frac{1}{16384} < \frac{1}{8\sqrt{2}} = d_0.$$

This establishes the effect, centering, and distance conclusions for every alternative realization in the finite-moment construction.

8. *Bounded construction.* Now take an admissible w and $v = (2, \alpha, \beta, \gamma)$. This v is admissible, and the assumed phase and rank conditions are exactly those used above. Under

$$u = 2b, \quad \varepsilon = \frac{1}{2}, \quad L = 1,$$

the input verification in the first step applies, and

$$\varepsilon Lu = b, \quad \varepsilon L^p = \varepsilon L^2 = \frac{1}{2} \leq 10.$$

Thus the null-membership calculation gives all restrictions of $H_0(v)$, and the alternative-membership calculation gives all restrictions of $\mathcal{M}_v$.

In either branch, the normalized table has outcome support

$$Y \in \{-1, 0, 1\}, \quad P(|Y| \leq 1) = 1.$$

Indeed, each conditional table atom has absolute outcome at most one, so its conditional probability of this event is one; integrating over λ preserves that value. This supplies [Assumption 10](#). Together with the previously established restrictions, it gives bounded null and alternative membership by [Definitions 7](#) and [9](#).

The table effect calculation uses $\varepsilon Lu = b$, so every bounded alternative has the same displayed effect formula. The tent centering and second-moment estimates depend on the ranks and coarse signs, and the distance calibration uses the retained amplitudes. Hence the centering identity, both distance bounds, and the strict comparison with d_0 also apply. Finally, the exact arm-moment calculation gives

$$\int_{\mathbb{R}} |y|^2 Q_{(1+\ell)/2, P}(dy \mid x) = \varepsilon L^2 = \frac{1}{2}, \quad \ell \in \{-1, 1\}, \quad x \in [0, 1].$$

The coefficient realization was arbitrary throughout, so all conclusions hold for every realization in both priors.

□

The support conclusion in [Lemma 6](#) applies to every coefficient realization. The rank-compatibility condition links the coarse effect resolution to the product of the two fine-scale amplitudes. The displayed effect is centered and has distance of order K^{-S} . In the finite-moment tuning, mark rarity and magnitude yield conditional p -moment exactly one; in the bounded tuning, unit mark magnitude and mark probability one half yield conditional second moment one half. Both tunings retain the same effect identity and distance bounds.

C.2 Augmentation and complete-record likelihood budgets

The component comparison retains the complete record through a common augmentation. Write $B_i = \mathbf{1}\{Y_i \neq 0\}$ for the mark indicator, and let $\mathcal{D}_\partial$ disclose coefficient pairs at the coarse-cell boundary nodes. The augmentation consists of the covariates, these indicators, and this boundary disclosure. Its common probability law is denoted by μ .

Conditional on the augmentation, use P_0 and P_1 for the two branch label laws. These are local conditional laws in the component comparison. Each label contains treatment and an outcome sign, including a retained sign when its mark indicator is zero. The intermediate conditional label law is denoted by Q , so that the intermediate augmented probability law is $\mu \otimes Q$. Recovering the observed outcome from the sign and mark indicator gives the original-record mixtures.

Conditioning on the augmentation fixes the allocation of covariates to fine and coarse cells and identifies which records carry nonzero marks. Boundary disclosure is part of this conditioning. The first and second conditional likelihood budgets, denoted by $\mathcal{A}$ and $\mathcal{B}$, bound the comparisons from P_0 to Q and from Q to P_1 , respectively. The following result averages these budgets over the same augmentation law and gives the resulting complete-record comparison.

Lemma 7. [lem:full-record-copula-component-bound] (Full-record copula component bound) *Let $v = (p, \alpha, \beta, \gamma)$ be any real tuple, and suppose:*

- **(Sample size and ranks.)** n, K, M are natural numbers, $n \geq 2$, K and M are powers of two, $M \geq 2$, $16M \leq K$, and $2^{40}n \leq K$.
- **(Amplitudes.)** $0 < a \leq 1/16$ and $0 < u \leq 1/16$.
- **(Mark parameters.)** $0 < \varepsilon < 1$ and $L > 0$.

Write π_0 and π_1 for the two copula-frame priors

$$\pi_\nu = \pi_{\nu;v,a,u,\varepsilon,L}^{K,M}, \quad \nu \in \{0, 1\},$$

and define their complete-record mixtures by

$$\mathbb{P}_\nu = \int P^{\otimes n} \pi_\nu(dP), \quad \nu \in \{0, 1\}.$$

Let λ denote Lebesgue probability measure on $[0, 1]$. Total variation, denoted by TV , is the supremum of the absolute difference of event probabilities. The directed chi-square divergence χ^2 is the integral of the squared Radon–Nikodym density minus one, for probability laws with the stated absolute continuity.

There exist a common augmentation law μ , conditional label kernels P_0, P_1, Q , and nonnegative measurable, integrable conditional budgets $\mathcal{A}, \mathcal{B}$ with the following properties. The augmentation coordinate is

$$((X_i)_{i=1}^n, (B_i)_{i=1}^n, \mathcal{D}_\partial), \quad B_i = \mathbf{1}\{Y_i \neq 0\},$$

where $\mathcal{D}_\partial$ records the sampled coefficient pairs (λ_j, η_j) exactly at nodes satisfying

$$jM \bmod K = 0, \quad 0 \leq j \leq K.$$

Its covariate marginal is $\lambda^{\otimes n}$, and its flag marginal is $\text{Bernoulli}(\varepsilon)^{\otimes n}$.

Each label consists of a binary treatment and a binary outcome sign, retaining the sign also when the mark flag is zero. Under branch ν , the joint law of the augmentation and these labels induced by the latent copula construction is $\mu \otimes P_\nu$; the disclosure equals the boundary restriction of the sampled coefficients almost surely. Under each branch, the $M/2$ coarse signs are independent fair signs, and their joint law with the augmentation is the product of their fair-sign law and μ .

For each branch, recovering the records by retaining X_i and the treatment label and setting the outcome to the signed magnitude L when $B_i = 1$, and to zero when $B_i = 0$, sends $\mu \otimes P_\nu$ to $\mathbb{P}_\nu$. Both augmented branch laws have augmentation marginal μ .

The intermediate augmented law $\mu \otimes Q$ is a probability law. For μ -almost every augmentation, P_0, P_1, Q are probability laws on the labels and satisfy

$$Q \ll P_0, \quad P_1 \ll Q, \quad \mathcal{A} \geq 0, \quad \mathcal{B} \geq 0, \\ 1 + \chi^2(Q \| P_0) \leq \exp(\mathcal{A}), \quad 1 + \chi^2(P_1 \| Q) \leq \exp(\mathcal{B}).$$

Here all three conditional laws and both budgets are evaluated at the full augmentation triple. Their expectations satisfy

$$\int \mathcal{A} d\mu \leq \frac{2^{38} a^4 u^4 \varepsilon^2 n^2}{K}, \quad \int \mathcal{B} d\mu \leq \frac{2^{56} a^4 u^4 \varepsilon^2 n^4}{MK^2}.$$

If

$$\frac{2^{38} a^4 u^4 \varepsilon^2 n^2}{K} + \frac{2^{56} a^4 u^4 \varepsilon^2 n^4}{MK^2} \leq 2^{-16},$$

then

$$\text{TV}(\mathbb{P}_0, \mathbb{P}_1) < \frac{1}{8}.$$

Proof of Lemma 7. 1. We first construct the common augmentation and the three label kernels. For $J \in \{K, M\}$, define the cell index, including its value at the right endpoint, by

$$\text{cell}_J(x) = \min\{J - 1, \lfloor Jx \rfloor\}, \quad x \in [0, 1].$$

Here $K > 0$, since $16M \leq K$ and $M \geq 2$. Define the boundary-node set and the disclosure map by

$$\mathbf{N}_\partial = \{j \in \{0, \dots, K\} : jM \bmod K = 0\}, \quad \mathbf{D}(\boldsymbol{\lambda}, \boldsymbol{\eta})_j = \begin{cases} (\lambda_j, \eta_j), & j \in \mathbf{N}_\partial, \\ \perp, & j \notin \mathbf{N}_\partial. \end{cases}$$

The symbol $\perp$ denotes an undisclosed pair. Thus the space of possible disclosures is

$$\Delta_K = (\{\perp\} \cup \{-1, 1\}^2)^{\{0, \dots, K\}},$$

and the augmentation space and a generic augmentation are

$$\Omega_{\text{aug}} = [0, 1]^n \times \{0, 1\}^n \times \Delta_K, \quad \omega = (\mathbf{X}, \mathbf{B}, \delta).$$

Connect observations i and i' when

$$\begin{aligned} \text{cell}_M(X_i) &= \text{cell}_M(X_{i'}), \\ \text{cell}_K(X_i) &= \text{cell}_K(X_{i'}) \quad \text{or} \quad \left[\begin{array}{l} |\text{cell}_K(X_i) - \text{cell}_K(X_{i'})| = 1, \\ \max\{\text{cell}_K(X_i), \text{cell}_K(X_{i'})\} \notin \mathbf{N}_\partial. \end{array} \right] \end{aligned}$$

Let $\mathcal{C}(\omega)$ be the collection of connected components of this graph, with connectivity including paths of length zero. It partitions $\{1, \dots, n\}$. For every subset $\mathbf{c} \subseteq \{1, \dots, n\}$, set

$$n_{\mathbf{c}} = |\mathbf{c}|, \quad k_{\mathbf{c}} = \sum_{i \in \mathbf{c}} B_i, \quad j(\mathbf{c}) = \max \left(\left\{ \left\lfloor \frac{\text{cell}_M(X_i)}{2} \right\rfloor : i \in \mathbf{c} \right\} \cup \{0\} \right).$$

The last convention defines $j(\emptyset) = 0$.

We represent the binary labels by their signs:

$$\mathbf{z} = (z_i^t, z_i^y)_{i=1}^n \in Z_n := (\{-1, 1\}^2)^n, \quad F_n = 4^{-n} \sum_{\mathbf{z} \in Z_n} \text{Dirac}_{\mathbf{z}}.$$

The treatment label is $(1 + z_i^t)/2$; the outcome-sign label is retained for both values of B_i . For $\nu \in \{0, 1\}$, a coarse-sign vector $\sigma \in \{-1, 1\}^{M/2}$, and a disclosure δ , define the conditional coefficient weights by

$$w_{\nu, \sigma, \delta}(\boldsymbol{\lambda}, \boldsymbol{\eta}) = \prod_{j=0}^K \begin{cases} \mathbf{1}\{(\lambda_j, \eta_j) = \delta_j\}, & \delta_j \neq \perp, \\ \frac{1 + \nu \kappa g_{\sigma}(j/K) \lambda_j \eta_j}{4}, & \delta_j = \perp, \end{cases} \quad \kappa = \frac{1}{16}.$$

For every real function Ψ on the finite coefficient space, write

$$\mathbb{E}_{\nu, \sigma, \delta}[\Psi] = \sum_{\boldsymbol{\lambda}, \boldsymbol{\eta} \in \{-1, 1\}^{K+1}} w_{\nu, \sigma, \delta}(\boldsymbol{\lambda}, \boldsymbol{\eta}) \Psi(\boldsymbol{\lambda}, \boldsymbol{\eta}).$$

Each node weight is nonnegative and sums to one: for an undisclosed node this follows from

$$|\kappa g_{\sigma}(j/K)| \leq \frac{1}{16}, \quad \sum_{\lambda_j, \eta_j \in \{-1, 1\}} \lambda_j \eta_j = 0.$$

Consequently,

$$w_{\nu, \sigma, \delta} \geq 0, \quad \sum_{\boldsymbol{\lambda}, \boldsymbol{\eta}} w_{\nu, \sigma, \delta}(\boldsymbol{\lambda}, \boldsymbol{\eta}) = 1.$$

Use the coordinate fields of [Algorithm 5](#), and define the record-label density by

$$r_{\nu, i} = 1 + z_i^t \xi(X_i) + B_i \{z_i^y v(X_i) + z_i^t z_i^y \zeta(X_i)\}.$$

Its dependence on the coefficient draw and σ is through those coordinate fields. On each fine cell, the two active frame coordinates satisfy

$$\sum_{j=0}^K f_j(x)^2 = 1, \quad \sum_{j=0}^K |f_j(x)| \leq \sqrt{2} \leq 2.$$

The same identities hold at cell endpoints. They imply

$$|\xi(x)| \leq \frac{1}{8}, \quad |v(x)| \leq \frac{1}{8}, \quad |\tilde{g}_{\sigma}(x)| \leq 1, \quad |t_{\nu}(x)| \leq \frac{au\kappa}{1-a^2} \leq \frac{1}{128}.$$

Indeed, $a^2 \leq 1/256$, and hence $1 - a^2 \geq 255/256 > 0$. Moreover,

$$|1 - \xi(x)^2| \leq 1, \quad |\zeta(x)| \leq \frac{1}{64} + \frac{1}{128} \leq \frac{1}{8}.$$

Thus, for every augmentation, coefficient draw, and label vector,

$$\frac{1}{2} \leq r_{\nu,i} \leq 2, \quad \frac{1}{4} \sum_{z_i^t, z_i^y \in \{-1,1\}} r_{\nu,i} = 1.$$

For $s_{\text{coarse}} \in \{-1, 1\}$, let $\sigma^{s_{\text{coarse}}}$ denote the constant coarse-sign vector:

$$\sigma_j^{s_{\text{coarse}}} = s_{\text{coarse}} \quad (1 \leq j \leq M/2).$$

Define, for every subset $\mathbf{c}$,

$$\mathbf{d}_{\mathbf{c}, \nu, s_{\text{coarse}}}(\omega, \mathbf{z}) = \mathbb{E}_{\nu, \sigma^{s_{\text{coarse}}}, \delta} \left[\prod_{i \in \mathbf{c}} r_{\nu,i} \right],$$

and abbreviate

$$\mathbf{d}_{\mathbf{c},0} = \mathbf{d}_{\mathbf{c},0,-1}, \quad \mathbf{d}_{\mathbf{c},+} = \mathbf{d}_{\mathbf{c},1,1}, \quad \mathbf{d}_{\mathbf{c},-} = \mathbf{d}_{\mathbf{c},1,-1}, \quad \mathbf{d}_{\mathbf{c},\text{av}} = \frac{\mathbf{d}_{\mathbf{c},+} + \mathbf{d}_{\mathbf{c},-}}{2}.$$

The null density is independent of the coarse signs. Empty products equal one, so all these component densities equal one when $\mathbf{c} = \emptyset$. Finite averaging and label normalization give

$$2^{-n_{\mathbf{c}}} \leq \mathbf{d}_{\mathbf{c}, \nu, s_{\text{coarse}}} \leq 2^{n_{\mathbf{c}}}, \quad \int \mathbf{d}_{\mathbf{c}, \nu, s_{\text{coarse}}} d\mathbf{F}_n = 1,$$

and the same bounds and normalization hold for $\mathbf{d}_{\mathbf{c},\text{av}}$.

Now define the three full-label densities and kernels by

$$\begin{aligned} \mathbf{f}_0(\omega, \mathbf{z}) &= \prod_{\mathbf{c} \in \mathcal{C}(\omega)} \mathbf{d}_{\mathbf{c},0}(\omega, \mathbf{z}), \\ \mathbf{f}_1(\omega, \mathbf{z}) &= 2^{-M/2} \sum_{\sigma \in \{-1,1\}^{M/2}} \mathbb{E}_{1,\sigma,\delta} \left[\prod_{i=1}^n r_{1,i} \right], \\ \mathbf{f}_{\text{av}}(\omega, \mathbf{z}) &= \prod_{\mathbf{c} \in \mathcal{C}(\omega)} \mathbf{d}_{\mathbf{c},\text{av}}(\omega, \mathbf{z}), \\ P_{\nu}(\omega, d\mathbf{z}) &= \mathbf{f}_{\nu}(\omega, \mathbf{z}) \mathbf{F}_n(d\mathbf{z}), \quad \nu \in \{0, 1\}, \\ Q(\omega, d\mathbf{z}) &= \mathbf{f}_{\text{av}}(\omega, \mathbf{z}) \mathbf{F}_n(d\mathbf{z}). \end{aligned}$$

These definitions apply to every augmentation, including disclosures outside the support of the augmentation law. Component densities depend only on labels in their component. Since the components partition the observations, their normalized products are normalized. The full coefficient average defining $\mathbf{f}_1$ is normalized by the record-label normalization above. Therefore $P_0(\omega), P_1(\omega), Q(\omega)$ are probability laws for every ω .

Let the disclosure law be the pushforward of independent fair coefficient pairs:

$$\mu_{\partial} = 4^{-(K+1)} \sum_{\lambda, \eta \in \{-1,1\}^{K+1}} \text{Dirac}_{\mathbf{D}(\lambda, \eta)}, \quad \mu = \lambda^{\otimes n} \otimes \text{Bernoulli}(\varepsilon)^{\otimes n} \otimes \mu_{\partial}.$$

It is a probability law with the required covariate and flag marginals, and

$$\delta_j \neq \perp \iff j \in \mathbf{N}_\partial \quad \text{for } \mu\text{-almost every } \omega.$$

The kernels are measurable: the graph is determined by finitely many measurable cell assignments, and all coefficient and label averages are finite. Their normalization gives

$$(\mu \otimes P_\nu) \circ \text{pr}_{\text{aug}}^{-1} = \mu, \quad \nu \in \{0, 1\}, \quad (\mu \otimes Q)(\Omega_{\text{aug}} \times \mathbf{Z}_n) = 1.$$

2. We establish the conditional comparisons. Write

$$K = 2^{k_f}, \quad M = 2^{k_c}, \quad k_f, k_c \in \mathbb{N}.$$

Since $M \leq K$ and $M \geq 2$,

$$k_c \leq k_f, \quad k_c \geq 1, \quad M \mid K, \quad 2 \mid M.$$

In particular,

$$\mathbf{N}_\partial = \{jK/M : 0 \leq j \leq M\}.$$

Fix an augmentation in the support of μ . Its disclosed coefficients are fixed. The undisclosed coefficient nodes used by a component are

$$\mathbf{N}_c = \{j \notin \mathbf{N}_\partial : f_j(X_i) \neq 0 \text{ for some } i \in c\}.$$

These node sets are disjoint for distinct components. Indeed, two observations sharing an undisclosed active node lie in the same fine interval or in neighboring fine intervals joined through that node, and therefore are connected. Each component lies in one coarse cell. Its coefficient coupling and its correction consequently depend on the single sign indexed by $j(c)$. Conditional coefficient independence then gives

$$\mathbb{E}_{\nu, \sigma, \delta} \left[\prod_{i=1}^n r_{\nu, i} \right] = \prod_{c \in \mathcal{C}(\omega)} d_{c, \nu, \sigma_{j(c)+1}}.$$

Define the occupied coarse-pair indices and their component groups by

$$\mathcal{J}(\omega) = \{j(c) : c \in \mathcal{C}(\omega)\}, \quad \mathcal{C}_j(\omega) = \{c \in \mathcal{C}(\omega) : j(c) = j\}.$$

The index $j(c)$ belongs to $\{0, \dots, M/2 - 1\}$. Averaging the independent fair coarse signs yields

$$f_1 = \prod_{j \in \mathcal{J}(\omega)} \frac{1}{2} \left\{ \prod_{c \in \mathcal{C}_j(\omega)} d_{c,+} + \prod_{c \in \mathcal{C}_j(\omega)} d_{c,-} \right\}.$$

For every augmentation and every subset c , define

$$\begin{aligned} \Delta_{c,e} &= d_{c,\text{av}} - d_{c,0}, & \Delta_{c,o} &= \frac{d_{c,+} - d_{c,-}}{2}, \\ \mathfrak{e}_c &= \int \frac{\Delta_{c,e}^2}{d_{c,0}} dF_n, & \mathfrak{o}_c &= \int \frac{\Delta_{c,o}^2}{d_{c,\text{av}}} dF_n. \end{aligned}$$

The denominators are strictly positive. Component normalization gives

$$\int \Delta_{c,e} dF_n = \int \Delta_{c,o} dF_n = 0, \quad \mathfrak{e}_c \geq 0, \quad \mathfrak{o}_c \geq 0.$$

Choose the conditional budgets

$$\mathcal{A}(\omega) = \sum_{c \in \mathcal{C}(\omega)} \mathfrak{e}_c, \quad \mathcal{B}(\omega) = \frac{1}{2} \sum_{c, c' \in \mathcal{C}(\omega)} \mathbf{1}\{c \neq c', j(c) = j(c')\} \mathfrak{o}_c \mathfrak{o}_{c'}.$$

For the first comparison, cancellation of the even discrepancy gives

$$\int \frac{d_{c,av}^2}{d_{c,0}} dF_n = 1 + \mathfrak{e}_c.$$

Integration over disjoint label blocks therefore gives

$$1 + \chi^2(Q(\omega) \| P_0(\omega)) = \prod_{c \in \mathcal{C}(\omega)} (1 + \mathfrak{e}_c) \leq \exp \left(\sum_{c \in \mathcal{C}(\omega)} \mathfrak{e}_c \right) = \exp(\mathcal{A}(\omega)).$$

For the second comparison, define

$$\vartheta_c = \frac{\Delta_{c,o}}{d_{c,av}}.$$

Under the component law with density $d_{c,av}$,

$$\int \vartheta_c d_{c,av} dF_n = 0, \quad \int \vartheta_c^2 d_{c,av} dF_n = \mathfrak{o}_c.$$

For two signs $s_{\text{coarse}}, s'_{\text{coarse}} \in \{-1, 1\}$, it follows that

$$\int (1 + s_{\text{coarse}} \vartheta_c)(1 + s'_{\text{coarse}} \vartheta_c) d_{c,av} dF_n = 1 + s_{\text{coarse}} s'_{\text{coarse}} \mathfrak{o}_c.$$

Using two independent copies of each fair coarse sign when squaring the likelihood ratio gives the exact identity

$$1 + \chi^2(P_1(\omega) \| Q(\omega)) = \prod_{j \in \mathcal{J}(\omega)} \frac{1}{2} \left\{ \prod_{c \in \mathcal{C}_j(\omega)} (1 + \mathfrak{o}_c) + \prod_{c \in \mathcal{C}_j(\omega)} (1 - \mathfrak{o}_c) \right\}.$$

Here is the exponential bound for each factor. For any finite collection $\mathcal{U}$ of components, define

$$\begin{aligned} H_e(\mathcal{U}) &= \frac{1}{2} \left\{ \prod_{c \in \mathcal{U}} (1 + \mathfrak{o}_c) + \prod_{c \in \mathcal{U}} (1 - \mathfrak{o}_c) \right\}, \\ H_o(\mathcal{U}) &= \frac{1}{2} \left\{ \prod_{c \in \mathcal{U}} (1 + \mathfrak{o}_c) - \prod_{c \in \mathcal{U}} (1 - \mathfrak{o}_c) \right\}, \\ D_o(\mathcal{U}) &= \sum_{c \in \mathcal{U}} \mathfrak{o}_c, \\ T_o(\mathcal{U}) &= \frac{1}{2} \sum_{\substack{c, c' \in \mathcal{U} \\ c \neq c'}} \mathfrak{o}_c \mathfrak{o}_{c'}. \end{aligned}$$

For the empty collection,

$$H_e = 1, \quad H_o = D_o = T_o = 0.$$

Upon adjoining $c_* \notin \mathcal{U}$, the recurrences are

$$\begin{aligned} H_e(\mathcal{U} \cup \{c_*\}) &= H_e(\mathcal{U}) + \mathfrak{o}_{c_*} H_o(\mathcal{U}), \\ H_o(\mathcal{U} \cup \{c_*\}) &= H_o(\mathcal{U}) + \mathfrak{o}_{c_*} H_e(\mathcal{U}), \\ T_o(\mathcal{U} \cup \{c_*\}) &= T_o(\mathcal{U}) + \mathfrak{o}_{c_*} D_o(\mathcal{U}). \end{aligned}$$

These recurrences preserve

$$H_e \geq 0, \quad H_o \geq 0, \quad H_o \leq D_o H_e.$$

For the last inequality, the old bound gives

$$H_o + \mathfrak{o}_{c_*} H_e \leq (D_o + \mathfrak{o}_{c_*}) H_e \leq (D_o + \mathfrak{o}_{c_*})(H_e + \mathfrak{o}_{c_*} H_o).$$

Induction, using $1 + x \leq e^x$, now gives

$$\begin{aligned} H_e(\mathcal{U} \cup \{c_*\}) &\leq H_e(\mathcal{U})(1 + \mathfrak{o}_{c_*} D_o(\mathcal{U})) \\ &\leq \exp(T_o(\mathcal{U}) + \mathfrak{o}_{c_*} D_o(\mathcal{U})). \end{aligned}$$

Applying this to each $\mathcal{C}_j(\omega)$ proves

$$1 + \chi^2(P_1(\omega) \| Q(\omega)) \leq \exp(\mathcal{B}(\omega)).$$

Finally, f_0 and f_{av} are strictly positive on the finite label alphabet. Hence

$$Q(\omega) \ll P_0(\omega), \quad P_1(\omega) \ll Q(\omega).$$

All these comparisons are conditional on the full triple ω .

3. We derive the component estimates used for occupancy counting. In this step, the finite density expressions are considered as algebraic functions of

$$a' \in (-1, 1), \quad u' \in \mathbb{R},$$

with the augmentation and coefficient weights fixed. Quantitative derivative bounds will be used on $[0, 1/16]^2$. Write $\Delta_{c,e}(a', u')$ and $\Delta_{c,o}(a', u')$ for the resulting discrepancies.

The entire propensity-coefficient marginal and the entire outcome-coefficient marginal are common to both branches, with disclosed coordinates pinned. Therefore

$$\Delta_{c,e}(a', 0) = \Delta_{c,o}(a', 0) = \Delta_{c,e}(0, u') = \Delta_{c,o}(0, u') = 0.$$

We also need the two first-order even cancellations.

To see them, the conditional coefficient means at an undisclosed node are

$$\mathbb{E}_{\nu, \sigma, \delta}[\eta_j \mid \boldsymbol{\lambda}] = \nu \kappa g_\sigma(j/K) \lambda_j, \quad \mathbb{E}_{\nu, \sigma, \delta}[\lambda_j \mid \boldsymbol{\eta}] = \nu \kappa g_\sigma(j/K) \eta_j.$$

At a disclosed node, these means are the corresponding pinned values. For constant coarse sign, the undisclosed means change sign when the coarse sign changes.

If $\mathbf{c}$ has exactly one marked observation i , its record product equals

$$\prod_{i' \in \mathbf{c}} r_{\nu, i'} = \prod_{i' \in \mathbf{c} \setminus \{i\}} (1 + z_{i'}^t \xi(X_{i'})) \\ \times [(1 + z_i^t \xi(X_i))(1 + z_i^y v(X_i)) + z_i^t z_i^y t_\nu(X_i)(1 - \xi(X_i)^2)].$$

Conditioning on the propensity coefficients, its outcome-field term is affine in the outcome coefficients. The branch-dependent conditional means of the undisclosed outcome coefficients, and the correction t_1 , are linear in the constant coarse sign. Averaging its two values cancels these changes. Disclosed-coordinate terms are common. Thus

$$\mathbf{d}_{\mathbf{c},+} + \mathbf{d}_{\mathbf{c},-} = 2\mathbf{d}_{\mathbf{c},0}, \quad \Delta_{\mathbf{c},e} = 0 \quad \text{if } k_{\mathbf{c}} \leq 1.$$

For zero marks, every record factor uses only the propensity coefficients, so

$$\Delta_{\mathbf{c},o} = 0 \quad \text{if } k_{\mathbf{c}} = 0.$$

At $a' = 0$, all undifferentiated record factors depend only on the outcome coefficients. Differentiating one factor produces a term linear in the propensity coefficients and a correction term linear in the coarse sign. Conditioning on the outcome coefficients and averaging the two constant coarse signs, using the conditional means above, therefore gives

$$\partial_{a'} \Delta_{\mathbf{c},e}(0, u') = 0.$$

For the outcome derivative, define the modified augmentation

$$\omega^{(i)} = (\mathbf{X}, (B_{i'} \mathbf{1}\{i' = i\})_{i'=1}^n, \delta).$$

Each record factor is affine in u' , so the product rule at $u' = 0$ gives

$$\partial_{u'} \Delta_{\mathbf{c},e}(a', 0; \omega) = \sum_{i \in \mathbf{c}} \left\{ \Delta_{\mathbf{c},e}(a', 1; \omega^{(i)}) - \Delta_{\mathbf{c},e}(a', 0; \omega) \right\} = 0.$$

Here evaluation at unit outcome amplitude extracts the coefficient of the affine marked factor; the algebraic functions are defined there. Each modified augmentation has at most one mark, so the preceding cancellation applies.

For the derivative bounds, define the coefficient fields and the rational correction factor by

$$\mathbf{F}_\lambda(x) = \sum_{j=0}^K f_j(x) \lambda_j, \quad \mathbf{F}_\eta(x) = \sum_{j=0}^K f_j(x) \eta_j, \quad c_x(a') = \frac{a'(1 - (a')^2 \mathbf{F}_\lambda(x)^2)}{1 - (a')^2}.$$

The frame identities give

$$|\mathbf{F}_\lambda|, |\mathbf{F}_\eta| \leq \sqrt{2} \leq 2, \quad |\mathbf{F}_\lambda \mathbf{F}_\eta| \leq 2.$$

Direct differentiation gives

$$c'_x(a') = 1 + (1 - \mathbf{F}_\lambda(x)^2) \frac{(a')^2(3 - (a')^2)}{(1 - (a')^2)^2}, \\ c''_x(a') = (1 - \mathbf{F}_\lambda(x)^2) \frac{2a'(3 + (a')^2)}{(1 - (a')^2)^3}.$$

On $0 \leq a' \leq 1/16$,

$$1 - (a')^2 \geq \frac{15}{16}, \quad |1 - F_\lambda^2| \leq 1,$$

and hence

$$|c_x| \leq \frac{1}{15}, \quad |c'_x| \leq 1 + \frac{3/256}{(15/16)^2} \leq 2, \quad |c''_x| \leq \frac{1/2}{(15/16)^3} \leq 1.$$

In this notation the interaction coordinate is

$$\zeta(x) = u' \{ a' F_\lambda(x) F_\eta(x) - \nu \kappa \tilde{g}_\sigma(x) c_x(a') \}.$$

The estimates needed for its derivatives are

$$\begin{aligned} |F_\lambda F_\eta - \nu \kappa \tilde{g}_\sigma c'_x| &\leq 4 + \frac{1}{8} = \frac{33}{8}, \\ |F_\lambda F_\eta - \nu \kappa \tilde{g}_\sigma c'_x| &\leq 2 + \frac{1}{8} = \frac{17}{8}, \\ |a' F_\lambda F_\eta - \nu \kappa \tilde{g}_\sigma c_x| &\leq \frac{1}{8} + \frac{1}{240} = \frac{31}{240}. \end{aligned}$$

The first, coarser bound is sufficient for the first propensity derivative. For every record factor on the amplitude rectangle,

$$\begin{aligned} |\partial_{a'} r_{\nu,i}| &\leq 2 + \frac{1}{16} \frac{33}{8} \leq 4, \\ |\partial_{a'}^2 r_{\nu,i}| &\leq \frac{1}{256} \leq 4, \\ |\partial_{u'} r_{\nu,i}| &\leq 2 + \frac{31}{240} \leq 4, \\ |\partial_{a'} \partial_{u'} r_{\nu,i}| &\leq \frac{17}{8} \leq 4, \\ |\partial_{a'}^2 \partial_{u'} r_{\nu,i}| &\leq \frac{1}{16} \leq 4, \quad \partial_{u'}^2 r_{\nu,i} = 0. \end{aligned}$$

Also $|r_{\nu,i}| \leq 2$ throughout this closed rectangle.

Assigning the four labelled differentiations to record factors gives at most n_c^4 terms in the fourth mixed product derivative. Each differentiated factor is bounded by four, and every remaining factor by two. Finite averaging therefore gives

$$|\partial_{a'}^2 \partial_{u'}^2 d_{c,\nu,s_{\text{coarse}}}| \leq 256 n_c^4 2^{n_c},$$

and the two-differentiation product rule similarly gives

$$|\partial_{a'} \partial_{u'} d_{c,\nu,s_{\text{coarse}}}| \leq 16 n_c^2 2^{n_c}.$$

These statements include the empty product, whose derivatives vanish. Taking the even and odd contrasts yields

$$|\partial_{a'}^2 \partial_{u'}^2 \Delta_{c,e}| \leq 512 n_c^4 2^{n_c}, \quad |\partial_{a'} \partial_{u'} \Delta_{c,o}| \leq 16 n_c^2 2^{n_c}.$$

The axis identities and first-order cancellations imply

$$\partial_{a'}^2 \Delta_{c,e}(a', 0) = \partial_{a'}^2 \partial_{u'} \Delta_{c,e}(a', 0) = 0.$$

Applying the second-order remainder formula first in outcome amplitude and then in propensity amplitude gives

$$\begin{aligned} |\Delta_{c,e}(a, u)| &\leq 512 \mathfrak{n}_c^4 2^{n_c} \int_0^a (a - a') da' \int_0^u (u - u') du' \\ &= 128 a^2 u^2 \mathfrak{n}_c^4 2^{n_c}. \end{aligned}$$

For the odd discrepancy, $\partial_{a'} \Delta_{c,o}(a', 0) = 0$. The mixed derivative bound and the mean-value inequality give, successively,

$$|\partial_{a'} \Delta_{c,o}(a', u)| \leq 16u \mathfrak{n}_c^2 2^{n_c}, \quad |\Delta_{c,o}(a, u)| \leq 16au \mathfrak{n}_c^2 2^{n_c}.$$

It remains to check singleton cancellation under the actual disclosure. Distinct boundary nodes are separated by $K/M \geq 16$ fine intervals, so at most one active frame node at any covariate is disclosed. Distinct undisclosed coefficient pairs are independent, with each individual coefficient having mean zero. Thus, for every fixed coarse-sign vector,

$$\mathbb{E}_{1,\sigma,\delta}[\xi(x)^2] = a^2,$$

$$\mathbb{E}_{1,\sigma,\delta}[\xi(x)v(x)] - \mathbb{E}_{0,\sigma,\delta}[\xi(x)v(x)] = a\kappa\tilde{g}_\sigma(x).$$

For the first identity, the diagonal terms sum to $a^2 \sum_j f_j(x)^2 = a^2$, and every cross term has an undisclosed mean-zero factor. For the second, the undisclosed same-node correlation is $\kappa g_\sigma(j/K)$; disclosed same-node terms are common, and $g_\sigma(j/K) = 0$ at disclosed nodes. Cross terms again vanish. The first moments of ξ and v are common to both branches, and

$$a\kappa\tilde{g}_\sigma(x) + t_1(x)(1 - a^2) = 0.$$

Consequently every singleton label density agrees with its null density, for either value of its coarse sign. These identities also hold at endpoints by the frame formulas. For the empty subset, normalization of the coefficient weights gives all component densities equal to one. Hence

$$\Delta_{c,e} = \Delta_{c,o} = 0 \quad \text{if } \mathfrak{n}_c \leq 1.$$

Squaring the discrepancy bounds and using the denominator lower bounds now gives, for μ -almost every augmentation and every component,

$$\mathfrak{e}_c \leq 2^{14} a^4 u^4 \mathfrak{n}_c^8 8^{n_c} \mathbf{1}\{\mathfrak{n}_c \geq 2, \mathfrak{k}_c \geq 2\},$$

$$\mathfrak{o}_c \leq 2^8 a^2 u^2 \mathfrak{n}_c^4 8^{n_c} \mathbf{1}\{\mathfrak{n}_c \geq 2, \mathfrak{k}_c \geq 1\}.$$

4. We bound the expectations of the budgets. Define the two occupancy scores by

$$S_{\text{one}}(\omega) = \sum_{c \in \mathcal{C}(\omega)} \mathfrak{n}_c^8 8^{n_c} \mathbf{1}\{\mathfrak{n}_c \geq 2, \mathfrak{k}_c \geq 2\},$$

$$S_{\text{pair}}(\omega) = \frac{1}{2} \sum_{\mathbf{c}, \mathbf{c}' \in \mathcal{C}(\omega)} n_{\mathbf{c}}^4 n_{\mathbf{c}'}^4 8^{n_{\mathbf{c}} + n_{\mathbf{c}'}} \\ \times \mathbf{1} \left\{ \begin{array}{l} \mathbf{c} \neq \mathbf{c}', \mathbf{j}(\mathbf{c}) = \mathbf{j}(\mathbf{c}'), \\ n_{\mathbf{c}}, n_{\mathbf{c}'} \geq 2, k_{\mathbf{c}}, k_{\mathbf{c}'} \geq 1 \end{array} \right\}.$$

The component estimates imply

$$\mathcal{A} \leq 2^{14} a^4 u^4 S_{\text{one}}, \quad \mathcal{B} \leq 2^{16} a^4 u^4 S_{\text{pair}} \quad \mu\text{-almost surely.}$$

Under μ , the fine-cell indices are independent uniform draws from $\{0, \dots, K-1\}$, and the flags are independent Bernoulli draws, independent of those indices. A connected component of size r_{occ} occupies consecutive fine cells spanning at most r_{occ} cells. For a specified observation subset of that size and a specified starting cell, the necessary occupancy event has probability at most

$$\left(\frac{r_{\text{occ}}}{K} \right)^{r_{\text{occ}}}.$$

A window extending past the final cell has fewer available cells and obeys the same bound. The probability of at least two marks in that subset is at most $r_{\text{occ}}^2 \varepsilon^2$. There are at most K starting cells and $\binom{n}{r_{\text{occ}}}$ observation subsets. Overcounting these necessary events yields

$$\int S_{\text{one}} d\mu \leq \varepsilon^2 K \sum_{r_{\text{occ}}=2}^n \binom{n}{r_{\text{occ}}} r_{\text{occ}}^{10} 8^{r_{\text{occ}}} \left(\frac{r_{\text{occ}}}{K} \right)^{r_{\text{occ}}}.$$

For two distinct components, their observation subsets are disjoint. Each requires a mark, so for sizes $r_{\text{occ}}, r'_{\text{occ}}$ the joint flag probability is at most $r_{\text{occ}} r'_{\text{occ}} \varepsilon^2$. A coarse pair contains $2K/M$ fine cells. Thus the number of ordered starting-cell choices in a common coarse pair is at most

$$\frac{M}{2} \left(\frac{2K}{M} \right)^2 = \frac{2K^2}{M}.$$

The number of disjoint subset choices is bounded by $\binom{n}{r_{\text{occ}}} \binom{n}{r'_{\text{occ}}}$. Using independence on the disjoint subsets, and overcounting by ordered pairs, gives

$$\int S_{\text{pair}} d\mu \leq \frac{2\varepsilon^2 K^2}{M} \left\{ \sum_{r_{\text{occ}}=2}^n \binom{n}{r_{\text{occ}}} r_{\text{occ}}^5 8^{r_{\text{occ}}} \left(\frac{r_{\text{occ}}}{K} \right)^{r_{\text{occ}}} \right\}^2.$$

When the collection of eligible pairs is empty, its score is zero and this bound still applies.

Define the occupancy ratio by

$$\zeta_{\text{occ}} = 8e \frac{n}{K}.$$

The factorial and binomial bounds

$$r_{\text{occ}}! \geq \left(\frac{r_{\text{occ}}}{e} \right)^{r_{\text{occ}}}, \quad \binom{n}{r_{\text{occ}}} \leq \frac{n^{r_{\text{occ}}}}{r_{\text{occ}}!}$$

imply, for every $r_{\text{occ}} \geq 2$ and every nonnegative integer d_{occ} ,

$$\binom{n}{r_{\text{occ}}} r_{\text{occ}}^{d_{\text{occ}}} 8^{r_{\text{occ}}} \left(\frac{r_{\text{occ}}}{K} \right)^{r_{\text{occ}}} \leq r_{\text{occ}}^{d_{\text{occ}}} \zeta_{\text{occ}}^{r_{\text{occ}}} \leq r_{\text{occ}}^{d_{\text{occ}}+1} \zeta_{\text{occ}}^{r_{\text{occ}}}.$$

The factorial inequality follows from the usual Stirling lower bound, whose square-root factor is at least one for positive integers. Extending the nonnegative finite sums gives

$$\begin{aligned}\int \mathcal{A} d\mu &\leq 2^{14} a^4 u^4 \varepsilon^2 K \sum_{r_{\text{occ}}=2}^{\infty} r_{\text{occ}}^{11} \zeta_{\text{occ}}^{r_{\text{occ}}}, \\ \int \mathcal{B} d\mu &\leq \frac{2^{17} a^4 u^4 \varepsilon^2 K^2}{M} \left\{ \sum_{r_{\text{occ}}=2}^{\infty} r_{\text{occ}}^6 \zeta_{\text{occ}}^{r_{\text{occ}}} \right\}^2.\end{aligned}$$

For $d_{\text{ser}} \in \{6, 11\}$, the elementary inequality $r_{\text{occ}} \leq 2^{r_{\text{occ}}-1}$ gives

$$\begin{aligned}\sum_{r_{\text{occ}}=2}^{\infty} r_{\text{occ}}^{d_{\text{ser}}} \zeta_{\text{occ}}^{r_{\text{occ}}} &\leq 2^{d_{\text{ser}}} \zeta_{\text{occ}}^2 \sum_{r_{\text{occ}}=2}^{\infty} (2^{d_{\text{ser}}} \zeta_{\text{occ}})^{r_{\text{occ}}-2} \\ &= \frac{2^{d_{\text{ser}}} \zeta_{\text{occ}}^2}{1 - 2^{d_{\text{ser}}} \zeta_{\text{occ}}} \leq 2^{d_{\text{ser}}+1} \zeta_{\text{occ}}^2.\end{aligned}$$

The geometric series converges because

$$0 \leq \zeta_{\text{occ}} \leq 24 \frac{n}{K}, \quad 2^{11} \zeta_{\text{occ}} \leq 2^{11} \cdot 24 \cdot 2^{-40} < \frac{1}{2}.$$

Here we used $e < 3$ and $2^{40}n \leq K$. Consequently,

$$\begin{aligned}\int \mathcal{A} d\mu &\leq 2^{26} 24^2 \frac{a^4 u^4 \varepsilon^2 n^2}{K} \leq \frac{2^{38} a^4 u^4 \varepsilon^2 n^2}{K}, \\ \int \mathcal{B} d\mu &\leq 2^{31} 24^4 \frac{a^4 u^4 \varepsilon^2 n^4}{MK^2} \leq \frac{2^{56} a^4 u^4 \varepsilon^2 n^4}{MK^2},\end{aligned}$$

since $2^{26} 24^2 < 2^{38}$ and $2^{31} 24^4 < 2^{56}$. The budgets are measurable finite sums. The uniform component denominator bounds and finite label alphabet make them bounded for fixed parameters, hence integrable under the probability law μ .

5. We identify the original-record projections. Define the recovery map by

$$\mathcal{O}_L(\omega, \mathbf{z}) = \left(X_i, \frac{1 + z_i^t}{2}, B_i z_i^y L \right)_{i=1}^n.$$

This map is measurable.

For a fixed σ , the coefficient law in [Algorithm 5](#) assigns mass

$$\prod_{j=0}^K \frac{1 + \nu \kappa g_{\sigma}(j/K) \lambda_j \eta_j}{4}$$

to a coefficient draw. At boundary nodes, $g_{\sigma}(j/K) = 0$. Its disclosure law is therefore μ_{∂} , and conditioning on a disclosure gives precisely the weights $w_{\nu, \sigma, \delta}$. In particular, for every function Ψ on the coefficient space,

$$\int \mathbb{E}_{\nu, \sigma, \delta}[\Psi] \mu_{\partial}(d\delta) = \sum_{\lambda, \eta} \left\{ \prod_{j=0}^K \frac{1 + \nu \kappa g_{\sigma}(j/K) \lambda_j \eta_j}{4} \right\} \Psi(\lambda, \eta).$$

For the null branch, the component factorization in [Item 2](#) identifies f_0 with the conditional full-record label average. For the alternative branch, that average is the definition of f_1 . Integrating the disclosure thus recovers the respective original coefficient averages.

For a fixed latent draw and covariate x , the flag and fair-label weights give

$$\Pr_{\nu} \left(A = \frac{1 + z^t}{2}, Y = z^y L \mid x \right) = \frac{\varepsilon}{4} \{ 1 + z^t \xi(x) + z^y v(x) + z^t z^y \zeta(x) \},$$

and, summing the retained outcome-sign label when the flag is zero,

$$\Pr_{\nu} \left(A = \frac{1 + z^t}{2}, Y = 0 \mid x \right) = \sum_{z^y \in \{-1, 1\}} \frac{1 - \varepsilon}{4} (1 + z^t \xi(x)) = \frac{1 - \varepsilon}{2} (1 + z^t \xi(x)).$$

These are exactly the categories of [Definition 29](#). Independent record generation conditional on the latent draw, followed by its prior average, therefore proves

$$(\mu \otimes P_0) \circ \mathbf{O}_L^{-1} = \mathbb{P}_0, \quad (\mu \otimes P_1) \circ \mathbf{O}_L^{-1} = \mathbb{P}_1.$$

6. Assume the sum of the two stated public budget bounds is at most 2^{-16} . We first obtain an envelope for conditional total variation.

On a finite alphabet, absolute continuity and Cauchy–Schwarz give

$$\text{TV}(Q(\omega), P_0(\omega)) \leq \frac{1}{2} \sqrt{\chi^2(Q(\omega) \| P_0(\omega))},$$

$$\text{TV}(P_1(\omega), Q(\omega)) \leq \frac{1}{2} \sqrt{\chi^2(P_1(\omega) \| Q(\omega))}.$$

Indeed, total variation is half the sum of absolute differences of the masses; writing those differences relative to the dominating law and applying Cauchy–Schwarz gives the displayed bounds.

For $0 \leq x_{\text{exp}} \leq 1/256$, comparison of the exponential series with the geometric series gives

$$e^{x_{\text{exp}}} \leq \frac{1}{1 - x_{\text{exp}}} \leq 1 + 2x_{\text{exp}},$$

where the second inequality follows after multiplication by $1 - x_{\text{exp}} > 0$. If $\mathcal{A}(\omega) + \mathcal{B}(\omega) \leq 1/256$, the conditional divergence bounds and the triangle inequality therefore yield

$$\begin{aligned} \text{TV}(P_0(\omega), P_1(\omega)) &\leq \frac{1}{2} \sqrt{2\mathcal{A}(\omega)} + \frac{1}{2} \sqrt{2\mathcal{B}(\omega)} \\ &\leq \sqrt{\mathcal{A}(\omega) + \mathcal{B}(\omega)} \leq \frac{1}{16}. \end{aligned}$$

The middle inequality follows by squaring and using $2\sqrt{\mathcal{A}\mathcal{B}} \leq \mathcal{A} + \mathcal{B}$. In the complementary case, $\mathcal{A}(\omega) + \mathcal{B}(\omega) > 1/256$, total variation between probability laws is at most one. Both cases consequently give

$$\text{TV}(P_0(\omega), P_1(\omega)) \leq \frac{1}{16} + 256(\mathcal{A}(\omega) + \mathcal{B}(\omega)) \quad \mu\text{-almost surely.}$$

For any measurable augmented event $\mathbf{E}_{\text{aug}} \subseteq \Omega_{\text{aug}} \times \mathbf{Z}_n$, define its label section by

$$\mathbf{E}_{\text{aug}, \omega} = \{\mathbf{z} : (\omega, \mathbf{z}) \in \mathbf{E}_{\text{aug}}\}.$$

The common augmentation marginal implies

$$\begin{aligned} & |(\mu \otimes P_0)(\mathbf{E}_{\text{aug}}) - (\mu \otimes P_1)(\mathbf{E}_{\text{aug}})| \\ & \leq \int |P_0(\omega)(\mathbf{E}_{\text{aug}, \omega}) - P_1(\omega)(\mathbf{E}_{\text{aug}, \omega})| \mu(d\omega) \\ & \leq \frac{1}{16} + 256 \left(\int \mathcal{A} d\mu + \int \mathcal{B} d\mu \right) \leq \frac{1}{16} + 256 \cdot 2^{-16} = \frac{17}{256}. \end{aligned}$$

Taking the supremum over augmented events proves

$$\text{TV}(\mu \otimes P_0, \mu \otimes P_1) \leq \frac{17}{256}.$$

For every measurable original-record event $\mathbf{E}_{\text{rec}}$, its inverse image under $\mathbf{O}_L$ is an augmented event. The projection identities from [Item 5](#) therefore give

$$\text{TV}(\mathbb{P}_0, \mathbb{P}_1) \leq \text{TV}(\mu \otimes P_0, \mu \otimes P_1) \leq \frac{17}{256} < \frac{1}{8}.$$

7. Finally, we verify that the augmentation is induced by the stipulated latent construction. Define its latent coefficient law by the masses

$$\mathbf{W}_\nu(\{\sigma, \boldsymbol{\lambda}, \boldsymbol{\eta}\}) = 2^{-M/2} \prod_{j=0}^K \frac{1 + \nu \kappa g_\sigma(j/K) \lambda_j \eta_j}{4}.$$

Define the joint latent augmented law by

$$\begin{aligned} & \mathbf{L}_\nu(d\sigma, d\boldsymbol{\lambda}, d\boldsymbol{\eta}, d\mathbf{X}, d\mathbf{B}, d\delta, d\mathbf{z}) \\ & = \mathbf{W}_\nu(d\sigma, d\boldsymbol{\lambda}, d\boldsymbol{\eta}) \lambda^{\otimes n}(d\mathbf{X}) \text{Bernoulli}(\varepsilon)^{\otimes n}(d\mathbf{B}) \text{Dirac}_{\mathbf{D}(\boldsymbol{\lambda}, \boldsymbol{\eta})}(d\delta) \left(\prod_{i=1}^n r_{\nu, i} \right) \mathbf{F}_n(d\mathbf{z}). \end{aligned}$$

The coefficient weights sum to one, and the conditional label product integrates to one. Thus

$$\mathbf{L}_\nu(\text{entire space}) = 1, \quad \mathbf{L}_\nu\{\delta = \mathbf{D}(\boldsymbol{\lambda}, \boldsymbol{\eta})\} = 1.$$

Conditioning its finite coefficient average on the disclosure, and then using the component factorization for branch zero and the full coefficient average for branch one, gives

$$\mathbf{L}_\nu \circ ((\sigma, \boldsymbol{\lambda}, \boldsymbol{\eta}, \omega, \mathbf{z}) \mapsto (\omega, \mathbf{z}))^{-1} = \mu \otimes P_\nu, \quad \nu \in \{0, 1\}.$$

There are $M + 1$ disclosed nodes, and the coarse tents vanish at all of them. Their coefficient pairs are therefore independent fair pairs conditional on every coarse-sign vector. Summing the undisclosed coefficient weights gives, for each $\sigma_* \in \{-1, 1\}^{M/2}$ and each actual boundary disclosure δ ,

$$\Pr_\nu\{\sigma = \sigma_*, \mathcal{D}_\partial = \delta\} = 2^{-M/2} 4^{-(M+1)}.$$

Define the fair coarse-sign law by

$$\mathbf{U}_\sigma = 2^{-M/2} \sum_{\sigma \in \{-1,1\}^{M/2}} \text{Dirac}_\sigma.$$

The covariates and flags are drawn independently of these latent variables, and integrating the normalized labels leaves their law unchanged. Hence

$$\mathbf{L}_\nu \circ ((\sigma, \boldsymbol{\lambda}, \boldsymbol{\eta}, \omega, \mathbf{z}) \mapsto (\sigma, \omega))^{-1} = \mathbf{U}_\sigma \otimes \mu.$$

This proves both the independent fair-sign assertion and its product relation with the full augmentation. Since $L > 0$, recovery also satisfies $B_i = \mathbf{1}\{Y_i \neq 0\}$. Together with the kernel, budget, projection, and total-variation properties already established, these are all the required conclusions. $\square$

Algorithm 6 (Branchwise converse priors $\pi_{0,n,v}, \pi_{1,n,v}$)

Inputs are the sample size n and the public tuple $v = (p, \alpha, \beta, \gamma)$; $q = (p-1)/p$, $F(p)$ is the exponent-comparison scalar, D_p is the rough-confounding exponent denominator, H_{fine} is the fine-rank multiplier, and $N_{\text{low}}(v)$ is the rough-construction threshold.

1. If $F(p) < 1$ and $n \geq N_{\text{low}}(v)$, select

$$K = 2^{\lceil \log_2(H_{\text{fine}} n^{2/D_p}) \rceil}, \quad M = 2^{\lfloor \log_2(K^{(\alpha+\beta)/\gamma}/16) \rfloor},$$

and set

$$a = \frac{K^{-\alpha}}{16}, \quad u = \frac{1}{16}, \quad \varepsilon_{n,v} = 16^{-1/q} K^{-\beta/q}, \quad L_{n,v} = \varepsilon_{n,v}^{-1/p}.$$

Select the finite copula-frame priors

$$\pi_{\nu,n,v} = \pi_{\nu;v,a,u,\varepsilon_{n,v},L_{n,v}}^{K,M}, \quad \nu \in \{0,1\}.$$

2. Otherwise, set

$$N = 2^{\lceil n^{2q/(2\gamma+q)} \rceil}, \quad h = \frac{1}{N}, \quad \varepsilon_{n,v} = h^{\gamma/q}, \quad L_{n,v} = h^{-\gamma/(p-1)}.$$

Select the opposite paired-tent null and alternative priors at this rank, cell width, rare-mark probability, and mark magnitude, denoted respectively by

$$\pi_{0,n,v}, \quad \pi_{1,n,v}.$$

3. Output the selected priors and their complete independent-record mixtures

$$(\pi_{0,n,v}, \pi_{1,n,v}), \quad \mathbb{P}_{\nu,n,v} = \int P^{\otimes n} \pi_{\nu,n,v}(dP), \quad \nu \in \{0,1\}.$$

The selected construction satisfies the stipulated balancing, support-law membership, true null-mixture denominator, complete occupancy and transition accounting, and bounded-scale comparison properties.

The common augmentation in [Lemma 7](#) includes the disclosed boundary coefficients, and the coarse signs remain independent of it under both branches. All three conditional label laws are normalized probability laws. The first expected budget has sample-size and rank factor n^2/K ; the second has factor $n^4/(MK^2)$. Both include the amplitude and mark factor $a^4u^4\varepsilon^2$. Their stated sum condition gives total variation below one eighth for the recovered complete-record mixtures, including the treatment labels and zero outcomes.

C.3 Selected priors and the rough lower bound

The selected converse construction in [Algorithm 6](#) now has its ingredients. For $F(p) < 1$ and sample size at least the public lower-construction threshold, it selects the copula-frame priors at its specified ranks and finite-moment tuning. The other branch uses the paired-tent package. This selection combines the rough-phase construction with witnesses available for every integer sample size at least two.

At moment order two, the bounded companion uses the bounded copula tuning in the large-sample rough branch and the signed-binary paired-tent package otherwise. Let $\mathcal{V}$ denote the admissible tuple domain stipulated in [Definition 1](#). The following definition fixes the bounded prior pair and its complete-record mixtures.

Definition 31. `[synth_1]` (Bounded converse priors) *Fix $v = (2, \alpha, \beta, \gamma) \in \mathcal{V}$, write $w = (\alpha, \beta, \gamma)$, and let $n \geq 2$. Use the phase functional and public constants of [Definition 15](#). For $\nu \in \{0, 1\}$, define the bounded prior pair $\pi_{\nu, n, v}^b$ as follows.*

If $F(2) < 1$ and $n \geq N_{\text{low}}(v)$, put $S = \alpha + \beta$, let K be the least power of two at least $H_{\text{fine}}n^{2/D_2}$, and set

$$M = 2^{\lfloor \log_2(K^{S/\gamma}/16) \rfloor}, \quad a = \frac{K^{-\alpha}}{16}, \quad b = \frac{K^{-\beta}}{256}.$$

Define

$$\pi_{\nu, n, v}^b = \pi_{\nu; v, a, 2b, 1/2, 1}^{K, M}, \quad \nu \in \{0, 1\},$$

using the finite coefficient draws and record-law pushforward of [Algorithm 5](#). This is the bounded tuning of [Lemma 6](#): it uses the same ranks and amplitudes a, b as the rough branch of [Algorithm 6](#), with outcome-coordinate amplitude $u = 2b$, mark probability $\varepsilon = 1/2$, and mark magnitude $L = 1$.

If $F(2) \geq 1$, or if $F(2) < 1$ and $n < N_{\text{low}}(v)$, take the signed-binary paired-tent package of [Lemma 5](#) at sample size n and tuple w . Writing P_0^b for its selected null law and π_1^b for its selected finite alternative prior, define

$$\pi_{0, n, v}^b = \delta_{P_0^b}, \quad \pi_{1, n, v}^b = \pi_1^b.$$

This supplies the signed-binary counterpart of the paired-tent branch and small-sample fallback in [Algorithm 6](#).

In either branch, the associated complete-record mixtures are

$$\mathbb{P}_{\nu, n, v}^b = \int P^{\otimes n} \pi_{\nu, n, v}^b(dP), \quad \nu \in \{0, 1\}.$$

The bounded prior pair $\pi_{\nu,n,v}^b$ in [Definition 31](#) is defined for every admissible moment-order-two tuple and every $n \geq 2$. Its complete-record mixture $\mathbb{P}_{\nu,n,v}^b$ uses the same draw-once convention as $\mathbb{P}_{\pi,n}$. The next result assembles support placement, likelihood control, and the testing-risk lower bound in the rough phase.

Lemma 8. [[lem:rough-full-record-sharp-lower](#)] (Sharp rough-phase record lower bound)
Use the selected converse priors and complete-record mixtures of [Algorithm 6](#), and the phase functional, exponents, multipliers, and scales of [Definition 15](#). Let λ denote Lebesgue probability measure on $[0, 1]$. Define the heterogeneity distance, model supremum, and witness distance by

$$d(P) = \left(\int_{[0,1]} \left(\tau_P(x) - \int_{[0,1]} \tau_P(z) d\lambda(z) \right)^2 d\lambda(x) \right)^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P), \quad d_0 = \frac{1}{8\sqrt{2}}.$$

For probability laws, TV denotes the supremum of the absolute differences between their probabilities of the same measurable event.

Suppose that:

- **(Admissibility.)** $v = (p, \alpha, \beta, \gamma)$ is admissible as in [Definition 1](#).
- **(Rough phase.)** $F(p) < 1$.
- **(Sample size.)** n is an integer with $n \geq 2$.

The finite priors $\pi_{0,n,v}$ and $\pi_{1,n,v}$ have nonnegative weights summing to one. Every positive-weight null support law belongs to $H_0(v)$ of [Definition 2](#), and every positive-weight alternative support law satisfies

$$P \in \mathcal{M}_v, \quad c_v n^{-E_4(p)} \leq d(P) < d_0.$$

Moreover,

$$d_0 \leq D_v, \quad \text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) < \frac{1}{2}, \quad B_n\left(v, c_v n^{-E_4(p)}\right) \geq \frac{2}{5},$$

with testing risk as in [Definition 5](#).

If $n \geq N_{\text{low}}(v)$, the selected fine and coarse ranks K, M of [Algorithm 6](#) satisfy the rank conditions of [Lemma 6](#). In particular, they are powers of two and satisfy

$$16M \leq K, \quad 2 \leq M \leq K^{S/\gamma}, \quad 2^{40}n \leq K.$$

With the selected rare-mark probability $\varepsilon_{n,v}$ of [Algorithm 6](#), the deterministic budget bounds are

$$\frac{2^{38}(K^{-\alpha}/16)^4(1/16)^4\varepsilon_{n,v}^2 n^2}{K} \leq \frac{2^{-10}}{H_{\text{fine}}},$$

$$\frac{2^{56}(K^{-\alpha}/16)^4(1/16)^4\varepsilon_{n,v}^2 n^4}{MK^2} \leq \frac{2^{13}}{H_{\text{fine}}^2}.$$

The same complete-record mixtures then satisfy

$$\text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) < \frac{1}{8}.$$

For the bounded experiment, suppose that:

- **(Admissibility.)** $w = (\alpha, \beta, \gamma)$ is admissible as in [Definition 7](#); write $v = (2, \alpha, \beta, \gamma)$.
- **(Rough phase.)** $F(2) < 1$.
- **(Sample size.)** n is an integer with $n \geq 2$.

Let $\pi_{0,n,v}^b$ and $\pi_{1,n,v}^b$ be the selected bounded converse priors of [Definition 31](#), and let $\mathbb{P}_{0,n,v}^b$ and $\mathbb{P}_{1,n,v}^b$ denote their respective complete-record mixtures. Both finite priors have nonnegative weights summing to one. Every positive-weight null support law belongs to $H_0^b(w)$ of [Definition 9](#), and every positive-weight alternative support law satisfies

$$P \in \mathcal{M}_w^b, \quad c_w^b n^{-E_4(2)} \leq d(P) < d_0.$$

Furthermore,

$$\text{TV}(\mathbb{P}_{0,n,v}^b, \mathbb{P}_{1,n,v}^b) < \frac{1}{2}, \quad B_n^b(w, c_w^b n^{-E_4(2)}) \geq \frac{2}{5},$$

with bounded testing risk as in [Definition 11](#).

Proof of [Lemma 8](#). Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$ with $F(p) < 1$, and an integer $n \geq 2$. Use the quantities of [Definition 15](#), in particular

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad D_p = 1 + 2\alpha + \frac{\beta}{q} + \frac{S}{2\gamma}.$$

Admissibility gives

$$1 < p \leq 2, \quad \alpha, \beta > 0, \quad \gamma \geq \frac{1}{4},$$

so

$$0 < q \leq \frac{1}{2}, \quad S > 0, \quad 2\gamma + q > 0, \quad D_p > 1.$$

We first establish the original-model conclusions, splitting at $N_{\text{low}}(v)$.

1. **The small-sample branch.** Suppose $n < N_{\text{low}}(v)$. Subtracting the two exponent formulas gives

$$\begin{aligned} E_4(p) - E_0(p) &= \frac{2S(2\gamma + q) - 2\gamma q D_p}{(2\gamma + q)D_p} \\ &= \frac{2\gamma q}{(2\gamma + q)D_p} \left(\frac{2S}{q} - 2\alpha - \frac{\beta}{q} + \frac{S}{2\gamma} - 1 \right) \\ &= \frac{2\gamma q}{(2\gamma + q)D_p} \{F(p) - 1\} < 0. \end{aligned}$$

The last identity uses $q^{-1} - 1 = (p-1)^{-1}$. Thus $E_0(p) - E_4(p) > 0$. The definition of c_v , together with $n \leq N_{\text{low}}(v)$, yields

$$\begin{aligned} c_v n^{-E_4(p)} &\leq c_0^e N_{\text{low}}(v)^{-(E_0(p) - E_4(p))} n^{-E_4(p)} \\ &\leq c_0^e n^{-(E_0(p) - E_4(p))} n^{-E_4(p)} = c_0^e n^{-E_0(p)}. \end{aligned}$$

The selected priors are the paired-tent priors of [Algorithm 6](#), at the even rank

$$N = 2 \left\lceil n^{2q/(2\gamma+q)} \right\rceil, \quad h = \frac{1}{N}, \quad \varepsilon_{n,v} = h^{\gamma/q}, \quad L_{n,v} = h^{-\gamma/(p-1)}.$$

We identify their canonical support laws explicitly. For $\sigma \in \{-1, 1\}^{N/2}$, define the paired-tent field by

$$g_\sigma^{\text{tent}}(x) = \begin{cases} \sigma_j b_\Delta(Nx - (2j - 2)), & x \in [(2j - 2)/N, (2j - 1)/N], \\ -\sigma_j b_\Delta(Nx - (2j - 1)), & x \in [(2j - 1)/N, 2j/N], \end{cases} \quad 1 \leq j \leq N/2,$$

where b_Δ is the triangular tent of [Algorithm 5](#); boundary values are zero. Let P_σ^{tent} be the law with uniform covariate, propensity $1/2$, and conditional outcome kernels

$$\begin{aligned} Q_{0, P_\sigma^{\text{tent}}}(\cdot \mid x) &= \delta_0, \\ Q_{1, P_\sigma^{\text{tent}}}(\cdot \mid x) &= (1 - \varepsilon_{n,v})\delta_0 \\ &\quad + \frac{\varepsilon_{n,v}}{2} \left(1 + \frac{g_\sigma^{\text{tent}}(x)}{16}\right) \delta_{L_{n,v}} + \frac{\varepsilon_{n,v}}{2} \left(1 - \frac{g_\sigma^{\text{tent}}(x)}{16}\right) \delta_{-L_{n,v}}. \end{aligned}$$

Here δ_y denotes a point mass at y . Define P_0 by the same kernels with the tent field replaced by zero. These are exactly the null and alternative laws selected by the paired-tent branch: its treated mean is

$$\tau_{P_\sigma^{\text{tent}}}(x) = \frac{\varepsilon_{n,v} L_{n,v}}{16} g_\sigma^{\text{tent}}(x) = \frac{h^\gamma}{16} g_\sigma^{\text{tent}}(x), \quad \tau_{P_0}(x) = 0,$$

since $q^{-1} - (p - 1)^{-1} = 1$. Thus the selected alternative prior and its complete-record mixture are

$$\pi_{1,n,v} = 2^{-N/2} \sum_{\sigma \in \{-1, 1\}^{N/2}} \delta_{P_\sigma^{\text{tent}}}, \quad \bar{P}_1^{(n)} = 2^{-N/2} \sum_{\sigma \in \{-1, 1\}^{N/2}} (P_\sigma^{\text{tent}})^{\otimes n}.$$

Before pushing forward to laws, each of the $2^{N/2}$ sign configurations has weight $2^{-N/2}$; hence

$$2^{-N/2} > 0, \quad 2^{N/2} 2^{-N/2} = 1.$$

Every null sign configuration produces the same zero-effect law P_0 just defined. Consequently,

$$\pi_{0,n,v} = \delta_{P_0}, \quad \mathbb{P}_{0,n,v} = P_0^{\otimes n}, \quad \mathbb{P}_{1,n,v} = \bar{P}_1^{(n)}.$$

Here the equality of priors is as probability measures on laws; the finite sign enumeration may retain repeated copies of P_0 .

We now certify these canonical laws directly. Ceiling bounds give

$$2n^{2q/(2\gamma+q)} \leq N \leq 4n^{2q/(2\gamma+q)}, \quad N \geq 2, \quad 0 < h \leq \frac{1}{2}.$$

Hence $0 < \varepsilon_{n,v} < 1$ and $L_{n,v} > 0$. The displayed kernels have nonnegative masses summing to one, and their arm moments satisfy

$$\int |y|^p Q_{0,P}(dy \mid x) = 0, \quad \int |y|^p Q_{1,P}(dy \mid x) = \varepsilon_{n,v} L_{n,v}^p = 1$$

for $P = P_0$ and every $P = P_\sigma^{\text{tent}}$. The propensity and baseline mean are constant, while

$$\|g_\sigma^{\text{tent}}\|_\infty \leq 1, \quad |g_\sigma^{\text{tent}}(x) - g_\sigma^{\text{tent}}(z)| \leq 8N|x - z|.$$

For the latter bound, each paired bump has Lipschitz constant at most $4N$, and at most two paired bumps contribute at the two endpoints. Combining this bound with the envelope bound gives, for $0 < \gamma \leq 1$,

$$h^\gamma |g_\sigma^{\text{tent}}(x) - g_\sigma^{\text{tent}}(z)| \leq 8|x - z|^\gamma.$$

Indeed, when $N|x - z| \leq 1$, use $N|x - z| \leq (N|x - z|)^\gamma$; when $N|x - z| > 1$, use the envelope bound $2 \leq 8(N|x - z|)^\gamma$. Consequently,

$$\|\tau_{P_\sigma^{\text{tent}}}\|_\infty \leq \frac{1}{16}, \quad |\tau_{P_\sigma^{\text{tent}}}(x) - \tau_{P_\sigma^{\text{tent}}}(z)| \leq \frac{1}{2}|x - z|^\gamma.$$

These calculations verify all restrictions of [Definitions 1](#) and [2](#), so

$$P_0 \in H_0(v), \quad P_\sigma^{\text{tent}} \in \mathcal{M}_v.$$

Opposite pairing and integration of the squared triangular tent give

$$\int g_\sigma^{\text{tent}} d\lambda = 0, \quad \int (g_\sigma^{\text{tent}})^2 d\lambda = \frac{1}{3}.$$

Thus the distance of each canonical alternative is

$$d(P_\sigma^{\text{tent}}) = \frac{h^\gamma}{16\sqrt{3}} \geq \frac{4^{-\gamma}}{16\sqrt{3}} n^{-2\gamma q/(2\gamma+q)} = c_0^e n^{-E_0(p)}, \quad d(P_\sigma^{\text{tent}}) \leq \frac{1}{16} < d_0.$$

To compare the canonical complete-record mixtures, define the one-record likelihood ratio on the full record space by

$$\ell_\sigma^{\text{tent}}(X, A, Y) = 1 + \frac{g_\sigma^{\text{tent}}(X)}{16} \begin{cases} 1, & A = 1, Y = L_{n,v}, \\ -1, & A = 1, Y = -L_{n,v}, \\ 0, & \text{otherwise.} \end{cases}$$

The conditional masses show directly that

$$\frac{dP_\sigma^{\text{tent}}}{dP_0} = \ell_\sigma^{\text{tent}} \quad P_0\text{-almost surely.}$$

For two sign configurations σ, σ' , integration over all record categories and then over the paired cells gives

$$\begin{aligned} \int g_\sigma^{\text{tent}} g_{\sigma'}^{\text{tent}} d\lambda &= \frac{2}{3N} \sum_{j=1}^{N/2} \sigma_j \sigma'_j, \\ \int \ell_\sigma^{\text{tent}} \ell_{\sigma'}^{\text{tent}} dP_0 &= 1 + \frac{\varepsilon_{n,v}}{768N} \sum_{j=1}^{N/2} \sigma_j \sigma'_j. \end{aligned}$$

Write $\chi^2(Q\|P) = \int (dQ/dP - 1)^2 dP$ for the directed chi-square divergence. Independence of the records and finite-mixture expansion yield

$$1 + \chi^2(\bar{P}_1^{(n)}\|P_0^{\otimes n}) = \mathbb{E}_{\sigma, \sigma'} \left[\left(1 + \frac{\varepsilon_{n,v}}{768N} \sum_{j=1}^{N/2} \sigma_j \sigma'_j \right)^n \right],$$

where both sign configurations are independent and uniform. The expression inside the power is positive. Using $(1 + \vartheta_{\text{exp}})^n \leq e^{n\vartheta_{\text{exp}}}$ for $\vartheta_{\text{exp}} > -1$, independence of the sign products, and $\cosh(\vartheta_{\text{exp}}) \leq e^{\vartheta_{\text{exp}}^2/2}$ for $\vartheta_{\text{exp}} \in \mathbb{R}$, we obtain

$$\begin{aligned} 1 + \chi^2(\overline{P}_1^{(n)} \| P_0^{\otimes n}) &\leq \left[\cosh\left(\frac{n\varepsilon_{n,v}}{768N}\right) \right]^{N/2} \\ &\leq \exp\left(\frac{n^2\varepsilon_{n,v}^2 h}{36 \cdot 16^4}\right). \end{aligned}$$

The selected rank calibrates this exponent because

$$n^2\varepsilon_{n,v}^2 h = n^2 N^{-(1+2\gamma/q)} \leq 1.$$

Here the lower rank bound implies $N \geq n^{2q/(2\gamma+q)}$, and $\frac{2q}{2\gamma+q}(1 + 2\gamma/q) = 2$. Since $e^{\vartheta_{\text{exp}}} \leq (1 - \vartheta_{\text{exp}})^{-1}$ for $0 \leq \vartheta_{\text{exp}} < 1$,

$$1 + \chi^2(\overline{P}_1^{(n)} \| P_0^{\otimes n}) \leq \exp\left(\frac{1}{36 \cdot 16^4}\right) < 2.$$

Cauchy–Schwarz therefore gives

$$\text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) \leq \frac{1}{2} \sqrt{\chi^2(\overline{P}_1^{(n)} \| P_0^{\otimes n})} < \frac{1}{2}.$$

The model-supremum bound is supplied by [Lemma 5](#):

$$d_0 \leq D_v.$$

Together with the multiplier comparison, these canonical calculations give, for every positive-weight alternative support law,

$$P \in \mathcal{M}_v, \quad c_v n^{-E_4(p)} \leq c_0^e n^{-E_0(p)} \leq d(P) < d_0.$$

Finally, for every size-valid test ϕ , averaging over its independent uniform seed and using the total-variation expectation bound gives

$$\begin{aligned} \sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq c_0^e n^{-E_0(p)}}} \{1 - \mathbb{R}_n(P, \phi)\} &\geq \int \{1 - \mathbb{R}_n(P, \phi)\} \pi_{1,n,v}(dP) \\ &\geq \frac{9}{10} - \text{TV}(P_0^{\otimes n}, \overline{P}_1^{(n)}) > \frac{2}{5}. \end{aligned}$$

Taking the infimum over size-valid tests proves the canonical risk bound

$$B_n\left(v, c_0^e n^{-E_0(p)}\right) \geq \frac{2}{5}.$$

For each test satisfying the size constraint, decreasing the separation enlarges the alternative set, so

$$\sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq c_0^e n^{-E_0(p)}}} \{1 - \mathbb{R}_n(P, \phi)\} \leq \sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq c_v n^{-E_4(p)}}} \{1 - \mathbb{R}_n(P, \phi)\}.$$

The positive-weight alternative supplied by the paired-tent construction makes both sets nonempty. Taking infima over the same size-valid tests and using its risk bound gives

$$\frac{2}{5} \leq B_n\left(v, c_0^e n^{-E_0(p)}\right) \leq B_n\left(v, c_v n^{-E_4(p)}\right).$$

Finally, $n < N_{\text{low}}(v)$ excludes the hypothesis $n \geq N_{\text{low}}(v)$ of the additional rank and activity conclusions.

2. **Ranks in the large-sample branch.** Suppose now $n \geq N_{\text{low}}(v)$. Define the selected fine and coarse ranks by

$$K = 2^{\lceil \log_2(H_{\text{fine}} n^{2/D_p}) \rceil}, \quad M = 2^{\lfloor \log_2(K^{S/\gamma}/16) \rfloor}.$$

Since $p-1 \leq 1$ and $q \leq 1/2$,

$$2S + \frac{S}{2\gamma} \leq F(p) < 1, \quad S < \frac{2\gamma}{4\gamma+1} \leq \gamma.$$

Also,

$$D_p - 1 = 2\alpha + \frac{\beta}{q} + \frac{S}{2\gamma} \leq F(p) < 1, \quad 1 < D_p < 2.$$

In particular $2/D_p \geq 1$, and therefore $n^{2/D_p} \geq n$. The ceiling inequalities

$$\log_2(H_{\text{fine}} n^{2/D_p}) \leq \left\lceil \log_2(H_{\text{fine}} n^{2/D_p}) \right\rceil \leq \log_2(H_{\text{fine}} n^{2/D_p}) + 1$$

give

$$H_{\text{fine}} n^{2/D_p} \leq K \leq 2H_{\text{fine}} n^{2/D_p}, \quad 2^{40} n \leq K.$$

The definition of the threshold gives

$$n \geq 32^{D_p \gamma / (2S)}, \quad K^{S/\gamma} \geq n^{2S/(D_p \gamma)} \geq 32.$$

Thus the coarse target $K^{S/\gamma}/16$ is at least 2. Applying the floor inequalities to its base-two logarithm gives

$$\frac{K^{S/\gamma}}{32} \leq M \leq \frac{K^{S/\gamma}}{16}, \quad M \geq 2.$$

Both ranks are powers of two. Since $K \geq 1$ and $S/\gamma < 1$,

$$16M \leq K^{S/\gamma} \leq K, \quad M \leq K^{S/\gamma}.$$

These are all the required rank-legality conditions.

3. **Normalization, support, and separation.** At these ranks, define the legality tuning and its copula priors by

$$a_{\text{leg}} = \frac{K^{-\alpha}}{16}, \quad b_{\text{leg}} = \frac{K^{-\beta}}{256}, \quad u = \frac{1}{16}, \quad \varepsilon = (16b_{\text{leg}})^{1/q}, \quad L_{\text{mark}} = \varepsilon^{-1/p},$$

$$\pi_\nu = \pi_{\nu; v, a_{\text{leg}}, u, \varepsilon, L_{\text{mark}}}^{K, M}, \quad \mathbb{P}_\nu = \int P^{\otimes n} \pi_\nu(dP), \quad \nu \in \{0, 1\}.$$

Because $K \geq 1$ and $\alpha, \beta > 0$,

$$0 < a_{\text{leg}} \leq \frac{1}{16}, \quad 0 < b_{\text{leg}} \leq \frac{1}{256}, \quad 0 < u \leq \frac{1}{16}, \quad 0 < \varepsilon < 1, \quad L_{\text{mark}} > 0.$$

For completeness, the coefficient weights in [Algorithm 5](#) are normalized as follows. The triangular tent satisfies

$$0 \leq b_{\Delta}(t_{\Delta}) = 2 \min\{t_{\Delta}, 1 - t_{\Delta}\} \leq 1 \quad (t_{\Delta} \in [0, 1]),$$

so $|g_{\sigma}(x)| \leq 1$ on $[0, 1]$. With $\kappa = 1/16$, each node-pair probability satisfies

$$\frac{1 + \nu \kappa g_{\sigma}(i/K) \lambda_i \eta_i}{4} \geq \frac{1 - \kappa}{4} > 0,$$

and cancellation of the sign product gives

$$\sum_{(\lambda_i, \eta_i) \in \{-1, 1\}^2} \frac{1 + \nu \kappa g_{\sigma}(i/K) \lambda_i \eta_i}{4} = 1.$$

Conditional independence therefore makes the sum of the fine-pair product weights equal to one for every coarse-sign configuration. Summing over the fair coarse signs then gives

$$\sum_{\sigma \in \{-1, 1\}^{M/2}} 2^{-M/2} = 1.$$

Thus both finite coefficient distributions, and their pushforwards π_0, π_1 , have nonnegative weights summing to one.

The rank conditions established above allow us to apply [Lemma 6](#). Every null support law belongs to $H_0(v)$, and every alternative support law satisfies

$$P \in \mathcal{M}_v, \quad 2^{-17} K^{-S} \leq d(P) < d_0.$$

The upper rounding bound and $S > 0$ imply

$$\begin{aligned} c_v n^{-E_4(p)} &\leq k_v n^{-E_4(p)} \\ &= 2^{-17} (2H_{\text{fine}})^{-S} n^{-2S/D_p} \\ &= 2^{-17} (2H_{\text{fine}} n^{2/D_p})^{-S} \\ &\leq 2^{-17} K^{-S} \leq d(P). \end{aligned}$$

This proves the required alternative placement for the copula pair.

4. **Activity calibration.** Define the deterministic first- and second-component activity bounds by

$$\mathcal{A}_{\text{det}} = \frac{2^{38} a_{\text{leg}}^4 u^4 \varepsilon^2 n^2}{K}, \quad \mathcal{B}_{\text{det}} = \frac{2^{56} a_{\text{leg}}^4 u^4 \varepsilon^2 n^4}{MK^2}.$$

In [Lemma 7](#), let μ be the common augmentation law and write $\mathcal{A}_{\text{cond}}, \mathcal{B}_{\text{cond}}$ for its conditional chi-square exponent budgets. The expectation bounds supplied there are

$$\int \mathcal{A}_{\text{cond}} d\mu \leq \mathcal{A}_{\text{det}}, \quad \int \mathcal{B}_{\text{cond}} d\mu \leq \mathcal{B}_{\text{det}}.$$

We calibrate the deterministic quantities on the right.

First, raising the lower fine-rank bound to $D_p > 0$ gives

$$K^{D_p} \geq H_{\text{fine}}^{D_p} n^2 \geq H_{\text{fine}} n^2.$$

Consequently,

$$n^2 K^{-D_p} \leq \frac{1}{H_{\text{fine}}}, \quad n^4 K^{-2D_p} = (n^2 K^{-D_p})^2 \leq \frac{1}{H_{\text{fine}}^2}.$$

The tuning gives the exact amplitude product

$$\varepsilon = 16^{-1/q} K^{-\beta/q}, \quad a_{\text{leg}}^4 u^4 \varepsilon^2 = 16^{-8-2/q} K^{-(4\alpha+2\beta/q)} \leq 2^{-48} K^{-(4\alpha+2\beta/q)},$$

where the last inequality follows from $q \leq 1/2$. Furthermore,

$$\frac{S}{2\gamma} \leq 2S = 2\alpha + 2\beta \leq 2\alpha + \frac{\beta}{q},$$

and hence

$$D_p \leq 1 + 4\alpha + \frac{2\beta}{q}.$$

As $K \geq 1$, these inequalities yield

$$\begin{aligned} \mathcal{A}_{\text{det}} &\leq 2^{-10} n^2 K^{-(1+4\alpha+2\beta/q)} \\ &\leq 2^{-10} n^2 K^{-D_p} \leq \frac{2^{-10}}{H_{\text{fine}}}. \end{aligned}$$

For the second budget, use the lower coarse-rank bound:

$$\begin{aligned} \mathcal{B}_{\text{det}} &\leq \frac{2^8 n^4 K^{-(2+4\alpha+2\beta/q)}}{M} \\ &\leq 2^{13} n^4 K^{-(2+4\alpha+2\beta/q+S/\gamma)} \\ &= 2^{13} n^4 K^{-2D_p} \leq \frac{2^{13}}{H_{\text{fine}}^2}. \end{aligned}$$

The exponent identity in the last line is

$$2D_p = 2 + 4\alpha + \frac{2\beta}{q} + \frac{S}{\gamma}.$$

Since $H_{\text{fine}} = 2^{40}$,

$$\mathcal{A}_{\text{det}} + \mathcal{B}_{\text{det}} \leq \frac{2^{-10}}{H_{\text{fine}}} + \frac{2^{13}}{H_{\text{fine}}^2} = 2^{-50} + 2^{-67} < 2^{-16}.$$

All sample-size, rank, amplitude, and mark hypotheses of [Lemma 7](#) have now been discharged. Its complete-record comparison gives

$$\text{TV}(\mathbb{P}_0, \mathbb{P}_1) < \frac{1}{8} < \frac{1}{2}.$$

5. **Testing risk and identification of the selected pair.** Let ϕ be any randomized test satisfying the size constraint in [Definition 5](#). Average over the independent public uniform seed U of [Definition 4](#), defining

$$\bar{\phi} : ([0, 1] \times \{0, 1\} \times \mathbb{R})^n \longrightarrow [0, 1], \quad \bar{\phi}(\mathcal{D}_n) = \mathbb{E}_U[\phi(\mathcal{D}_n, U)].$$

Finite-mixture linearity and null support give

$$\int \bar{\phi} d\mathbb{P}_0 = \int R_n(P, \phi) \pi_0(dP) \leq \frac{1}{10}.$$

For a measurable function taking values in $[0, 1]$, the expectation difference is bounded by total variation. Thus

$$\left| \int \bar{\phi} d\mathbb{P}_1 - \int \bar{\phi} d\mathbb{P}_0 \right| \leq \text{TV}(\mathbb{P}_0, \mathbb{P}_1).$$

Using alternative support and normalization,

$$\begin{aligned} \sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq c_v n^{-E_4(p)}}} \{1 - R_n(P, \phi)\} &\geq \int \{1 - R_n(P, \phi)\} \pi_1(dP) \\ &= 1 - \int \bar{\phi} d\mathbb{P}_1 \\ &\geq \frac{9}{10} - \text{TV}(\mathbb{P}_0, \mathbb{P}_1). \end{aligned}$$

Normalization ensures that at least one alternative weight is positive, so the alternative set is nonempty; zero-weight terms contribute zero. The zero test satisfies the size constraint, so the infimum is also over a nonempty set. Taking that infimum gives

$$B_n(v, c_v n^{-E_4(p)}) \geq \frac{9}{10} - \text{TV}(\mathbb{P}_0, \mathbb{P}_1) > \frac{2}{5}.$$

It remains to identify this copula pair with the selected pair. In the present branch, the formulas in [Algorithm 6](#) satisfy

$$\varepsilon_{n,v} = 16^{-1/q} K^{-\beta/q} = \left(16 \frac{K^{-\beta}}{256}\right)^{1/q} = \varepsilon, \quad L_{n,v} = \varepsilon_{n,v}^{-1/p} = L_{\text{mark}}.$$

Therefore

$$\pi_{\nu,n,v} = \pi_\nu, \quad \mathbb{P}_{\nu,n,v} = \mathbb{P}_\nu, \quad \nu \in \{0, 1\}.$$

For the selected pair, the conditional budgets supplied by [Lemma 7](#) consequently satisfy

$$\int \mathcal{A}_{\text{cond}} d\mu \leq \mathcal{A}_{\text{det}} \leq \frac{2^{-10}}{H_{\text{fine}}}, \quad \int \mathcal{B}_{\text{cond}} d\mu \leq \mathcal{B}_{\text{det}} \leq \frac{2^{13}}{H_{\text{fine}}^2}.$$

The calibrated activity bounds apply to the deterministic right-hand sides of the component expectation estimates. Finally, the model witness supplied by [Lemma 5](#) gives

$$d_0 \leq D_v.$$

Together with the preceding calculations, this establishes normalization, support, separation, the model-supremum bound, testing risk, total variation, rank legality, the fine-rank bound, and the deterministic activity calibration in the large-sample branch.

6. **The bounded small-sample branch.** Fix an admissible $w = (\alpha, \beta, \gamma)$, put

$$v = (2, \alpha, \beta, \gamma), \quad c_w^b = c_v, \quad E_0(2) = \frac{2\gamma}{4\gamma + 1},$$

and suppose $F(2) < 1$. This v is admissible. If $n < N_{\text{low}}(v)$, the exponent and multiplier calculation at $p = 2$ gives

$$c_w^b n^{-E_4(2)} \leq c_0^e n^{-E_0(2)}.$$

The bounded fallback in [Definition 31](#) selects the signed-binary paired-tent package of [Lemma 5](#):

$$\begin{aligned} \pi_{0,n,v}^b &= \delta_{P_0^b}, & \pi_{1,n,v}^b &= \pi_1^b, \\ \mathbb{P}_{0,n,v}^b &= (P_0^b)^{\otimes n}, & \mathbb{P}_{1,n,v}^b &= \overline{P}_{1,b}^{(n)}. \end{aligned}$$

The point mass and the finite alternative prior are normalized. Signed-binary support implies $|Y| \leq 1$, so their support laws satisfy

$$P_0^b \in H_0^b(w),$$

and, on every positive-weight alternative,

$$P \in \mathcal{M}_w^b, \quad c_w^b n^{-E_4(2)} \leq c_0^e n^{-E_0(2)} \leq d(P) < d_0.$$

The same paired-tent bound supplies

$$\text{TV}(\mathbb{P}_{0,n,v}^b, \mathbb{P}_{1,n,v}^b) < \frac{1}{2}, \quad B_n^b(w, c_0^e n^{-E_0(2)}) \geq \frac{2}{5}.$$

For every bounded-null size-valid test, the alternative-set inclusion gives

$$\sup_{\substack{P \in \mathcal{M}_w^b \\ d(P) \geq c_0^e n^{-E_0(2)}}} \{1 - R_n(P, \phi)\} \leq \sup_{\substack{P \in \mathcal{M}_w^b \\ d(P) \geq c_w^b n^{-E_4(2)}}} \{1 - R_n(P, \phi)\}.$$

The paired-tent prior has a positive weight, ensuring nonempty alternative sets. Taking infima yields

$$B_n^b(w, c_w^b n^{-E_4(2)}) \geq B_n^b(w, c_0^e n^{-E_0(2)}) \geq \frac{2}{5}.$$

7. **The bounded large-sample branch.** Suppose $n \geq N_{\text{low}}(v)$. Evaluate the rank formulas and amplitudes $a_{\text{leg}}, b_{\text{leg}}$ at $v = (2, \alpha, \beta, \gamma)$. The rank calculation gives

$$K = 2^{\lceil \log_2(H_{\text{fine}} n^{2/D_2}) \rceil}, \quad M = 2^{\lfloor \log_2(K^{S/\gamma}/16) \rfloor},$$

$$H_{\text{fine}} n^{2/D_2} \leq K \leq 2H_{\text{fine}} n^{2/D_2}, \quad 2^{40} n \leq K,$$

$$\frac{K^{S/\gamma}}{32} \leq M \leq \frac{K^{S/\gamma}}{16}, \quad M \geq 2, \quad 16M \leq K, \quad M \leq K^{S/\gamma}.$$

Use the separate bounded tuning

$$a_{\text{leg}} = \frac{K^{-\alpha}}{16}, \quad b_{\text{leg}} = \frac{K^{-\beta}}{256}, \quad u_b = 2b_{\text{leg}} = \frac{K^{-\beta}}{128}, \quad \varepsilon_b = \frac{1}{2}, \quad L_b = 1,$$

and define its priors and complete-record mixtures by

$$\pi_\nu^{\text{b,frame}} = \pi_{\nu;v,a_{\text{leg}},u_{\text{b}},\varepsilon_{\text{b}},L_{\text{b}}}^{K,M}, \quad \mathbb{P}_\nu^{\text{b,frame}} = \int P^{\otimes n} \pi_\nu^{\text{b,frame}}(dP), \quad \nu \in \{0,1\}.$$

Here

$$0 < a_{\text{leg}} \leq \frac{1}{16}, \quad 0 < u_{\text{b}} \leq \frac{1}{128} \leq \frac{1}{16}, \quad 0 < \varepsilon_{\text{b}} < 1, \quad L_{\text{b}} > 0.$$

The coefficient weights depend on the ranks and the copula coupling, so the product-weight calculation proves normalization for both bounded priors as well. The bounded clause of [Lemma 6](#) supplies

$$\begin{aligned} P &\in H_0^{\text{b}}(w) \quad \text{on null support,} \\ P &\in \mathcal{M}_w^{\text{b}}, \quad 2^{-17}K^{-S} \leq d(P) < d_0 \quad \text{on alternative support.} \end{aligned}$$

The upper fine-rank bound gives

$$c_w^{\text{b}} n^{-E_4(2)} \leq k_v n^{-E_4(2)} = 2^{-17}(2H_{\text{fine}} n^{2/D_2})^{-S} \leq 2^{-17}K^{-S}.$$

For this tuning, the exact amplitude product is

$$a_{\text{leg}}^4 u_{\text{b}}^4 \varepsilon_{\text{b}}^2 = 2^{-46}K^{-4S}.$$

Thus the deterministic component budgets, denoted by $\mathcal{A}_{\text{b}}$ and $\mathcal{B}_{\text{b}}$, are

$$\mathcal{A}_{\text{b}} = 2^{-8}n^2K^{-(1+4S)}, \quad \mathcal{B}_{\text{b}} = \frac{2^{10}n^4K^{-(2+4S)}}{M}.$$

At $p = 2$,

$$D_2 = 1 + 2S + \frac{S}{2\gamma} \leq 1 + 4S,$$

because $\gamma \geq 1/4$. Raising the fine-rank bound to D_2 , and then squaring the resulting reciprocal bound, gives

$$n^2K^{-D_2} \leq \frac{1}{H_{\text{fine}}}, \quad n^4K^{-2D_2} \leq \frac{1}{H_{\text{fine}}^2}.$$

It follows that

$$\mathcal{A}_{\text{b}} \leq 2^{-8}n^2K^{-D_2} \leq \frac{2^{-8}}{H_{\text{fine}}},$$

and, using $M \geq K^{S/\gamma}/32$,

$$\begin{aligned} \mathcal{B}_{\text{b}} &\leq 2^{15}n^4K^{-(2+4S+S/\gamma)} \\ &= 2^{15}n^4K^{-2D_2} \leq \frac{2^{15}}{H_{\text{fine}}^2}. \end{aligned}$$

Consequently,

$$\mathcal{A}_{\text{b}} + \mathcal{B}_{\text{b}} \leq \frac{2^{-8}}{H_{\text{fine}}} + \frac{2^{15}}{H_{\text{fine}}^2} = 2^{-48} + 2^{-65} < 2^{-16}.$$

Applying [Lemma 7](#), with the verified bounded parameters, gives

$$\text{TV}(\mathbb{P}_0^{\text{b,frame}}, \mathbb{P}_1^{\text{b,frame}}) < \frac{1}{8} < \frac{1}{2}.$$

For a test satisfying the bounded-null size constraint, seed averaging, null support, and the total-variation expectation bound give

$$\begin{aligned} \int \bar{\phi} d\mathbb{P}_0^{\text{b,frame}} &\leq \frac{1}{10}, \\ \sup_{\substack{P \in \mathcal{M}_w^{\text{b}} \\ d(P) \geq c_w^{\text{b}} n^{-E_4(2)}}} \{1 - \mathbf{R}_n(P, \phi)\} &\geq 1 - \int \bar{\phi} d\mathbb{P}_1^{\text{b,frame}} \\ &\geq \frac{9}{10} - \text{TV}(\mathbb{P}_0^{\text{b,frame}}, \mathbb{P}_1^{\text{b,frame}}). \end{aligned}$$

Both priors are normalized, so their alternative support is nonempty; the zero test again satisfies the size constraint. Taking the infimum therefore yields

$$B_n^{\text{b}}(w, c_w^{\text{b}} n^{-E_4(2)}) \geq \frac{9}{10} - \text{TV}(\mathbb{P}_0^{\text{b,frame}}, \mathbb{P}_1^{\text{b,frame}}) > \frac{2}{5}.$$

Finally, the selection in [Definition 31](#) identifies

$$\pi_{\nu,n,v}^{\text{b}} = \pi_{\nu}^{\text{b,frame}}, \quad \mathbb{P}_{\nu,n,v}^{\text{b}} = \mathbb{P}_{\nu}^{\text{b,frame}}, \quad \nu \in \{0, 1\}.$$

This transfers normalization, support, separation, total variation, and the testing-risk bound to the selected bounded pair, completing the bounded clause. $\square$

Under $F(p) < 1$, [Lemma 8](#) gives the rough separation exponent for every $n \geq 2$. Above the public lower-construction threshold, the selected ranks also meet the component-comparison conditions and give the stronger total-variation bound of one eighth. The deterministic budget bounds control the upper bounds on the expectations of $\mathcal{A}$ and $\mathcal{B}$ in [Lemma 7](#). The bounded conclusion uses its own outcome tuning at $p = 2$, preserving the rough exponent and the finite-sample risk guarantee.

C.4 Tail essentiality and the moment-order-two face

The final result combines the rough and paired-tent branches with attainment. Its bounded-prior comparison quantifies over normalized finite priors in the convention fixed above: all support restrictions apply to positive-weight laws, and each mixture draws its latent law once for the complete sample. For each fixed admissible tuple with moment order below two, the comparison has a tuple-dependent eventual threshold. On the moment-order-two face, the selected bounded constructions give witnesses for every $n \geq 2$.

Theorem 3. `[thm:tail-essential-full-record-converse]` (Tail essentiality and bounded witnesses) *Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$ as in [Definition 1](#), and write $w = (\alpha, \beta, \gamma)$. Use the scales, constants, thresholds, and total tests of [Definition 15](#), and the selected priors, mark magnitudes, and complete-record mixtures of [Algorithm 6](#).*

Let λ denote Lebesgue probability measure on $[0, 1]$. Define the heterogeneity distance and its model supremum by

$$d(P) = \left\{ \int_{[0,1]} \left(\tau_P(x) - \int_{[0,1]} \tau_P(z) \lambda(dz) \right)^2 \lambda(dx) \right\}^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P).$$

For probability laws, TV denotes the supremum of the absolute differences of their probabilities over measurable events.

A normalized finite prior on original-record laws is

$$\pi = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad k \geq 1, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1.$$

Its positive-weight support and complete-record mixture are defined by

$$\text{supp}_+(\pi) = \{P_j : \omega_j > 0\}, \quad \mathbb{P}_{\pi,n} = \sum_{j=1}^k \omega_j P_j^{\otimes n}.$$

For every integer $n \geq 2$, both $\pi_{0,n,v}$ and $\pi_{1,n,v}$ are normalized finite priors, and

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}) &\subseteq H_0(v), \\ \text{supp}_+(\pi_{1,n,v}) &\subseteq \{P \in \mathcal{M}_v : c_v \rho_n(v) \leq d(P) < D_v\}, \\ 0 &< c_v \rho_n(v) < D_v, \\ \text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) &< \frac{1}{2}, \quad B_n(v, c_v \rho_n(v)) \geq \frac{2}{5}, \end{aligned}$$

where the classes and testing risk are those of [Definitions 1, 2 and 5](#). For every positive-weight law in either prior, each conditional outcome arm assigns probability one to $\{0, -L_{n,v}, L_{n,v}\}$ for λ -almost every covariate value. Moreover,

$$L_{n,v} \longrightarrow \infty \quad (n \longrightarrow \infty).$$

Use the matched bounded model $\mathcal{M}_w^b$ and null $H_0^b(w)$ of [Definitions 7 and 9](#). If $p < 2$, then

$$\frac{\rho_n(v)}{\rho_n^b(w)} \longrightarrow \infty \quad (n \longrightarrow \infty).$$

For each such fixed v , there exists an integer $N(v) \geq 2$ such that, for every integer $n \geq N(v)$ and every pair of normalized finite priors π_0, π_1 ,

$$\left. \begin{aligned} \text{supp}_+(\pi_0) &\subseteq H_0^b(w), \\ \text{supp}_+(\pi_1) &\subseteq \{P \in \mathcal{M}_w^b : c_v \rho_n(v) \leq d(P)\} \end{aligned} \right\} \implies \text{TV}(\mathbb{P}_{\pi_0,n}, \mathbb{P}_{\pi_1,n}) \geq \frac{1}{2}.$$

If $p = 2$, the following conclusions hold for every integer $n \geq 2$. Let $\pi_{\nu,n,v}^b$, for $\nu \in \{0, 1\}$, denote the selected bounded witness priors: the bounded rough priors of [Lemma 8](#) when $F(2) < 1$ and $n \geq N_{\text{low}}(v)$, and the signed-binary paired-tent priors of [Lemma 5](#) otherwise. Define

$$\mathbb{P}_{\nu,n,v}^b = \mathbb{P}_{\pi_{\nu,n,v}^b,n}, \quad \nu \in \{0, 1\}.$$

Both priors are normalized and have nonempty positive-weight supports. Their positive-weight masses satisfy

$$\sum_{P \in \text{supp}_+(\pi_{\nu,n,v}^b)} \pi_{\nu,n,v}^b(\{P\}) = 1, \quad \nu \in \{0, 1\}.$$

Let $d_0 > 0$ be the fixed witness distance of [Lemma 1](#). The support, total-variation, and risk conclusions are

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}^b) &\subseteq H_0^b(w), & \text{supp}_+(\pi_{0,n,v}^b) &\subseteq H_0(v), \\ \text{supp}_+(\pi_{1,n,v}^b) &\subseteq \{P \in \mathcal{M}_w^b : c_v \rho_n(v) \leq d(P) < d_0\}, \\ \text{supp}_+(\pi_{1,n,v}^b) &\subseteq \{P \in \mathcal{M}_v : c_v \rho_n(v) \leq d(P)\}, \\ \text{TV}(\mathbb{P}_{0,n,v}^b, \mathbb{P}_{1,n,v}^b) &< \frac{1}{2}, \\ \frac{2}{5} &\leq B_n^b(w, c_v \rho_n(v)) \leq B_n(v, c_v \rho_n(v)), \end{aligned}$$

with bounded risk defined in [Definition 11](#).

Define the bounded-model distance supremum by

$$D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P).$$

On this face, the scales, multipliers, thresholds, and distance suprema agree:

$$\begin{aligned} \rho_n(v) &= \rho_n^b(w), & c_v &= c_w^b, & C_v &= C_w^b, \\ N_v &= N_w^b, & D_v &= D_w^b. \end{aligned}$$

The capped critical radii of [Definitions 6, 13](#) and [14](#) satisfy

$$\begin{aligned} c_v \rho_n(v) &\leq r_n^*(v) \leq C_v \rho_n(v), \\ c_w^b \rho_n^b(w) &\leq r_n^{*,b}(w) = r_n^{*,\text{bin}}(w) \leq C_w^b \rho_n^b(w). \end{aligned}$$

Let $R_n(P, \phi)$ denote the expected rejection probability under n independent original records and the public randomization of [Definition 4](#). The total tests satisfy, for every law in the indicated null class,

$$\begin{aligned} R_n(P, \phi_{n,v}^*) &\leq 0.0075 & (P \in H_0(v)), \\ R_n(P, \phi_{n,w}^{*,b}) &\leq 0.0075 & (P \in H_0^b(w)). \end{aligned}$$

Whenever $C_v \rho_n(v) < D_v$, both separated alternative classes are nonempty:

$$\begin{aligned} \{P \in \mathcal{M}_v : C_v \rho_n(v) \leq d(P)\} &\neq \emptyset, \\ \{P \in \mathcal{M}_w^b : C_w^b \rho_n^b(w) \leq d(P)\} &\neq \emptyset. \end{aligned}$$

Under this same condition, every law in the corresponding separated class satisfies

$$\begin{aligned} 1 - R_n(P, \phi_{n,v}^*) &\leq 0.0075 & (P \in \mathcal{M}_v, C_v \rho_n(v) \leq d(P)), \\ 1 - R_n(P, \phi_{n,w}^{*,b}) &\leq 0.0075 & (P \in \mathcal{M}_w^b, \\ C_w^b \rho_n^b(w) &\leq d(P)). \end{aligned}$$

For $n \geq N_v$, the common upper separation satisfies

$$\begin{aligned} C_v \rho_n(v) &\leq \frac{d_0}{2}, \\ C_v \rho_n(v) &< D_v, & C_w^b \rho_n^b(w) &< D_w^b. \end{aligned}$$

At $n = 2, 3$, both total tests have zero rejection probability at every sample:

$$\phi_{n,v}^*(z) = 0, \quad \phi_{n,w}^{*,b}(z) = 0 \quad \text{for every } z.$$

The distinction in [Theorem 3](#) concerns the outcome distributions that sustain close full-record comparisons at the finite-moment separation scale. The selected finite-moment witnesses have growing mark magnitudes and total variation below one half for every $n \geq 2$. For each fixed tuple with $p < 2$, the bounded scale is asymptotically smaller, and every normalized finite bounded-prior pair satisfying the stated support restrictions has total variation at least one half once $n \geq N(v)$. This threshold depends on the fixed tuple.

At $p = 2$, the bounded copula construction and the signed-binary paired-tent construction together supply normalized, nonempty bounded witnesses for every sample size. The common scales and constants then connect these lower comparisons to the attained upper bounds. The power conclusions apply when the common upper separation is below the model distance supremum, a condition guaranteed by the public threshold N_v ; the capped-radius conclusions retain the conventions of [Definitions 6, 13 and 14](#).

D Supplied-propensity proofs and comparison algebra

The supplied-propensity results combine inverse weighting, removal of the constant effect, and a quadratic comparison of independent blocks. The construction in [Algorithm 2](#) applies over the continuous-carrier class of [Definition 22](#). Its upper bounds and the common-propensity lower witnesses give the matching separation guarantees in [Theorems 5 and 6](#). The comparison algebra then distinguishes the value of supplied assignment information on the original model from the cost of outcome tails at matched primitive smoothness.

D.1 Inverse weighting and the public error bounds

Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$ and a law $P \in \mathcal{M}_v^{e,br}$, as defined in [Definition 22](#). Throughout this section, λ denotes Lebesgue probability measure on $[0, 1]$. The conditional-mean identity in [Theorem 5](#) identifies the mean effect through the untruncated inverse-propensity score. The treatment probability cancels its inverse weight separately in each arm, leaving the difference between the treated and baseline conditional means. This applies the inverse-probability weighting principle of [Horvitz and Thompson \[1952, pp. 665–666\]](#) to the conditional mean contrast.

Outcome clipping makes this identity usable under the conditional moment envelope. The score R_i , cutoff T , and public bounds b, Λ are those of [Algorithm 2](#):

$$b = 20T^{1-p}, \quad \Lambda = \frac{80T^{2-p}}{s}, \quad s = \lfloor n/2 \rfloor.$$

The bias bound accounts for clipping in both treatment arms. The covariance bound combines the clipped second-moment envelope with the bounded inverse weights and averaging over an evaluation block. These calculations use uniform design, overlap, and the conditional moment restriction in [Assumptions 1, 2 and 8](#).

The statistical roles of the two bounds are distinct. Increasing the cutoff reduces clipping bias through T^{1-p} , while the covariance allowance grows through T^{2-p} . At $p = 2$, the latter factor equals one. Effect smoothness in [Assumption 5](#) controls histogram approximation of the mean contrast. Thus the broader-class attainment result depends on the moment order and effect smoothness, with continuous propensity and arm-mean representatives supplied by [Definition 22](#).

D.2 Removing the unknown constant

The histogram representation in [Definition 19](#) makes the constant-effect direction explicit. Let

$$u = J^{-1/2}(1, \dots, 1)^T, \quad C_J = I_J - uu^T,$$

where I_J is the $J \times J$ identity matrix and C_J is the orthogonal projection onto the complement of the constant direction. In the notation of [Algorithm 2](#), u is a unit vector and

$$C_J F_J(x) = F_J(x) - u.$$

Consequently, the centered block vectors and their inner product have the equivalent representations

$$V_b = \frac{1}{s} \sum_{i \in \mathcal{I}_b} C_J F_J(X_i) R_i, \quad b \in \{1, 2\},$$

and

$$\langle V_1, V_2 \rangle = \frac{1}{s^2} \sum_{i \in \mathcal{I}_1} \sum_{k \in \mathcal{I}_2} \{\Pi_J(X_i, X_k) - 1\} R_i R_k.$$

These are representations of the statistic already specified in [Algorithm 2](#).

Constant centering targets the distance of the effect from its average. Every unknown constant in the null of [Definition 23](#) occupies the same removed direction. Clipping bias remains covered by the public bound b , while effect smoothness governs the approximation of the remaining heterogeneity. The two disjoint evaluation blocks provide independent score vectors, so their quadratic comparison measures the squared population signal subject to the stated bias and covariance allowances.

The construction also specifies the test on its full supplied-function domain. For an arbitrary supplied continuous function f , define its overlap-clipped version by

$$\tilde{f}(x) = \max\{1/4, \min\{f(x), 3/4\}\}.$$

The score $R_i(f)$ and block vector $V_b(f)$ denote the expressions in [Algorithm 2](#) evaluated with $\tilde{f}$ in place of $\tilde{e}$. This convention bounds the denominators throughout the test domain. For a legal true supplied propensity, it agrees with the propensity itself, as required by the rejection expectation in [Definition 24](#).

D.3 Calibration, attainment, and common-propensity witnesses

The threshold in [Algorithm 2](#) is

$$2b^2 + 1024\sqrt{J} \Lambda.$$

Its first term accommodates the population discrepancy caused by clipping under a constant effect. The second supplies the quadratic fluctuation allowance. The resulting calibration in [Theorem 5](#) controls rejection probability by 0.01 uniformly over the entire broader-class null for every $n \geq 2$. At the smallest sample sizes, the public tuning gives $J = 1$, for which constant centering makes the statistic identically zero.

The tuning coordinates clipping, projection, and quadratic variability. With the target scale $a = s^{-E_0}$ from [Algorithm 2](#), the cutoff and rank are upper dyadic roundings of

$$a^{-1/(p-1)} \quad \text{and} \quad a^{-1/\gamma},$$

respectively. Their common separation exponent is

$$E_0(p) = \frac{2\gamma q}{2\gamma + q}, \quad q = \frac{p-1}{p},$$

as defined in [Definition 15](#). The attained scale is $\rho_n^e(v) = n^{-E_0(p)}$. At moment order two,

$$E_0(2) = \frac{2\gamma}{4\gamma + 1}.$$

This agrees numerically with the familiar one-dimensional quadratic-testing exponent in the Gaussian Sobolev setting of [Ingster and Sapatinas \[2009, Section 6.1, pp. 249–250\]](#), where the stated smoothness condition is strictly above $1/4$. The supplied-propensity guarantees here have the conditional-moment and continuous-carrier scope specified in [Theorem 5](#), including $\gamma = 1/4$.

The lower and upper multipliers c_v^e, C_v^e and threshold N_v^e are defined in [Theorem 5](#). Their radius bounds hold for every $n \geq 2$. The power guarantee applies when

$$C_v^e \rho_n^e(v) < D_v^{e, \text{br}}.$$

The public threshold places this upper separation at most $d_0/2$, where $d_0 = 1/(8\sqrt{2})$ is the positive witness distance in that result. The lower bound $D_v^{e, \text{br}} \geq d_0$ then supplies a nonempty separation range.

The common-propensity witnesses connect this attainment result to both supplied experiments. In [Theorem 5](#), the null law and every positive-weight alternative law belong to the original and broader classes and share the propensity $x \mapsto 1/2$. Their complete-record distributions have total variation below one half. Supplying that same function preserves the comparison underlying the lower error bound.

Class inclusion transfers the broader-class calibration and power guarantees to the original model:

$$\mathcal{M}_v \subseteq \mathcal{M}_v^{e, \text{br}}, \quad H_0(v) \subseteq H_0^{e, \text{br}}(v),$$

using [Definitions 1, 2, 22](#) and [23](#). The witnesses also lie within the original class, where [Theorem 6](#) supplies the corresponding necessity statement. Its multipliers c_v^e, C_v^e have the same displayed values as their broader-class counterparts; its power condition uses the original supremum D_v .

D.4 The preservation boundary on the original class

Hold $w = (\alpha, \beta, \gamma)$ fixed when evaluating the exponents in [Definition 15](#). The comparison identity in [Theorem 4](#) is

$$E_4(p) - E_0(p) = \frac{2\gamma q}{(2\gamma + q)D_p} \{F(p) - 1\}.$$

Its positive prefactor makes $F(p) = 1$ the equality boundary between the two separation exponents. The original-model exponent is their minimum.

The nonnegative comparison exponent $G(v)$, defined in [Theorem 7](#), therefore has the representations

$$G(v) = E_0(p) - E(p) = \frac{2\gamma q}{(2\gamma + q)D_p} \max\{0, 1 - F(p)\}.$$

The same-class radius ratio is bounded on both sides by positive multiples of $n^{G(v)}$, with the exact multipliers given in that result. For $F(p) \geq 1$, this exponent equals zero and the ratio

remains bounded over all $n \geq 2$. Equality points $F(p) = 1$ are included. For $F(p) < 1$, the positive exponent quantifies the polynomial gain from supplied assignment information. This yields precisely the preservation region in [Definition 28](#) and [Theorem 7](#).

D.5 The finite-moment and bounded comparison

The bounded comparison evaluates the original-record exponent at moment order two while preserving w . For presentation, let $\delta_0(p)$ and $\delta_4(p)$ denote the within-mechanism exponent differences. The formulas in [Definition 15](#) give

$$\delta_0(p) := E_0(2) - E_0(p) = \frac{4\gamma^2(2-p)}{p(4\gamma+1)(2\gamma+q)},$$

and

$$\delta_4(p) := E_4(2) - E_4(p) = \frac{2S\beta(1/q-2)}{D_2D_p}, \quad D_2 = 1 + 2S + \frac{S}{2\gamma}.$$

Both differences are strictly positive for $p < 2$ and vanish at $p = 2$.

At fixed smoothness, $F(p) \geq F(2)$. The selected exponent difference consequently has three compatible representations:

$$E(2) - E(p) = \begin{cases} \delta_0(p), & F(2) \geq 1, \\ \delta_4(p) + \frac{2\gamma q}{(2\gamma+q)D_p} \{F(p) - 1\}, & F(p) \geq 1 > F(2), \\ \delta_4(p), & F(p) < 1. \end{cases}$$

At $F(p) = 1$, the additional term in the middle expression vanishes. At $F(2) = 1$, the bounded exponents coincide, so the adjoining expressions agree. These equality points preserve the exponent comparison across the change in the rate-determining mechanism.

The matched capped-radius inequalities in [Theorem 4](#) attach the factor $n^{E(2)-E(p)}$ to $\Delta_n(v)$, the ratio defined in [Definition 16](#). Strict positivity for $p < 2$ gives the diverging ratio characterized by [Definition 17](#). At $p = 2$, the exponent difference is zero and the ratio remains uniformly bounded over sample sizes, as characterized by [Definition 18](#). The phase equality $F(p) = 1$ and the moment-order equality $p = 2$ thus describe distinct comparisons.

All these statements retain the finite-sample multipliers and saturation conventions of the cited radius results. Their local uniformity is expressed through common positive lower multipliers, finite upper multipliers, and finite bounds on the public power thresholds over compact subsets of the admissible parameter domains, as stipulated in [Theorems 4 to 7](#). These uniform constants accompany the parameter-dependent powers, including at the stated equality boundaries.

Theorem 4. [[thm:heavy-tail-comparison](#)] (Heavy-tail critical-radius comparison) *Let the admissible exponent domains be*

$$\begin{aligned} \mathcal{V} &= \{(p, \alpha, \beta, \gamma) \in \mathbb{R}^4 : 1 < p \leq 2, 0 < \alpha \leq 1, 0 < \beta \leq 1, 1/4 \leq \gamma \leq 1\}, \\ \mathcal{W} &= \{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : 0 < \alpha \leq 1, 0 < \beta \leq 1, 1/4 \leq \gamma \leq 1\}. \end{aligned}$$

For $v = (p, \alpha, \beta, \gamma) \in \mathcal{V}$, write $w = (\alpha, \beta, \gamma)$. Use the public scales, multipliers, thresholds, and original-record tests defined in [Definition 15](#), and the models, null classes, risks, and capped radii

defined in [Definitions 1, 2, 5 to 7, 9, 11, 13 and 14](#). The matched radius ratio from [Definition 16](#) is

$$\Delta_n(v) = \frac{r_n^*(v)}{r_n^{*,b}(w)}.$$

Here λ is Lebesgue probability measure on $[0, 1]$, and the population distance of the mean effect from its average is

$$d(P) = \left(\int_0^1 \left| \tau_P(x) - \int_0^1 \tau_P(z) d\lambda(z) \right|^2 d\lambda(x) \right)^{1/2}.$$

The maximal distances over the respective models are

$$D_v = \sup_{P \in \mathcal{M}_v} d(P), \quad D_w^b = \sup_{P \in \mathcal{M}_w^b} d(P),$$

and $d_0 > 0$ is the fixed witness distance supplied by [Lemma 1](#). For the original-record experiment in [Definition 4](#), write the rejection expectation as

$$R_n(P, \phi) = \mathbb{E}_{S_{n,P}}[\phi].$$

The following assertions hold.

- **(Matched radii and lower separations.)** For every $v \in \mathcal{V}$, $c_v > 0$ and $C_v > 0$. For every integer $n \geq 2$, both scales are positive and

$$\begin{aligned} c_v \rho_n(v) &\leq r_n^*(v) \leq C_v \rho_n(v), \\ c_w^b \rho_n^b(w) &\leq r_n^{*,b}(w) = r_n^{*,\text{bin}}(w) \leq C_w^b \rho_n^b(w). \end{aligned}$$

The lower separations satisfy

$$0 < c_v \rho_n(v) < D_v, \quad 0 < c_w^b \rho_n^b(w) < D_w^b,$$

and the corresponding minimax type-II errors satisfy

$$B_n(v, c_v \rho_n(v)) > \frac{1}{10}, \quad B_n^b(w, c_w^b \rho_n^b(w)) > \frac{1}{10}.$$

- **(Size and conditional power.)** For every $v \in \mathcal{V}$ and every integer $n \geq 2$,

$$\begin{aligned} R_n(P, \phi_{n,v}^*) &\leq \frac{1}{10} \quad \text{for every } P \in H_0(v), \\ R_n(P, \phi_{n,w}^{*,b}) &\leq \frac{1}{10} \quad \text{for every } P \in H_0^b(w). \end{aligned}$$

If $C_v \rho_n(v) < D_v$, then

$$1 - R_n(P, \phi_{n,v}^*) \leq \frac{1}{10}$$

for every $P \in \mathcal{M}_v$ satisfying $d(P) \geq C_v \rho_n(v)$. If $C_w^b \rho_n^b(w) < D_w^b$, then

$$1 - R_n(P, \phi_{n,w}^{*,b}) \leq \frac{1}{10}$$

for every $P \in \mathcal{M}_w^b$ satisfying $d(P) \geq C_w^b \rho_n^b(w)$.

- **(Vanishing scales and public thresholds.)** For every $v \in \mathcal{V}$,

$$\rho_n(v) \longrightarrow 0, \quad \rho_n^b(w) \longrightarrow 0 \quad \text{as } n \longrightarrow \infty.$$

For every natural number n ,

$$\begin{aligned} n \geq N_v &\implies C_v \rho_n(v) < d_0, \\ n \geq N_w^b &\implies C_w^b \rho_n^b(w) < d_0. \end{aligned}$$

- **(Phase identities and radius comparison.)** Use the supplied-propensity exponent $E_0(p)$, rough-confounding exponent $E_4(p)$, phase-comparison scalar $F(p)$, and denominator D_p from [Definition 15](#), with the smoothness tuple w fixed. With

$$q = \frac{p-1}{p}, \quad E(p) = \min\{E_0(p), E_4(p)\},$$

every $v \in \mathcal{V}$ satisfies

$$E_4(p) - E_0(p) = \frac{2\gamma q}{(2\gamma + q)D_p} (F(p) - 1),$$

and

$$F(p) \geq 1 \implies E(p) = E_0(p), \quad F(p) < 1 \implies E(p) = E_4(p).$$

With

$$D_2 = 1 + 2S + \frac{S}{2\gamma}, \quad F_2 = 2S + \frac{S}{2\gamma},$$

the exponent difference, with evaluation at 2 preserving w , is

$$E(2) - E(p) = \begin{cases} \frac{4\gamma^2(2-p)}{(4\gamma+1)(2\gamma p + p - 1)}, & F_2 \geq 1, \\ E_4(2) - E_0(p), & F_2 < 1 \leq F(p), \\ \frac{2S\beta(2-p)}{(p-1)D_p D_2}, & F(p) < 1. \end{cases}$$

For every integer $n \geq 2$,

$$\frac{c_v}{C_w^b} n^{E(2)-E(p)} \leq \Delta_n(v) \leq \frac{C_v}{c_w^b} n^{E(2)-E(p)}, \quad 1 \leq \Delta_n(v).$$

- **(Strict and equality regions.)** The strict-enlargement and bounded-ratio regions of [Definitions 17](#) and [18](#) are exactly

$$\mathcal{T} = \{v \in \mathcal{V} : p < 2\}, \quad \mathcal{E} = \{v \in \mathcal{V} : p = 2\}.$$

There exists a set $U \subseteq \mathbb{R}^4$, open in the product topology of the four public exponents, such that

$$U \cap \mathcal{V} \neq \emptyset, \quad U \cap \mathcal{V} \subseteq \mathcal{T}.$$

For every $v \in \mathcal{V}$ with $p < 2$,

$$E(p) < E(2), \quad \frac{\rho_n(v)}{\rho_n^b(w)} \longrightarrow \infty \quad \text{as } n \longrightarrow \infty.$$

- **(Tail-essential complete-record priors.)** For every $v \in \mathcal{V}$ with $p < 2$, the selected mark magnitudes from [Algorithm 6](#) satisfy

$$L_{n,v} \longrightarrow \infty \quad \text{as } n \longrightarrow \infty.$$

For every integer $n \geq 2$, the selected priors $\pi_{0,n,v}$ and $\pi_{1,n,v}$ from [Algorithm 6](#) are normalized finite probability priors. Every law receiving positive null-prior weight belongs to $H_0(v)$, and every law receiving positive alternative-prior weight belongs to $\mathcal{M}_v$ and satisfies

$$c_v \rho_n(v) \leq d(P) < D_v.$$

For every law receiving positive weight in either prior, each conditional arm law assigns probability one to the three-point mark set: writing $Q_{a,P}(dy \mid x)$ for the conditional outcome probability kernel in arm a ,

$$Q_{a,P}(\{0, -L_{n,v}, L_{n,v}\} \mid x) = 1 \quad \text{for } a \in \{0, 1\} \text{ and } \lambda\text{-almost every } x.$$

The complete-record mixtures defined in [Algorithm 6](#) satisfy

$$\text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) < \frac{1}{2},$$

where total variation is the supremum of the absolute difference in event probabilities.

For each such v , there exists a natural-number threshold $N(v)$ such that, for every $n \geq N(v)$, every pair of normalized finite probability priors supported respectively on

$$H_0^b(w) \quad \text{and} \quad \{P \in \mathcal{M}_w^b : c_v \rho_n(v) \leq d(P)\}$$

has complete-record mixture total variation at least $1/2$, in the sense of [Theorem 3](#).

- **(Compact uniformity.)** For every compact set of admissible public tuples, there exist real constants c, C, N , with $c > 0$, such that

$$c \leq c_v, \quad C_v \leq C, \quad N_v \leq N$$

for every tuple in that set. Likewise, for every compact set of admissible smoothness tuples, there exist real constants c, C, N , with $c > 0$, such that

$$c \leq c_w^b, \quad C_w^b \leq C, \quad N_w^b \leq N$$

for every smoothness tuple in that set.

Theorem 5. [thm:oracle-frontier-broad-class] (Supplied-propensity broader-class frontier) For every admissible public tuple $v = (p, \alpha, \beta, \gamma)$ of [Definition 1](#), consider the broader model and its constant-effect null in [Definitions 22](#) and [23](#), with testing risk and capped critical radius as in [Definitions 24](#) and [25](#). Put

$$q = \frac{p-1}{p}, \quad E_0 = \frac{2\gamma q}{2\gamma + q}, \quad \rho_n^e(v) = n^{-E_0}, \quad d_0 = \frac{1}{8\sqrt{2}},$$

and define the lower multiplier, upper multiplier, and public sample-size threshold by

$$c_v^e = \frac{4^{-\gamma}}{16\sqrt{3}}, \quad C_v^e = 3(120 + 128\sqrt{160\sqrt{2}}), \quad N_v^e = \left\lceil \max \left\{ 2, \left(\frac{2C_v^e}{d_0} \right)^{1/E_0} \right\} \right\rceil.$$

Let λ denote Lebesgue probability measure on $[0, 1]$, and write the heterogeneity distance as

$$d(P) = \left\{ \int_{[0,1]} \left(\tau_P(x) - \int_{[0,1]} \tau_P(z) \lambda(dz) \right)^2 \lambda(dx) \right\}^{1/2}.$$

Then

$$d_0 \leq D_v^{\text{e,br}} \leq \frac{1}{2}, \quad \frac{1}{64\sqrt{3}} \leq c_v^{\text{e}}, \quad 0 < C_v^{\text{e}}.$$

For every integer $n \geq 2$, the following guarantees hold.

- **(Critical radius.)**

$$c_v^{\text{e}} \rho_n^{\text{e}}(v) \leq r_n^{*,\text{e,br}}(v) \leq C_v^{\text{e}} \rho_n^{\text{e}}(v), \quad B_n^{\text{e,br}}(v, c_v^{\text{e}} \rho_n^{\text{e}}(v)) \geq \frac{2}{5}.$$

- **(Calibration and power.)** The supplied-propensity test $\phi_{n,v}^{*,\text{e}}$ of [Algorithm 2](#) satisfies

$$\mathbb{R}_n^{\text{e}}(P, \phi_{n,v}^{*,\text{e}}) \leq 0.01 \quad \text{for every } P \in H_0^{\text{e,br}}(v).$$

Here $\mathbb{R}_n^{\text{e}}(P, \phi^{\text{e}})$ is the expected rejection probability when the test receives the entire true continuous propensity e_P , together with the independent original-record sample and public randomization. If

$$C_v^{\text{e}} \rho_n^{\text{e}}(v) < D_v^{\text{e,br}},$$

then every $P \in \mathcal{M}_v^{\text{e,br}}$ with $d(P) \geq C_v^{\text{e}} \rho_n^{\text{e}}(v)$ satisfies

$$1 - \mathbb{R}_n^{\text{e}}(P, \phi_{n,v}^{*,\text{e}}) \leq 0.01.$$

- **(Finite lower-bound witnesses.)** There exist a continuous function f , a null law P_0 , and a normalized finite probability prior

$$f(x) = \frac{1}{2} \quad (x \in [0, 1]), \quad \pi_1 = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1,$$

with at least one strictly positive weight, such that

$$P_0 \in H_0(v) \cap H_0^{\text{e,br}}(v), \quad e_{P_0} = f.$$

Here $H_0(v)$ is the original-model null of [Definition 2](#). For every j with $\omega_j > 0$,

$$P_j \in \mathcal{M}_v \cap \mathcal{M}_v^{\text{e,br}}, \quad e_{P_j} = f, \quad c_v^{\text{e}} \rho_n^{\text{e}}(v) \leq d(P_j) < d_0,$$

where $\mathcal{M}_v$ is defined in [Definition 1](#). Define the complete-record alternative mixture by

$$\overline{P}_1^{(n)} = \sum_{j=1}^k \omega_j P_j^{\otimes n}.$$

With total variation defined as the supremum of absolute event-probability differences,

$$\text{TV}(P_0^{\otimes n}, \overline{P}_1^{(n)}) < \frac{1}{2}.$$

For every allowed randomized supplied-propensity test ϕ^e satisfying

$$R_n^e(P, \phi^e) \leq \frac{1}{10} \quad \text{for every } P \in H_0^{e, \text{br}}(v),$$

there is an index j with $\omega_j > 0$ such that

$$1 - R_n^e(P_j, \phi^e) \geq \frac{2}{5}.$$

For every integer $n \geq N_v^e$,

$$C_v^e \rho_n^e(v) \leq \frac{d_0}{2}.$$

For each $P \in \mathcal{M}_v^{e, \text{br}}$, let $Q_{a,P}(dy \mid x)$ denote the conditional outcome probability kernel in treatment arm a . The untruncated inverse-propensity score has conditional mean equal to the canonical effect: for λ -almost every x ,

$$e_P(x) \int_{\mathbb{R}} \frac{y}{e_P(x)} Q_{1,P}(dy \mid x) + (1 - e_P(x)) \int_{\mathbb{R}} \frac{-y}{1 - e_P(x)} Q_{0,P}(dy \mid x) = \tau_P(x).$$

The constants and exponent are locally uniform on compact subsets of the admissible public parameter domain, equipped with the product topology on $(p, \alpha, \beta, \gamma)$.

Theorem 6. [thm:oracle-frontier-resolved] (Supplied-propensity frontier and attainment)
Let $\mathcal{V}$ denote the admissible domain of public tuples $v = (p, \alpha, \beta, \gamma)$ in [Definition 1](#), equipped with the topology inherited from $\mathbb{R}^4$. For $v \in \mathcal{V}$, define

$$q = \frac{p-1}{p}, \quad E_0 = \frac{2\gamma q}{2\gamma + q}, \quad \rho_n^e(v) = n^{-E_0}, \quad c_v^e = \frac{4^{-\gamma}}{16\sqrt{3}}, \quad C_v^e = 3 \left(120 + 128\sqrt{160\sqrt{2}} \right).$$

Use the supplied-propensity rule $\phi_{n,v}^{*,e}$ of [Algorithm 2](#), and the risk and capped radius of [Definitions 26](#) and [27](#). Write $R_n^e(P, \phi^e)$ for the expected rejection probability when the entire true continuous propensity e_P is supplied to the test.

Let λ be Lebesgue probability measure on $[0, 1]$, and define

$$d(P) = \left\{ \int_0^1 \left(\tau_P(x) - \int_0^1 \tau_P(z) d\lambda(z) \right)^2 d\lambda(x) \right\}^{1/2}, \quad D_v = \sup_{P \in \mathcal{M}_v} d(P).$$

Let $d_0 > 0$ be the fixed distance supplied by the legal model witness, and set

$$N_v^e = \left\lceil \max \left\{ 2, \left(\frac{2C_v^e}{d_0} \right)^{1/E_0} \right\} \right\rceil.$$

Then

$$\frac{1}{64\sqrt{3}} \leq c_v^e, \quad 0 < C_v^e.$$

For every integer $n \geq 2$,

$$c_v^e \rho_n^e(v) \leq r_n^{*,e}(v) \leq C_v^e \rho_n^e(v), \quad B_n^e(v, c_v^e \rho_n^e(v)) \geq \frac{2}{5}.$$

The attaining rule satisfies

$$R_n^e(P, \phi_{n,v}^{*,e}) \leq 0.01 \quad (P \in H_0(v)).$$

Whenever $C_v^e \rho_n^e(v) < D_v$, it also satisfies

$$1 - R_n^e(P, \phi_{n,v}^{*,e}) \leq 0.01 \quad (P \in \mathcal{M}_v, \quad C_v^e \rho_n^e(v) \leq d(P)).$$

For each such v, n , there exist a null law $P_0 \in H_0(v)$ and a normalized finite prior

$$\pi_1 = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1, \quad \exists j : \omega_j > 0,$$

such that every positive-weight support law satisfies

$$P_j \in \mathcal{M}_v, \quad e_{P_j} = e_{P_0}, \quad c_v^e \rho_n^e(v) \leq d(P_j) < D_v.$$

Define its complete-record mixture by

$$\mathbb{P}_{\pi_1, n} = \sum_{j=1}^k \omega_j P_j^{\otimes n},$$

and let TV denote the supremum of absolute event-probability differences. These laws satisfy

$$\text{TV}(P_0^{\otimes n}, \mathbb{P}_{\pi_1, n}) < \frac{1}{2}.$$

For every supplied-propensity test ϕ^e satisfying

$$R_n^e(P, \phi^e) \leq \frac{1}{10} \quad (P \in H_0(v)),$$

there is an index j with

$$\omega_j > 0, \quad 1 - R_n^e(P_j, \phi^e) \geq \frac{2}{5}.$$

For every integer $n \geq N_v^e$,

$$C_v^e \rho_n^e(v) \leq \frac{d_0}{2}.$$

For $P \in \mathcal{M}_v$, let $Q_{a,P}(dy \mid x)$ denote the conditional outcome probability kernel in treatment arm a . The untruncated inverse-propensity score identifies the original mean effect: for λ -almost every x ,

$$e_P(x) \int_{\mathbb{R}} \frac{y}{e_P(x)} Q_{1,P}(dy \mid x) + (1 - e_P(x)) \int_{\mathbb{R}} \frac{-y}{1 - e_P(x)} Q_{0,P}(dy \mid x) = \tau_P(x).$$

The frontier constants and nonempty-power thresholds are locally uniform over admissible tuples: on every compact subset of $\mathcal{V}$, the lower multipliers admit a common positive lower bound, the upper multipliers admit a common finite upper bound, and the thresholds admit a common finite upper bound, including at $p = 2$ and $\gamma = 1/4$.

Theorem 7. [thm:exact-propensity-preservation-boundary] (Exact propensity preservation boundary) Let $\mathcal{V}$ be the domain of public tuples $v = (p, \alpha, \beta, \gamma)$ satisfying:

- (*Moment order.*) $1 < p \leq 2$.
- (*Propensity smoothness.*) $0 < \alpha \leq 1$.
- (*Baseline smoothness.*) $0 < \beta \leq 1$.
- (*Effect smoothness.*) $1/4 \leq \gamma \leq 1$.

For every $v \in \mathcal{V}$, use the quantities and original-model radius multipliers c_v, C_v of [Definition 15](#), with

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad D_p = 1 + 2\alpha + \frac{\beta}{q} + \frac{S}{2\gamma},$$

$$E_0(p) = \frac{2\gamma q}{2\gamma + q}, \quad E_4(p) = \frac{2S}{D_p}, \quad F(p) = \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{S}{2\gamma}.$$

Define the comparison exponent and the supplied-propensity radius multipliers by

$$G(v) = \max\{0, E_0(p) - E_4(p)\}, \quad c_v^e = \frac{4^{-\gamma}}{16\sqrt{3}}, \quad C_v^e = 3(120 + 128\sqrt{160\sqrt{2}}).$$

Then

$$G(v) = \frac{2\gamma q}{(2\gamma + q)D_p} \max\{0, 1 - F(p)\}.$$

For the unknown- and supplied-propensity experiments on the same class $\mathcal{M}_v$ of [Definition 1](#), the capped critical radii of [Definitions 6](#) and [27](#) satisfy, for every integer $n \geq 2$,

$$\frac{c_v}{C_v^e} n^{G(v)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} n^{G(v)}.$$

If $F(p) = 1$, then

$$0 < \frac{c_v}{C_v^e}, \quad 0 < \frac{C_v}{c_v^e},$$

and, for every integer $n \geq 2$,

$$\frac{c_v}{C_v^e} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e}.$$

If $F(p) < 1$, then

$$\lim_{n \rightarrow \infty} \frac{r_n^*(v)}{r_n^{*,e}(v)} = +\infty,$$

and, for every integer $n \geq 2$,

$$\frac{c_v}{C_v^e} n^{E_0(p) - E_4(p)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} n^{E_0(p) - E_4(p)}.$$

The preservation region of [Definition 28](#) is exactly

$$\mathcal{R} = \left\{ v \in \mathcal{V} : \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{\alpha + \beta}{2\gamma} \geq 1 \right\}.$$

Moreover, for every compact subset K of $\mathcal{V}$, in the product topology on the four real coordinates, there exist real constants c, C , with $c > 0$, such that, for every $v \in K$,

$$c \leq \frac{c_v}{C_v^e}, \quad 64\sqrt{3} C_v \leq C.$$

E Verification note

The theorem statements and their proofs are machine-checked in Lean 4. The anchors attached to mathematical environments connect the displayed definitions and results to their corresponding formal declarations. Checking establishes the stated conclusions from the hypotheses and dependencies of each result, preserving its parameter domain, quantifiers, constants, and sample-size conditions.

The restrictions defining the statistical models, including those in [Definitions 1, 7, 8](#) and [22](#), are stipulated hypotheses. The verified conclusions concern deductions under these restrictions, including calibration, attainment, lower bounds, and separation-radius comparisons. Where a result uses a published conclusion as a dependency, its theorem-local verification-scope footnote identifies that conclusion and records whether the source proof was formalized. These local disclosures specify the dependency boundary alongside the checked statements and proofs.

F Proofs of the main results

Proof of [Proposition 1](#). Fix the admissible tuple $v = (p, \alpha, \beta, \gamma)$ and its matched smoothness tuple $w = (\alpha, \beta, \gamma)$.

1. We first establish the integrated conversion identity. Write

$$\mathcal{O} = [0, 1] \times \{0, 1\} \times \mathbb{R}$$

for the Borel record space. For a law P in either $\mathcal{M}_v$ or $\mathcal{M}_{(2, \alpha, \beta, \gamma)}$, let $Q_{a_{\text{arm}}, P}(\cdot \mid x)$ be its conditional outcome probability kernel in arm $a_{\text{arm}} \in \{0, 1\}$. Thus

$$\begin{aligned} P(dx, \{a_{\text{arm}}\}, dy) &= \lambda(dx) e_P(x)^{a_{\text{arm}}} (1 - e_P(x))^{1-a_{\text{arm}}} Q_{a_{\text{arm}}, P}(dy \mid x), \\ m_{a_{\text{arm}}, P}(x) &= m_{0, P}(x) + a_{\text{arm}} \tau_P(x) = \int y Q_{a_{\text{arm}}, P}(dy \mid x) \quad \text{for } \lambda\text{-almost every } x. \end{aligned}$$

The mean bounds give

$$|m_{0, P}(x)| \leq \frac{1}{2}, \quad |m_{1, P}(x)| \leq |m_{0, P}(x)| + |\tau_P(x)| \leq 1.$$

Define the signed-binary conversion $\mathcal{C}_{\text{bin}}(P)$ by

$$\begin{aligned} Q_{a_{\text{arm}}, P}^{\text{bin}}(dy \mid x) &= \frac{1 + m_{a_{\text{arm}}, P}(x)}{2} \delta_1(dy) + \frac{1 - m_{a_{\text{arm}}, P}(x)}{2} \delta_{-1}(dy), \\ \mathcal{C}_{\text{bin}}(P)(dx, \{a_{\text{arm}}\}, dy) &= \lambda(dx) e_P(x)^{a_{\text{arm}}} (1 - e_P(x))^{1-a_{\text{arm}}} Q_{a_{\text{arm}}, P}^{\text{bin}}(dy \mid x). \end{aligned}$$

The two arm weights are nonnegative and sum to one, and

$$\int y Q_{a_{\text{arm}}, P}^{\text{bin}}(dy \mid x) = m_{a_{\text{arm}}, P}(x), \quad \int |y|^2 Q_{a_{\text{arm}}, P}^{\text{bin}}(dy \mid x) = 1.$$

Consequently the conversion has signed-binary support and retains the uniform design, overlap, smoothness restrictions, and mean bounds. In particular,

$$\begin{aligned} \mathcal{C}_{\text{bin}}(P) &\in \mathcal{M}_w^{\text{bin}}, \quad (\mathcal{C}_{\text{bin}}(P))_{X, A} = P_{X, A}, \\ e_{\mathcal{C}_{\text{bin}}(P)} &= e_P, \quad m_{0, \mathcal{C}_{\text{bin}}(P)} = m_{0, P}, \quad \tau_{\mathcal{C}_{\text{bin}}(P)} = \tau_P. \end{aligned}$$

Substitution into the definitions of distance and constancy yields

$$d(\mathcal{C}_{\text{bin}}(P)) = d(P),$$

and

$$\begin{aligned} \exists c \in [-1/2, 1/2] \forall x, \tau_{\mathcal{C}_{\text{bin}}(P)}(x) = c \\ \iff \exists c \in [-1/2, 1/2] \forall x, \tau_P(x) = c. \end{aligned}$$

To define record randomization on the whole record space, set

$$\ell_1(y) = \max\{-1, \min\{1, y\}\} \quad (y \in \mathbb{R})$$

and define the probability kernel

$$\mathsf{K}_{\text{sign}}((x, a_{\text{obs}}, y), \cdot) = \frac{1 + \ell_1(y)}{2} \delta_{(x, a_{\text{obs}}, 1)} + \frac{1 - \ell_1(y)}{2} \delta_{(x, a_{\text{obs}}, -1)}, \quad (x, a_{\text{obs}}, y) \in \mathcal{O}.$$

For $P \in \mathcal{M}_w^b$, uniform design and fixed overlap transfer the observed outcome bound to both conditional arm kernels:

$$Q_{a_{\text{arm}}, P}([-1, 1] \mid x) = 1 \quad \text{for } \lambda\text{-almost every } x, \text{ for both arms.}$$

Indeed, disintegrating $P(|Y| > 1) = 0$ gives

$$\int [(1 - e_P(x))Q_{0, P}(\{|y| > 1\} \mid x) + e_P(x)Q_{1, P}(\{|y| > 1\} \mid x)] \lambda(dx) = 0.$$

Both summands are nonnegative, and both arm weights are at least 1/4. Each conditional probability therefore vanishes almost everywhere. Since $\ell_1(y) = y$ on $[-1, 1]$, for $b_{\text{sign}} \in \{-1, 1\}$,

$$\int \frac{1 + b_{\text{sign}} \ell_1(y)}{2} Q_{a_{\text{arm}}, P}(dy \mid x) = \frac{1 + b_{\text{sign}} m_{a_{\text{arm}}, P}(x)}{2} \quad \text{for } \lambda\text{-almost every } x.$$

Integrating the two atomic weights and then the treatment mixture proves the one-record identity

$$\int_{\mathcal{O}} \mathsf{K}_{\text{sign}}(o, \cdot) P(do) = \mathcal{C}_{\text{bin}}(P).$$

Now let $n \in \mathbb{N}$, let ϕ be a measurable $[0, 1]$ -valued test, and write its input as

$$\mathbf{o} = (o_i)_{i=1}^n \in \mathcal{O}^n, \quad o_i = (x_i, a_{\text{obs}, i}, y_i), \quad u_{\text{seed}} \in [0, 1].$$

Define its integrated rejection map by

$$\tilde{\phi}(\mathbf{o}, u_{\text{seed}}) = \sum_{\mathbf{b}_{\text{sign}} \in \{-1, 1\}^n} \phi(((x_i, a_{\text{obs}, i}, b_{\text{sign}, i}))_{i=1}^n, u_{\text{seed}}) \prod_{i=1}^n \frac{1 + b_{\text{sign}, i} \ell_1(y_i)}{2}, \quad \mathbf{b}_{\text{sign}} = (b_{\text{sign}, i})_{i=1}^n.$$

Its weights are nonnegative and satisfy

$$\sum_{\mathbf{b}_{\text{sign}} \in \{-1, 1\}^n} \prod_{i=1}^n \frac{1 + b_{\text{sign}, i} \ell_1(y_i)}{2} = \prod_{i=1}^n \left(\frac{1 + \ell_1(y_i)}{2} + \frac{1 - \ell_1(y_i)}{2} \right) = 1.$$

Thus $\tilde{\phi}$ is measurable and takes values in $[0, 1]$. For $n = 0$, the sign-vector set contains the single empty vector and the empty product is one, so the same definition applies.

The finite sum is the integral of ϕ against the independent product of the record kernels. Independence and the one-record identity imply

$$\int_{\mathcal{O}^n} \bigotimes_{i=1}^n \mathsf{K}_{\text{sign}}(o_i, \cdot) P^{\otimes n}(d\mathbf{o}) = \mathcal{C}_{\text{bin}}(P)^{\otimes n}.$$

This follows by factorization on measurable rectangles; the empty product gives the identity for $n = 0$. Hence, for every fixed u_{seed} ,

$$\int_{\mathcal{O}^n} \tilde{\phi}(\mathbf{o}, u_{\text{seed}}) P^{\otimes n}(d\mathbf{o}) = \int_{\mathcal{O}^n} \phi(\mathbf{o}, u_{\text{seed}}) \mathcal{C}_{\text{bin}}(P)^{\otimes n}(d\mathbf{o}).$$

Integrating in the public seed and using the experiment of [Definition 4](#) gives

$$\mathsf{R}_n(P, \tilde{\phi}) = \mathsf{R}_n(\mathcal{C}_{\text{bin}}(P), \phi).$$

Boundedness of the tests justifies the integral exchanges. On bounded records, the kernel uses exactly the probabilities $(1 + Y)/2$ and $(1 - Y)/2$ stated in the proposition.

2. We next compare the binary and bounded testing risks. In the infimum–supremum comparisons, a supremum of type-II errors over an empty alternative set is assigned the value zero. The test

$$\phi_{\text{zero}}(\mathbf{o}, u_{\text{seed}}) = 0$$

is level-valid for every null class. The sets of level-valid tests are therefore nonempty, and

$$0 \leq 1 - \mathsf{R}_n(P, \phi) \leq 1$$

bounds every individual type-II error and every corresponding worst-case error.

Signed-binary support implies bounded support. Keeping the remaining restrictions and the constancy condition gives

$$\mathcal{M}_w^{\text{bin}} \subseteq \mathcal{M}_w^{\text{b}}, \quad H_0^{\text{bin}}(w) \subseteq H_0^{\text{b}}(w).$$

Every bounded-null level-valid test is consequently binary-null level-valid, with

$$\sup_{\substack{P \in \mathcal{M}_w^{\text{bin}} \\ d(P) \geq r}} \{1 - \mathsf{R}_n(P, \phi)\} \leq \sup_{\substack{P \in \mathcal{M}_w^{\text{b}} \\ d(P) \geq r}} \{1 - \mathsf{R}_n(P, \phi)\}.$$

For a nonempty binary alternative set, this is the supremum comparison under inclusion; for an empty binary alternative set, it follows from the zero convention and nonnegativity. Taking infima gives

$$B_n^{\text{bin}}(w, r) \leq B_n^{\text{b}}(w, r).$$

Conversely, take any binary-null level-valid test ϕ . For every $P \in H_0^{\text{b}}(w)$, the conversion belongs to $H_0^{\text{bin}}(w)$, so the preceding identity gives

$$\mathsf{R}_n(P, \tilde{\phi}) = \mathsf{R}_n(\mathcal{C}_{\text{bin}}(P), \phi) \leq \frac{1}{10}.$$

For a bounded alternative P with $d(P) \geq r$, its conversion is a binary alternative with the same distance, and

$$1 - R_n(P, \tilde{\phi}) = 1 - R_n(\mathcal{C}_{\text{bin}}(P), \phi) \leq \sup_{\substack{P \in \mathcal{M}_w^{\text{bin}} \\ d(P) \geq r}} \{1 - R_n(P, \phi)\}.$$

If the bounded alternative set is nonempty, taking its supremum preserves this inequality; the binary supremum is bounded above by one. If the bounded alternative set is empty, its worst-case error is zero and the binary worst-case error is nonnegative. Thus in either case,

$$\sup_{\substack{P \in \mathcal{M}_w^{\text{b}} \\ d(P) \geq r}} \{1 - R_n(P, \tilde{\phi})\} \leq \sup_{\substack{P \in \mathcal{M}_w^{\text{bin}} \\ d(P) \geq r}} \{1 - R_n(P, \phi)\}.$$

The bounded-model infimum is at most the left-hand side for each such ϕ . Taking the binary-model infimum proves

$$B_n^{\text{b}}(w, r) \leq B_n^{\text{bin}}(w, r), \quad B_n^{\text{bin}}(w, r) = B_n^{\text{b}}(w, r).$$

3. For the remaining model inclusion, take $P \in \mathcal{M}_w^{\text{b}}$. The conditional support established above and the nonnegative moment exponent supplied by admissibility give

$$\int |y|^p Q_{a_{\text{arm}}, P}(dy | x) \leq \int 1 Q_{a_{\text{arm}}, P}(dy | x) = 1 \leq 10 \quad \text{for } \lambda\text{-almost every } x.$$

All the other primitive restrictions are identical at the matched smoothness tuple. Therefore

$$\mathcal{M}_w^{\text{bin}} \subseteq \mathcal{M}_w^{\text{b}} \subseteq \mathcal{M}_v, \quad H_0^{\text{b}}(w) \subseteq H_0(v).$$

4. The distance suprema are equal because the attained distance sets are equal. The binary-to-bounded inclusion gives one direction, and converting each bounded law gives the other:

$$\{d(P) : P \in \mathcal{M}_w^{\text{bin}}\} = \{d(P) : P \in \mathcal{M}_w^{\text{b}}\}.$$

For comparison with the full model, convert each bounded law. Its converted arm kernels also satisfy

$$\int |y|^p Q_{a_{\text{arm}}, P}^{\text{bin}}(dy | x) = 1 \leq 10,$$

so the conversion belongs to $\mathcal{M}_v$ and has the same distance. Conversely, the conversion of every $P \in \mathcal{M}_v$ belongs to the binary model, hence to the bounded model, and retains $d(P)$. Thus

$$\{d(P) : P \in \mathcal{M}_w^{\text{b}}\} = \{d(P) : P \in \mathcal{M}_v\}.$$

Taking suprema yields

$$D_w^{\text{bin}} = D_w^{\text{b}} = D_v.$$

Finally, [Lemma 1](#) supplies the boundedness of the full-model distance set and the witness lower bound

$$d_0 \leq D_v \leq \frac{1}{2}.$$

5. A test level-valid for $H_0(v)$ is level-valid for $H_0^b(w)$, and model inclusion gives

$$\sup_{\substack{P \in \mathcal{M}_w^b \\ d(P) \geq r}} \{1 - R_n(P, \phi)\} \leq \sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq r}} \{1 - R_n(P, \phi)\}.$$

For a nonempty bounded alternative set, this follows by inclusion; for an empty set, it follows from the zero convention and nonnegativity of the full-model error. Taking infima over full-null level-valid tests, and allowing all bounded-null level-valid tests on the left, proves

$$B_n^b(w, r) \leq B_n(v, r).$$

Together with the equality already established, this gives, for $n \geq 2$ and $0 < r < D_v$,

$$B_n^{\text{bin}}(w, r) = B_n^b(w, r) \leq B_n(v, r).$$

6. We establish strict positivity of the binary capped radius by a small cosine alternative. For $t_{\text{cos}} \in [0, 1]$, define the deterministic law and its conversion by

$$\begin{aligned} P_{\text{cos}, t_{\text{cos}}}^{\text{det}} : \quad X &\sim \lambda, \quad \Pr(A = 1 \mid X = x) = \frac{1}{2}, \quad Y = A \frac{t_{\text{cos}}}{8} \cos(2\pi X), \\ P_{\text{cos}, t_{\text{cos}}} &= \mathcal{C}_{\text{bin}}(P_{\text{cos}, t_{\text{cos}}}^{\text{det}}). \end{aligned}$$

Their primitive functions are

$$e_{P_{\text{cos}, t_{\text{cos}}}}(x) = \frac{1}{2}, \quad m_{0, P_{\text{cos}, t_{\text{cos}}}}(x) = 0, \quad \tau_{P_{\text{cos}, t_{\text{cos}}}}(x) = \frac{t_{\text{cos}}}{8} \cos(2\pi x).$$

The propensity and baseline are constant. For the effect, the cosine increment bound and $|x - z| \leq 1$ imply

$$\begin{aligned} |\tau_{P_{\text{cos}, t_{\text{cos}}}}(x)| &\leq \frac{t_{\text{cos}}}{8} \leq \frac{1}{2}, \\ |\tau_{P_{\text{cos}, t_{\text{cos}}}}(x) - \tau_{P_{\text{cos}, t_{\text{cos}}}}(z)| &\leq t_{\text{cos}} \frac{2\pi}{8} |x - z| \leq t_{\text{cos}} 20 |x - z|^\gamma \leq 20 |x - z|^\gamma. \end{aligned}$$

The deterministic outcomes have absolute value at most $1/8$, so their conditional second moments are at most one. These bounds verify membership of the deterministic law in $\mathcal{M}_{(2, \alpha, \beta, \gamma)}$; conversion then gives

$$P_{\text{cos}, t_{\text{cos}}} \in \mathcal{M}_w^{\text{bin}}, \quad P_{\text{cos}, 0} \in H_0^{\text{bin}}(w).$$

Moreover,

$$\int \tau_{P_{\text{cos}, t_{\text{cos}}}} d\lambda = 0, \quad \int \tau_{P_{\text{cos}, t_{\text{cos}}}}^2 d\lambda = \frac{t_{\text{cos}}^2}{128}, \quad d(P_{\text{cos}, t_{\text{cos}}}) = t_{\text{cos}} d_0.$$

For probability measures $\mu_{\text{tv}}, \nu_{\text{tv}}$ on a common measurable space $\mathcal{Z}_{\text{tv}}$, use

$$\text{TV}(\mu_{\text{tv}}, \nu_{\text{tv}}) = \sup_{E_{\text{tv}} \text{ measurable in } \mathcal{Z}_{\text{tv}}} |\mu_{\text{tv}}(E_{\text{tv}}) - \nu_{\text{tv}}(E_{\text{tv}})|.$$

The conditional distribution of (A, Y) under the cosine law is

$$\begin{aligned}\kappa_{\cos, t_{\cos}}(x, \cdot) &= \frac{1 + \tau_{P_{\cos, t_{\cos}}}(x)}{4} \delta_{(1,1)} + \frac{1 - \tau_{P_{\cos, t_{\cos}}}(x)}{4} \delta_{(1,-1)} \\ &\quad + \frac{1}{4} \delta_{(0,1)} + \frac{1}{4} \delta_{(0,-1)}.\end{aligned}$$

For a measurable record event $E_{\text{tv}} \subseteq \mathcal{O}$, define its section by

$$E_{\text{tv}, x} = \{(a_{\text{arm}}, y) : (x, a_{\text{arm}}, y) \in E_{\text{tv}}\}.$$

Subtracting the four displayed masses gives

$$\begin{aligned}\kappa_{\cos, 0}(x, E_{\text{tv}, x}) - \kappa_{\cos, t_{\cos}}(x, E_{\text{tv}, x}) \\ = -\frac{\tau_{P_{\cos, t_{\cos}}}(x)}{4} [\mathbf{1}_{E_{\text{tv}, x}}(1, 1) - \mathbf{1}_{E_{\text{tv}, x}}(1, -1)].\end{aligned}$$

The absolute difference of these two indicators is at most one. Therefore

$$|\kappa_{\cos, 0}(x, E_{\text{tv}, x}) - \kappa_{\cos, t_{\cos}}(x, E_{\text{tv}, x})| \leq \frac{|\tau_{P_{\cos, t_{\cos}}}(x)|}{4} \leq \frac{t_{\cos}}{16}.$$

Integration over the common uniform design gives

$$\begin{aligned}|P_{\cos, 0}(E_{\text{tv}}) - P_{\cos, t_{\cos}}(E_{\text{tv}})| &\leq \int |\kappa_{\cos, 0}(x, E_{\text{tv}, x}) - \kappa_{\cos, t_{\cos}}(x, E_{\text{tv}, x})| \lambda(dx) \\ &\leq \frac{t_{\cos}}{16},\end{aligned}$$

and hence

$$\text{TV}(P_{\cos, 0}, P_{\cos, t_{\cos}}) \leq \frac{t_{\cos}}{16}.$$

We use the expectation bound for any measurable $f_{\text{tv}} : \mathcal{Z}_{\text{tv}} \rightarrow [0, 1]$. The layer-cake identity is

$$\int f_{\text{tv}} d\eta_{\text{tv}} = \int_0^1 \eta_{\text{tv}} \{z_{\text{tv}} : f_{\text{tv}}(z_{\text{tv}}) \geq h_{\text{tv}}\} dh_{\text{tv}}, \quad \eta_{\text{tv}} \in \{\mu_{\text{tv}}, \nu_{\text{tv}}\}.$$

Subtracting these identities, taking absolute values, and bounding each event-probability difference gives

$$\left| \int f_{\text{tv}} d\mu_{\text{tv}} - \int f_{\text{tv}} d\nu_{\text{tv}} \right| \leq \text{TV}(\mu_{\text{tv}}, \nu_{\text{tv}}).$$

Inserting the mixed product $P_{\cos, 0} \otimes P_{\cos, t_{\cos}}^{\otimes n_{\text{prod}}}$ between the two product laws, applying the triangle inequality, and integrating event sections gives

$$\begin{aligned}\text{TV}\left(P_{\cos, 0}^{\otimes (n_{\text{prod}}+1)}, P_{\cos, t_{\cos}}^{\otimes (n_{\text{prod}}+1)}\right) \\ \leq \text{TV}(P_{\cos, 0}, P_{\cos, t_{\cos}}) + \text{TV}\left(P_{\cos, 0}^{\otimes n_{\text{prod}}}, P_{\cos, t_{\cos}}^{\otimes n_{\text{prod}}}\right), \quad n_{\text{prod}} \in \mathbb{N}.\end{aligned}$$

For the first term, the integrated section probability is a $[0, 1]$ -valued function, so the expectation bound applies; for the second, each section difference is bounded by the product

total variation. At $n_{\text{prod}} = 0$, both product laws are the point mass on the empty sample. Induction therefore yields

$$\text{TV} \left(P_{\cos,0}^{\otimes n}, P_{\cos,t_{\cos}}^{\otimes n} \right) \leq n \text{TV}(P_{\cos,0}, P_{\cos,t_{\cos}}) \leq \frac{nt_{\cos}}{16}.$$

For the fixed $n \geq 2$, choose

$$t_{\text{small}} = \frac{8}{100n}, \quad 0 < t_{\text{small}} \leq 1.$$

The preceding product bound becomes

$$\text{TV} \left(P_{\cos,0}^{\otimes n}, P_{\cos,t_{\text{small}}}^{\otimes n} \right) \leq \frac{nt_{\text{small}}}{16} = \frac{1}{200}.$$

For a binary-null level-valid test ϕ , define its seed average by

$$\bar{\phi}(\mathbf{o}) = \int_{[0,1]} \phi(\mathbf{o}, u_{\text{seed}}) \lambda(du_{\text{seed}}).$$

It is measurable, satisfies $0 \leq \bar{\phi} \leq 1$, and obeys

$$\mathbf{R}_n(P, \phi) = \int_{\mathcal{O}^n} \bar{\phi} dP^{\otimes n}.$$

Applying the expectation bound to this average and using null validity gives

$$\begin{aligned} 1 - \mathbf{R}_n(P_{\cos,t_{\text{small}}}, \phi) &\geq \frac{9}{10} - \text{TV} \left(P_{\cos,0}^{\otimes n}, P_{\cos,t_{\text{small}}}^{\otimes n} \right) \\ &\geq \frac{179}{200} > \frac{1}{10}. \end{aligned}$$

Whenever $0 < r \leq t_{\text{small}}d_0$, this cosine law belongs to the separated alternative set. Taking its error as a lower bound for the worst-case error, and then taking the infimum over level-valid tests, proves

$$B_n^{\text{bin}}(w, r) \geq \frac{179}{200} > \frac{1}{10} \quad (0 < r \leq t_{\text{small}}d_0).$$

Thus every successful separation in the binary capped-radius set exceeds $t_{\text{small}}d_0$. Its sentinel also satisfies

$$t_{\text{small}}d_0 \leq d_0 \leq D_w^{\text{bin}}.$$

The set is nonempty because it contains the sentinel. Taking its infimum, as defined in [Definition 14](#), gives

$$0 < t_{\text{small}}d_0 \leq r_n^{*,\text{bin}}(w).$$

7. Finally, the common cap and the risk equality give identical sets in the definitions of the binary and bounded radii:

$$\begin{aligned} &\left\{ r \in (0, D_w^{\text{bin}}) : B_n^{\text{bin}}(w, r) \leq \frac{1}{10} \right\} \cup \{D_w^{\text{bin}}\} \\ &= \left\{ r \in (0, D_w^{\text{b}}) : B_n^{\text{b}}(w, r) \leq \frac{1}{10} \right\} \cup \{D_w^{\text{b}}\}. \end{aligned}$$

Their infima are therefore equal:

$$r_n^{*,\text{bin}}(w) = r_n^{*,\text{b}}(w).$$

The risk ordering also gives

$$\begin{aligned} & \left\{ r \in (0, D_v) : B_n(v, r) \leq \frac{1}{10} \right\} \cup \{D_v\} \\ & \subseteq \left\{ r \in (0, D_v) : B_n^{\text{b}}(w, r) \leq \frac{1}{10} \right\} \cup \{D_v\}. \end{aligned}$$

Both sets contain D_v and are bounded below by zero, since $D_v \geq d_0 > 0$. Taking infima reverses this inclusion. By [Definitions 6](#) and [13](#), it follows that

$$r_n^{*,\text{b}}(w) \leq r_n^*(v).$$

Combining this with the positivity just proved establishes

$$0 < r_n^{*,\text{bin}}(w) = r_n^{*,\text{b}}(w) \leq r_n^*(v).$$

The conversion properties and the identity for every $n \in \mathbb{N}$ were established in the first step. □

Proof of [Theorem 1](#). Fix an admissible tuple v and an integer $n \geq 2$. Throughout, write

$$M := J_c, \quad B := \mathfrak{b}_{n,v}.$$

These quantities have the values prescribed in [Algorithm 1](#), including its small-sample branch.

1. First,

$$B \geq 0, \quad M \geq 1.$$

For $n < 4$, these follow from $B = 0$ and $M = 1$. For $n \geq 4$, every summand in the formula for B is nonnegative: the clipping cutoffs and ranks are positive, and the increment weights a_j are nonnegative. This also covers an empty increment sum. The coarse rank is a nonnegative integer power of two, hence is at least one.

2. Let $P \in \mathcal{M}_v$. By [Lemma 3](#), the affine coefficients in

$$Z_b(c) = Z_{b,0} - cZ_{b,A}, \quad b \in \{1, 2\},$$

have square-integrable scalar projections, and, for all $u, z \in \mathbb{R}^M$,

$$\begin{aligned} \text{Var}(u^T Z_{b,0}) &\leq \Lambda_{n,v} \|u\|_2^2, \\ \text{Var}(u^T Z_{b,A}) &\leq \Lambda_{n,v} \|u\|_2^2, \\ |\text{Cov}(u^T Z_{b,0}, z^T Z_{b,A})| &\leq \Lambda_{n,v} \|u\|_2 \|z\|_2. \end{aligned}$$

Taking the coordinate directions and summing their second moments proves square integrability of each coefficient vector.

Define the sample event

$$\mathcal{E}_{\text{prof}}(P) := \left\{ \text{sample} : \forall |c| \leq \frac{1}{2}, \quad |W(c) - \|\theta_P(c)\|_2^2| \leq 128 \left(\sqrt{\Lambda_{n,v}} \|\theta_P(c)\|_2 + \sqrt{M} \Lambda_{n,v} \right) \right\}.$$

By [Lemma 4](#), this event is measurable and

$$P^{\otimes n}(\mathcal{E}_{\text{prof}}(P)) \geq 0.9925.$$

The same result supplies, when $\Lambda_{n,v} = 0$, the simultaneous almost-sure identity

$$W(c) = \|\theta_P(c)\|_2^2 \quad \text{for every } |c| \leq \frac{1}{2}.$$

3. Suppose $P \in H_0(v)$. By [Definition 2](#), there is a constant c_0 satisfying

$$c_0 \in [-\frac{1}{2}, \frac{1}{2}], \quad \tau_P(x) = c_0 \quad \text{for every } x \in [0, 1].$$

If $n \geq 4$, [Lemma 2](#) gives

$$\|\theta_P(c_0)\|_2 \leq B.$$

If $n < 4$, the zero-score branch of [Algorithm 1](#) gives

$$\theta_P(c) = 0 \quad \text{for every } c \in \mathbb{R}, \quad B = 0,$$

so the same bound holds. Since norms are nonnegative and c_0 is an admissible candidate,

$$\inf_{|c| \leq 1/2} \|\theta_P(c)\|_2 \leq \|\theta_P(c_0)\|_2 \leq B.$$

4. We next establish the size bound for $n \geq 4$. The covariance ledger is strictly positive. Indeed, the construction gives

$$s_n = \lfloor n/2 \rfloor \geq 2, \quad T_0 \geq 1, \quad V_0 = 32T_0^{2-p} > 0, \quad L_1 \geq 16\sqrt{V_0} > 0.$$

Consequently,

$$\Lambda_{n,v} = \frac{L_1^2}{s_n} + \frac{2L_2^2}{s_n(s_n - 1)} > 0.$$

On $\mathcal{E}_{\text{prof}}(P)$, define

$$u_{c_0} := \|\theta_P(c_0)\|_2.$$

Then $0 \leq u_{c_0} \leq B$, and the profiling inequality implies

$$W(c_0) \leq u_{c_0}^2 + 128\sqrt{\Lambda_{n,v}} u_{c_0} + 128\sqrt{M} \Lambda_{n,v}.$$

Completing the square gives

$$0 \leq \left(B - 64\sqrt{\Lambda_{n,v}} \right)^2, \quad 128\sqrt{\Lambda_{n,v}} B \leq B^2 + 4096\Lambda_{n,v}.$$

Also, $M \geq 1$ implies

$$\sqrt{M} \geq 1, \quad \Lambda_{n,v} \leq \sqrt{M} \Lambda_{n,v}.$$

Thus

$$\begin{aligned} W(c_0) &\leq B^2 + 128\sqrt{\Lambda_{n,v}} B + 128\sqrt{M} \Lambda_{n,v} \\ &\leq 2B^2 + 4096\Lambda_{n,v} + 128\sqrt{M} \Lambda_{n,v} \\ &\leq 2B^2 + 4224\sqrt{M} \Lambda_{n,v} \\ &\leq 2B^2 + 65536\sqrt{M} \Lambda_{n,v}. \end{aligned}$$

5. We justify the comparison with the computed profile minimum and the rounded cutoff. At a fixed sample, define

$$Z_{b,0} := Z_b(0), \quad Z_{b,A} := Z_b(0) - Z_b(1), \quad b \in \{1, 2\}.$$

The affine score construction gives $Z_b(c) = Z_{b,0} - cZ_{b,A}$. Define its quadratic coefficients by

$$\begin{aligned} \omega_0 &:= \langle Z_{1,0}, Z_{2,0} \rangle, \\ \omega_1 &:= -\langle Z_{1,A}, Z_{2,0} \rangle - \langle Z_{1,0}, Z_{2,A} \rangle, \\ \omega_2 &:= \langle Z_{1,A}, Z_{2,A} \rangle. \end{aligned}$$

Expansion yields

$$W(c) = \omega_0 + \omega_1 c + \omega_2 c^2.$$

Define the smaller endpoint value and the computed profile value by

$$\begin{aligned} \omega_{\text{end}} &:= \min \left\{ \omega_0 - \frac{\omega_1}{2} + \frac{\omega_2}{4}, \omega_0 + \frac{\omega_1}{2} + \frac{\omega_2}{4} \right\}, \\ W_{\min} &:= \begin{cases} \min \left\{ \omega_{\text{end}}, \omega_0 - \frac{\omega_1^2}{4\omega_2} \right\}, & \omega_2 > 0, \quad -\frac{1}{2} \leq -\frac{\omega_1}{2\omega_2} \leq \frac{1}{2}, \\ \omega_{\text{end}}, & \text{otherwise.} \end{cases} \end{aligned}$$

The quotient in the first branch is used under $\omega_2 > 0$.

For every $|c| \leq 1/2$,

$$W_{\min} \leq W(c).$$

To see this in the first branch, complete the square:

$$W(c) - \left(\omega_0 - \frac{\omega_1^2}{4\omega_2} \right) = \omega_2 \left(c + \frac{\omega_1}{2\omega_2} \right)^2 \geq 0.$$

If $\omega_2 > 0$ and the vertex lies to the left of the interval, then $\omega_1 > \omega_2$, and

$$W(c) - W(-\frac{1}{2}) = (c + \frac{1}{2})[\omega_1 + \omega_2(c - \frac{1}{2})] \geq 0.$$

If the vertex lies to the right, then $\omega_1 < -\omega_2$, and

$$W(c) - W(\frac{1}{2}) = (\frac{1}{2} - c)[- \omega_1 - \omega_2(c + \frac{1}{2})] \geq 0.$$

Finally, if $\omega_2 \leq 0$, then $c^2 \leq 1/4$. For $\omega_1 \geq 0$,

$$W(c) - W(-\frac{1}{2}) = \omega_1(c + \frac{1}{2}) + \omega_2(c^2 - \frac{1}{4}) \geq 0,$$

whereas for $\omega_1 < 0$,

$$W(c) - W(\frac{1}{2}) = \omega_1(c - \frac{1}{2}) + \omega_2(c^2 - \frac{1}{4}) \geq 0.$$

These cases include a constant or linear statistic.

For the cutoff, define

$$x_{\text{cut}} := 2B^2 + 65536\sqrt{M} \Lambda_{n,v} > 0, \quad k_{\text{cut}} := \lceil \log_2 x_{\text{cut}} \rceil.$$

The defining ceiling inequalities give

$$\log_2 x_{\text{cut}} \leq k_{\text{cut}} < \log_2 x_{\text{cut}} + 1.$$

Exponentiating and using the cutoff formula therefore yields

$$2B^2 + 65536\sqrt{M} \Lambda_{n,v} \leq h_{n,v} \leq 4B^2 + 131072\sqrt{M} \Lambda_{n,v}.$$

Combining these bounds with the preceding step, on $\mathcal{E}_{\text{prof}}(P)$,

$$W_{\min} \leq W(c_0) \leq 2B^2 + 65536\sqrt{M} \Lambda_{n,v} \leq h_{n,v}.$$

The strict rejection rule in [Algorithm 1](#) consequently gives

$$\psi_{n,v}(\text{sample}, \text{seed}) = 0 \quad \text{on } \mathcal{E}_{\text{prof}}(P) \times [0, 1].$$

Define the experiment measure and its good event by

$$\mathbb{P}_{\text{exp}} := P^{\otimes n} \otimes \lambda, \quad \mathcal{E}_{\text{exp}}(P) := \mathcal{E}_{\text{prof}}(P) \times [0, 1].$$

Independence and the unit mass of λ imply

$$\mathbb{P}_{\text{exp}}(\mathcal{E}_{\text{exp}}(P)) = P^{\otimes n}(\mathcal{E}_{\text{prof}}(P)).$$

The test takes values in $[0, 1]$, so pointwise

$$0 \leq \psi_{n,v} \leq \mathbf{1}_{\mathcal{E}_{\text{exp}}(P)^c}.$$

Integrating this inequality gives

$$\begin{aligned} R_n(P, \psi_{n,v}) &\leq \mathbb{P}_{\text{exp}}(\mathcal{E}_{\text{exp}}(P)^c) \\ &= 1 - P^{\otimes n}(\mathcal{E}_{\text{prof}}(P)) \\ &\leq 0.0075. \end{aligned}$$

For $n < 4$, the zero-test branch gives directly

$$R_n(P, \psi_{n,v}) = 0 \leq 0.0075.$$

6. Now assume

$$n \geq 4, \quad R_{\text{att}}(n, v) < D_v, \quad P \in \mathcal{M}_v, \quad d(P) \geq R_{\text{att}}(n, v).$$

Work on $\mathcal{E}_{\text{prof}}(P)$. The computed profile value is attained: there exists

$$c_{\min} \in [-\frac{1}{2}, \frac{1}{2}], \quad W(c_{\min}) = W_{\min}.$$

Indeed, the smaller endpoint value is attained at one of the two endpoints. In the vertex branch, choose that endpoint if its value is at most the vertex value; otherwise choose

$$c_{\text{vert}} := -\frac{\omega_1}{2\omega_2}, \quad W(c_{\text{vert}}) = \omega_0 - \frac{\omega_1^2}{4\omega_2}.$$

Here $\omega_2 > 0$ and the branch condition places c_{vert} in the candidate interval. In the remaining branch, choose the endpoint attaining ω_{end} .

Define

$$u_{\min} := \|\theta_P(c_{\min})\|_2.$$

By [Lemma 2](#),

$$u_{\min} \geq \frac{3}{16}d(P) - 16M^{-\gamma} - B.$$

Substituting the assumed separation and its displayed definition gives

$$\begin{aligned} u_{\min} &\geq \frac{3}{16}R_{\text{att}}(n, v) - 16M^{-\gamma} - B \\ &= 4B + 1024M^{1/4}\sqrt{\Lambda_{n,v}}. \end{aligned}$$

We derive the scalar power bound. Since $M \geq 1$ and $\Lambda_{n,v} > 0$,

$$M^{1/4} \geq 1, \quad \left(M^{1/4}\sqrt{\Lambda_{n,v}}\right)^2 = \sqrt{M}\Lambda_{n,v} > 0.$$

Thus

$$u_{\min} \geq 256\sqrt{\Lambda_{n,v}} \geq 0, \quad 128\sqrt{\Lambda_{n,v}}u_{\min} \leq \frac{1}{2}u_{\min}^2.$$

Squaring the signal bound, with all its terms nonnegative, also gives

$$\begin{aligned} u_{\min}^2 &\geq \left(4B + 1024M^{1/4}\sqrt{\Lambda_{n,v}}\right)^2 \\ &\geq 16B^2 + 1048576\sqrt{M}\Lambda_{n,v}. \end{aligned}$$

The lower side of the profiling inequality now yields

$$\begin{aligned} W(c_{\min}) &\geq u_{\min}^2 - 128\sqrt{\Lambda_{n,v}}u_{\min} - 128\sqrt{M}\Lambda_{n,v} \\ &\geq \frac{1}{2}u_{\min}^2 - 128\sqrt{M}\Lambda_{n,v} \\ &\geq 8B^2 + 524160\sqrt{M}\Lambda_{n,v} \\ &> 4B^2 + 131072\sqrt{M}\Lambda_{n,v}. \end{aligned}$$

The last inequality is strict because $\sqrt{M}\Lambda_{n,v} > 0$.

Using the upper cutoff bound and attainment,

$$h_{n,v} \leq 4B^2 + 131072\sqrt{M}\Lambda_{n,v} < W(c_{\min}) = W_{\min}.$$

Hence

$$\psi_{n,v} = 1 \quad \text{on } \mathcal{E}_{\text{exp}}(P), \quad 0 \leq 1 - \psi_{n,v} \leq \mathbf{1}_{\mathcal{E}_{\text{exp}}(P)^c}.$$

Measurability and boundedness make the test integrable under the probability measure $\mathbb{P}_{\text{exp}}$. Integrating the last inequality gives

$$\begin{aligned} 1 - R_n(P, \psi_{n,v}) &= \int (1 - \psi_{n,v}) d\mathbb{P}_{\text{exp}} \\ &\leq 1 - \mathbb{P}_{\text{exp}}(\mathcal{E}_{\text{exp}}(P)) \\ &= 1 - P^{\otimes n}(\mathcal{E}_{\text{prof}}(P)) \\ &\leq 0.0075. \end{aligned}$$

7. Finally, if $n = 2$ or $n = 3$, then $n < 4$, and the explicit small-sample branch gives

$$\psi_{n,v}(\text{sample}, \text{seed}) = 0 \quad \text{for every sample and seed.}$$

□

Proof of Theorem 2. 1. Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$. Its parameter restrictions give

$$1 < p \leq 2, \quad 0 < \alpha, \beta \leq 1, \quad \frac{1}{4} \leq \gamma \leq 1.$$

Using the quantities in the statement, write

$$D_p = 1 + 2\alpha + \frac{\beta}{q} + \frac{S}{2\gamma}.$$

Then

$$0 < q \leq \frac{1}{2}, \quad S > 0, \quad t \geq 0, \quad D_p > 1, \quad E_0 > 0, \quad E_4 > 0.$$

In particular,

$$0 < E \leq \gamma \leq 1, \quad \frac{E}{S} < 2, \quad \frac{E}{p-1} < 1.$$

Indeed, the three upper bounds follow from

$$\frac{E}{\gamma} \leq \frac{2q}{2\gamma + q} \leq 1, \quad \frac{E}{S} \leq \frac{2}{D_p} < 2, \quad \frac{E}{p-1} \leq \frac{2\gamma}{p(2\gamma + q)} < 1.$$

The first uses $q \leq 1/2 \leq 2\gamma$, and the last uses $p > 1$ and $q > 0$.

For every positive argument of the geometric-series factor,

$$0 < 2^{-x} < 1, \quad A(x) > 0.$$

Since $\alpha > 0$ and $\lambda_0 > 0$, it follows that $B_* \geq 0$. Thus

$$C_v^{\text{att}} = 16\{16 + 5B_* + 1024\sqrt{A_*}\} \geq 256 > 0.$$

2. The rule named $\psi_{n,v}$ in Theorem 1 is precisely $\phi_{n,v}^*$. That result supplies, for every integer $n \geq 2$,

$$\mathbf{R}_n(P, \phi_{n,v}^*) \leq 0.0075 \quad (P \in H_0(v)).$$

It also supplies power, for $n \geq 4$, at the calibrated separation

$$R_{\text{att}}(n, v) = \frac{16}{3} \left\{ 16M^{-\gamma} + 5B + 1024M^{1/4}\sqrt{\Lambda_{n,v}} \right\}, \quad B = \mathbf{b}_{n,v},$$

provided $R_{\text{att}}(n, v) < D_v$.

The smallest sample sizes can be handled directly. By Lemma 1,

$$d_0 \leq D_v \leq \frac{1}{2}.$$

For $n \in \{2, 3\}$, the bound $E \leq 1$ gives

$$n^{-E} \geq n^{-1} \geq \frac{1}{3}, \quad D_v \leq \frac{1}{2} < \frac{256}{3} \leq C_v^{\text{att}} n^{-E}.$$

Consequently,

$$C_v^{\text{att}} n^{-E} < D_v \implies n \geq 4 \quad (n \geq 2).$$

The zero rule at $n = 2, 3$ has the asserted size by the same calibration result. We next establish the deterministic estimates needed to compare $R_{\text{att}}(n, v)$ with the stated public level.

3. Let $n \geq 4$, and use the tuning of [Algorithm 1](#) and [Definition 20](#). Explicitly,

$$s = \lfloor n/2 \rfloor \geq 2, \quad a = s^{-E} \in (0, 1], \quad M = \mathcal{R}(a^{-1/\gamma}), \quad K = \mathcal{R}(\max\{M, a^{-1/S}\}).$$

Upper dyadic rounding obeys

$$x_{\text{dyad}} \leq \mathcal{R}(x_{\text{dyad}}) \leq 2x_{\text{dyad}} \quad (x_{\text{dyad}} \geq 1), \quad \mathcal{R}(2^j) = 2^j \quad (j \in \mathbb{N}_0),$$

by the lower and upper ceiling bounds for $\log_2 x_{\text{dyad}}$. It is also increasing. Hence

$$a^{-1/\gamma} \leq M \leq 2a^{-1/\gamma}, \quad K \geq a^{-1/S},$$

and therefore

$$M^{-\gamma} \leq a, \quad K^{-S} \leq a.$$

The bounds $E/\gamma \leq 1$, $E/S < 2$, and $E/(p-1) < 1$ give

$$a^{-1/\gamma} = s^{E/\gamma} \leq s, \quad a^{-1/S} = s^{E/S} \leq s^2, \quad 1 \leq a^{-1/(p-1)} = s^{E/(p-1)} \leq s.$$

Thus $M \leq 2s$. For the fine rank, if $a^{-1/S} \leq M$, dyadic idempotence gives $K = M \leq 2s \leq 2s^2$; if $M < a^{-1/S}$, rounding gives $K \leq 2a^{-1/S} \leq 2s^2$. In either case,

$$M \leq 2s, \quad K \leq 2s^2.$$

For the increment-cutoff bounds, the public quantities in either score branch are

$$L = \log_2(K/M) \in \mathbb{N}_0, \quad R_j = 2^j M \quad (0 \leq j \leq L),$$

$$T_j = \mathcal{R} \left(\max \left\{ 1, \min \left\{ a^{-1/(p-1)}, \left[\frac{R_j^{-\alpha}}{a(R_j/K)^{\lambda_0}} \right]^{1/(p-1)} \right\} \right\} \right) \quad (1 \leq j \leq L).$$

The single-cutoff score uses the main cutoff

$$T_0 = \mathcal{R}(a^{-1/(p-1)}).$$

Since the argument defining each T_j lies between 1 and $a^{-1/(p-1)}$, rounding and monotonicity yield

$$1 \leq T_0 \leq 2s, \quad 1 \leq T_j \leq T_0 \quad (1 \leq j \leq L).$$

These statements also cover $L = 0$, when the increment index set is empty.

4. We now bound the bias ledger. The main-cutoff estimate is

$$T_0^{1-p} \leq \left(a^{-1/(p-1)}\right)^{1-p} = a.$$

In the single-cutoff branch, this and $K^{-S} \leq a$ immediately give

$$B = 400K^{-S} + 20T_0^{1-p} \leq 420a \leq B_*a.$$

For the multilevel branch, $S < \gamma$ and $\alpha < q$. Define the positive local tail targets by

$$z_{\text{tail},j} = \frac{R_j^{-\alpha}}{a(R_j/K)^{\lambda_0}} > 0 \quad (1 \leq j \leq L).$$

The cutoff formula gives

$$T_j \geq \min \left\{ a^{-1/(p-1)}, z_{\text{tail},j}^{1/(p-1)} \right\}.$$

If the minimum is the first term, its $(1-p)$ -power is a ; if it is the second term, that power is

$$\left(z_{\text{tail},j}^{1/(p-1)}\right)^{1-p} = z_{\text{tail},j}^{-1} = a(R_j/K)^{\lambda_0} R_j^{\alpha}.$$

Since $1-p < 0$, both cases imply

$$T_j^{1-p} \leq a + a(R_j/K)^{\lambda_0} R_j^{\alpha}.$$

The increment ranks satisfy $M \geq 1$ and $R_L \leq K$. Forward and backward geometric summation therefore give, for every $r_{\text{geom}} > 0$,

$$\begin{aligned} \sum_{j=1}^L R_{j-1}^{-r_{\text{geom}}} &\leq \sum_{j=0}^{L-1} 2^{-jr_{\text{geom}}} \leq A(r_{\text{geom}}), \\ \sum_{j=1}^L (R_j/K)^{r_{\text{geom}}} &\leq \sum_{j=0}^{L-1} 2^{-jr_{\text{geom}}} \leq A(r_{\text{geom}}), \\ \sum_{j=1}^L R_j^{r_{\text{geom}}} &\leq K^{r_{\text{geom}}} A(r_{\text{geom}}). \end{aligned}$$

For the second inequality, use $R_j/K \leq 2^{-(L-j)}$; the third follows by factoring out $K^{r_{\text{geom}}}$. Each bound remains valid for an empty sum.

Using $a_j = 40R_{j-1}^{-\alpha}$, $R_j = 2R_{j-1}$, and $2^\alpha \leq 2$, we obtain

$$\begin{aligned} a_j T_j^{1-p} &\leq 40a R_{j-1}^{-\alpha} + 40a 2^\alpha (R_j/K)^{\lambda_0} \\ &\leq 80a \left\{ R_{j-1}^{-\alpha} + (R_j/K)^{\lambda_0} \right\}. \end{aligned}$$

Summation consequently gives

$$\sum_{j=1}^L a_j T_j^{1-p} \leq 80a \{A(\alpha) + A(\lambda_0)\}.$$

Substituting into the multilevel bias ledger proves

$$\begin{aligned} B &= 400K^{-S} + 20T_0^{1-p} + 10 \sum_{j=1}^L a_j T_j^{1-p} \\ &\leq [420 + 800\{A(\alpha) + A(\lambda_0)\}] a = B_* a. \end{aligned}$$

Thus this bias estimate holds in both score branches.

5. Consider first the single-cutoff covariance branch,

$$S \geq \gamma \quad \text{or} \quad \alpha \geq q.$$

If $S \geq \gamma$, then $E \leq \gamma \leq S$. If $\alpha \geq q$, then

$$E \leq E_0 \leq q \leq \alpha < S.$$

Thus $a^{-1/S} = s^{E/S} \leq s$. The same two cases for fine-rank rounding used above now give

$$K \leq 2s.$$

Since $0 \leq 2 - p \leq 1$ and $T_0 \leq 2a^{-1/(p-1)}$, the main second-moment quantity satisfies

$$V_0 = 32T_0^{2-p} \leq 32 \cdot 2^{2-p} a^{-t} \leq 64a^{-t}.$$

The single-cutoff formulas $L_1 = 16\sqrt{V_0}$ and $L_2 = 4\sqrt{KV_0}$ yield the exact identity

$$\Lambda_{n,v} = \frac{256V_0}{s} + \frac{32KV_0}{s(s-1)}.$$

Because $s \geq 2$ and $K \leq 2s$,

$$32K \leq 64s \leq 128(s-1).$$

It follows that

$$\Lambda_{n,v} \leq \frac{384V_0}{s} \leq \frac{24576a^{-t}}{s}.$$

The tail-channel identity and the choice $E \leq E_0$ give

$$E_0 \left(2 + t + \frac{1}{2\gamma} \right) = 1, \quad E \left(2 + t + \frac{1}{2\gamma} \right) \leq 1.$$

Hence

$$\frac{a^{-1/(2\gamma)} a^{-t}}{s} = s^{E(1/(2\gamma)+t)-1} \leq s^{-2E} = a^2.$$

Finally, $M \leq 2a^{-1/\gamma}$ implies

$$\begin{aligned} \sqrt{M} \Lambda_{n,v} &\leq 24576\sqrt{2} \frac{a^{-1/(2\gamma)} a^{-t}}{s} \\ &\leq 24576\sqrt{2} a^2 \leq A_* a^2, \end{aligned}$$

where the last inequality follows from the displayed definition of A_* .

6. Continue with the multilevel branch $S < \gamma$, $\alpha < q$. Here $a^{-1/\gamma} \leq a^{-1/S}$. If $a^{-1/S} \leq M$, then

$$K = M \leq 2a^{-1/\gamma} \leq 2a^{-1/S};$$

if $M < a^{-1/S}$, rounding gives the same final bound. Thus

$$K \leq 2a^{-1/S}, \quad aK^\alpha \leq 2^\alpha a^{1-\alpha/S} = 2^\alpha a^{\beta/S} \leq 2.$$

The refinement depth satisfies

$$2^L \leq R_L \leq K \leq 2a^{-1/S}, \quad L \log 2 \leq \log 2 - \frac{\log a}{S}.$$

The exponential tangent inequality, applied at $-\log a - 1$, gives

$$-\log a \leq \exp(-\log a - 1) = a^{-1} \exp(-1), \quad -a \log a \leq \exp(-1).$$

Consequently,

$$aL \leq a + \frac{-a \log a}{S \log 2} \leq 1 + \frac{1}{\exp(1)S \log 2} = D_*.$$

The bound $T_j^{1-p} \leq a + a(R_j/K)^{\lambda_0} R_j^\alpha$ can be rewritten using

$$a(R_j/K)^{\lambda_0} R_j^\alpha = aK^\alpha (R_j/K)^{\alpha+\lambda_0}.$$

Geometric summation, $aK^\alpha \leq 2$, and the fact that A decreases on positive arguments now give

$$\begin{aligned} \sum_{j=1}^L T_j^{1-p} &\leq aL + aK^\alpha A(\alpha + \lambda_0) \\ &\leq D_* + 2A(\alpha). \end{aligned}$$

The same geometric estimates supply

$$\sum_{j=1}^L a_j \leq 80A(\alpha),$$

and, from $d_j = 110R_{j-1}^{-s_0} + 20T_j^{1-p}$,

$$\sum_{j=1}^L d_j \leq 110A(s_0) + 20D_* + 40A(\alpha).$$

Moreover, $T_j \leq T_0$ and $2 - p \geq 0$ imply

$$\sqrt{V_j} \leq \sqrt{V_0}, \quad \sum_{j=1}^L a_j \sqrt{V_j} \leq 80A(\alpha) \sqrt{V_0}.$$

Since $T_0 \geq 1$, we also have $\sqrt{V_0} \geq 1$. Inserting these bounds into the singleton ledger gives

$$\begin{aligned} L_1 &= 16\sqrt{V_0} + 8 \sum_{j=1}^L (d_j + a_j \sqrt{V_j}) \\ &\leq \{16 + 880A(s_0) + 160D_* + 960A(\alpha)\} \sqrt{V_0} = L_* \sqrt{V_0}. \end{aligned}$$

Using $V_0 \leq 64a^{-t}$, we obtain

$$L_1^2 \leq 64L_*^2 a^{-t}.$$

Together with $a^{-1/(2\gamma)}a^{-t}/s \leq a^2$, this proves

$$\sqrt{M} \frac{L_1^2}{s} \leq \sqrt{2} 64L_*^2 \frac{a^{-1/(2\gamma)}a^{-t}}{s} \leq \sqrt{2} 64L_*^2 a^2.$$

7. Still in the multilevel branch, define the backward summability exponent by

$$\eta = 1 - (\alpha + \lambda_0)t.$$

Because $\alpha < q$ and $t \geq 0$,

$$\eta \geq 1 - (q + \lambda_0)t = \frac{(p-1)(p+6)}{4p} = \eta_0 > 0.$$

Also,

$$0 \leq \alpha t = 1 - \eta - \lambda_0 t < 1.$$

The positive-power cutoff estimate follows from

$$T_j \leq 2 \max\{1, z_{\text{tail},j}^{1/(p-1)}\}.$$

Indeed, using $2^{2-p} \leq 2$ gives

$$V_j = 32T_j^{2-p} \leq 64 \max\{1, z_{\text{tail},j}^t\},$$

and therefore

$$\sqrt{V_j} \leq \sqrt{64} \left(1 + z_{\text{tail},j}^{t/2}\right).$$

Multiplying by $\sqrt{R_j}$ and expanding the local target yields

$$\sqrt{R_j V_j} \leq \sqrt{64} \left\{ \sqrt{R_j} + a^{-t/2} K^{\lambda_0 t/2} R_j^{\eta/2} \right\}.$$

Backward geometric summation implies

$$\sum_{j=1}^L \sqrt{R_j} \leq \sqrt{K} A(1/2), \quad \sum_{j=1}^L R_j^{\eta/2} \leq K^{\eta/2} A(\eta/2) \leq K^{\eta/2} A(\eta_0/2).$$

The fine-rank amplitude estimate also gives

$$\begin{aligned} \sqrt{K} &= (aK^\alpha)^{t/2} a^{-t/2} K^{(1-\alpha t)/2} \\ &\leq 2^{t/2} a^{-t/2} K^{(1-\alpha t)/2}. \end{aligned}$$

Since $\lambda_0 t + \eta = 1 - \alpha t$, summing the guarded increment bound proves

$$\begin{aligned} \sum_{j=1}^L \sqrt{R_j V_j} &\leq \sqrt{64} \left\{ 2^{t/2} A(1/2) + A(\eta_0/2) \right\} a^{-t/2} K^{(1-\alpha t)/2} \\ &= G_* a^{-t/2} K^{(1-\alpha t)/2}. \end{aligned}$$

Now

$$0 < \frac{1 - \alpha t}{2} \leq \frac{1}{2}, \quad t + \frac{1 - \alpha t}{S} = \frac{1 + \beta t}{S}.$$

Thus $K \leq 2a^{-1/S}$ converts the last bound into

$$\sum_{j=1}^L \sqrt{R_j V_j} \leq \sqrt{2} G_* a^{-(1+\beta t)/(2S)}.$$

Using the definition of L_2 and $(x + y)^2 \leq 2x^2 + 2y^2$ for its two nonnegative summands, we get

$$\begin{aligned} L_2^2 &= 16 \left(\sqrt{M V_0} + \sum_{j=1}^L \sqrt{R_j V_j} \right)^2 \\ &\leq 32 M V_0 + 32 \left(\sum_{j=1}^L \sqrt{R_j V_j} \right)^2 \\ &\leq 4096 s a^{-t} + 64 G_*^2 a^{-(1+\beta t)/S}. \end{aligned}$$

Here the last line uses $M \leq 2s$ and $V_0 \leq 64a^{-t}$. Since $s(s-1) \geq s^2/2$, it follows that

$$\frac{2L_2^2}{s(s-1)} \leq \frac{16384a^{-t}}{s} + \frac{256G_*^2 a^{-(1+\beta t)/S}}{s^2}.$$

For the second noise channel, direct algebra gives

$$2 + t = \frac{1}{q}, \quad 2S + 1 + \beta t + \frac{S}{2\gamma} = D_p,$$

and hence

$$E_4 \left(2 + \frac{1 + \beta t}{S} + \frac{1}{2\gamma} \right) = 2.$$

Using $E \leq E_4$, we obtain

$$\frac{a^{-1/(2\gamma)} a^{-(1+\beta t)/S}}{s^2} = s^{E(1/(2\gamma) + (1+\beta t)/S) - 2} \leq s^{-2E} = a^2.$$

Combining this with the tail-channel balance gives

$$\sqrt{M} \frac{2L_2^2}{s(s-1)} \leq \sqrt{2} \{16384 + 256G_*^2\} a^2.$$

Adding the bound $\sqrt{M} L_1^2/s \leq \sqrt{2} 64L_*^2 a^2$ therefore proves

$$\begin{aligned} \sqrt{M} \Lambda_{n,v} &= \sqrt{M} \left\{ \frac{L_1^2}{s} + \frac{2L_2^2}{s(s-1)} \right\} \\ &\leq \sqrt{2} \{64L_*^2 + 16384 + 256G_*^2\} a^2 \\ &\leq A_* a^2. \end{aligned}$$

This includes $L = 0$, since all the increment sums then vanish. Together with the single-cutoff bound $\sqrt{M} \Lambda_{n,v} \leq 24576\sqrt{2} a^2 \leq A_* a^2$, the covariance estimate holds in both branches.

8. The preceding estimates now give

$$M^{-\gamma} \leq a, \quad B \leq B_* a, \quad M^{1/4} \sqrt{\Lambda_{n,v}} = \sqrt{\sqrt{M} \Lambda_{n,v}} \leq \sqrt{A_*} a.$$

Consequently,

$$R_{\text{att}}(n, v) \leq \frac{16}{3} \{16 + 5B_* + 1024\sqrt{A_*}\} a.$$

The relation $n \leq 3s$, together with $0 < E \leq 1$, implies

$$a = s^{-E} \leq 3^E n^{-E} \leq 3n^{-E}.$$

Thus

$$R_{\text{att}}(n, v) \leq C_v^{\text{att}} n^{-E}.$$

If $C_v^{\text{att}} n^{-E} < D_v$, the bound $D_v \leq C_v^{\text{att}} n^{-E}$ for $n \in \{2, 3\}$ ensures $n \geq 4$. The last inequality then gives

$$R_{\text{att}}(n, v) < D_v, \quad d(P) \geq C_v^{\text{att}} n^{-E} \implies d(P) \geq R_{\text{att}}(n, v).$$

The power assertion of [Theorem 1](#) therefore yields

$$1 - R_n(P, \phi_{n,v}^*) \leq 0.0075$$

for every stipulated alternative.

To obtain the capped-radius bound, observe that $C_v^{\text{att}} n^{-E} > 0$ and that both error bounds are below $1/10$. When $C_v^{\text{att}} n^{-E} < D_v$, the explicit test is admissible in [Definition 5](#), so

$$B_n(v, C_v^{\text{att}} n^{-E}) \leq 0.0075 \leq \frac{1}{10}.$$

For an empty alternative set its worst-error supremum has value zero, giving the same inequality. The defining set in [Definition 6](#) therefore contains $C_v^{\text{att}} n^{-E}$. When $D_v \leq C_v^{\text{att}} n^{-E}$, its sentinel D_v instead gives

$$r_n^*(v) \leq D_v \leq C_v^{\text{att}} n^{-E}.$$

The defining set is nonempty and bounded below by zero in both cases. Hence

$$r_n^*(v) \leq C_v^{\text{att}} n^{-E} \quad (n \geq 2).$$

9. Let $n \geq N_v^{\text{att}}$. By the ceiling definition,

$$n \geq \max \left\{ 4, \left(\frac{2C_v^{\text{att}}}{d_0} \right)^{1/E} \right\}.$$

All bases are positive, and $-E < 0$. Taking this negative power therefore gives

$$n^{-E} \leq \left[\left(\frac{2C_v^{\text{att}}}{d_0} \right)^{1/E} \right]^{-E} = \left(\frac{2C_v^{\text{att}}}{d_0} \right)^{-1}.$$

Thus

$$C_v^{\text{att}} n^{-E} \leq \frac{d_0}{2} < d_0,$$

which proves the witness-threshold assertion. The bounds $M \leq 2s$, $K \leq 2s^2$, $1 \leq T_0 \leq 2s$, and $1 \leq T_j \leq T_0$ for $1 \leq j \leq L$ give the tuning assertions.

10. For compact uniformity, let

$$\mathcal{K} \subseteq \mathcal{V}$$

be compact in the product topology. At every admissible tuple,

$$p - 1 > 0, \quad S > 0, \quad \gamma > 0, \quad q > 0, \quad D_p > 0, \quad s_0 > 0, \quad \lambda_0 > 0, \quad \eta_0 > 0.$$

In particular, every geometric denominator appearing in the constants satisfies

$$1 - 2^{-x} > 0 \quad (x \in \{\alpha, \lambda_0, s_0, 1/2, \eta_0/2\}).$$

The displayed formulas consequently show that $E(v)$ and C_v^{att} are continuous at every admissible tuple: their operations are sums, products, quotients with nonzero denominators, minima, square roots, and real powers with positive bases.

Define the real threshold target on the admissible domain by

$$\Xi(v) = \max \left\{ 4, \left(\frac{2C_v^{\text{att}}}{d_0} \right)^{1/E(v)} \right\} \quad (v \in \mathcal{V}).$$

Since $E(v) > 0$, $C_v^{\text{att}} > 0$, and $d_0 > 0$, this function is continuous there as well.

If $\mathcal{K}$ is nonempty, compactness supplies tuples $v_C, v_E, v_N \in \mathcal{K}$ satisfying

$$C_{v_C}^{\text{att}} = \max_{v \in \mathcal{K}} C_v^{\text{att}}, \quad E(v_E) = \min_{v \in \mathcal{K}} E(v), \quad \Xi(v_N) = \max_{v \in \mathcal{K}} \Xi(v).$$

Choose

$$C_{\mathcal{K}} = C_{v_C}^{\text{att}}, \quad N_{\mathcal{K}} = \Xi(v_N) + 1, \quad e_{\mathcal{K}} = E(v_E) > 0.$$

The ceiling bound gives, for every $v \in \mathcal{K}$,

$$C_v^{\text{att}} \leq C_{\mathcal{K}}, \quad N_v^{\text{att}} = \lceil \Xi(v) \rceil < \Xi(v) + 1 \leq N_{\mathcal{K}}, \quad e_{\mathcal{K}} \leq E(v).$$

If $\mathcal{K}$ is empty, choose

$$(C_{\mathcal{K}}, N_{\mathcal{K}}, e_{\mathcal{K}}) = (0, 0, 1);$$

the universal bounds are then vacuous.

11. Finally, let $w = (\alpha, \beta, \gamma) \in \mathcal{W}$, and define its matched admissible tuple by

$$v_b = (2, \alpha, \beta, \gamma) \in \mathcal{V}.$$

At this tuple,

$$q = \frac{1}{2}, \quad t = 0, \quad S = \alpha + \beta,$$

so substitution into the two exponent formulas gives

$$E_0(v_b) = \frac{2\gamma}{4\gamma + 1}, \quad E_4(v_b) = \frac{2S}{1 + 2S + S/(2\gamma)}.$$

Taking their minimum proves

$$E(v_b) = E_b.$$

The definitions in [Definitions 7, 9 and 15](#) give

$$\mathcal{M}_w^b \subseteq \mathcal{M}_{v_b}, \quad H_0^b(w) \subseteq H_0(v_b), \quad \phi_{n,w}^{*,b} = \phi_{n,v_b}^*,$$

and

$$C_{(2,w)}^{\text{att}} n^{-E_b} = C_{v_b}^{\text{att}} n^{-E(v_b)} > 0.$$

Moreover, [Proposition 1](#) supplies the exact cap identity

$$D_w^b = D_{v_b} \geq d_0.$$

The original-model size bound therefore applies to every bounded null law. If

$$C_{(2,w)}^{\text{att}} n^{-E_b} < D_w^b,$$

the cap identity makes the original-model power condition hold at v_b . Thus every bounded-model law with distance at least this level satisfies

$$1 - \mathbf{R}_n(P, \phi_{n,w}^{*,b}) \leq 0.0075.$$

When this positive level is below D_w^b , the resulting bounded-model risk bound is

$$B_n^b(w, C_{(2,w)}^{\text{att}} n^{-E_b}) \leq 0.0075 \leq \frac{1}{10},$$

so the level belongs to the defining set of [Definition 13](#). When it is at least D_w^b , the sentinel gives the radius bound directly. Therefore, for every $n \geq 2$,

$$r_n^{*,b}(w) \leq C_{(2,w)}^{\text{att}} n^{-E_b}.$$

□

Proof of [Theorem 3](#). Fix an admissible tuple $v = (p, \alpha, \beta, \gamma)$, and put $w = (\alpha, \beta, \gamma)$. Throughout, use the quantities and selected constructions of [Definition 15](#) and [Algorithm 6](#).

1. Fix $n \geq 2$, and first suppose $F(p) < 1$. By [Lemma 8](#), both selected priors are normalized, and

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}) &\subseteq H_0(v), \\ \text{supp}_+(\pi_{1,n,v}) &\subseteq \{P \in \mathcal{M}_v : c_v n^{-E_4(p)} \leq d(P) < d_0\}, \\ d_0 \leq D_v, \quad \text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) &< \frac{1}{2}, \quad B_n(v, c_v n^{-E_4(p)}) \geq \frac{2}{5}. \end{aligned}$$

These conclusions cover every $n \geq 2$, including the paired-tent selection below $N_{\text{low}}(v)$.

The scale identification follows by subtracting the two exponents:

$$\begin{aligned} E_4(p) - E_0(p) &= \frac{2S(2\gamma + q) - 2\gamma q D_p}{(2\gamma + q) D_p} \\ &= \frac{2\gamma q}{(2\gamma + q) D_p} \{F(p) - 1\}. \end{aligned}$$

Admissibility gives $q > 0$, $S > 0$, $\gamma > 0$, and $D_p > 0$. Thus $F(p) < 1$ implies

$$E(p) = E_4(p), \quad \rho_n(v) = n^{-E_4(p)}.$$

We also verify the arm support at the selected magnitude. When $n \geq N_{\text{low}}(v)$, the rank conditions are supplied by [Lemma 8](#), and the tuning agrees with [Lemma 6](#):

$$b = \frac{K^{-\beta}}{256}, \quad \varepsilon_{n,v} = (16b)^{1/q} = 16^{-1/q} K^{-\beta/q}, \quad L_{n,v} = \varepsilon_{n,v}^{-1/p}.$$

The normalized table in [Definition 29](#) assigns its outcome mass to $0, -L_{n,v}, L_{n,v}$. Conditioning on either treatment arm therefore retains all its mass on those three points. When $n < N_{\text{low}}(v)$, the selected paired-tent construction has the same arm-support conclusion by [Lemma 5](#). Consequently, in either selection,

$$Q_{a,P}(\{0, -L_{n,v}, L_{n,v}\} \mid x) = 1$$

for each $a \in \{0, 1\}$, for λ -almost every x , and for every positive-weight law in either prior. Finally, normalization and nonnegative weights ensure that the alternative prior has a positive-weight law. Choosing such a law P gives

$$c_v \rho_n(v) \leq d(P) < d_0 \leq D_v.$$

Moreover, $N_{\text{low}}(v) \geq 2$, $k_v > 0$, and $c_0^e > 0$, so the defining minimum for c_v is positive. Since $n > 0$, also $\rho_n(v) > 0$. Hence

$$0 < c_v \rho_n(v) < D_v,$$

and every positive-weight alternative law has distance strictly below D_v . This proves the first group of conclusions in the rough phase.

2. Suppose instead $F(p) \geq 1$. The exponent-difference identity from the preceding step gives

$$E(p) = E_0(p), \quad c_v = c_0^e, \quad c_v \rho_n(v) = c_0^e n^{-E_0(p)}.$$

The selector of [Algorithm 6](#) chooses the paired-tent priors for every $n \geq 2$. Their finite sign distributions are normalized. All null sign configurations give the same zero-effect null law P_0 , so

$$\pi_{0,n,v} = \delta_{P_0}, \quad \mathbb{P}_{0,n,v} = P_0^{\otimes n}.$$

By [Lemma 5](#),

$$\begin{aligned} P_0 &\in H_0(v), \\ \text{supp}_+(\pi_{1,n,v}) &\subseteq \{P \in \mathcal{M}_v : c_0^e n^{-E_0(p)} \leq d(P) < d_0\}, \\ d_0 &\leq D_v, \quad \text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) < \frac{1}{2}, \\ B_n(v, c_0^e n^{-E_0(p)}) &\geq \frac{2}{5}. \end{aligned}$$

That construction also gives the required three-point conditional arm support at $L_{n,v}$. Its alternative prior has a positive-weight law; for any such law,

$$c_v \rho_n(v) \leq d(P) < d_0 \leq D_v.$$

Positivity of c_0^e and $n^{-E_0(p)}$ yields $0 < c_v \rho_n(v) < D_v$. Substituting the displayed scale identity proves the first group of conclusions in this phase as well. The equality case $F(p) = 1$ is included here.

3. We next prove divergence of the selected magnitude, again splitting at $F(p) < 1$. In that phase, for all $n \geq N_{\text{low}}(v)$, the selected fine rank satisfies

$$K \geq H_{\text{fine}} n^{2/D_p} \longrightarrow \infty,$$

and moment normalization gives

$$L_{n,v} = \left(16^{-1/q} K^{-\beta/q}\right)^{-1/p} = 16^{1/(pq)} K^{\beta/(pq)}.$$

Both $2/D_p$ and $\beta/(pq)$ are positive, so $L_{n,v} \rightarrow \infty$. The finitely many preceding selections leave this limit unchanged.

When $F(p) \geq 1$, the paired-tent formula applies throughout:

$$N = 2 \left\lceil n^{2q/(2\gamma+q)} \right\rceil \geq n^{2q/(2\gamma+q)} \longrightarrow \infty, \quad L_{n,v} = N^{\gamma/(p-1)}.$$

Here $2q/(2\gamma+q) > 0$ and $\gamma/(p-1) > 0$, proving the same divergence.

4. Assume $p < 2$. At the matched smoothness tuple, write

$$D_2 = 1 + 2S + \frac{S}{2\gamma}.$$

The identity $q^{-1} = 2 + (2-p)/(p-1)$ gives

$$D_p = D_2 + \frac{\beta(2-p)}{p-1}.$$

Direct subtraction then yields the two strictly positive differences

$$E_0(2) - E_0(p) = \frac{4\gamma^2(2-p)}{(4\gamma+1)(2\gamma p + p - 1)} > 0,$$

$$E_4(2) - E_4(p) = \frac{2S\beta(2-p)}{(p-1)D_p D_2} > 0.$$

All denominators are positive by admissibility. Since $E(p)$ is the minimum of its two component exponents,

$$E(p) \leq E_0(p) < E_0(2), \quad E(p) \leq E_4(p) < E_4(2).$$

Thus the exponent gap, defined on this face by

$$\delta_v = E(2) - E(p),$$

is strictly positive. For every positive integer n ,

$$\frac{\rho_n(v)}{\rho_n^b(w)} = n^{\delta_v} \longrightarrow \infty.$$

5. Continue with $p < 2$. Since $c_v > 0$, the preceding limit permits an integer $N_{\text{dom}}(v) \in \mathbb{N}$ such that

$$\frac{\rho_n(v)}{\rho_n^b(w)} \geq \frac{C_w^b}{c_v} \quad \text{for every integer } n \geq N_{\text{dom}}(v) \text{ with } n > 0.$$

Multiplying by the positive bounded scale and by c_v gives

$$C_w^b \rho_n^b(w) \leq c_v \rho_n(v).$$

Define the required threshold by

$$N(v) = \max\{2, N_{\text{dom}}(v), N_w^b\}.$$

For $n \geq N(v)$, [Theorem 2](#), applied at $(2, w)$, and [Proposition 1](#) give

$$C_w^b \rho_n^b(w) < d_0 \leq D_w^b.$$

The bounded total test therefore satisfies

$$R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad (P \in H_0^b(w)),$$

and

$$1 - R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad (P \in \mathcal{M}_w^b, d(P) \geq c_v \rho_n(v)).$$

Its size is below $1/10$, so

$$B_n^b(w, c_v \rho_n(v)) \leq 0.0075.$$

Let π_0, π_1 be any normalized finite priors satisfying the bounded support conditions in the statement. We establish the two-prior inequality used to compare their mixtures. For any randomized test ϕ , define its seed average on the complete record space by

$$f_\phi : ([0, 1] \times \{0, 1\} \times \mathbb{R})^n \longrightarrow [0, 1], \quad f_\phi(\mathcal{D}_n) = \int_{[0,1]} \phi(\mathcal{D}_n, U) \lambda(dU).$$

Bounded product integration and finite-sum linearity give, for either prior,

$$\int f_\phi d\mathbb{P}_{\pi_\nu, n} = \int R_n(P, \phi) \pi_\nu(dP), \quad \nu \in \{0, 1\}.$$

The bounded-function form of the total-variation inequality gives

$$\left| \int f_\phi d\mathbb{P}_{\pi_1, n} - \int f_\phi d\mathbb{P}_{\pi_0, n} \right| \leq \text{TV}(\mathbb{P}_{\pi_0, n}, \mathbb{P}_{\pi_1, n}).$$

If ϕ has size at most $1/10$ on the bounded null, normalization and the null support condition imply

$$\int R_n(P, \phi) \pi_0(dP) \leq \frac{1}{10}.$$

The alternative support condition and normalization consequently yield

$$\begin{aligned} \sup_{\substack{P \in \mathcal{M}_w^b \\ d(P) \geq c_v \rho_n(v)}} \{1 - R_n(P, \phi)\} &\geq \int \{1 - R_n(P, \phi)\} \pi_1(dP) \\ &\geq \frac{9}{10} - \text{TV}(\mathbb{P}_{\pi_0, n}, \mathbb{P}_{\pi_1, n}). \end{aligned}$$

Weights equal to zero contribute zero in these averages; every positive-weight law satisfies the stipulated support condition. Normalization ensures a positive-weight alternative law, and the zero rejection map is an admissible test. Taking the infimum over null-sized tests therefore gives

$$\frac{9}{10} - \text{TV}(\mathbb{P}_{\pi_0,n}, \mathbb{P}_{\pi_1,n}) \leq B_n^b(w, c_v \rho_n(v)).$$

Combining this with the attained risk bound proves

$$\text{TV}(\mathbb{P}_{\pi_0,n}, \mathbb{P}_{\pi_1,n}) \geq 0.8925 \geq \frac{1}{2}.$$

The threshold depends on v , and the argument applies to every normalized finite prior pair in the statement.

6. Now assume $p = 2$, and fix $n \geq 2$. Then $v = (2, \alpha, \beta, \gamma)$. We first verify the original-model support conclusions for the selected bounded priors, splitting at $F(2) < 1$.

In the rough phase, [Lemma 8](#) supplies

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}^b) &\subseteq H_0^b(w), \\ \text{supp}_+(\pi_{1,n,v}^b) &\subseteq \{P \in \mathcal{M}_w^b : c_v n^{-E_4(2)} \leq d(P) < d_0\}. \end{aligned}$$

By the phase identity, $\rho_n(v) = n^{-E_4(2)}$. The model inclusion of [Proposition 1](#) preserves the effect, and the null constancy condition is the same in both models. Hence

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}^b) &\subseteq H_0(v), \\ \text{supp}_+(\pi_{1,n,v}^b) &\subseteq \{P \in \mathcal{M}_v : c_v \rho_n(v) \leq d(P)\}. \end{aligned}$$

In the complementary phase $F(2) \geq 1$, the selected bounded priors are the signed-binary paired-tent priors. To identify their null support, define the balanced deterministic zero-effect law by

$$P_{\text{zero}}(dx, dA, dy) = \lambda(dx) \frac{\delta_0 + \delta_1}{2}(dA) \delta_0(dy).$$

It has propensity $1/2$, baseline mean zero, and effect zero. Its smoothness conditions hold because these functions are constant, and its raw moment is zero. Its signed-binary conversion from [Proposition 1](#) is

$$P_{\text{zero}}^{\text{bin}}(dx, dA, dy) = \lambda(dx) \frac{\delta_0 + \delta_1}{2}(dA) \frac{\delta_{-1} + \delta_1}{2}(dy).$$

This conversion preserves the means and effect, and its conditional second moment is one. Thus $P_{\text{zero}}^{\text{bin}} \in H_0(v)$. Every selected null sign configuration gives precisely this law, so

$$\text{supp}_+(\pi_{0,n,v}^b) \subseteq \{P_{\text{zero}}^{\text{bin}}\} \subseteq H_0(v).$$

For the alternatives, [Lemma 5](#) and [Proposition 1](#) give original-model membership and

$$d(P) \geq c_0^e n^{-E_0(2)} = c_v \rho_n(v) \quad (P \in \text{supp}_+(\pi_{1,n,v}^b)).$$

This proves the original-model support conclusions in both phases.

7. The bounded risk lower bound follows from the same phase split. When $F(2) < 1$, [Lemma 8](#) gives

$$B_n^b(w, c_v n^{-E_4(2)}) \geq \frac{2}{5}.$$

When $F(2) \geq 1$, [Lemma 5](#) gives

$$B_n^b(w, c_0^e n^{-E_0(2)}) \geq \frac{2}{5}.$$

The phase identities identify each displayed separation with $c_v \rho_n(v)$. The risk comparison in [Proposition 1](#) consequently yields

$$\frac{2}{5} \leq B_n^b(w, c_v \rho_n(v)) \leq B_n(v, c_v \rho_n(v)).$$

8. We assemble the remaining second-moment conclusions. Evaluation at $v = (2, w)$, together with [Proposition 1](#), gives

$$\begin{aligned} \rho_n(v) &= \rho_n^b(w), & c_v &= c_w^b, & C_v &= C_w^b, & N_v &= N_w^b, \\ D_v &= D_w^b. \end{aligned}$$

For $F(2) < 1$, the selected-prior conclusions of [Lemma 8](#) apply for every $n \geq 2$, covering the bounded rough construction above its threshold and the signed-binary paired-tent construction below it. For $F(2) \geq 1$, use the signed-binary conclusions of [Lemma 5](#). The signed-binary model inclusion of [Proposition 1](#) supplies bounded membership, with null constancy retained. In either case, both selected bounded priors are normalized and

$$\begin{aligned} \text{supp}_+(\pi_{0,n,v}^b) &\subseteq H_0^b(w), \\ \text{supp}_+(\pi_{1,n,v}^b) &\subseteq \{P \in \mathcal{M}_w^b : c_v \rho_n(v) \leq d(P) < d_0\}, \\ \text{TV}(\mathbb{P}_{0,n,v}^b, \mathbb{P}_{1,n,v}^b) &< \frac{1}{2}, & B_n^b(w, c_v \rho_n(v)) &\geq \frac{2}{5}. \end{aligned}$$

For any finite representation of either prior, nonnegative weights summing to one satisfy

$$\sum_{j: \omega_j > 0} \omega_j = \sum_j \omega_j = 1.$$

Indeed, every omitted weight is zero. At least one weight is positive, and combining weights of coincident laws preserves their sum. Therefore

$$\text{supp}_+(\pi_{\nu,n,v}^b) \neq \emptyset, \quad \sum_{P \in \text{supp}_+(\pi_{\nu,n,v}^b)} \pi_{\nu,n,v}^b(\{P\}) = 1, \quad \nu \in \{0, 1\}.$$

Choose a positive-weight alternative law P . Its placement, and the witness lower bound from [Proposition 1](#), give

$$0 < c_v \rho_n(v) \leq d(P) < d_0 \leq D_w^b.$$

9. We obtain the lower capped-radius bound first in the bounded model. Every successful bounded separation satisfies

$$r_{\text{succ}} \in (0, D_w^b), \quad B_n^b(w, r_{\text{succ}}) \leq \frac{1}{10}.$$

If $r_{\text{succ}} < c_v \rho_n(v)$, containment of the separated alternative classes gives

$$B_n^b(w, c_v \rho_n(v)) \leq B_n^b(w, r_{\text{succ}}) \leq \frac{1}{10},$$

contradicting the lower bound $2/5$. The cap D_w^b also exceeds $c_v \rho_n(v)$. Thus every member of the nonempty set defining the capped infimum is at least $c_v \rho_n(v)$, and

$$c_v \rho_n(v) \leq r_n^{*,b}(w).$$

Now [Proposition 1](#) gives

$$c_v \rho_n(v) \leq r_n^{*,b}(w) = r_n^{*,\text{bin}}(w) \leq r_n^*(v).$$

The upper capped-radius bounds and both size bounds follow from [Theorem 2](#):

$$r_n^*(v) \leq C_v \rho_n(v), \quad r_n^{*,b}(w) \leq C_w^b \rho_n^b(w),$$

$$R_n(P, \phi_{n,v}^*) \leq 0.0075 \quad (P \in H_0(v)), \quad R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad (P \in H_0^b(w)).$$

Substituting the matching scale and multiplier identities proves both asserted two-sided radius bounds.

10. Suppose $C_v \rho_n(v) < D_v$. The matching identities give

$$C_w^b \rho_n^b(w) < D_w^b.$$

To prove nonemptiness in the bounded model, suppose its separated class were empty. Every bounded-model law would then satisfy

$$d(P) < C_w^b \rho_n^b(w).$$

The bounded model is nonempty, as witnessed by a positive-weight law from the selected alternative prior. The displayed bound would imply

$$D_w^b \leq C_w^b \rho_n^b(w),$$

contradicting the strict inequality. Hence there is a bounded-model law with

$$d(P) \geq C_w^b \rho_n^b(w) = C_v \rho_n(v).$$

Its original-model membership follows from [Proposition 1](#), proving nonemptiness of both separated classes.

Under this same strict-separation condition, [Theorem 2](#) supplies

$$1 - R_n(P, \phi_{n,v}^*) \leq 0.0075 \quad (P \in \mathcal{M}_v, d(P) \geq C_v \rho_n(v)),$$

and

$$1 - R_n(P, \phi_{n,w}^{*,b}) \leq 0.0075 \quad (P \in \mathcal{M}_w^b, d(P) \geq C_w^b \rho_n^b(w)).$$

11. Finally, let $n \geq N_v$. Positivity of the component exponents implies $E(2) > 0$, and [Theorem 2](#) gives $C_v > 0$. The threshold formula yields

$$n \geq \max \left\{ 4, \left(\frac{2C_v}{d_0} \right)^{1/E(2)} \right\}.$$

Since raising positive numbers to the power $-E(2)$ reverses their order,

$$\begin{aligned} C_v \rho_n(v) &= C_v n^{-E(2)} \\ &\leq C_v \left\{ \left(\frac{2C_v}{d_0} \right)^{1/E(2)} \right\}^{-E(2)} = \frac{d_0}{2}. \end{aligned}$$

Also, the witness-threshold conclusion of [Theorem 2](#) gives

$$C_v \rho_n(v) < d_0 \leq D_v.$$

The matching identities therefore imply

$$C_w^b \rho_n^b(w) < D_w^b.$$

At $n = 2, 3$, the small-sample branch of [Algorithm 1](#) sets the total rule to zero. Evaluating that same branch at $v = (2, w)$ gives, for every sample z ,

$$\phi_{n,v}^*(z) = 0, \quad \phi_{n,w}^{*,b}(z) = 0.$$

This completes all conclusions on the second-moment face.

□

Proof of [Theorem 4](#). 1. Fix an admissible public tuple and its matched smoothness tuple:

$$v = (p, \alpha, \beta, \gamma) \in \mathcal{V}, \quad w = (\alpha, \beta, \gamma), \quad v_2 = (2, \alpha, \beta, \gamma) \in \mathcal{V}.$$

Use the quantities of [Definition 15](#), in particular

$$q = \frac{p-1}{p}, \quad S = \alpha + \beta, \quad E(p) = \min\{E_0(p), E_4(p)\}.$$

Admissibility gives

$$p-1 > 0, \quad 0 < q \leq \frac{1}{2}, \quad S > 0, \quad \gamma > 0, \quad 2\gamma + q > 0, \quad D_p > 0.$$

Indeed, $q \leq 1/2$ follows by multiplying by $p > 0$ and using $p \leq 2$; every term added to 1 in D_p is positive. Consequently,

$$E_0(p) > 0, \quad E_4(p) > 0, \quad E(p) > 0.$$

The same conclusions apply at v_2 .

We first establish the phase selection needed for the lower bounds. Since

$$pq = p-1, \quad \frac{1}{q} - 1 = \frac{1}{p-1},$$

subtracting the two channel exponents gives

$$\begin{aligned} E_4(p) - E_0(p) &= \frac{2S(2\gamma + q) - 2\gamma q D_p}{(2\gamma + q)D_p} \\ &= \frac{2\gamma q}{(2\gamma + q)D_p} \left(\frac{2S}{q} - 2\alpha - \frac{\beta}{q} + \frac{S}{2\gamma} - 1 \right) \\ &= \frac{2\gamma q}{(2\gamma + q)D_p} (F(p) - 1). \end{aligned}$$

The coefficient multiplying $F(p) - 1$ is strictly positive. Thus

$$\begin{aligned} F(p) \geq 1 &\implies E_0(p) \leq E_4(p) \implies E(p) = E_0(p), \\ F(p) < 1 &\implies E_4(p) < E_0(p) \implies E(p) = E_4(p). \end{aligned}$$

These implications also hold with $p = 2$.

The definition of the lower multiplier gives

$$N_{\text{low}}(v) \geq 2, \quad c_0^e > 0, \quad k_v > 0, \quad c_v > 0,$$

because every power occurring in either branch has a positive base. Applying this argument at v_2 , and applying [Theorem 2](#) at both tuples, yields

$$c_w^b > 0, \quad C_v > 0, \quad C_w^b > 0.$$

For every $n \geq 2$, positivity of real powers also gives

$$\rho_n(v) > 0, \quad \rho_n^b(w) > 0.$$

2. Fix an integer $n \geq 2$. We obtain the bounded lower bound by splitting according to the phase at v_2 .

If $F(2) < 1$, then

$$E(2) = E_4(2), \quad c_w^b \rho_n^b(w) = c_w^b n^{-E_4(2)}.$$

The bounded conclusion of [Lemma 8](#) supplies normalized finite alternative witnesses and

$$B_n^b(w, c_w^b \rho_n^b(w)) \geq \frac{2}{5}.$$

A normalized finite prior has a positive-weight atom: otherwise every nonnegative weight would be zero, contradicting that their sum is one. Choosing such an alternative atom gives

$$\exists P \in \mathcal{M}_w^b : \quad c_w^b \rho_n^b(w) \leq d(P).$$

If $F(2) \geq 1$, the public definitions and the phase selection give

$$E(2) = E_0(2) = \frac{2\gamma}{4\gamma + 1}, \quad c_w^b = c_0^e = \frac{4^{-\gamma}}{16\sqrt{3}}.$$

The signed-binary witnesses in [Lemma 5](#) therefore give the same risk bound and a positive-weight alternative satisfying

$$P \in \mathcal{M}_w^{\text{bin}} \subseteq \mathcal{M}_w^b, \quad c_w^b \rho_n^b(w) \leq d(P),$$

where the inclusion follows from [Proposition 1](#). Hence, in both cases,

$$\exists P \in \mathcal{M}_w^b : \quad c_w^b \rho_n^b(w) \leq d(P), \quad B_n^b(w, c_w^b \rho_n^b(w)) \geq \frac{2}{5}.$$

We also verify that this bounded lower separation lies below the bounded cap. For any admissible tuple, the branch $F(p) \geq 1$ gives $c_v = c_0^e$. In the other branch, the phase identity gives $E_0(p) - E_4(p) > 0$, so

$$N_{\text{low}}(v)^{-(E_0(p)-E_4(p))} \leq 1, \quad c_v \leq c_0^e N_{\text{low}}(v)^{-(E_0(p)-E_4(p))} \leq c_0^e.$$

Since $E(p) > 0$ and $n \geq 2$,

$$0 < \rho_n(v) \leq 1, \quad 0 < c_v \rho_n(v) \leq c_0^e \leq \frac{1}{16\sqrt{3}} < \frac{1}{8\sqrt{2}} = d_0.$$

The strict numerical inequality follows from $\sqrt{2} < 2\sqrt{3}$, obtained by squaring these positive quantities. Applying this bound at v_2 , and using $d_0 \leq D_w^b$ from [Proposition 1](#), gives

$$0 < c_w^b \rho_n^b(w) < d_0 \leq D_w^b.$$

For the original model, [Theorem 3](#) supplies

$$0 < c_v \rho_n(v) < D_v, \quad B_n(v, c_v \rho_n(v)) \geq \frac{2}{5},$$

and a normalized finite alternative prior. Choosing a positive-weight atom as above gives

$$\exists P \in \mathcal{M}_v : \quad c_v \rho_n(v) \leq d(P) < D_v.$$

We now pass from these failures to the capped radii. For $0 < r_1 \leq r_2 < D_v$, with a nonempty alternative class at r_2 , inclusion of the alternative classes gives, for every admissible test,

$$\sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq r_2}} \{1 - \mathbf{R}_n(P, \phi)\} \leq \sup_{\substack{P \in \mathcal{M}_v \\ d(P) \geq r_1}} \{1 - \mathbf{R}_n(P, \phi)\}.$$

The errors lie in $[0, 1]$, and the null-size constraint is unchanged. Taking infima over tests therefore yields

$$B_n(v, r_2) \leq B_n(v, r_1).$$

The identical argument gives bounded-risk monotonicity.

If a successful separation $\tilde{r} \in (0, D_v)$ satisfied $\tilde{r} < c_v \rho_n(v)$, monotonicity and its success condition would imply

$$\frac{2}{5} \leq B_n(v, c_v \rho_n(v)) \leq B_n(v, \tilde{r}) \leq \frac{1}{10},$$

a contradiction. The sentinel D_v also exceeds $c_v \rho_n(v)$. Thus every element of the nonempty set defining the capped infimum in [Definition 6](#) is at least $c_v \rho_n(v)$. The bounded argument uses its positive-weight witness and the legal separation established above. Consequently,

$$c_v \rho_n(v) \leq r_n^*(v), \quad c_w^b \rho_n^b(w) \leq r_n^{*,b}(w).$$

The displayed risk bounds also imply both asserted strict inequalities against $1/10$.

3. By [Theorem 2](#), the public tests satisfy

$$r_n^*(v) \leq C_v \rho_n(v), \quad r_n^{*,b}(w) \leq C_w^b \rho_n^b(w).$$

Together with [Proposition 1](#) and the preceding lower bounds, this gives

$$\begin{aligned} c_v \rho_n(v) &\leq r_n^*(v) \leq C_v \rho_n(v), \\ c_w^b \rho_n^b(w) &\leq r_n^{*,b}(w) = r_n^{*,\text{bin}}(w) \leq C_w^b \rho_n^b(w). \end{aligned}$$

The same attainment result supplies

$$\begin{aligned} R_n(P, \phi_{n,v}^*) &\leq 0.0075 < \frac{1}{10} & (P \in H_0(v)), \\ R_n(P, \phi_{n,w}^{*,b}) &\leq 0.0075 < \frac{1}{10} & (P \in H_0^b(w)). \end{aligned}$$

Under the respective conditions

$$C_v \rho_n(v) < D_v, \quad C_w^b \rho_n^b(w) < D_w^b,$$

it supplies, respectively,

$$\begin{aligned} 1 - R_n(P, \phi_{n,v}^*) &\leq 0.0075 < \frac{1}{10} & (P \in \mathcal{M}_v, d(P) \geq C_v \rho_n(v)), \\ 1 - R_n(P, \phi_{n,w}^{*,b}) &\leq 0.0075 < \frac{1}{10} & (P \in \mathcal{M}_w^b, d(P) \geq C_w^b \rho_n^b(w)). \end{aligned}$$

The bounded radius is positive by its lower bound. Dividing the matched bounds therefore gives

$$\frac{c_v \rho_n(v)}{C_w^b \rho_n^b(w)} \leq \Delta_n(v) \leq \frac{C_v \rho_n(v)}{c_w^b \rho_n^b(w)}.$$

For $n > 0$, the law of real powers gives

$$\frac{\rho_n(v)}{\rho_n^b(w)} = \frac{n^{-E(p)}}{n^{-E(2)}} = n^{E(2)-E(p)}.$$

Hence

$$\frac{c_v}{C_w^b} n^{E(2)-E(p)} \leq \Delta_n(v) \leq \frac{C_v}{c_w^b} n^{E(2)-E(p)}.$$

Finally, [Proposition 1](#) gives

$$0 < r_n^{*,b}(w) \leq r_n^*(v), \quad 1 \leq \Delta_n(v).$$

4. Since $E(p) > 0$ and $E(2) > 0$, the definitions of the scales give

$$\rho_n(v) = n^{-E(p)} \longrightarrow 0, \quad \rho_n^b(w) = n^{-E(2)} \longrightarrow 0.$$

The public thresholds are those of [Definition 15](#), evaluated at v and v_2 . Their formulas in [Theorem 2](#) give

$$N_v \geq 4, \quad N_w^b \geq 4.$$

Its witness-threshold conclusion therefore applies to every natural number satisfying the corresponding threshold and yields

$$\begin{aligned} n \geq N_v &\implies C_v \rho_n(v) < d_0, \\ n \geq N_w^b &\implies C_w^b \rho_n^b(w) < d_0. \end{aligned}$$

5. We make the exponent difference explicit, always preserving w when evaluating at 2. Define

$$D_2 = 1 + 2S + \frac{S}{2\gamma}, \quad F_2 = 2S + \frac{S}{2\gamma}.$$

Substitution of $q = (p - 1)/p$ gives

$$D_p = D_2 + \frac{\beta(2 - p)}{p - 1}, \quad F(p) = F_2 + \frac{(2\alpha + \beta)(2 - p)}{p - 1} \geq F_2.$$

For the supplied-propensity channel, a common denominator yields

$$\begin{aligned} E_0(2) - E_0(p) &= \frac{2\gamma}{4\gamma + 1} - \frac{2\gamma(p - 1)}{2\gamma p + p - 1} \\ &= \frac{4\gamma^2(2 - p)}{(4\gamma + 1)(2\gamma p + p - 1)}. \end{aligned}$$

Both denominators are positive. For the rough channel, the identity for D_p gives

$$\begin{aligned} E_4(2) - E_4(p) &= \frac{2S}{D_2} - \frac{2S}{D_p} \\ &= \frac{2S(D_p - D_2)}{D_2 D_p} \\ &= \frac{2S\beta(2 - p)}{(p - 1)D_p D_2}. \end{aligned}$$

Combining these identities with the phase rules $E(p) = E_0(p)$ for $F(p) \geq 1$ and $E(p) = E_4(p)$ for $F(p) < 1$, applied also at 2, gives the exhaustive expression

$$E(2) - E(p) = \begin{cases} \frac{4\gamma^2(2 - p)}{(4\gamma + 1)(2\gamma p + p - 1)}, & F_2 \geq 1, \\ E_4(2) - E_0(p), & F_2 < 1 \leq F(p), \\ \frac{2S\beta(2 - p)}{(p - 1)D_p D_2}, & F(p) < 1. \end{cases}$$

Indeed, in the first case $F(p) \geq F_2 \geq 1$, so both minimum exponents select the supplied-propensity channel. In the second case the bounded exponent selects the rough channel and the finite-moment exponent selects the supplied-propensity channel. In the third case $F_2 \leq F(p) < 1$, so both select the rough channel.

If $p < 2$, the two channel differences displayed above are strictly positive. Therefore

$$E(p) \leq E_0(p) < E_0(2), \quad E(p) \leq E_4(p) < E_4(2),$$

and hence

$$E(p) < \min\{E_0(2), E_4(2)\} = E(2).$$

Using the scale definitions in [Definition 15](#), we obtain

$$\frac{\rho_n(v)}{\rho_n^b(w)} = n^{E(2) - E(p)} \longrightarrow \infty.$$

6. Suppose $p < 2$. Apply [Theorem 3](#) to the selected priors and magnitudes of [Algorithm 6](#). It gives

$$L_{n,v} \longrightarrow \infty.$$

For every $n \geq 2$, both selected priors are normalized finite probability priors, and their positive-weight laws satisfy

$$\begin{aligned} \pi_{0,n,v}(\{P\}) > 0 &\implies P \in H_0(v), \\ \pi_{1,n,v}(\{P\}) > 0 &\implies P \in \mathcal{M}_v, \quad c_v \rho_n(v) \leq d(P) < D_v. \end{aligned}$$

For each positive-weight law in either prior, their conditional outcome kernels satisfy

$$Q_{a,P}(\{0, -L_{n,v}, L_{n,v}\} \mid x) = 1$$

for $a \in \{0, 1\}$ and λ -almost every x . The same result supplies the complete-record bound

$$\text{TV}(\mathbb{P}_{0,n,v}, \mathbb{P}_{1,n,v}) < \frac{1}{2}.$$

For any normalized finite probability prior π , write its complete-record mixture as

$$\mathbb{P}_{\pi,n} = \int P^{\otimes n} \pi(dP).$$

The bounded-prior conclusion of the cited result gives a threshold $N(v) \in \mathbb{N}$ such that, for every $n \geq N(v)$ and every pair of normalized finite probability priors π_0, π_1 ,

$$\left. \begin{aligned} \pi_0(\{P\}) > 0 &\implies P \in H_0^b(w), \\ \pi_1(\{P\}) > 0 &\implies P \in \mathcal{M}_w^b, \quad c_v \rho_n(v) \leq d(P) \end{aligned} \right\} \implies \text{TV}(\mathbb{P}_{\pi_0,n}, \mathbb{P}_{\pi_1,n}) \geq \frac{1}{2}.$$

These are precisely the asserted support, magnitude, and complete-record mixture conclusions.

7. For $p < 2$, the lower ratio bound and the strictly positive exponent difference give

$$0 < \frac{c_v}{C_w^b}, \quad \frac{c_v}{C_w^b} n^{E(2)-E(p)} \longrightarrow \infty, \quad \Delta_n(v) \longrightarrow \infty.$$

The definition in [Definition 17](#) includes admissibility and $p < 2$. The preceding divergence supplies its remaining condition, so

$$\mathcal{T} = \{v \in \mathcal{V} : p < 2\}.$$

To characterize [Definition 18](#), suppose that an admissible tuple has a bounded radius ratio. Thus there exists $C_\Delta \in \mathbb{R}$ such that

$$\Delta_n(v) \leq C_\Delta \quad (n \geq 2).$$

If $p \neq 2$, admissibility implies $p < 2$, and the divergence just established supplies $N_\Delta \in \mathbb{N}$ with

$$n \geq N_\Delta \implies \Delta_n(v) > C_\Delta.$$

At

$$n_\Delta = \max\{2, N_\Delta\},$$

the two bounds would give

$$C_\Delta < \Delta_{n_\Delta}(v) \leq C_\Delta,$$

a contradiction. Thus boundedness implies $p = 2$. Conversely, when $p = 2$, evaluation at 2 preserves the entire tuple, and the upper ratio bound reduces to

$$\Delta_n(v) \leq \frac{C_v}{c_w^b} \quad (n \geq 2).$$

Consequently,

$$\mathcal{E} = \{v \in \mathcal{V} : p = 2\}.$$

8. Define the ambient set

$$\mathcal{U} = \{(p, \alpha, \beta, \gamma) \in \mathbb{R}^4 : p < 2\}.$$

The coordinate map to p is continuous, so $\mathcal{U}$ is open in the product topology. The tuple

$$v_o = (3/2, 1/2, 1/2, 1/2)$$

satisfies every admissibility inequality and belongs to $\mathcal{U}$. Thus

$$v_o \in \mathcal{U} \cap \mathcal{V}, \quad \mathcal{U} \cap \mathcal{V} \neq \emptyset.$$

The characterization of $\mathcal{T}$ gives

$$\mathcal{U} \cap \mathcal{V} \subseteq \mathcal{T},$$

which supplies the required open set.

9. It remains to prove compact uniformity. For each admissible tuple define the auxiliary positive base and lower envelope by

$$\begin{aligned} \Theta(v) &= \max \left\{ 2, 32^{D_p \gamma / (2S)} \right\} + 1, \\ \underline{c}_{\text{env}}(v) &= \min \left\{ c_0^e, k_v, c_0^e \Theta(v)^{-|E_0(p) - E_4(p)|} \right\}. \end{aligned}$$

Every entry in this minimum is positive, so

$$\underline{c}_{\text{env}}(v) > 0.$$

The ceiling bound gives

$$2 \leq N_{\text{low}}(v) \leq \Theta(v).$$

Since the exponent $-|E_0(p) - E_4(p)|$ is nonpositive, and since $N_{\text{low}}(v) \geq 1$, monotonicity in the base and then in the exponent gives

$$\begin{aligned} \Theta(v)^{-|E_0(p) - E_4(p)|} &\leq N_{\text{low}}(v)^{-|E_0(p) - E_4(p)|} \\ &\leq N_{\text{low}}(v)^{-(E_0(p) - E_4(p))}. \end{aligned}$$

On $F(p) \geq 1$, the envelope is at most $c_0^e = c_v$. On $F(p) < 1$, it is at most each of

$$k_v, \quad c_0^e N_{\text{low}}(v)^{-(E_0(p) - E_4(p))}.$$

The branchwise definition therefore gives, throughout the admissible domain,

$$0 < \underline{c}_{\text{env}}(v) \leq c_v.$$

The envelope is continuous at every admissible tuple. Indeed, the coordinate maps are continuous; the denominators defining q, D_p, E_0, E_4 are nonzero there; and every variable power in the envelope has a positive base. Absolute value, maximum, minimum, and the remaining arithmetic operations preserve continuity.

Let $\mathcal{K} \subseteq \mathcal{V}$ be compact and nonempty. The continuous envelope attains a minimum at some $v_{\mathcal{K}} \in \mathcal{K}$. Define

$$c_{\mathcal{K}} = \underline{c}_{\text{env}}(v_{\mathcal{K}}) = \min_{v \in \mathcal{K}} \underline{c}_{\text{env}}(v) > 0.$$

Then

$$c_{\mathcal{K}} \leq \underline{c}_{\text{env}}(v) \leq c_v \quad (v \in \mathcal{K}).$$

The compact-uniformity conclusion of [Theorem 2](#) supplies real constants $C_{\mathcal{K}}, N_{\mathcal{K}}$ such that

$$C_v \leq C_{\mathcal{K}}, \quad N_v \leq N_{\mathcal{K}} \quad (v \in \mathcal{K}).$$

For the empty compact set, the choices

$$c_{\mathcal{K}} = 1, \quad C_{\mathcal{K}} = 0, \quad N_{\mathcal{K}} = 0$$

satisfy all requirements vacuously.

Finally, let $\mathcal{K}_b \subseteq \mathcal{W}$ be compact. Its embedded parameter set is

$$\mathcal{K}_2 = \{(2, \alpha, \beta, \gamma) : (\alpha, \beta, \gamma) \in \mathcal{K}_b\}.$$

The embedding $w \mapsto (2, w)$ is continuous, so $\mathcal{K}_2$ is compact, and all its tuples are admissible. Applying the preceding result to $\mathcal{K}_2$, and using the public identities

$$c_w^b = c_{(2,w)}, \quad C_w^b = C_{(2,w)}, \quad N_w^b = N_{(2,w)},$$

gives

$$c_{\mathcal{K}_2} \leq c_w^b, \quad C_w^b \leq C_{\mathcal{K}_2}, \quad N_w^b \leq N_{\mathcal{K}_2} \quad (w \in \mathcal{K}_b),$$

with $c_{\mathcal{K}_2} > 0$. This establishes both compact-uniformity assertions. □

Proof of [Theorem 5](#). Fix an admissible tuple v . We use the constants and exponent defined in the statement.

1. Comparing the restrictions in [Definitions 1, 2, 22](#) and [23](#) gives

$$\mathcal{M}_v \subseteq \mathcal{M}_v^{\text{e,br}}, \quad H_0(v) \subseteq H_0^{\text{e,br}}(v).$$

The law displayed in [Lemma 1](#) therefore supplies a member of the broader model with distance d_0 .

For any $P \in \mathcal{M}_v^{\text{e,br}}$, put

$$\bar{\tau}_P = \int_{[0,1]} \tau_P(x) \lambda(dx).$$

Expanding the centered square and using the effect cap gives

$$d(P)^2 = \int_{[0,1]} \tau_P(x)^2 \lambda(dx) - \bar{\tau}_P^2 \leq \int_{[0,1]} \tau_P(x)^2 \lambda(dx) \leq \frac{1}{4}.$$

Thus the set of broader-model distances is nonempty and bounded above, and

$$d_0 \leq D_v^{\text{e,br}} \leq \frac{1}{2}.$$

Furthermore, $\gamma \leq 1$ implies

$$4^{-\gamma} \geq 4^{-1} = \frac{1}{4}, \quad c_v^{\text{e}} = \frac{4^{-\gamma}}{16\sqrt{3}} \geq \frac{1}{64\sqrt{3}}, \quad C_v^{\text{e}} > 0.$$

2. Fix an integer $n \geq 2$, and define

$$r_{\text{low}} = c_v^{\text{e}} \rho_n^{\text{e}}(v).$$

By [Lemma 5](#), there are a null law $P_0 \in H_0(v)$ and a normalized finite prior

$$\pi_1 = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1,$$

with at least one positive weight, such that

$$e_{P_0}(x) = \frac{1}{2}, \quad \omega_j > 0 \implies P_j \in \mathcal{M}_v, \quad e_{P_j}(x) = \frac{1}{2}, \quad r_{\text{low}} \leq d(P_j) < d_0.$$

Define the supplied function and the complete-record mixture by

$$f = e_{P_0}, \quad \bar{P}_1^{(n)} = \sum_{j=1}^k \omega_j P_j^{\otimes n}.$$

The function f is continuous and satisfies $f(x) = 1/2$ everywhere. The same cited result gives

$$\text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}) < \frac{1}{2}.$$

The inclusions already established place P_0 in the broader null and every positive-weight P_j in the broader model, with $e_{P_j} = f$.

Admissibility gives $q > 0$, $\gamma > 0$, and hence $E_0 > 0$. Consequently,

$$0 < r_{\text{low}} \leq d(P_j) < d_0 \leq D_v^{\text{e,br}} \quad (\omega_j > 0).$$

In particular, r_{low} is a legal separation with a nonempty alternative class.

3. Let ϕ^{e} be an allowed supplied-propensity test satisfying the broader-null size constraint. Freeze its function input at f , and average its public randomization:

$$\bar{\phi}_f(\mathcal{D}_n) = \int_0^1 \phi^{\text{e}}(f, \mathcal{D}_n, U) dU, \quad 0 \leq \bar{\phi}_f(\mathcal{D}_n) \leq 1.$$

Joint measurability makes this a measurable function of the sample. Since all positive-weight witnesses have supplied function f , integration against the finite mixture gives

$$\int \bar{\phi}_f dP_0^{\otimes n} = R_n^e(P_0, \phi^e), \quad \int \bar{\phi}_f d\bar{P}_1^{(n)} = \sum_{j=1}^k \omega_j R_n^e(P_j, \phi^e).$$

For a measurable function taking values in $[0, 1]$, integrating its level-set probability differences bounds the difference of its expectations by total variation. Therefore

$$\left| \sum_{j=1}^k \omega_j R_n^e(P_j, \phi^e) - R_n^e(P_0, \phi^e) \right| \leq \text{TV}(P_0^{\otimes n}, \bar{P}_1^{(n)}),$$

and hence

$$\sum_{j=1}^k \omega_j R_n^e(P_j, \phi^e) < \frac{1}{10} + \frac{1}{2} = \frac{3}{5}.$$

If every positive-weight atom had type-II error strictly below $2/5$, then its rejection expectation would exceed $3/5$. Zero-weight atoms contribute zero, so normalization would imply

$$\sum_{j=1}^k \omega_j R_n^e(P_j, \phi^e) \geq \frac{3}{5} \sum_{j=1}^k \omega_j = \frac{3}{5},$$

a contradiction. Thus

$$\exists j : \quad \omega_j > 0, \quad 1 - R_n^e(P_j, \phi^e) \geq \frac{2}{5}.$$

The feasible test class is nonempty: the jointly measurable test

$$\phi_0^e(f', \mathcal{D}_n, U) = 0 \quad (f' \in C([0, 1]), \mathcal{D}_n \text{ an } n\text{-record sample}, U \in [0, 1])$$

has rejection expectation zero under every law. For every allowed test and every law,

$$0 \leq R_n^e(P, \phi^e) \leq 1.$$

Thus the alternative errors are bounded above by one. The positive-weight atom just obtained belongs to the alternative class at r_{low} , so

$$\sup_{\substack{P \in \mathcal{M}_v^{\text{e}, \text{br}} \\ d(P) \geq r_{\text{low}}}} \{1 - R_n^e(P, \phi^e)\} \geq \frac{2}{5}.$$

Taking the infimum over feasible tests in [Definition 24](#) proves

$$B_n^{\text{e}, \text{br}}(v, r_{\text{low}}) \geq \frac{2}{5}.$$

This also proves the required error witness for the particular prior constructed above.

4. For every separation r with $0 < r \leq r_{\text{low}}$, the alternative class at r_{low} is contained in the alternative class at r . Both are nonempty, their error suprema are bounded, and the feasible test class is unchanged. Therefore

$$B_n^{\text{e,br}}(v, r) \geq B_n^{\text{e,br}}(v, r_{\text{low}}) \geq \frac{2}{5} > \frac{1}{10}.$$

Every successful separation in the set defining [Definition 25](#) is consequently above r_{low} ; its sentinel $D_v^{\text{e,br}}$ is also above r_{low} . Taking the infimum gives

$$r_n^{*,\text{e,br}}(v) \geq r_{\text{low}}.$$

5. We now establish calibration and power directly on the broader model. Use the construction in [Algorithm 2](#), with

$$s = \lfloor n/2 \rfloor, \quad a = s^{-E_0}, \quad T = 2^{\lceil \log_2(a^{-1/(p-1)}) \rceil}, \quad J = 2^{\lceil \log_2(a^{-1/\gamma}) \rceil},$$

$$b = 20T^{1-p}, \quad \Lambda = \frac{80T^{2-p}}{s}.$$

Here

$$s \geq 1, \quad 0 < a \leq 1, \quad T \geq 1, \quad J \geq 1, \quad b \geq 0, \quad \Lambda > 0.$$

For a broader-model law P , overlap makes the clipped supplied propensity equal to e_P .

The elementary clipping inequalities are

$$|y - \ell_T(y)| \leq |y|^p T^{1-p}, \quad \ell_T(y)^2 \leq |y|^p T^{2-p}.$$

Indeed, when $|y| \leq T$, the first difference vanishes and

$$y^2 = |y|^p |y|^{2-p} \leq |y|^p T^{2-p}.$$

When $|y| > T$, the clipping remainder is at most $|y|$, and

$$|y| = |y|^p |y|^{1-p} \leq |y|^p T^{1-p}, \quad \ell_T(y)^2 \leq T^2 = T^p T^{2-p} \leq |y|^p T^{2-p}.$$

The raw moment restriction gives first-moment integrability in both arms, using $|y| \leq 1 + |y|^p$. Integrating the two clipping inequalities yields, for λ -almost every x ,

$$\left| \int_{\mathbb{R}} (y - \ell_T(y)) Q_{\iota_{\text{arm}}, P}(dy \mid x) \right| \leq 10T^{1-p}, \quad \int_{\mathbb{R}} \ell_T(y)^2 Q_{\iota_{\text{arm}}, P}(dy \mid x) \leq 10T^{2-p} \quad (\iota_{\text{arm}} \in \{0, 1\}).$$

Define the clipped conditional score mean by

$$g_{T,P}(x) = \int_{\mathbb{R}} \ell_T(y) Q_{1,P}(dy \mid x) - \int_{\mathbb{R}} \ell_T(y) Q_{0,P}(dy \mid x).$$

Cancellation of the arm probabilities against the inverse-propensity denominators identifies this function as the conditional mean of R_i . The conditional arm-mean identities then give

$$|g_{T,P}(x) - \tau_P(x)| \leq 20T^{1-p} = b$$

almost everywhere. The conditional second moment of the score satisfies

$$\begin{aligned} \mathbb{E}(R_i^2 \mid X_i = x) &= \frac{\int \ell_T(y)^2 Q_{1,P}(dy \mid x)}{e_P(x)} + \frac{\int \ell_T(y)^2 Q_{0,P}(dy \mid x)}{1 - e_P(x)} \\ &\leq 40T^{2-p} + 40T^{2-p} = 80T^{2-p}. \end{aligned}$$

6. Define the constant-removing matrix and the sample expectation notation by

$$u = J^{-1/2}(1, \dots, 1)^T, \quad C_J = \text{Id}_{\mathbb{R}^J} - uu^T,$$

$$\mathbb{P}_{n,P} = P^{\otimes n}, \quad \mathbb{E}_{n,P}[\cdot] = \int (\cdot) dP^{\otimes n}.$$

The histogram cells in [Definition 19](#) are disjoint and have Lebesgue mass $1/J$. Consequently,

$$\int_{[0,1]} F_J(x) F_J(x)^T \lambda(dx) = \text{Id}_{\mathbb{R}^J}, \quad u^T F_J(x) = 1, \quad C_J F_J(x) = F_J(x) - u.$$

For every $z_{\text{dir}} \in \mathbb{R}^J$,

$$\|C_J z_{\text{dir}}\|^2 = \|z_{\text{dir}}\|^2 - (u^T z_{\text{dir}})^2 \leq \|z_{\text{dir}}\|^2.$$

Combining these identities with the conditional score second-moment bound gives

$$\begin{aligned} \mathbb{E}_{n,P} \left[(z_{\text{dir}}^T (F_J(X_i) - u) R_i)^2 \right] &\leq 80T^{2-p} \int_{[0,1]} (z_{\text{dir}}^T C_J F_J(x))^2 \lambda(dx) \\ &= 80T^{2-p} \|C_J z_{\text{dir}}\|^2 \\ &\leq 80T^{2-p} \|z_{\text{dir}}\|^2. \end{aligned}$$

The two disjoint evaluation blocks have equal size and consist of independent records. Their common mean and centered vectors are

$$\mu_P = \mathbb{E}_{n,P} V_1 = \mathbb{E}_{n,P} V_2 = C_J \int_{[0,1]} F_J(x) g_{T,P}(x) \lambda(dx), \quad Z_{\iota_{\text{block}},P} = V_{\iota_{\text{block}}} - \mu_P \quad (\iota_{\text{block}} \in \{1, 2\}).$$

Variance additivity within a block and the preceding single-record bound imply

$$\mathbb{E}_{n,P} [(z_{\text{dir}}^T Z_{\iota_{\text{block}},P})^2] = \text{Var}_{n,P}(z_{\text{dir}}^T V_{\iota_{\text{block}}}) \leq \Lambda \|z_{\text{dir}}\|^2.$$

The centered vectors are independent and have mean zero.

Define the quadratic statistic by

$$W_P = \langle V_1, V_2 \rangle.$$

Block independence gives

$$\mathbb{E}_{n,P} W_P = \|\mu_P\|^2, \quad W_P - \|\mu_P\|^2 = \langle \mu_P, Z_{1,P} \rangle + \langle \mu_P, Z_{2,P} \rangle + \langle Z_{1,P}, Z_{2,P} \rangle.$$

All three cross-product expectations in the square of this expansion vanish:

$$\begin{aligned} \mathbb{E}_{n,P} [\langle \mu_P, Z_{1,P} \rangle \langle \mu_P, Z_{2,P} \rangle] &= 0, \\ \mathbb{E}_{n,P} [\langle \mu_P, Z_{1,P} \rangle \langle Z_{1,P}, Z_{2,P} \rangle] &= 0, \\ \mathbb{E}_{n,P} [\langle \mu_P, Z_{2,P} \rangle \langle Z_{1,P}, Z_{2,P} \rangle] &= 0. \end{aligned}$$

The first equality uses independence and both zero means; the other two follow by integrating first over the block appearing linearly.

For the remaining quadratic term, independence and the directional second-moment bound give

$$\begin{aligned}\mathbb{E}_{n,P}[\langle Z_{1,P}, Z_{2,P} \rangle^2] &\leq \Lambda \mathbb{E}_{n,P} \|Z_{1,P}\|^2 \\ &= \Lambda \sum_{j=1}^J \mathbb{E}_{n,P} (Z_{1,P})_j^2 \leq J\Lambda^2.\end{aligned}$$

Thus

$$\text{Var}_{n,P}(W_P) \leq 2\Lambda \|\mu_P\|^2 + J\Lambda^2.$$

All quantities have finite second moments, since the clipped scores and finite histogram features are bounded.

Define the good event by

$$\mathcal{G}_{n,P} = \left\{ \mathcal{D}_n : |W_P - \|\mu_P\|^2| \leq 16(\sqrt{\Lambda} \|\mu_P\| + \sqrt{J} \Lambda) \right\}.$$

Its tolerance is strictly positive. Applying the second-moment tail bound gives

$$\begin{aligned}\mathbb{P}_{n,P}(\mathcal{G}_{n,P}^c) &\leq \frac{2\Lambda \|\mu_P\|^2 + J\Lambda^2}{256(\sqrt{\Lambda} \|\mu_P\| + \sqrt{J} \Lambda)^2} \\ &\leq \frac{1}{128} < 0.01.\end{aligned}$$

The last inequality uses the nonnegative cross term in the denominator.

7. Suppose $P \in H_0^{\text{e,br}}(v)$, and choose its null constant:

$$c_{\text{null},P} \in [-1/2, 1/2], \quad \tau_P(x) = c_{\text{null},P} \quad (x \in [0, 1]).$$

The histogram coefficient of this constant is $c_{\text{null},P}u$, which C_J annihilates. Hence

$$\mu_P = C_J \int_{[0,1]} F_J(x) (g_{T,P}(x) - c_{\text{null},P}) \lambda(dx).$$

The clipping bound and cell masses imply

$$\left| \left(\int F_J(x) (g_{T,P}(x) - c_{\text{null},P}) \lambda(dx) \right)_j \right| \leq \sqrt{J} \frac{b}{J} = \frac{b}{\sqrt{J}}.$$

Summing the squared coordinate bounds and using contraction by C_J gives

$$\|\mu_P\|^2 \leq \sum_{j=1}^J \frac{b^2}{J} = b^2.$$

On $\mathcal{G}_{n,P}$,

$$W_P \leq b^2 + 16\sqrt{\Lambda} b + 16\sqrt{J} \Lambda \leq 2b^2 + 80\sqrt{J} \Lambda \leq 2b^2 + 1024\sqrt{J} \Lambda.$$

Here

$$16\sqrt{\Lambda} b \leq b^2 + 64\Lambda$$

follows from $(b - 8\sqrt{\Lambda})^2 \geq 0$, and $\sqrt{J} \geq 1$.

The deterministic rule in [Algorithm 2](#) rejects precisely when W_P exceeds its displayed cutoff. Its rejection event is therefore contained in $\mathcal{G}_{n,P}^c$, and averaging the independent public randomization leaves that probability unchanged. Thus

$$\mathbf{R}_n^e(P, \phi_{n,v}^{*,e}) \leq 0.01 \quad (P \in H_0^{e,\text{br}}(v)).$$

8. For a general broader-model law P , define its histogram coefficients and their associated constant by

$$\theta_{\text{hist},P} = \int_{[0,1]} F_J(x) \tau_P(x) \lambda(dx), \quad c_{\text{hist},P} = \frac{1}{\sqrt{J}} \sum_{j=1}^J (\theta_{\text{hist},P})_j, \quad \theta_{\text{cen},P} = C_J \theta_{\text{hist},P}.$$

Subtracting the constant coefficient gives

$$\theta_{\text{cen},P} = \int F_J(x) (\tau_P(x) - c_{\text{hist},P}) \lambda(dx) = \int F_J(x) ((\Pi_J \tau_P)(x) - c_{\text{hist},P}) \lambda(dx).$$

The second equality holds because histogram projection preserves every cell integral.

Effect Hölder smoothness bounds the deviation from a cell average:

$$|\tau_P(x) - (\Pi_J \tau_P)(x)| \leq 20J^{-\gamma}, \quad \int (\tau_P - \Pi_J \tau_P)^2 d\lambda \leq (20J^{-\gamma})^2.$$

Indeed, every point in the same cell lies within distance $1/J$, so averaging the Hölder bound over that cell proves the pointwise inequality.

The histogram projection is constant on each cell, and the residual has integral zero on each cell. Its residual is therefore orthogonal to the projection and to constants. Expanding the square, and using the histogram feature isometry, yields

$$\int (\tau_P - c_{\text{hist},P})^2 d\lambda = \|\theta_{\text{cen},P}\|^2 + \int (\tau_P - \Pi_J \tau_P)^2 d\lambda.$$

Also, expansion around the mean gives

$$\int (\tau_P - c_{\text{hist},P})^2 d\lambda = d(P)^2 + (\bar{\tau}_P - c_{\text{hist},P})^2 \geq d(P)^2.$$

Consequently,

$$d(P)^2 \leq \|\theta_{\text{cen},P}\|^2 + (20J^{-\gamma})^2 \leq (\|\theta_{\text{cen},P}\| + 20J^{-\gamma})^2,$$

and hence

$$d(P) - 20J^{-\gamma} \leq \|\theta_{\text{cen},P}\|.$$

Finally,

$$\mu_P - \theta_{\text{cen},P} = C_J \int F_J(x) (g_{T,P}(x) - \tau_P(x)) \lambda(dx).$$

The same cellwise coefficient bound used for the null, now applied to $|g_{T,P} - \tau_P| \leq b$, gives

$$\|\mu_P - \theta_{\text{cen},P}\| \leq b.$$

The reverse triangle inequality therefore proves the required population signal bound:

$$d(P) - 20J^{-\gamma} - b \leq \|\mu_P\|.$$

9. If

$$d(P) \geq 20J^{-\gamma} + 5b + 128\sqrt{\sqrt{J}\Lambda},$$

the preceding population bound gives

$$\|\mu_P\| \geq 4b + 128\sqrt{\sqrt{J}\Lambda}.$$

Since $J \geq 1$,

$$\sqrt{\Lambda} \leq \sqrt{\sqrt{J}\Lambda}, \quad 16\sqrt{\Lambda} \|\mu_P\| \leq \frac{\|\mu_P\|^2}{2}.$$

On $\mathcal{G}_{n,P}$, it follows that

$$\begin{aligned} W_P &\geq \|\mu_P\|^2 - 16\sqrt{\Lambda} \|\mu_P\| - 16\sqrt{J}\Lambda \\ &\geq \frac{\|\mu_P\|^2}{2} - 16\sqrt{J}\Lambda \\ &\geq 8b^2 + 8176\sqrt{J}\Lambda \\ &> 2b^2 + 1024\sqrt{J}\Lambda. \end{aligned}$$

The penultimate inequality expands $(4b + 128\sqrt{\sqrt{J}\Lambda})^2/2$ and discards its nonnegative cross term. The last inequality is strict because $\Lambda > 0$. Thus the rule rejects throughout $\mathcal{G}_{n,P}$, proving

$$1 - R_n^e(P, \phi_{n,v}^{*,e}) \leq 0.01$$

at the displayed sufficient separation.

10. To express this separation in the public rate, define the variance-growth exponent

$$t = \frac{2-p}{p-1} \geq 0.$$

Direct algebra gives

$$2+t = \frac{1}{q}, \quad E_0\left(2+t+\frac{1}{2\gamma}\right) = 1.$$

Upward dyadic rounding costs at most a factor of two:

$$a^{-1/(p-1)} \leq T \leq 2a^{-1/(p-1)}, \quad a^{-1/\gamma} \leq J \leq 2a^{-1/\gamma}.$$

The lower rounding bounds and the negative bias exponents imply

$$b = 20T^{1-p} \leq 20a, \quad J^{-\gamma} \leq a.$$

Because $0 \leq 2-p \leq 1$, the upper rounding bounds imply

$$T^{2-p} \leq 2a^{-t}, \quad \sqrt{J} \leq \sqrt{2}a^{-1/(2\gamma)}.$$

The exponent balance now gives

$$s^{-1}a^{-t-1/(2\gamma)} = a^2,$$

and consequently

$$\sqrt{J} \Lambda = \frac{80\sqrt{J} T^{2-p}}{s} \leq 160\sqrt{2} a^2.$$

Combining these estimates yields

$$20J^{-\gamma} + 5b + 128\sqrt{\sqrt{J} \Lambda} \leq (120 + 128\sqrt{160\sqrt{2}})a.$$

For every $n \geq 2$, the block size satisfies $n \leq 3s$. Moreover,

$$0 < E_0 = \frac{2\gamma q}{2\gamma + q} \leq 1,$$

so

$$a = s^{-E_0} \leq 3^{E_0} n^{-E_0} \leq 3\rho_n^e(v).$$

It follows that

$$20J^{-\gamma} + 5b + 128\sqrt{\sqrt{J} \Lambda} \leq C_v^e \rho_n^e(v).$$

Thus every broader-model law at or above the nominal upper separation has the asserted type-II bound, in particular under the stated gate $C_v^e \rho_n^e(v) < D_v^{\text{e,br}}$. All preceding estimates apply when $s = 1$ or $J = 1$; positivity of s, T, J, Λ suffices throughout.

11. Define

$$r_{\text{up}} = C_v^e \rho_n^e(v) > 0.$$

If $r_{\text{up}} < D_v^{\text{e,br}}$, calibration makes $\phi_{n,v}^{*,\text{e}}$ feasible at level 1/10, and its power bound gives

$$B_n^{\text{e,br}}(v, r_{\text{up}}) \leq \sup_{\substack{P \in \mathcal{M}_v^{\text{e,br}} \\ d(P) \geq r_{\text{up}}}} \{1 - \mathbb{R}_n^e(P, \phi_{n,v}^{*,\text{e}})\} \leq 0.01 \leq \frac{1}{10}.$$

For a nonempty alternative class, this supremum is bounded between zero and one; the empty-class convention assigns it zero and yields the same upper estimate. Here nonemptiness also follows from $r_{\text{up}} < D_v^{\text{e,br}}$: otherwise every model distance would be strictly below r_{up} , making r_{up} an upper bound for their supremum. Thus r_{up} belongs to the successful-radius set.

If $D_v^{\text{e,br}} \leq r_{\text{up}}$, the sentinel in [Definition 25](#) instead gives

$$r_n^{*,\text{e,br}}(v) \leq D_v^{\text{e,br}} \leq r_{\text{up}}.$$

The defining set is nonempty because it contains the sentinel, and it is bounded below by zero. In both cases,

$$r_n^{*,\text{e,br}}(v) \leq r_{\text{up}}.$$

Together with the lower bound, this proves the radius inequalities. The witnesses constructed above have all the required model memberships, common function input, distance bounds, normalization, total variation comparison, and positive-weight error witness.

12. For $n \geq N_v^e$, the ceiling definition implies

$$n \geq \left(\frac{2C_v^e}{d_0} \right)^{1/E_0}.$$

The base and E_0 are positive. Raising this inequality to the negative power $-E_0$ therefore gives

$$C_v^e n^{-E_0} \leq C_v^e \left[\left(\frac{2C_v^e}{d_0} \right)^{1/E_0} \right]^{-E_0} = C_v^e \left(\frac{2C_v^e}{d_0} \right)^{-1} = \frac{d_0}{2}.$$

13. For $P \in \mathcal{M}_v^{\text{e,br}}$, the conditional arm-mean identities hold simultaneously for λ -almost every x :

$$\int_{\mathbb{R}} y Q_{0,P}(dy | x) = m_{0,P}(x), \quad \int_{\mathbb{R}} y Q_{1,P}(dy | x) = m_{0,P}(x) + \tau_P(x).$$

Overlap gives $e_P(x) > 0$ and $1 - e_P(x) > 0$. Pulling the constant denominators out of the integrals and cancelling them yields

$$\begin{aligned} e_P(x) \int_{\mathbb{R}} \frac{y}{e_P(x)} Q_{1,P}(dy | x) + (1 - e_P(x)) \int_{\mathbb{R}} \frac{-y}{1 - e_P(x)} Q_{0,P}(dy | x) \\ = \int_{\mathbb{R}} y Q_{1,P}(dy | x) - \int_{\mathbb{R}} y Q_{0,P}(dy | x) = \tau_P(x). \end{aligned}$$

The conditional first moments are finite almost everywhere by the raw moment bound.

14. Finally, the coordinate maps $v \mapsto p$ and $v \mapsto \gamma$ are continuous in the product topology. On the admissible domain,

$$p > 0, \quad q = \frac{p-1}{p} > 0, \quad 2\gamma + q > 0.$$

Hence q and E_0 are continuous there, and E_0 is everywhere positive.

For a compact subset $\mathcal{K}$ of that domain, define

$$E_{\min, \mathcal{K}} = \begin{cases} \min_{v \in \mathcal{K}} E_0(v), & \mathcal{K} \neq \emptyset, \\ 1, & \mathcal{K} = \emptyset. \end{cases}$$

In the nonempty case, continuity and compactness supply an attained minimum, whose value is positive by admissibility. In the empty case the assertions are vacuous. Thus

$$E_{\min, \mathcal{K}} > 0, \quad E_{\min, \mathcal{K}} \leq E_0(v) \quad (v \in \mathcal{K}).$$

Together with the global lower-multiplier bound and the fixed finite upper multiplier,

$$c_v^e \geq \frac{1}{64\sqrt{3}}, \quad C_v^e = 3(120 + 128\sqrt{160\sqrt{2}}),$$

this proves the asserted local uniformity.

□

Proof of Theorem 6. Fix $v \in \mathcal{V}$. Admissibility gives

$$1 < p \leq 2, \quad \frac{1}{4} \leq \gamma \leq 1, \quad 0 < q = \frac{p-1}{p} \leq \frac{1}{2}, \quad 2\gamma + q > 0, \quad E_0 > 0.$$

We use the constants and scales in the statement.

1. Since $-1 \leq -\gamma$ and the base 4 exceeds one,

$$4^{-1} \leq 4^{-\gamma}, \quad \frac{1}{64\sqrt{3}} = \frac{4^{-1}}{16\sqrt{3}} \leq c_v^e.$$

Also,

$$C_v^e = 3 \left(120 + 128\sqrt{160\sqrt{2}} \right) > 0.$$

2. Fix an integer $n \geq 2$, and define the lower separation by

$$\varrho_- = c_v^e \rho_n^e(v).$$

By Lemma 5, there are a null law $P_0 \in H_0(v)$ and a finite prior

$$\pi_1 = \sum_{j=1}^k \omega_j \delta_{P_j}, \quad \omega_j \geq 0, \quad \sum_{j=1}^k \omega_j = 1, \quad \exists j : \omega_j > 0.$$

For every positive-weight support law, that result supplies

$$P_j \in \mathcal{M}_v, \quad e_{P_j}(x) = \frac{1}{2} = e_{P_0}(x) \quad (x \in [0, 1]), \quad \varrho_- \leq d(P_j) < d_0 \leq D_v.$$

It also supplies

$$\mathbb{P}_{\pi_1, n} = \sum_{j=1}^k \omega_j P_j^{\otimes n}, \quad \text{TV}(P_0^{\otimes n}, \mathbb{P}_{\pi_1, n}) < \frac{1}{2}, \quad B_n^e(v, \varrho_-) \geq \frac{2}{5}.$$

Choose an index $j_+ \in \{1, \dots, k\}$ with $\omega_{j_+} > 0$. Positivity of the lower multiplier and of n^{-E_0} yields

$$0 < \varrho_- \leq d(P_{j_+}) < D_v.$$

Thus the alternatives at separation ϱ_- are nonempty. For every $0 < \varrho \leq \varrho_-$, their inclusion in the alternatives at separation ϱ gives

$$B_n^e(v, \varrho_-) \leq B_n^e(v, \varrho).$$

The size constraint is the same in both infima. Consequently, every successful separation below D_v in Definition 27 is at least ϱ_- , because

$$B_n^e(v, \varrho) \geq \frac{2}{5} > \frac{1}{10} \quad (0 < \varrho \leq \varrho_-).$$

The additional candidate D_v also exceeds ϱ_- . Taking the infimum therefore gives

$$c_v^e \rho_n^e(v) = \varrho_- \leq r_n^{*,e}(v).$$

3. We next establish the null calibration of the rule in [Algorithm 2](#). Use its tuning quantities

$$s = \lfloor n/2 \rfloor, \quad a = s^{-E_0}, \quad T = 2^{\lceil \log_2(a^{-1/(p-1)}) \rceil}, \quad J = 2^{\lceil \log_2(a^{-1/\gamma}) \rceil},$$

$$b = 20T^{1-p}, \quad \Lambda = \frac{80T^{2-p}}{s}.$$

Here $s \geq 1$, $0 < a \leq 1$, $T \geq 1$, $J \geq 1$, $b \geq 0$, and $\Lambda > 0$. With F_J from [Definition 19](#), define

$$u = J^{-1/2}(1, \dots, 1)^T, \quad C_J = I_J - uu^T.$$

The cells have Lebesgue measure $1/J$, so

$$\int_0^1 F_J(x) F_J(x)^T d\lambda(x) = I_J, \quad \int_0^1 F_J(x) d\lambda(x) = u, \quad C_J F_J(x) = F_J(x) - u.$$

Thus C_J is a symmetric orthogonal projection and

$$\|C_J \xi_{\text{dir}}\|_2 \leq \|\xi_{\text{dir}}\|_2 \quad (\xi_{\text{dir}} \in \mathbb{R}^J).$$

For $P \in \mathcal{M}_v$, define the clipped conditional mean and second moment on $[0, 1]$ by

$$g_{T,P}(x) = \int_{\mathbb{R}} \ell_T(y) Q_{1,P}(dy | x) - \int_{\mathbb{R}} \ell_T(y) Q_{0,P}(dy | x),$$

$$m_{2,T,P}(x) = \frac{\int_{\mathbb{R}} \ell_T(y)^2 Q_{1,P}(dy | x)}{e_P(x)} + \frac{\int_{\mathbb{R}} \ell_T(y)^2 Q_{0,P}(dy | x)}{1 - e_P(x)}.$$

These functions are finite and measurable, since clipping is bounded and overlap makes both denominators positive.

For every $y \in \mathbb{R}$,

$$|y - \ell_T(y)| \leq |y|^p T^{1-p}, \quad \ell_T(y)^2 \leq |y|^p T^{2-p}.$$

Indeed, on $|y| \leq T$, the clipping remainder vanishes and $|y|^{2-p} \leq T^{2-p}$; on $|y| > T$, the remainder is at most $|y|$ and the clipped square equals T^2 . The moment restriction in [Assumption 8](#) therefore gives, for each $a' \in \{0, 1\}$ and almost every x ,

$$\left| \int_{\mathbb{R}} (y - \ell_T(y)) Q_{a',P}(dy | x) \right| \leq 10T^{1-p}, \quad \int_{\mathbb{R}} \ell_T(y)^2 Q_{a',P}(dy | x) \leq 10T^{2-p}.$$

The same moment restriction ensures integrability of y , using $|y| \leq 1 + |y|^p$. The arm-mean identities now yield

$$g_{T,P}(x) - \tau_P(x) = \int_{\mathbb{R}} (y - \ell_T(y)) Q_{0,P}(dy | x) - \int_{\mathbb{R}} (y - \ell_T(y)) Q_{1,P}(dy | x),$$

and hence

$$|g_{T,P}(x) - \tau_P(x)| \leq b, \quad m_{2,T,P}(x) \leq 10T^{2-p} \left(\frac{1}{e_P(x)} + \frac{1}{1 - e_P(x)} \right) \leq 80T^{2-p}$$

almost everywhere. These are respectively the conditional mean and second moment bounds for R_i , since the supplied true propensity agrees with its overlap-clipped version.

Define the common block mean and the two histogram coefficient vectors by

$$\mu_P = \mathbb{E}_{P^{\otimes n}}[V_1] = \mathbb{E}_{P^{\otimes n}}[V_2] = C_J \theta_{T,P},$$

$$\theta_{T,P} = \int_0^1 g_{T,P}(x) F_J(x) d\lambda(x), \quad \theta_{\tau,P} = \int_0^1 \tau_P(x) F_J(x) d\lambda(x).$$

The equality of the block means follows by averaging s identically distributed records in each block.

For a null law, define its constant effect by

$$c_P = \tau_P(0), \quad \tau_P(x) = c_P \quad (x \in [0, 1]).$$

Because $C_J u = 0$,

$$\mu_P = C_J \int_0^1 (g_{T,P}(x) - c_P) F_J(x) d\lambda(x).$$

For every histogram coordinate j ,

$$\left| \left(\int_0^1 (g_{T,P} - c_P) F_J d\lambda \right)_j \right| = \left| \sqrt{J} \int_{I_{j,J}} (g_{T,P} - c_P) d\lambda \right| \leq \frac{b}{\sqrt{J}}.$$

Summing the squared coordinate bounds and applying the contraction C_J gives

$$\|\mu_P\|_2^2 \leq \sum_{j=1}^J \frac{b^2}{J} = b^2, \quad \|\mu_P\|_2 \leq b.$$

For any model law and any $\xi_{\text{dir}} \in \mathbb{R}^J$, the conditional second moment bound and the feature isometry give

$$\mathbb{E}_P \left[\langle \xi_{\text{dir}}, C_J F_J(X) R \rangle^2 \right] \leq 80T^{2-p} \int_0^1 \langle C_J \xi_{\text{dir}}, F_J(x) \rangle^2 d\lambda(x) \leq 80T^{2-p} \|\xi_{\text{dir}}\|_2^2.$$

Here the single-record score is defined by

$$R = \frac{A\ell_T(Y)}{e_P(X)} - \frac{(1-A)\ell_T(Y)}{1 - e_P(X)}.$$

Independence within a block gives, for $b' \in \{1, 2\}$,

$$\text{Var}_{P^{\otimes n}}(\langle \xi_{\text{dir}}, V_{b'} \rangle) = \frac{1}{s} \text{Var}_P(\langle \xi_{\text{dir}}, C_J F_J(X) R \rangle) \leq \Lambda \|\xi_{\text{dir}}\|_2^2.$$

Define

$$\Delta_{b',P} = V_{b'} - \mu_P \quad (b' \in \{1, 2\}), \quad W_e = \langle V_1, V_2 \rangle.$$

The centered vectors are independent and have mean zero. In particular,

$$\mathbb{E}_{P^{\otimes n}}[W_e] = \|\mu_P\|_2^2,$$

$$W_e - \|\mu_P\|_2^2 = \langle \mu_P, \Delta_{1,P} \rangle + \langle \mu_P, \Delta_{2,P} \rangle + \langle \Delta_{1,P}, \Delta_{2,P} \rangle.$$

The expectations of all three cross products in the square of this sum vanish: the product of the two linear terms factors into their zero means, and conditioning on the other block makes each linear–bilinear cross product zero. To bound the remaining bilinear square, let

$$\mathbf{e}^{(j)} = (\mathbf{1}_{\{j'=j\}})_{j'=1}^J \quad (j = 1, \dots, J)$$

be the coordinate unit vectors. Independence and the directional second moment bounds imply

$$\mathbb{E}_{P^{\otimes n}} [\langle \Delta_{1,P}, \Delta_{2,P} \rangle^2] \leq \Lambda \mathbb{E}_{P^{\otimes n}} \|\Delta_{2,P}\|_2^2 = \Lambda \sum_{j=1}^J \mathbb{E}_{P^{\otimes n}} [\langle \mathbf{e}^{(j)}, \Delta_{2,P} \rangle^2] \leq J\Lambda^2.$$

Consequently,

$$\text{Var}_{P^{\otimes n}}(W_e) \leq 2\Lambda \|\mu_P\|_2^2 + J\Lambda^2.$$

All the required moments are finite because the clipped score and the finite-rank features are bounded.

The squared-deviation inequality now gives

$$\begin{aligned} P^{\otimes n} \left(|W_e - \|\mu_P\|_2^2| > 16\{\sqrt{\Lambda}\|\mu_P\|_2 + \sqrt{J}\Lambda\} \right) \\ \leq \frac{2\Lambda \|\mu_P\|_2^2 + J\Lambda^2}{256\{\sqrt{\Lambda}\|\mu_P\|_2 + \sqrt{J}\Lambda\}^2} \leq \frac{2}{256} < 0.01. \end{aligned}$$

The denominator is positive since $J \geq 1$ and $\Lambda > 0$. Define the good event by

$$\mathcal{G}_P = \left\{ |W_e - \|\mu_P\|_2^2| \leq 16\{\sqrt{\Lambda}\|\mu_P\|_2 + \sqrt{J}\Lambda\} \right\}.$$

Thus $P^{\otimes n}(\mathcal{G}_P^c) \leq 0.01$.

For a null law, on $\mathcal{G}_P$ we have

$$W_e \leq b^2 + 16\sqrt{\Lambda}b + 16\sqrt{J}\Lambda \leq 2b^2 + 80\sqrt{J}\Lambda \leq 2b^2 + 1024\sqrt{J}\Lambda.$$

The middle inequality follows from

$$16\sqrt{\Lambda}b \leq b^2 + 64\Lambda, \quad \Lambda \leq \sqrt{J}\Lambda,$$

the first being equivalent to $(b - 8\sqrt{\Lambda})^2 \geq 0$. The rejection event is therefore contained in $\mathcal{G}_P^c$. Since the rule ignores the public randomization variable,

$$\mathbb{R}_n^e(P, \phi_{n,v}^{*,e}) = P^{\otimes n} \left(W_e > 2b^2 + 1024\sqrt{J}\Lambda \right) \leq 0.01 \quad (P \in H_0(v)).$$

These calculations also apply when $s = 1$; when $J = 1$, $C_J = 0$ and both centered block vectors vanish.

4. We establish the alternative population bound before applying the same concentration event. Define

$$\bar{\tau}_P = \int_0^1 \tau_P(x) d\lambda(x), \quad c_{P,J} = \frac{1}{\sqrt{J}} \sum_{j=1}^J (\theta_{\tau,P})_j, \quad h_{P,J}(x) = \tau_P(x) - c_{P,J}.$$

The coefficient formula for constants and the definition of C_J give

$$C_J \theta_{\tau, P} = \int_0^1 h_{P, J}(x) F_J(x) d\lambda(x).$$

On each cell, the histogram projection is the cell average. Hence

$$\tau_P(x) - (\Pi_J \tau_P)(x) = J \int_{I_{j, J}} \{\tau_P(x) - \tau_P(z)\} d\lambda(z) \quad (x \in I_{j, J}),$$

and [Assumption 5](#) gives

$$|\tau_P(x) - (\Pi_J \tau_P)(x)| \leq 20J^{-\gamma}, \quad \int_0^1 \{\tau_P - \Pi_J \tau_P\}^2 d\lambda \leq (20J^{-\gamma})^2.$$

Subtracting a constant commutes with Π_J . Its cellwise residual has integral zero on every cell, and is therefore orthogonal to the cellwise constant projection. The feature isometry then yields

$$\int_0^1 h_{P, J}^2 d\lambda = \|C_J \theta_{\tau, P}\|_2^2 + \int_0^1 \{\tau_P - \Pi_J \tau_P\}^2 d\lambda.$$

Expanding the distance to the constant $c_{P, J}$ also gives

$$\int_0^1 h_{P, J}^2 d\lambda = d(P)^2 + (\bar{\tau}_P - c_{P, J})^2 \geq d(P)^2.$$

It follows that

$$d(P)^2 \leq \|C_J \theta_{\tau, P}\|_2^2 + (20J^{-\gamma})^2 \leq (\|C_J \theta_{\tau, P}\|_2 + 20J^{-\gamma})^2,$$

so

$$d(P) - 20J^{-\gamma} \leq \|C_J \theta_{\tau, P}\|_2.$$

The same cellwise coefficient bound used for the null, now applied to $g_{T, P} - \tau_P$, gives

$$\|\mu_P - C_J \theta_{\tau, P}\|_2 = \left\| C_J \int_0^1 (g_{T, P} - \tau_P) F_J d\lambda \right\|_2 \leq b.$$

Combining this with the preceding lower bound and the reverse triangle inequality yields

$$d(P) - 20J^{-\gamma} - b \leq \|\mu_P\|_2.$$

Define the noise scale by

$$\zeta_{\text{noise}} = \sqrt{\sqrt{J} \Lambda} > 0.$$

If

$$d(P) \geq 20J^{-\gamma} + 5b + 128\zeta_{\text{noise}},$$

then

$$\|\mu_P\|_2 \geq 4b + 128\zeta_{\text{noise}}.$$

Since $\sqrt{\Lambda} \leq \zeta_{\text{noise}}$, this implies

$$16\sqrt{\Lambda} \|\mu_P\|_2 \leq \frac{\|\mu_P\|_2^2}{2}.$$

On $\mathcal{G}_P$, therefore,

$$\begin{aligned} W_e &\geq \|\mu_P\|_2^2 - 16\sqrt{\Lambda} \|\mu_P\|_2 - 16\zeta_{\text{noise}}^2 \\ &\geq \frac{\|\mu_P\|_2^2}{2} - 16\zeta_{\text{noise}}^2 \\ &\geq 8b^2 + 8176\zeta_{\text{noise}}^2 > 2b^2 + 1024\zeta_{\text{noise}}^2. \end{aligned}$$

Here the third inequality uses

$$\frac{(4b + 128\zeta_{\text{noise}})^2}{2} \geq 8b^2 + 8192\zeta_{\text{noise}}^2,$$

and the last is strict because $\zeta_{\text{noise}} > 0$. Thus failure to reject is contained in $\mathcal{G}_P^c$, proving

$$1 - \mathbf{R}_n^e(P, \phi_{n,v}^{*,e}) \leq 0.01$$

under the displayed separation condition.

5. We now bound that sufficient separation by the scale in the statement. Define the variance-growth exponent by

$$t = \frac{2-p}{p-1} \geq 0.$$

Both dyadic targets are at least one, and upward rounding gives

$$a^{-1/(p-1)} \leq T \leq 2a^{-1/(p-1)}, \quad a^{-1/\gamma} \leq J \leq 2a^{-1/\gamma}.$$

Since $1-p < 0$ and $-\gamma < 0$,

$$b = 20T^{1-p} \leq 20a, \quad J^{-\gamma} \leq a.$$

Also $0 \leq 2-p \leq 1$, so

$$T^{2-p} \leq 2^{2-p} a^{-t} \leq 2a^{-t}, \quad \sqrt{J} \leq \sqrt{2} a^{-1/(2\gamma)}.$$

Substitution of the definitions of q and E_0 gives the balance identity

$$E_0 \left(2 + t + \frac{1}{2\gamma} \right) = 1.$$

Because $a = s^{-E_0}$, this identity implies

$$\frac{a^{-t-1/(2\gamma)}}{s} = a^2.$$

Consequently,

$$\sqrt{J} \Lambda \leq 160\sqrt{2} \frac{a^{-t-1/(2\gamma)}}{s} = 160\sqrt{2} a^2, \quad \zeta_{\text{noise}} \leq \sqrt{160\sqrt{2}} a.$$

For every $n \geq 2$, $n \leq 3s$. Moreover,

$$2\gamma q \leq 2\gamma + q, \quad 0 < E_0 \leq 1, \quad 3^{E_0} \leq 3.$$

Monotonicity of the negative power therefore gives

$$a = s^{-E_0} \leq 3^{E_0} n^{-E_0} \leq 3\rho_n^e(v).$$

Combining the approximation, clipping, and covariance bounds yields

$$\begin{aligned} 20J^{-\gamma} + 5b + 128\zeta_{\text{noise}} &\leq \left(120 + 128\sqrt{160\sqrt{2}}\right) a \\ &\leq C_v^e \rho_n^e(v). \end{aligned}$$

Thus, whenever $C_v^e \rho_n^e(v) < D_v$, every model law at distance at least $C_v^e \rho_n^e(v)$ has type-II error at most 0.01.

6. Define the upper separation by

$$\varrho_+ = C_v^e \rho_n^e(v) > 0.$$

The witnesses already give $D_v > 0$. If $\varrho_+ < D_v$, the calibrated rule satisfies the size constraint 1/10, and its alternative errors give

$$\sup_{P \in \mathcal{M}_v: d(P) \geq \varrho_+} \{1 - R_n^e(P, \phi_{n,v}^{*,e})\} \leq 0.01 \leq \frac{1}{10}.$$

For an empty alternative index set, the real-valued supremum is assigned the value 0, so the same bound applies in that case. All worst-error suprema are nonnegative, and the admissible test family contains the calibrated rule. Taking its value in the infimum of [Definition 26](#) therefore shows

$$B_n^e(v, \varrho_+) \leq \frac{1}{10}.$$

Hence ϱ_+ is a candidate in [Definition 27](#), and

$$r_n^{*,e}(v) \leq \varrho_+.$$

If instead $D_v \leq \varrho_+$, the candidate D_v gives

$$r_n^{*,e}(v) \leq D_v \leq \varrho_+.$$

Both cases establish the upper radius bound.

7. Retain the prior and null law already constructed, and let ϕ^e satisfy the stipulated size bound on $H_0(v)$. Define its seed-averaged map with the common propensity supplied by

$$\bar{\phi}_{P_0}^e(\mathcal{D}_n) = \int_0^1 \phi^e(e_{P_0}, \mathcal{D}_n, u_{\text{seed}}) d\lambda(u_{\text{seed}}), \quad 0 \leq \bar{\phi}_{P_0}^e \leq 1.$$

This is a measurable function on the complete-record sample space. Integrating event-probability differences over its level sets gives

$$\left| \int \bar{\phi}_{P_0}^e d\mathbb{P}_{\pi_1, n} - \int \bar{\phi}_{P_0}^e dP_0^{\otimes n} \right| \leq \text{TV}(P_0^{\otimes n}, \mathbb{P}_{\pi_1, n}).$$

Indeed, each integral equals the integral over $\vartheta_{\text{lev}} \in [0, 1]$ of the probability of the measurable level set $\{\bar{\phi}_{P_0}^e > \vartheta_{\text{lev}}\}$.

Every positive-weight support law has propensity e_{P_0} . Finite linearity of integration and the null size bound thus yield

$$\sum_{j=1}^k \omega_j \mathbf{R}_n^e(P_j, \phi^e) \leq \mathbf{R}_n^e(P_0, \phi^e) + \text{TV}(P_0^{\otimes n}, \mathbb{P}_{\pi_1, n}) < \frac{1}{10} + \frac{1}{2} = \frac{3}{5}.$$

If every positive-weight index had type-II error below $2/5$, its rejection probability would exceed $3/5$. The zero-weight terms vanish, and normalization would then imply

$$\sum_{j=1}^k \omega_j \mathbf{R}_n^e(P_j, \phi^e) \geq \frac{3}{5} \sum_{j=1}^k \omega_j = \frac{3}{5},$$

a contradiction. Therefore

$$\exists j : \quad \omega_j > 0, \quad 1 - \mathbf{R}_n^e(P_j, \phi^e) \geq \frac{2}{5}.$$

This proves the asserted individual error witness for the same common-propensity prior.

8. For $n \geq N_v^e$, the definition of the ceiling gives

$$n \geq \max \left\{ 2, \left(\frac{2C_v^e}{d_0} \right)^{1/E_0} \right\}.$$

Since $E_0 > 0$, raising the lower bound for n to the power $-E_0$ reverses the inequality:

$$n^{-E_0} \leq \left\{ \left(\frac{2C_v^e}{d_0} \right)^{1/E_0} \right\}^{-E_0} = \left(\frac{2C_v^e}{d_0} \right)^{-1}.$$

Multiplication by $C_v^e > 0$ proves

$$C_v^e \rho_n^e(v) \leq \frac{d_0}{2}.$$

9. Let $P \in \mathcal{M}_v$. On the common almost-everywhere set where the two conditional arm-mean identities hold,

$$\int_{\mathbb{R}} y Q_{0,P}(dy | x) = m_{0,P}(x), \quad \int_{\mathbb{R}} y Q_{1,P}(dy | x) = m_{0,P}(x) + \tau_P(x).$$

Overlap gives $e_P(x) > 0$ and $1 - e_P(x) > 0$ at every x . Pulling the constant denominators outside the arm integrals and cancelling them therefore yields

$$\begin{aligned} e_P(x) \int_{\mathbb{R}} \frac{y}{e_P(x)} Q_{1,P}(dy | x) + (1 - e_P(x)) \int_{\mathbb{R}} \frac{-y}{1 - e_P(x)} Q_{0,P}(dy | x) \\ = \int_{\mathbb{R}} y Q_{1,P}(dy | x) - \int_{\mathbb{R}} y Q_{0,P}(dy | x) = \tau_P(x). \end{aligned}$$

This is the claimed identity for the untruncated score.

10. Finally, let $\mathcal{K} \subseteq \mathcal{V}$ be compact. The coordinate functions p and γ are continuous in the inherited topology. Since $p > 1$ and $2\gamma + (p-1)/p > 0$, the function

$$v \mapsto E_0(v) = \frac{2\gamma((p-1)/p)}{2\gamma + (p-1)/p}$$

is continuous and positive throughout $\mathcal{V}$. If $\mathcal{K}$ is nonempty, it attains a positive minimum; if $\mathcal{K}$ is empty, choose the value one. Thus there is

$$\eta_{\mathcal{K}} > 0, \quad \eta_{\mathcal{K}} \leq E_0(v) \quad (v \in \mathcal{K}).$$

The multiplier bounds already proved give

$$c_v^e \geq \frac{1}{64\sqrt{3}}, \quad C_v^e = 3 \left(120 + 128\sqrt{160\sqrt{2}} \right) \quad (v \in \mathcal{K}).$$

Using $d_0 = 1/(8\sqrt{2})$ from [Lemma 5](#), the base $2C_v^e/d_0$ exceeds one and is independent of v . Therefore

$$N_v^e \leq \left\lceil \max \left\{ 2, \left(\frac{2C_v^e}{d_0} \right)^{1/\eta_{\mathcal{K}}} \right\} \right\rceil \quad (v \in \mathcal{K}).$$

This supplies a common finite threshold bound. Positivity of the denominators and continuity of the exponent also hold at $p = 2$ and $\gamma = 1/4$, so the same compact-uniform argument includes those faces.

□

Proof of [Theorem 7](#). 1. Fix $v = (p, \alpha, \beta, \gamma) \in \mathcal{V}$, and use the quantities of [Definition 15](#). By [Theorems 4](#) and [6](#),

$$c_v > 0, \quad C_v > 0, \quad c_v^e \geq \frac{1}{64\sqrt{3}} > 0, \quad C_v^e > 0.$$

Consequently,

$$\frac{c_v}{C_v^e} > 0, \quad \frac{C_v}{c_v^e} > 0.$$

2. We establish the identities for the comparison exponent. Admissibility gives

$$p-1 > 0, \quad p > 0, \quad q > 0, \quad \gamma > 0, \quad D_p > 0,$$

and hence

$$\frac{2\gamma q}{(2\gamma + q)D_p} > 0.$$

The phase identity and channel selection in [Theorem 4](#) give

$$E_4(p) - E_0(p) = \frac{2\gamma q}{(2\gamma + q)D_p} (F(p) - 1),$$

$$F(p) \geq 1 \implies E(p) = E_0(p), \quad F(p) < 1 \implies E(p) = E_4(p).$$

If $F(p) \geq 1$, the positive prefactor implies

$$E_0(p) - E_4(p) \leq 0.$$

Thus, using the definition of $G(v)$,

$$G(v) = 0 = E_0(p) - E(p) = \frac{2\gamma q}{(2\gamma + q)D_p} \max\{0, 1 - F(p)\}.$$

If $F(p) < 1$, the same identity implies

$$E_0(p) - E_4(p) = \frac{2\gamma q}{(2\gamma + q)D_p} (1 - F(p)) > 0.$$

Channel selection now gives

$$G(v) = E_0(p) - E_4(p) = E_0(p) - E(p) = \frac{2\gamma q}{(2\gamma + q)D_p} \max\{0, 1 - F(p)\}.$$

Together, these cases establish

$$G(v) = E_0(p) - E(p), \quad G(v) = \frac{2\gamma q}{(2\gamma + q)D_p} \max\{0, 1 - F(p)\}.$$

In particular,

$$F(p) \geq 1 \implies G(v) = 0.$$

3. For every integer $n \geq 2$, the two frontier conclusions in [Theorems 4](#) and [6](#) supply

$$c_v \rho_n(v) \leq r_n^*(v) \leq C_v \rho_n(v), \quad c_v^e \rho_n^e(v) \leq r_n^{*,e}(v) \leq C_v^e \rho_n^e(v).$$

Since $n > 0$, both power scales are positive:

$$\rho_n(v) = n^{-E(p)} > 0, \quad \rho_n^e(v) = n^{-E_0(p)} > 0.$$

The lower bounds therefore imply

$$r_n^*(v) > 0, \quad r_n^{*,e}(v) > 0.$$

Dividing the original-radius lower bound by the supplied-radius upper bound, and the original-radius upper bound by the supplied-radius lower bound, yields

$$\frac{c_v \rho_n(v)}{C_v^e \rho_n^e(v)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v \rho_n(v)}{c_v^e \rho_n^e(v)}.$$

The scale quotient satisfies

$$\frac{\rho_n(v)}{\rho_n^e(v)} = \frac{n^{-E(p)}}{n^{-E_0(p)}} = n^{E_0(p) - E(p)} = n^{G(v)}.$$

Consequently, for every integer $n \geq 2$,

$$\frac{c_v}{C_v^e} n^{G(v)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} n^{G(v)}.$$

4. If $F(p) = 1$, the exponent identity gives $G(v) = 0$. Substituting $n^0 = 1$ into the quotient bounds gives

$$\frac{c_v}{C_v^e} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} \quad (n \geq 2).$$

Both bounding constants are strictly positive by [Theorems 4 and 6](#).

5. Suppose $F(p) < 1$. The exponent identity gives

$$G(v) = \frac{2\gamma q}{(2\gamma + q)D_p}(1 - F(p)) > 0.$$

A positive power of the natural sample size tends to infinity, and multiplication by the positive lower multiplier preserves this limit:

$$n^{G(v)} \longrightarrow +\infty, \quad \frac{c_v}{C_v^e} n^{G(v)} \longrightarrow +\infty.$$

For all sufficiently large natural numbers n , we have $n \geq 2$, so the quotient lower bound gives

$$\frac{c_v}{C_v^e} n^{G(v)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)}.$$

Comparison with the diverging lower bound proves

$$\frac{r_n^*(v)}{r_n^{*,e}(v)} \longrightarrow +\infty.$$

Moreover, channel selection gives $E(p) = E_4(p)$, and hence

$$G(v) = E_0(p) - E(p) = E_0(p) - E_4(p).$$

Substitution into the quotient bounds proves

$$\frac{c_v}{C_v^e} n^{E_0(p) - E_4(p)} \leq \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} n^{E_0(p) - E_4(p)} \quad (n \geq 2).$$

6. We characterize the preservation region of [Definition 28](#). Suppose first that $v \in \mathcal{R}$. There is a real upper bound b_{sup} satisfying

$$\frac{r_n^*(v)}{r_n^{*,e}(v)} \leq b_{\text{sup}} \quad \text{for every integer } n \geq 2.$$

If $F(p) < 1$, the divergence just proved supplies an integer $n \geq 2$ such that

$$b_{\text{sup}} < \frac{r_n^*(v)}{r_n^{*,e}(v)} \leq b_{\text{sup}},$$

a contradiction. Thus $F(p) \geq 1$.

Conversely, if $v \in \mathcal{V}$ and $F(p) \geq 1$, then $G(v) = 0$, so

$$\frac{r_n^*(v)}{r_n^{*,e}(v)} \leq \frac{C_v}{c_v^e} \quad \text{for every integer } n \geq 2.$$

This finite upper bound places v in $\mathcal{R}$. Expanding $F(p)$ therefore gives

$$\mathcal{R} = \left\{ v \in \mathcal{V} : \frac{2\alpha}{p-1} + \frac{\beta}{q} + \frac{\alpha + \beta}{2\gamma} \geq 1 \right\}.$$

7. Finally, let $\mathcal{K} \subseteq \mathcal{V}$ be an arbitrary compact set. The compact-uniformity conclusion of [Theorem 4](#) supplies real constants $\underline{c}_{\mathcal{K}}, \overline{C}_{\mathcal{K}}$ satisfying

$$\underline{c}_{\mathcal{K}} > 0, \quad \underline{c}_{\mathcal{K}} \leq c_v, \quad C_v \leq \overline{C}_{\mathcal{K}} \quad (v \in \mathcal{K}).$$

The supplied-propensity upper multiplier is independent of v and positive. Define

$$c_{\text{pres}, \mathcal{K}} = \frac{\underline{c}_{\mathcal{K}}}{3(120 + 128\sqrt{160\sqrt{2}})}, \quad C_{\text{pres}, \mathcal{K}} = 64\sqrt{3}\overline{C}_{\mathcal{K}}.$$

Then $c_{\text{pres}, \mathcal{K}} > 0$, and for every $v \in \mathcal{K}$,

$$c_{\text{pres}, \mathcal{K}} \leq \frac{c_v}{C_v^e}, \quad 64\sqrt{3}C_v \leq C_{\text{pres}, \mathcal{K}}.$$

These are the required compact-uniform constants. □

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